1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

REV	ECN	DESCRIPTION OF REVISION	CK APPD	DATE
				2012-02-15

SCHEM,MLB_KEPLER_2PHASE,J31

FRB & RISK RAMP 02/15/12

Schematic / PCB #'s

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
051-9585	1	SCHEM,MLB_KEPLER_2PHASE,J31	SCH	CRITICAL	
820-3330	1	PCBF,MLB_KEPLER_2PHASE,J31	PCB	CRITICAL	

DRAWING
ABBREV=DRAWING
TITLE=MLB

LAST_MODIFIED=Wed Feb 15 20:30:03 2012

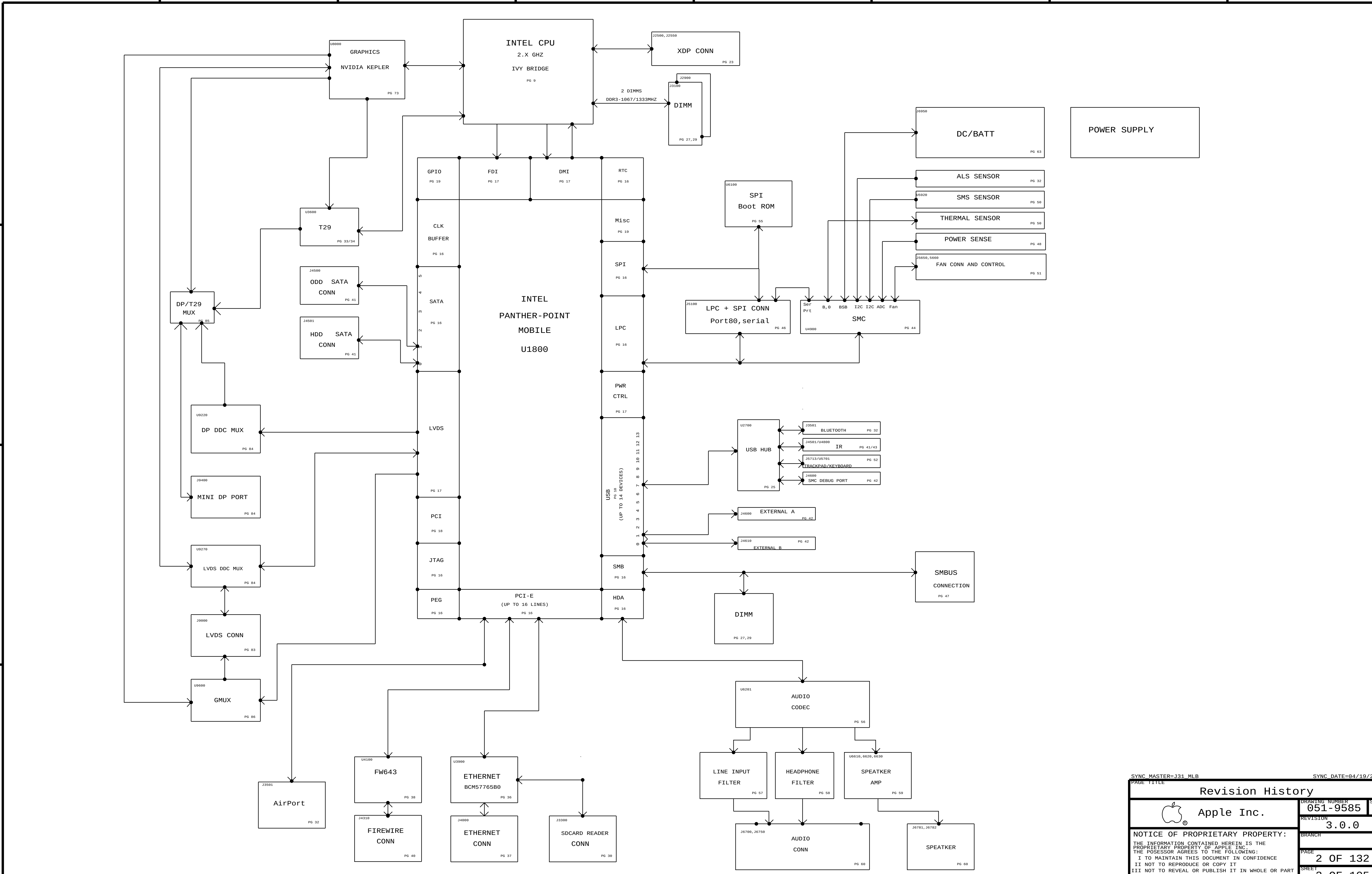
DRAWING TITLE
SCHEM,MLB_KEPLER,J31

Apple Inc.

DRAWING NUMBER 051-9585 SIZE D

REVISION 3.0.0

BRANCH



SYNC_MASTER=J31_MLB SYNC_DATE=04/19/2011

PAGE TITLE: Revision History

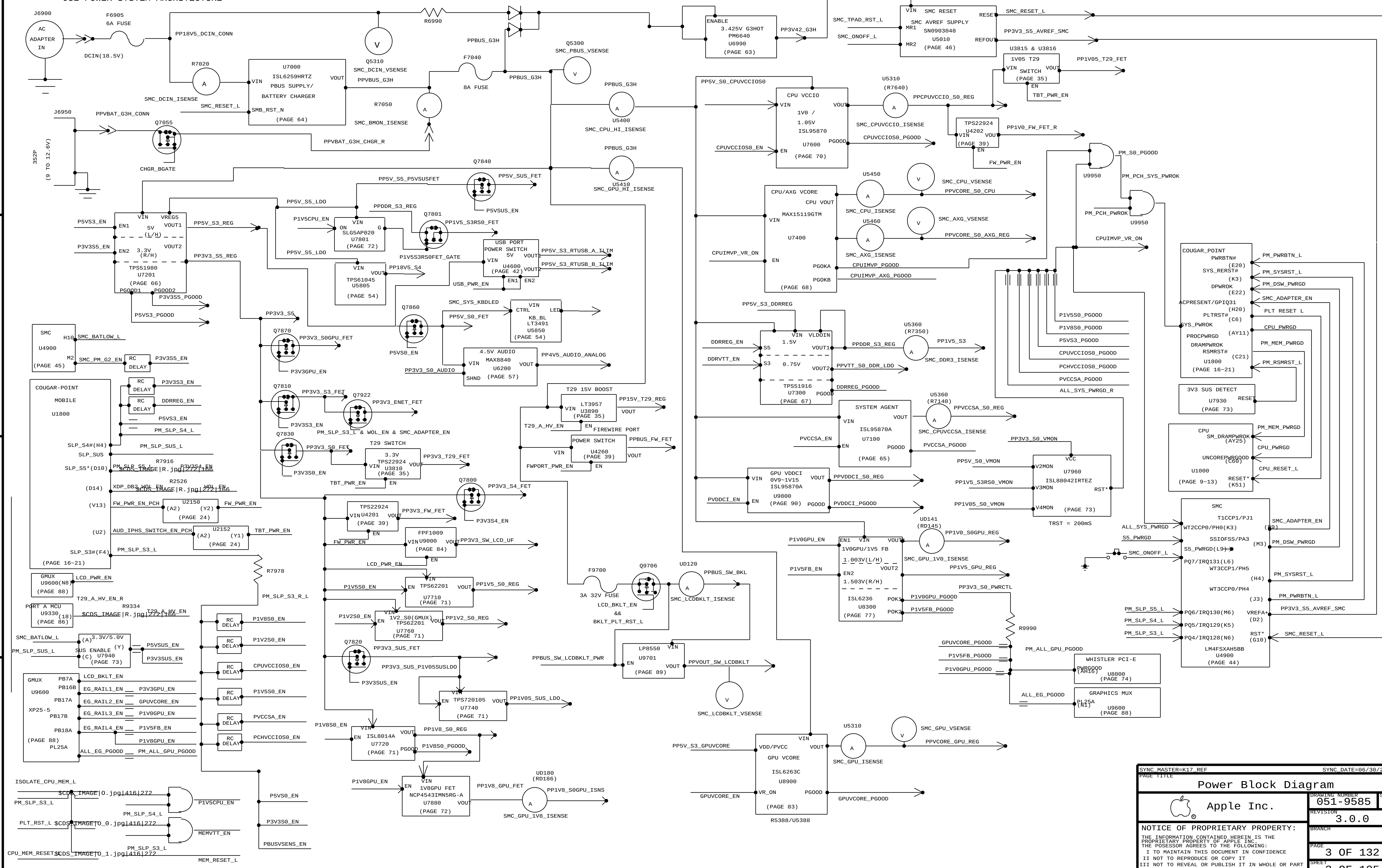
Apple Inc.

DRAWING NUMBER: 051-9585 SIZE: D

REVISION: 3.0.0

BRANCH

J31 POWER SYSTEM ARCHITECTURE



SYNC_MASTER=K17_REF SYNC_DATE=06/30/2009

PAGE TITLE
Power Block Diagram

DRAWING NUMBER 051-9585 SIZE D

Apple Inc.

REVISION 3.0.0

BRANCH

D C B A

SYNC_MASTER=MASTER SYNC_DATE=MASTER

PAGE TITLE

Revision History

Apple Inc.

DRAWING NUMBER	051-9585
SIZE	D
REVISION	3.0.0
BRANCH	
PAGE	4 OF 132
SHEET	4 OF 105

BOM VARIANTS - FSB

BOM NUMBER	BOM NAME	BOM OPTIONS
639-3860	PCBA,MLB_2P,FSB,2.3,FOX,512_HYN,REN,J31,F327	J31_CMNPTS,SODIMM:FOXCONN,CPU:2_3GHZ,FB_512_HYNIX,FET:REN,DEVEL_BOM,GPUDEC:EXP,EEEE:F327
639-3861	PCBA,MLB_2P,FSB,2.3,MOL,512_SAM,FAIR,J31,F32C	J31_CMNPTS,SODIMM:MOLEX,CPU:2_3GHZ,FB_512_SAMSUNG,FET:FAIR,DEVEL_BOM,GPUDEC:EXP,EEEE:F32C
639-3862	PCBA,MLB_2P,FSB,2.6,MOL,1G_HY,FAIR,J31,F325	J31_CMNPTS,SODIMM:MOLEX,CPU:2_6GHZ,FB_1G_HYNIX_A_DIE,FET:FAIR,DEVEL_BOM,GPUDEC:EXP,EEEE:F325
639-3863	PCBA,MLB_2P,FSB,2.6,FOX,1G_SAM,REN,J31,F324	J31_CMNPTS,SODIMM:FOXCONN,CPU:2_6GHZ,FB_1G_SAMSUNG,FET:REN,DEVEL_BOM,GPUDEC:EXP,EEEE:F324
639-3864	PCBA,MLB_2P,FSB,2.7,FOX,1G_HY,REN,J31,F328	J31_CMNPTS,SODIMM:FOXCONN,CPU:2_7GHZ,FB_1G_HYNIX_A_DIE,FET:REN,DEVEL_BOM,GPUDEC:EXP,EEEE:F328
639-3865	PCBA,MLB_2P,FSB,2.7,MOL,1G_SAM,FAIR,J31,F329	J31_CMNPTS,SODIMM:MOLEX,CPU:2_7GHZ,FB_1G_SAMSUNG,FET:FAIR,DEVEL_BOM,GPUDEC:EXP,EEEE:F329
607-9557	CMN PTS,PCBA,MLB_KEPLER,J31	J31_COMMON
085-4620	J31 MLB_KEP_2P DEVELOPMENT BOM	J31_DEVEL:PVT

SUB BOMS

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
085-4620	1	J31 MLB_KEP_2P DEVELOPMENT BOM	DEVEL	CRITICAL	DEVEL_BOM
607-9557	1	CMN PTS,PCBA,MLB_KEP_2P,J31	CMNPTS	CRITICAL	J31_CMNPTS

BOM GROUPS

BOM GROUP	BOM OPTIONS
J31_COMMON	ALTERNATE,COMMON,J31_COMMON1,J31_COMMON2,J31_PROGPARTS,J31_PROGPARTS1,UVGLUE_J31,J31_PVT
J31_COMMON1	CPUMEM_S0,RAMCFG_SLOT,USBHUB2513B,HUB_3NONREM,SMC_PACKAGE:PROD,MOJO:YES,TBTHV:P15V,SKIP_5V3V3:AUDIBLE
J31_COMMON2	BTPWR:S4,TPAD:Z2,T29:YES,TBTBST:Y,SDRV_PD,SDRVI2C:MCU,T29_DP_HPD:ALL_OR,LPCPLUS_R:YES,MEM_VDD_SEL:GPIO15,GPU:2P
J31_PROGPARTS	GMUX_PROG,IR_PROG,TPAD_PROG:FSB,ENETROM_PROG:FSB,T29ROM:PROG,T29MCU:PROG
J31_PROGPARTS1	SMC_PROG:RR,BOOTROM_PROG:FSB
J31_PVT	VREF:PROD,XDP,XDP_CPU:BPM,BKLT:PROD,LOADISNS:NO,XWLOADISNS:NO
J31_DEVEL:ENG	DDRVREF_DAC,VREF:ENG_M3,IVB_PPT_XDP,GMUX_JTAG_CONN,LPCPLUS_CONN:YES,BKLT:ENG,S0PGOOD_ISL,CPURIPPLE_ENG,LOADISNS:YES,XWLOADISNS:YES,DEBUG_ADC
J31_DEVEL:FSB	DDRVREF_DAC,VREF:ENG_M3,IVB_PPT_XDP,LPCPLUS_CONN:YES,BKLT:PROD,S0PGOOD_ISL,LOADISNS:YES,XWLOADISNS:NO
J31_DEVEL:PVT	LPCPLUS_CONN:YES,XDP_CONN_CPU
IVB_PPT_XDP	XDP,XDP_CONN_PCH,XDP_CONN_CPU,XDP_CPU:BPM,XDP_PCH

BOM GROUP	BOM OPTIONS
VREF:PROD	VREFDQ:M1_M3,VREFCA:LDO
VREF:ENG_M3	VREFDQ:M1_M3,VREFCA:LDO_DAC
VREF:ENG_LDO	VREFDQ:M1_DAC,VREFCA:LDO_DAC

Module Parts

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
337S4266	1	IC,CPU,IVB,S,R0MP,PRQ,E1,2.3,45W,4+2,1.2,6M,BGA	U1000	CRITICAL	CPU:2_3GHZ
337S4267	1	IC,CPU,IVB,S,R0MM,PRQ,E1,2.6,45W,4+2,1.25,6M,BGA	U1000	CRITICAL	CPU:2_6GHZ
337S4268	1	IC,CPU,IVB,S,R0MK,PRQ,E1,2.7,45W,4+2,1.25,8M,BGA	U1000	CRITICAL	CPU:2_7GHZ
337S4269	1	IC,PCH,PPT,C1,SLJ8C,PRQ,BD82HM77	U1800	CRITICAL	
337S4239	1	IC,GPU,NV GK107-GTX-QS-A2	U8000	CRITICAL	
338S1072	1	IC,ASSP,LIGHTRIDGE,PRQ,S LJJY,FCBGA,15X15MM,C1	U3600	CRITICAL	T29:YES
338S0753	1	IC,FW643-E,1394B PHY/OHCI LINK/PCI-E,12	U4100	CRITICAL	
353S3055	1	IC,PI3VEDP212,X2 DISPLAYPORT 2:1 MUX,QFN	U9390	CRITICAL	
333S0619	4	IC,SGRAM,GDDR5,32MX32.1.25GHz,G-DIE,HF	U8400,U8450,U8500,U8550	CRITICAL	FB_512_SAMSUNG
333S0620	4	IC,SDRAM,GDDR5,32MX32,1.5GHz,VEGA 44NM,B-DIE	U8400,U8450,U8500,U8550	CRITICAL	FB_512_HYNIX
333S0631	4	IC,SGRAM,GDDR5,64MX32,5GBPS,D-DIE,HF	U8400,U8450,U8500,U8550	CRITICAL	FB_1G_SAMSUNG
333S0630	4	IC,SGRAM,GDDR5,64MX32,5GBPS,A-DIE,HF	U8400,U8450,U8500,U8550	CRITICAL	FB_1G_HYNIX_A_DIE
333S0609	4	IC,SGRAM,GDDR5,64MX32,4.2GBPS,M-DIE,HF	U8400,U8450,U8500,U8550	CRITICAL	FB_1G_HYNIX_M_DIE
725-1479	1	MLB LOCTITE UV EB CPU,PCH,T29,GPU,K91	UV_GLUE_J31	CRITICAL	UVGLUE_J31
516S0806	1	CONN,204P,SODIMM,SOCKET,DDR3,RAM,BGA,FOXCONN	J3100	CRITICAL	SODIMM:FOXCONN
516-0246	1	CONN,204P,SODIMM,SOCKET,DDR3,p=0.6M,FOXCONN	J2900	CRITICAL	SODIMM:FOXCONN
516S0805	1	CONN,204P,SODIMM,SOCKET,DDR3,RAM,BGA,MOLEX	J3100	CRITICAL	SODIMM:MOLEX
516-0245	1	CONN,204P,SODIMM,SOCKET,DDR3,p=0.6M,MOLEX	J2900	CRITICAL	SODIMM:MOLEX
516S0805	1	CONN,204P,SODIMM,SOCKET,DDR3,RAM,BGA,MOLEX	J3100	CRITICAL	SODIMM:HYBRID
516-0246	1	CONN,204P,SODIMM,SOCKET,DDR3,p=0.6M,FOXCONN	J2900	CRITICAL	SODIMM:HYBRID
376S0964	2	RJK0225	Q7330,Q8360	CRITICAL	FET:REN
376S0965	2	RJK0225	Q7335,Q8361	CRITICAL	FET:REN
376S0979	2	FDMC0225	Q7330,Q8360	CRITICAL	FET:FAIR
376S0874	2	FDMC0202S	Q7335,Q8361	CRITICAL	FET:FAIR
376S0826	1	FET,N-CH,30V,3.6MOHM,LF,HF,RJK0332DPB	Q7030	CRITICAL	FET:REN
376S0617	1	FET,N-CH,30V,30A,6.7MOHM,RJK0305DPB	Q7035	CRITICAL	FET:REN
376S0917	1	FET,N-CH,30V,3.6MOHM,LF,HF,FDMS0355S	Q7030	CRITICAL	FET:FAIR
376S1018	1	FET,N-CH,30V,14A,13MOHM,FDMS0349	Q7035	CRITICAL	FET:FAIR

Bar Code Labels / EEEE #'s

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
826-4393	1	LBL,P/N LABEL,PCB,28MM X 6 MM	[EEEE_F327]	CRITICAL	EEEE:F327
826-4393	1	LBL,P/N LABEL,PCB,28MM X 6 MM	[EEEE_F32C]	CRITICAL	EEEE:F32C
826-4393	1	LBL,P/N LABEL,PCB,28MM X 6 MM	[EEEE_F325]	CRITICAL	EEEE:F325
826-4393	1	LBL,P/N LABEL,PCB,28MM X 6 MM	[EEEE_F324]	CRITICAL	EEEE:F324
826-4393	1	LBL,P/N LABEL,PCB,28MM X 6 MM	[EEEE_F328]	CRITICAL	EEEE:F328
826-4393	1	LBL,P/N LABEL,PCB,28MM X 6 MM	[EEEE_F329]	CRITICAL	EEEE:F329

Alternate Parts

(Alternate) (Primary)

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
157S0058	157S0055		ALL	Delta alt to TDK Magnetics
152S0896	152S0518		ALL	MAG LAYERS ALT TO CYNTEC
155S0457	155S0329		ALL	MAG LAYERS ALT TO MURATA
353S2805	353S2603		ALL	Fairchild wafer option
128S0264	128S0257		ALL	Sanyo alt to Kemet
128S0303	128S0282		ALL	Panasonic alt to Sanyo
353S3085	353S1658		ALL	ST Micro alt to LT
376S0972	376S0612		ALL	ROHM alt to Toshiba N-FET
376S0855	376S0613		ALL	Diodes alt to Toshiba dual N-FET
138S0676	138S0691		ALL	Murata alt to Samsung cap
138S0652	138S0648		ALL	Samsung / Murata alt for Taiyo Yuden
138S0681	138S0638		ALL	Taiyo Yuden alt for Samsung
152S0685	152S0796		ALL	Dale/Vishay/TDK alt for Cyntec
376S0977	376S0859		ALL	Diodes alt for Rohm
353S2592	353S3199		ALL	U6201 AUDIO CODEC OLD REV IS ALTERNATE FOR NEW REV
335S0550	335S0777		ALL	add 4K byte as alternative to 2K
371S0709	371S0652		ALL	NXP alternate for pin diodes
138S0671	138S0673		ALL	Taiyo Yuden alt for Murata 10 uF caps
514-0788	514-0671		ALL	Acon (with liteon) alt to Acon
155S0578	155S0367		ALL	Tayo Yuden alt to Murata inductors
138S0681	138S0638		ALL	Tayo Yuden alt to Samsung caps
138S0671	138S0673		ALL	Tayo Yuden alt to Murata caps
155S0625	155S0559		ALL	Murata alt to TDK cm mode filter
376S0777	376S0761		ALL	AON alternate to Siliconix
157S0084	157S0055		ALL	TDK alternate for ethernet transformer
353S3312	353S3055		ALL	NXP alternate to Pericom DP mux
376S0958	376S0953		ALL	For Q7260, Fairchild alt to Ren.
376S1053	376S0604		ALL	Radar 10562726
371S0713	371S0558		ALL	Radar 10562508
128S0311	128S0329		ALL	Radar 10257464
127S0134	127S0111		ALL	Radar 10360888
127S0127	127S0090		ALL	Radar 10382328
197S0431	197S0432		ALL	RADAR 10670230
197S0434	197S0343		ALL	RADAR 10739227
197S0435	197S0343		ALL	RADAR 10739227

Programmables - All Builds

PSOC

341S3099	1	IC,TP PSOC,K9x,DVT,PVT,J31	U5701	CRITICAL	TPAD_PROG:PROTO0
341S3351	1	IC,TP PSOC,PROTO1,J31	U5701	CRITICAL	TPAD_PROG:PROTO1
341S3227	1	IC,TP PSOC,PROTO2,PROTO3-Z2,J31	U5701	CRITICAL	TPAD_PROG:PROTO3
341S3489	1	IC,TP PSOC,PIB,J31	U5701	CRITICAL	TPAD_PROG:PIB
341S3522	1	IC,TP PSOC,FSB,J31	U5701	CRITICAL	TPAD_PROG:FSB

341S2830	1	IC,CPLD,LATTICE,GMUX,K91/K91F,J31	U9600	CRITICAL	GMUX_PROG
336S0042	1	IC,PLD,LATTICE,LFXP2-5E-5,132 BALL CSBGA	U9600	CRITICAL	GMUX_BLANK
341S2384	1	IR,ENCORE II,CY7C63833-LFXC	U4800	CRITICAL	IR_PROG
341S3430	1	IC,T29 EEPROM,LR,J30/J31	U3690	CRITICAL	T29ROM:PROG
335S0777	1	IC,EEPROM,SERIAL,8KB,SOIC	U3690	CRITICAL	T29ROM:BLANK
341S3365	1	IC,PROGRMD,LPC1112A,T29 PORT MCU,PVT,HVQFN25,J31	U9330	CRITICAL	T29MCU:PROG
337S3997	1	IC,MCU,32B,LPC1112A,16KB/2KB,HVQFN25	U9330	CRITICAL	T29MCU:BLANK
335S0852	1	IC,GPUROM,J31,BLANK	U8701	CRITICAL	GPUROM:BLANK

ETHERNET ROM

335S0663	1	IC,FLASH,SERIAL,SPI,1MBIT,2V7,8P,SOIC	U3990	CRITICAL	ENETROM_BLANK
341S3096	1	IC,ENET ROM,1MBIT,DVT,PVT,K90i/K91x,J31	U3990	CRITICAL	ENETROM_PROG:PROTO3
341S3492	1	IC,PRGRMD,ENET,SPI ROM,FSB,J30/J31	U3990	CRITICAL	ENETROM_PROG:FSB

SMC

338S0895	1	IC,SMC,HS8/2117,9MMX9MM,TLP	U4900	CRITICAL	SMC_BLANK
341S3258	1	IC,SMC,DEVELOPMENT-PROTO0,J31	U4900	CRITICAL	SMC_PROG:PROTO0
341S3294	1	IC,SMC,DEVELOPMENT-PROTO1,J31	U4900	CRITICAL	SMC_PROG:PROTO1
341S3401	1	IC,EXTERNAL,PROTO2,PROTO3,J31	U4900	CRITICAL	SMC_PROG:PROTO3
341S3481	1	IC,SMC,EXTERNAL,PIB,V2.1A83,A3,J31	U4900	CRITICAL	SMC_PROG:A3_PIB
341S3296	1	IC,SMC,EXTERNAL,FSB,V2.1A143,J31	U4900	CRITICAL	SMC_PROG:FSB
341S3297	1	IC,SMC,EXTERNAL,RISKRAMP,J31	U4900	CRITICAL	SMC_PROG:RR

EFI ROM

335S0740	1	64 MBIT SPI SERIAL DUAL I/O FLASH	U6100	CRITICAL	BOOTROM_BLANK
341S3257	1	IC,EFI,ROM,PROTO0, J31	U6100	CRITICAL	BOOTROM_PROG:PROTO0
341S3344	1	IC,EFI,ROM,PROTO1, J31	U6100	CRITICAL	BOOTROM_PROG:PROTO1
341S3419	1	IC,EFI,ROM,PROTO2,J31	U6100	CRITICAL	BOOTROM_PROG:PROTO2
341S3454	1	IC,EFI,ROM,PROTO3,J31	U6100	CRITICAL	BOOTROM_PROG:PROTO3
341S3510	1	IC,EFI,ROM,POST-PIB,J31	U6100	CRITICAL	BOOTROM_PROG:PIB2
341S3476	1	IC,EFI,ROM,FSB,J31	U6100	CRITICAL	BOOTROM_PROG:FSB

PD Parts

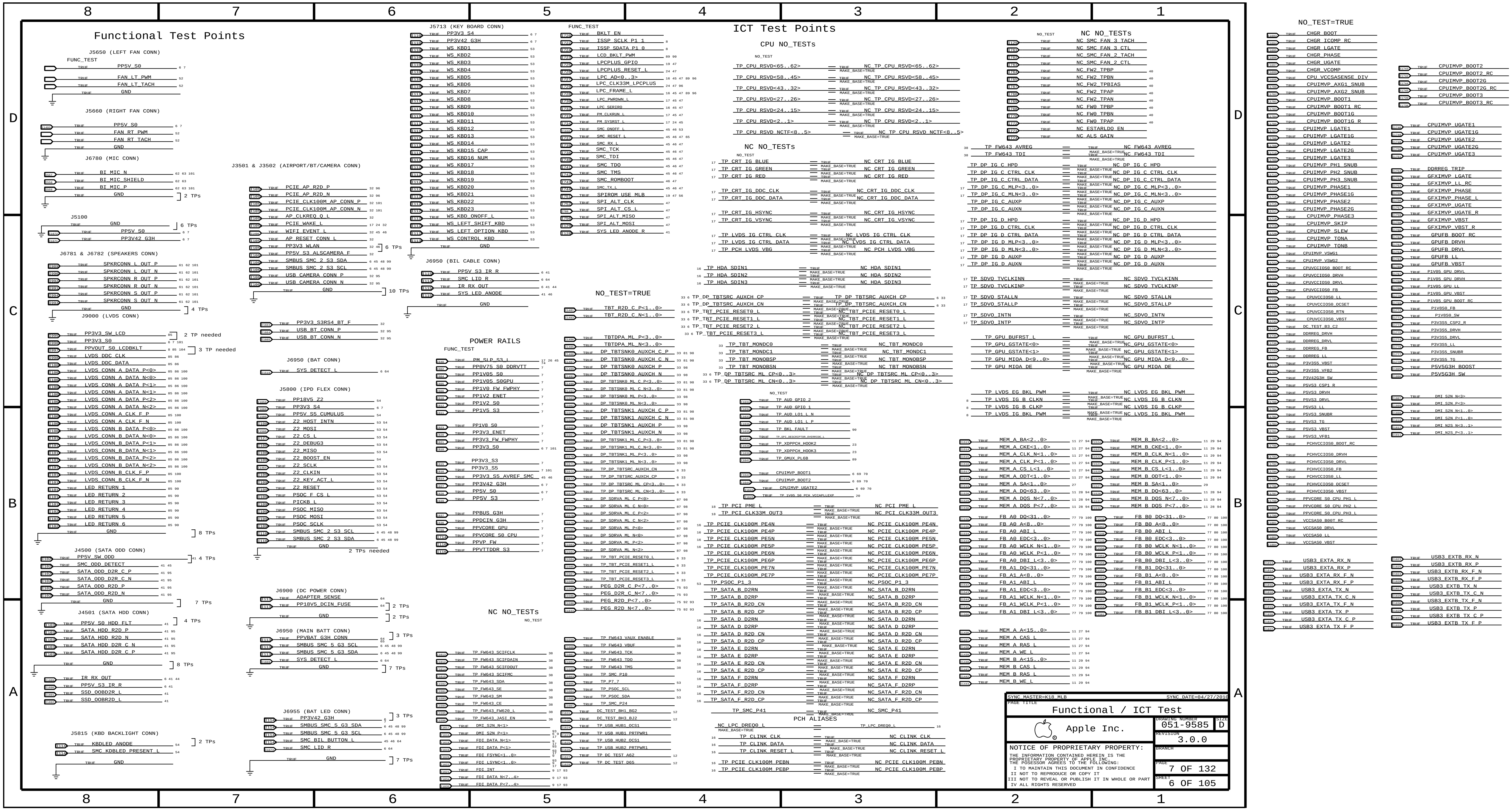
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
452-1708	2	SCR,M1.6X0.35X6.0,D4,HO.3,BLK,M97	SODIMM_SCREW1,SODIMM_SCREW2	CRITICAL	
452-1708	2	SCR,M1.6X0.35X6.0,D4,HO.3,BLK,M97	SODIMM_SCREW3,SODIMM_SCREW4	CRITICAL	
725-1607	1	INSULATOR,GPU,J31	GPU_INSULATOR	CRITICAL	

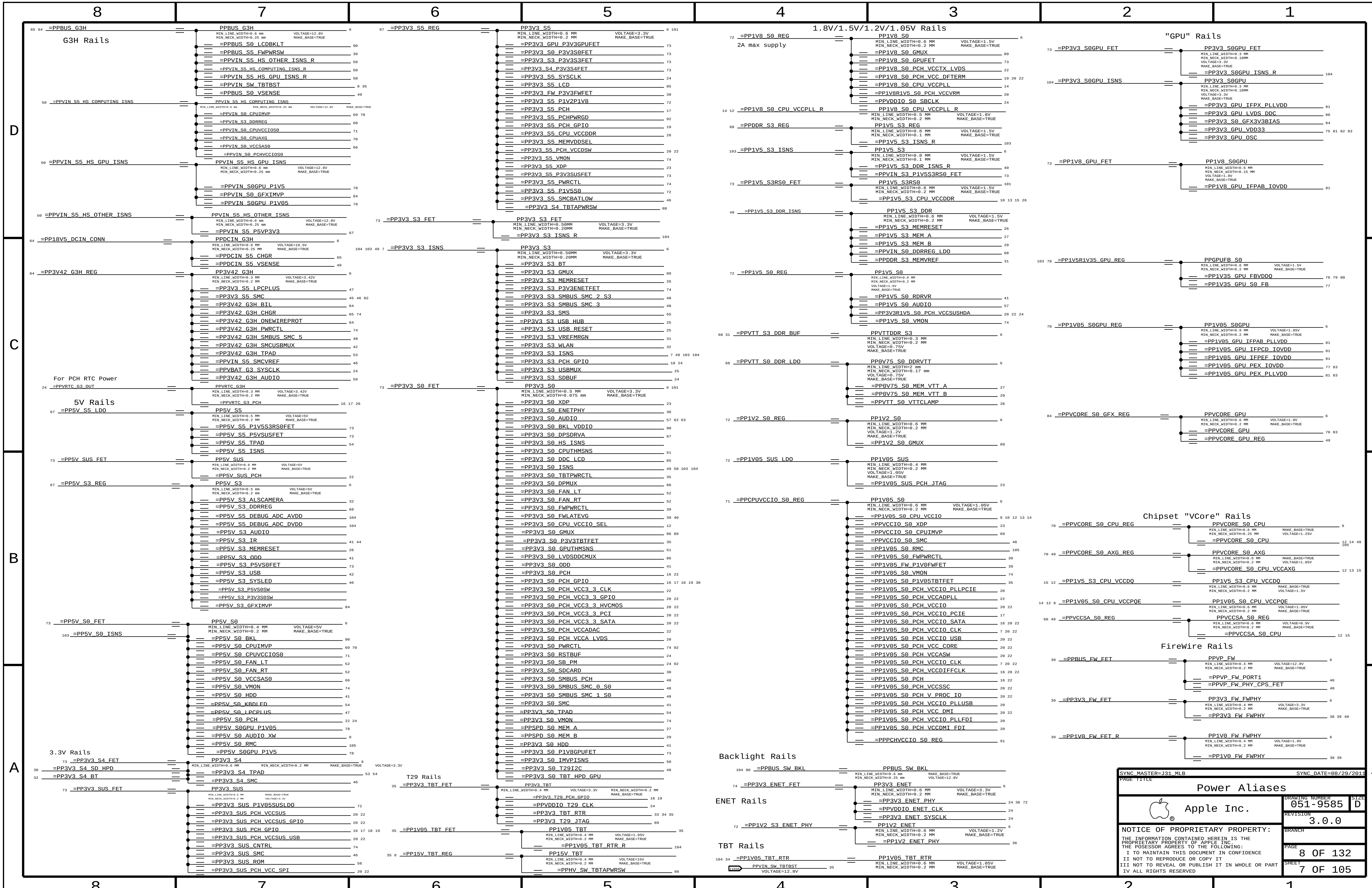
SYNC_MASTER=K17_REF SYNC_DATE=05/28/2009

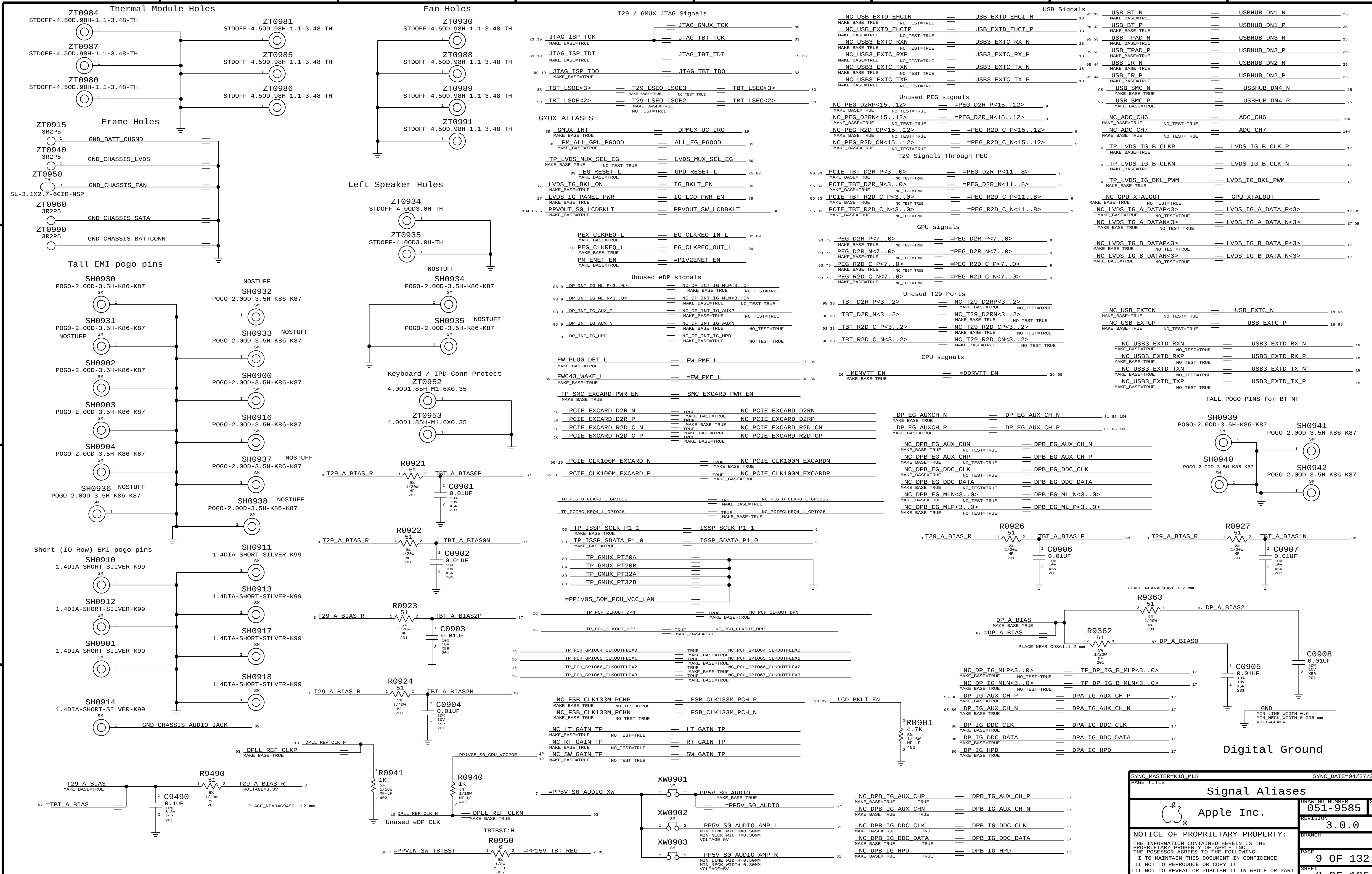
PAGE TITLE
BOM Configuration

Apple Inc.

DRAWING NUMBER 051-9585 SIZE D
REVISION 3.0.0
BRANCH


Functional Test Points
FUNC_TEST
J5650 (LEFT FAN CONN)
J5660 (RIGHT FAN CONN)
J6780 (MIC CONN)
J5100
J6781 & J6782 (SPEAKERS CONN)
J9000 (LVDS CONN)
J4500 (SATA ODD CONN)
J4501 (SATA HDD CONN)
J5815 (KBD BACKLIGHT CONN)
J3501 & J3502 (AIRPORT/BT/CAMERA CONN)
J6950 (BAT CONN)
J5800 (IPD FLEX CONN)
J6900 (DC POWER CONN)
J6950 (MAIN BATT CONN)
J6955 (BAT LED CONN)
J5713 (KEY BOARD CONN)
J6950 (BIL CABLE CONN)
POWER RAILS
NC NO_TESTs
NO_TEST=TRUE
ICT Test Points
CPU NO_TESTs
NC NO_TESTs
PCH ALIASES
2 TP needed
3 TP needed
10 TPs
2 TPs needed
SYNC_MASTER=K18_MLB
SYNC_DATE=04/27/2010
PAGE TITLE
Functional / ICT Test
Apple Inc.
DRAWING NUMBER
051-9585
SIZE
D
REVISION
3.0.0
NOTICE OF PROPRIETARY PROPERTY:
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC.
THE POSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED
BRANCH
PAGE
7 OF 132
SHEET
6 OF 105


G3H Rails
For PCH RTC Power
5V Rails
3.3V Rails
T29 Rails
1.8V/1.5V/1.2V/1.05V Rails
2A max supply
Backlight Rails
ENET Rails
TBT Rails
"GPU" Rails
Chipset "VCore" Rails
FireWire Rails
SYNC_MASTER=J31_MLB
SYNC_DATE=08/29/2011
PAGE TITLE
Power Aliases
Apple Inc.
DRAWING NUMBER
051-9585
SIZE
D
REVISION
3.0.0
BRANCH
PAGE
8 OF 132
SHEET
7 OF 105
NOTICE OF PROPRIETARY PROPERTY:
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC.
THE POSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED



SYNC_MASTER=K18_MLB SYNC_DATE=04/27/2010

PAGE TITLE

Signal Aliases

Apple Inc.

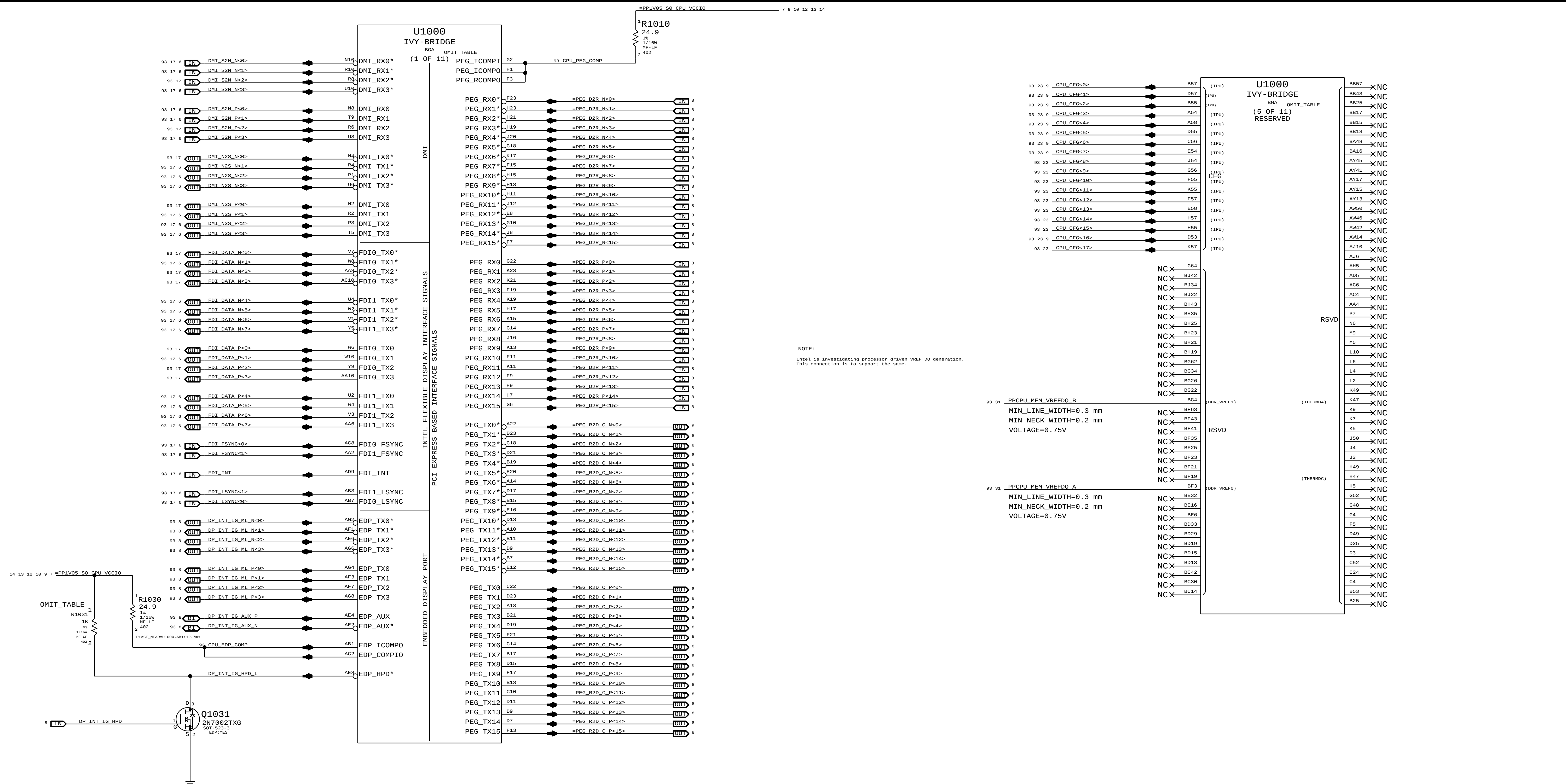
DRAWING NUMBER 051-9585 SIZE D

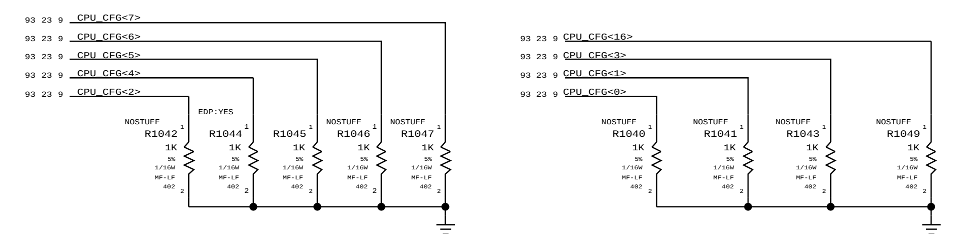
REVISION 3.0.0

BRANCH

PAGE 9 OF 132

SHEET 8 OF 105

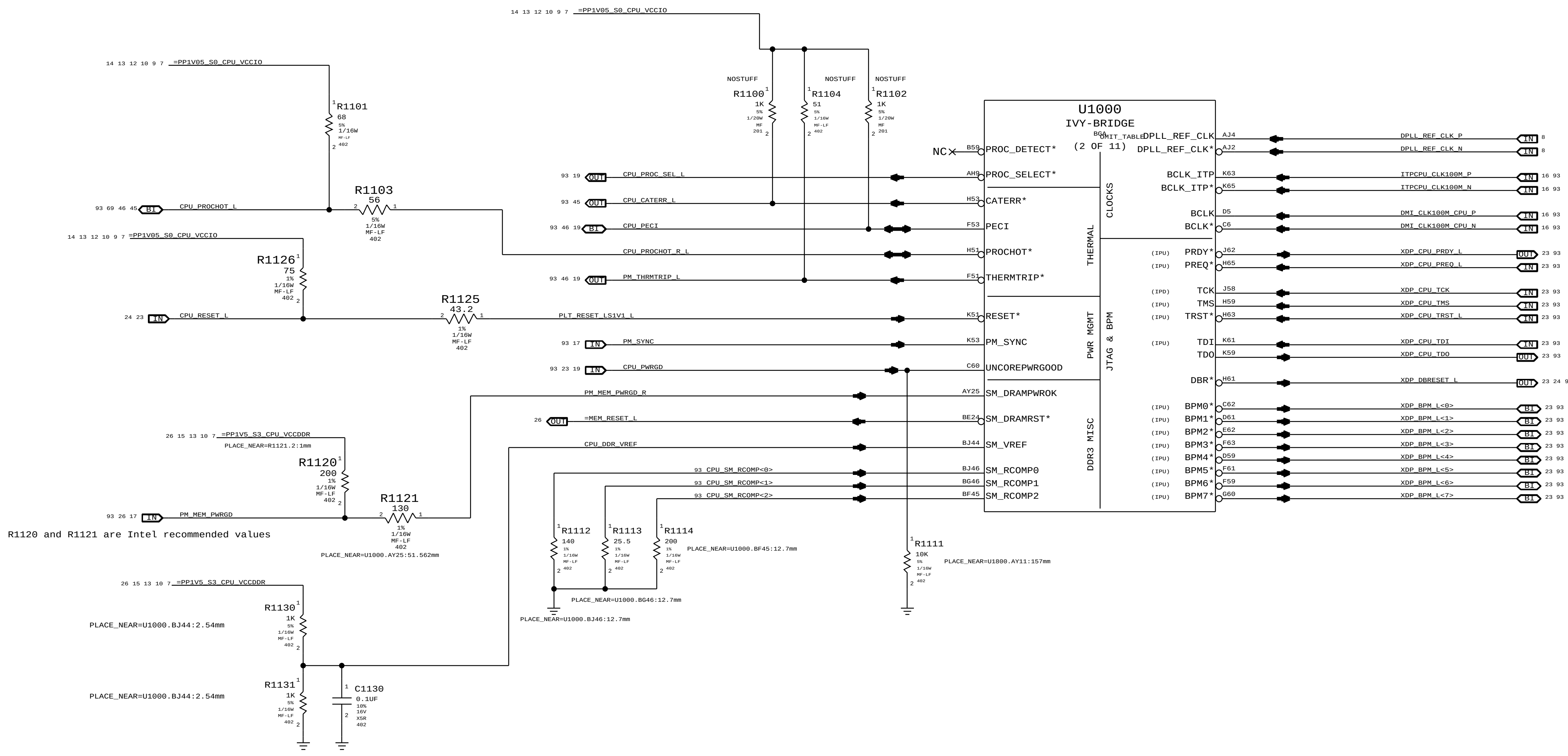




PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
116S0066	1	RES,MTL FILM,1/16W,1K,0402,SMD,LF	R1031		EDP:YES
116S0090	1	RES,MTL FILM,1/16W,10K,0402,SMD,LF	R1031		EDP:NO

CFG [7] :PEG DEFER TRAINING 1 = (DEFAULT) IMMEDIATELY AFTER xxRESETB 0 = WAIT FOR BIOS
CFG [6:5] :PCIE BIFURCATION 11 = 1 X16 (DEFAULT) 10 = 2 X8 01 = RSVD 00 = X8, X4, X4
CFG [4] :eDP ENABLE/DISABLE 1 = DISABLED 0 = ENABLED
CFG [3] :PCIE x4 LANE REVERSAL 1 = NORMAL OPERATION 0 = LANES REVERSED
CFG [2] :PCIE x16 LANE REVERSAL 1 = NORMAL OPERATION 0 = LANES REVERSED

PAGE TITLE
CPU DMI/PEG/FDI/RSVD
Apple Inc.
DRAWING NUMBER 051-9585 SIZE D
REVISION 3.0.0
BRANCH



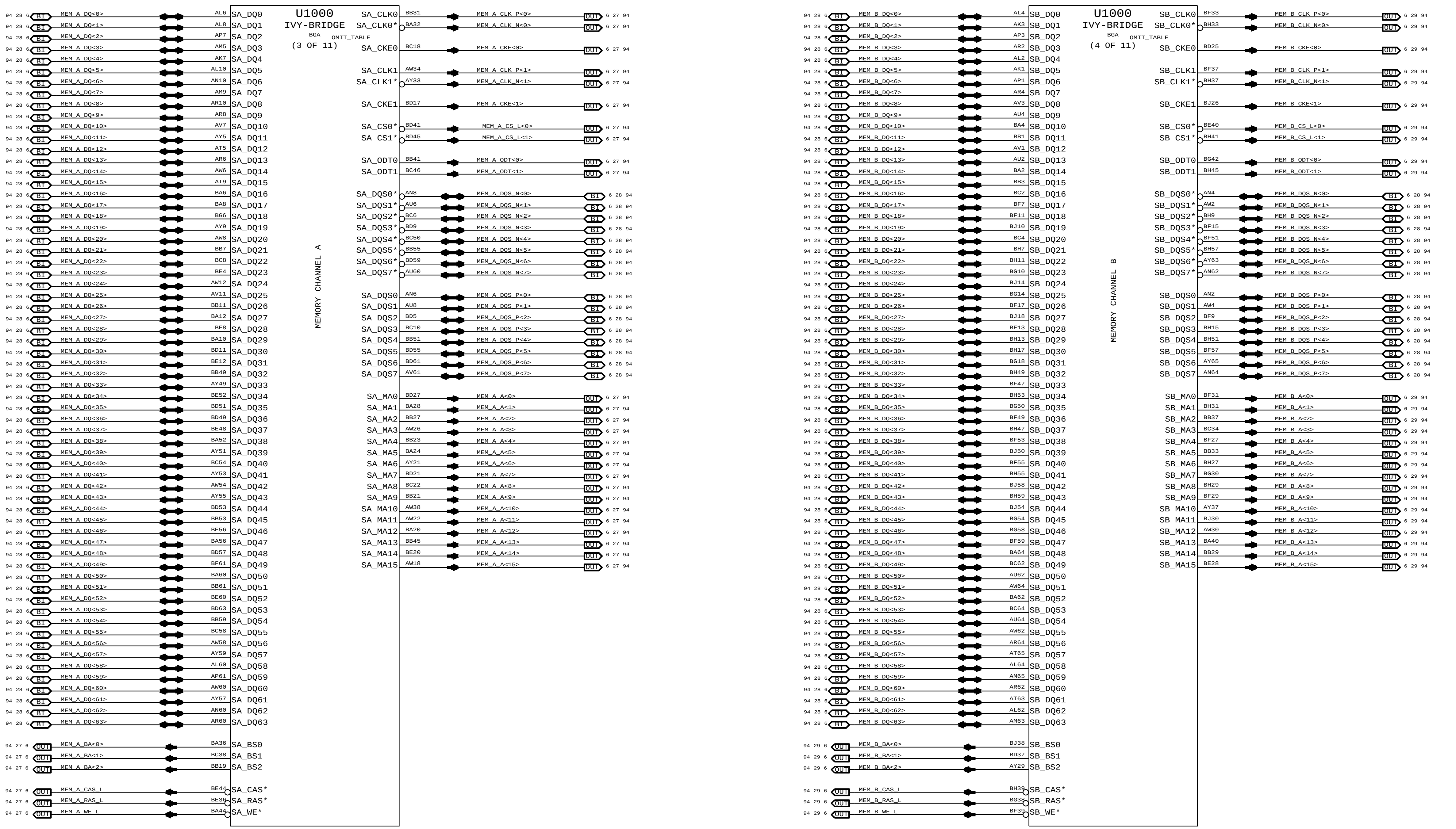
CPU CLOCK/MISC/JTAG

Apple Inc.

DRAWING NUMBER 051-9585 SIZE D

REVISION 3.0.0

BRANCH

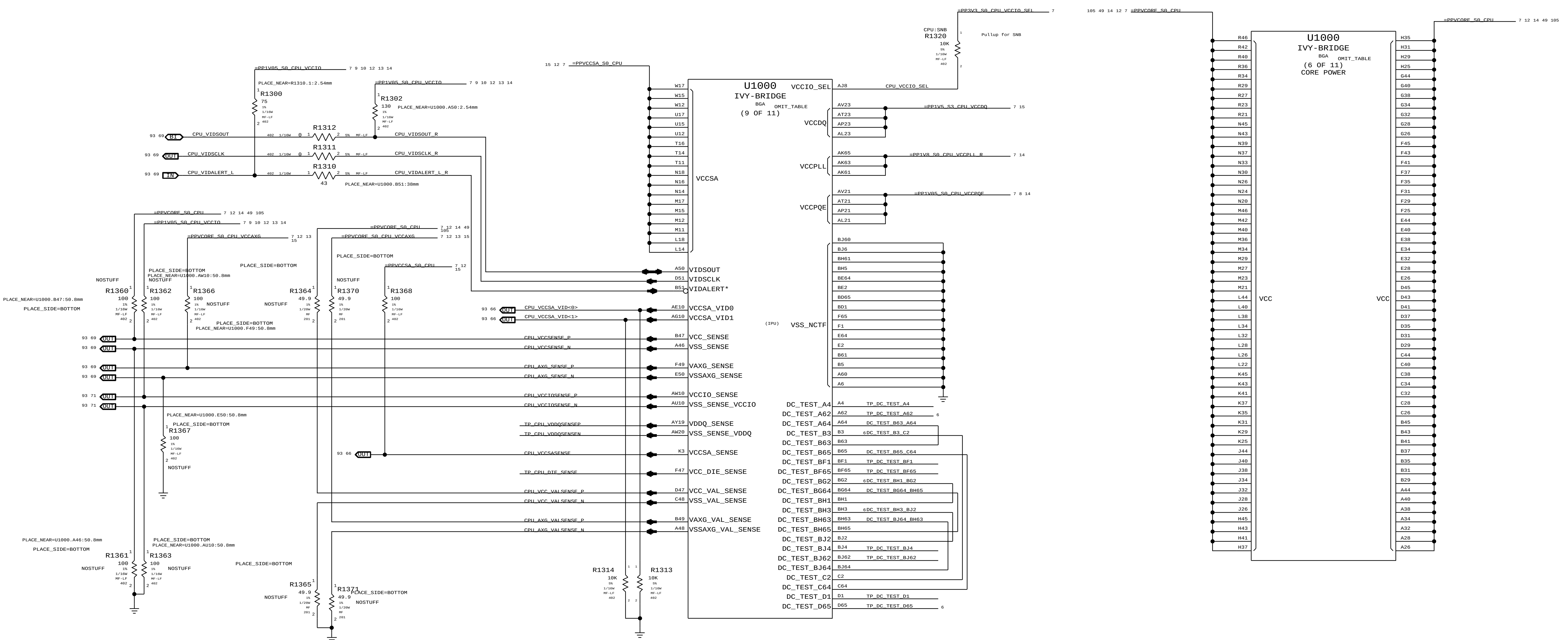


SYNC_DATE=06/15/2010

PAGE TITLE
CPU DDR3 INTERFACES

Apple Inc.

DRAWING NUMBER 051-9585
SIZE D
REVISION 3.0.0
BRANCH
PAGE 12 OF 132
SHEET 11 OF 105



NOTE: Intel validation sense lines per doc 439028 rev1.0
HR_PPDG sections 6.2.1 and 6.3.1.

SYNC_MASTER=K92_MLB SYNC_DATE=08/03/2010

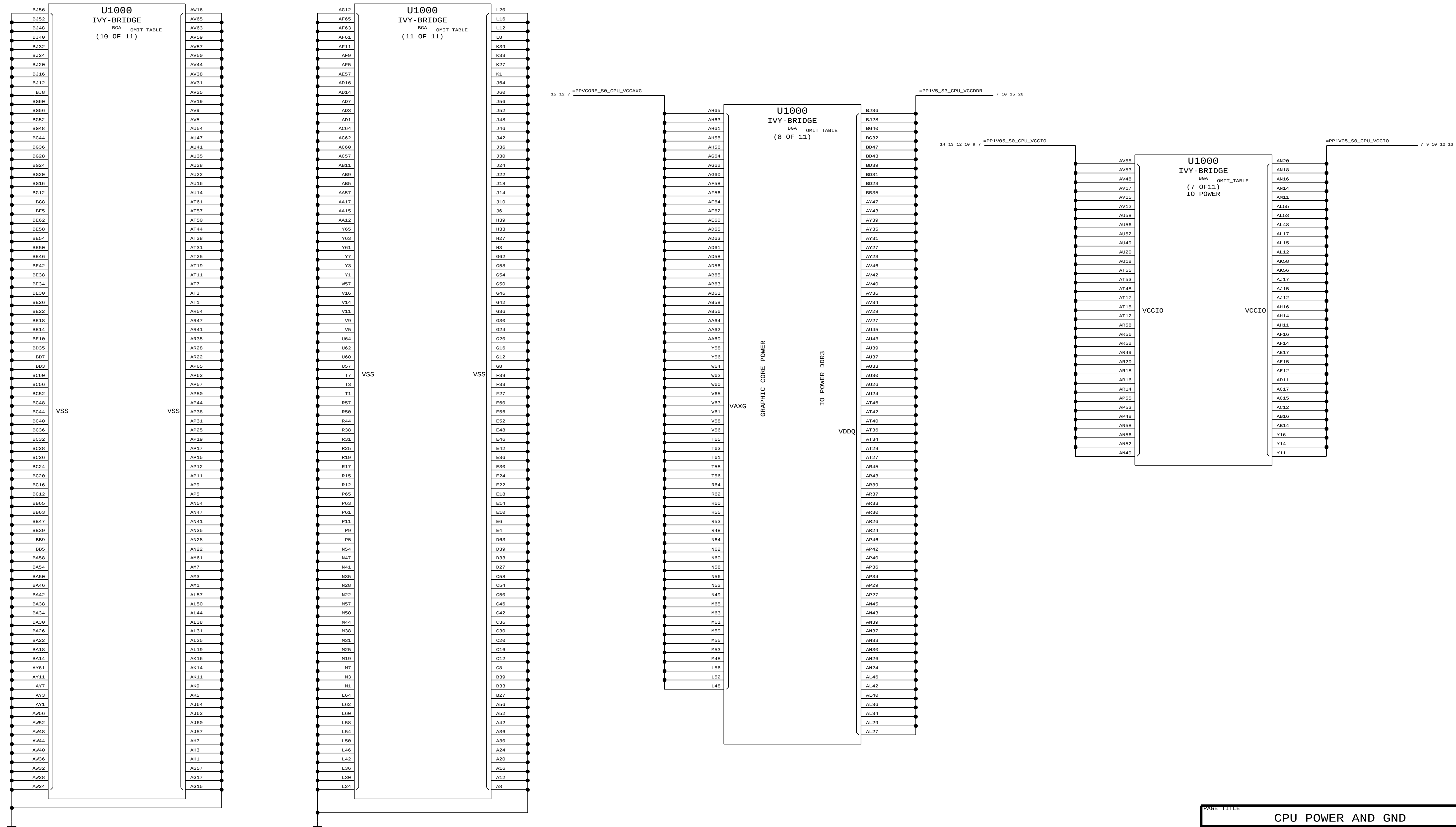
PAGE TITLE
CPU POWER

Apple Inc.

DRAWING NUMBER 051-9585 SIZE D

REVISION 3.0.0

BRANCH



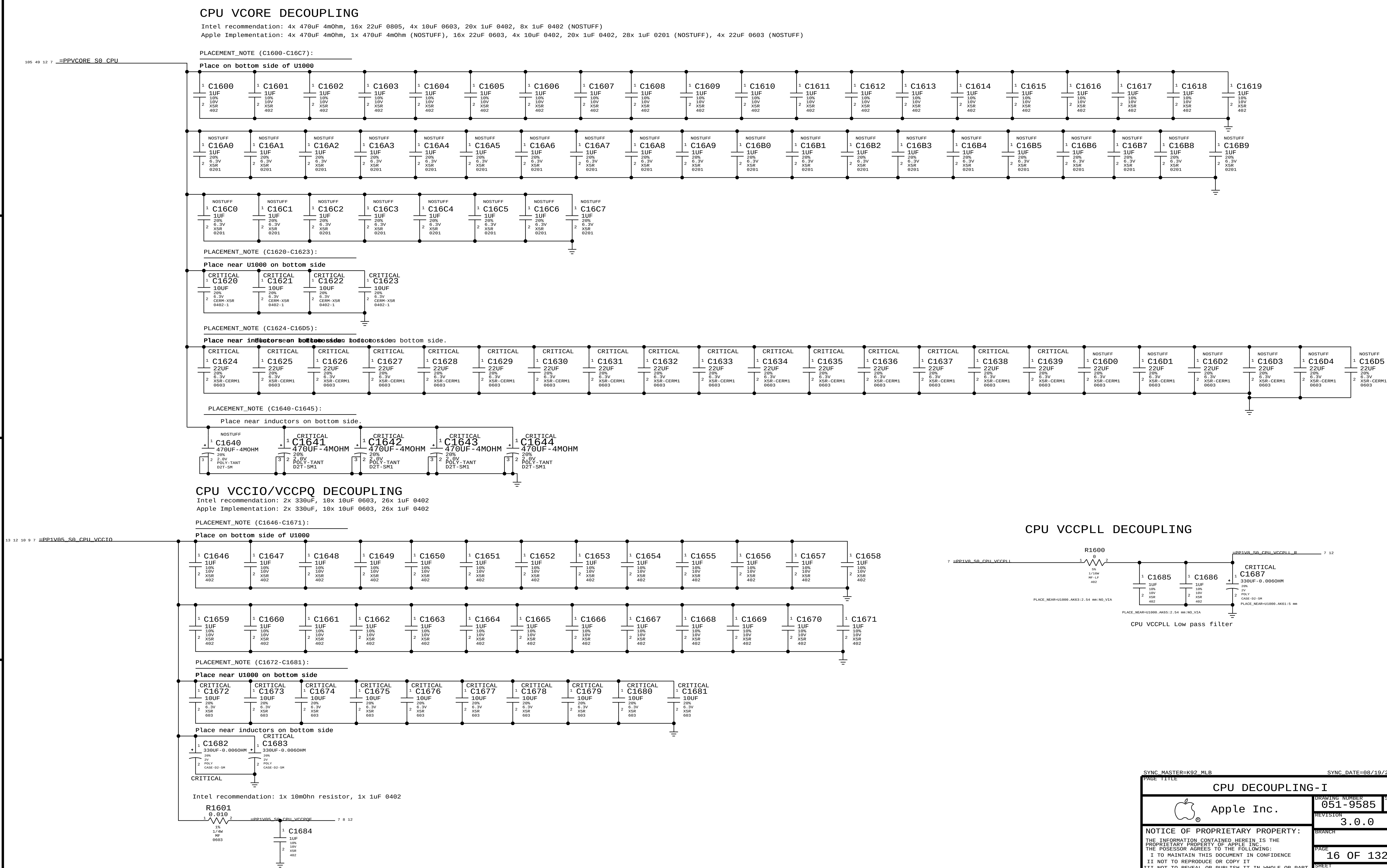
PAGE TITLE
CPU POWER AND GND

Apple Inc.

DRAWING NUMBER 051-9585 SIZE D

REVISION 3.0.0

BRANCH



SYNC_MASTER=K92_MLB SYNC_DATE=08/19/2010

PAGE TITLE: CPU DECOUPLING-I

Apple Inc.

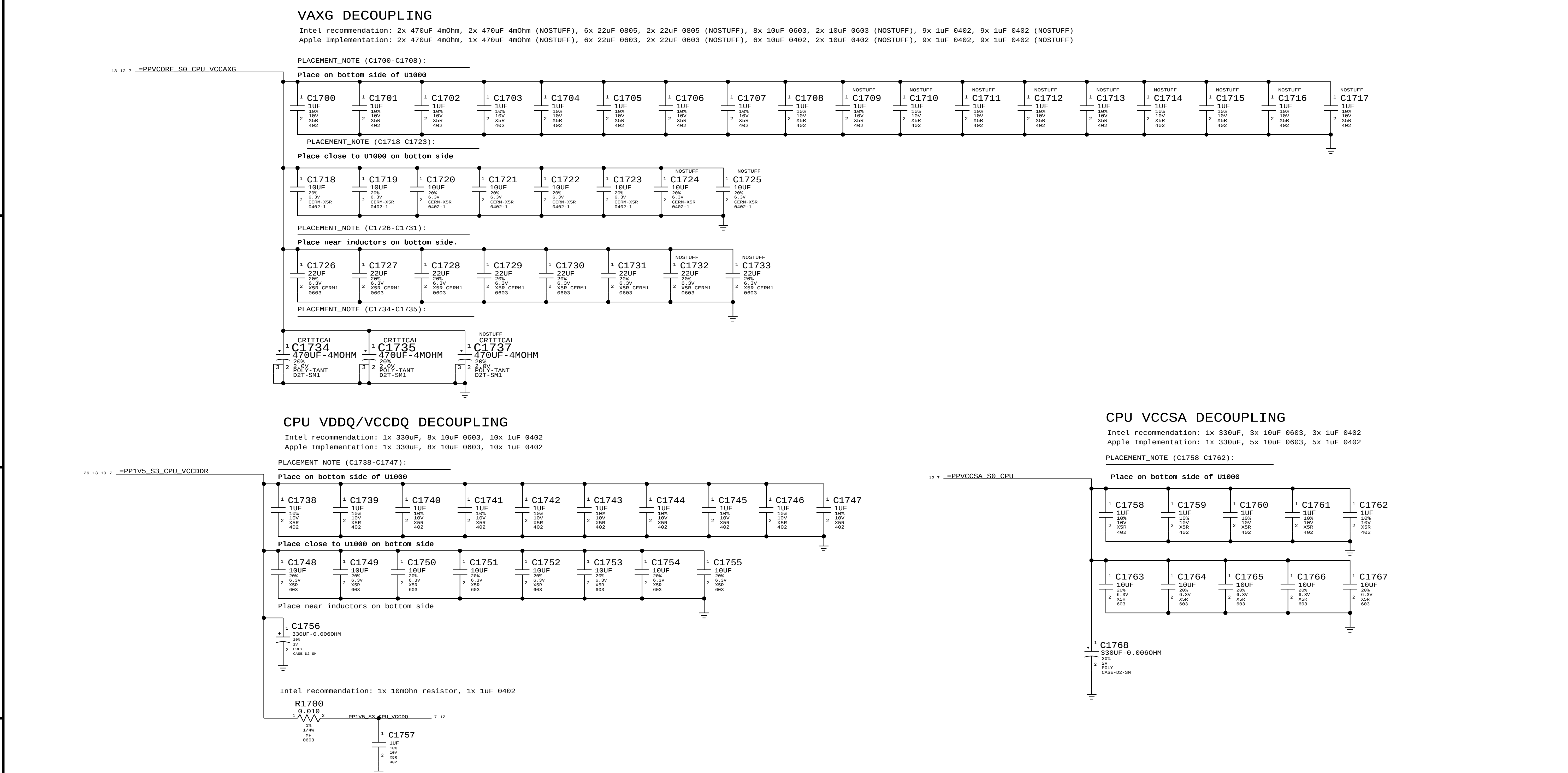
DRAWING NUMBER: 051-9585 SIZE: D

REVISION: 3.0.0

BRANCH

PAGE: 16 OF 132

SHEET: 14 OF 105



SYNC_MASTER=K92_MLB SYNC_DATE=08/19/2010

PAGE TITLE: CPU DECOUPLING-II

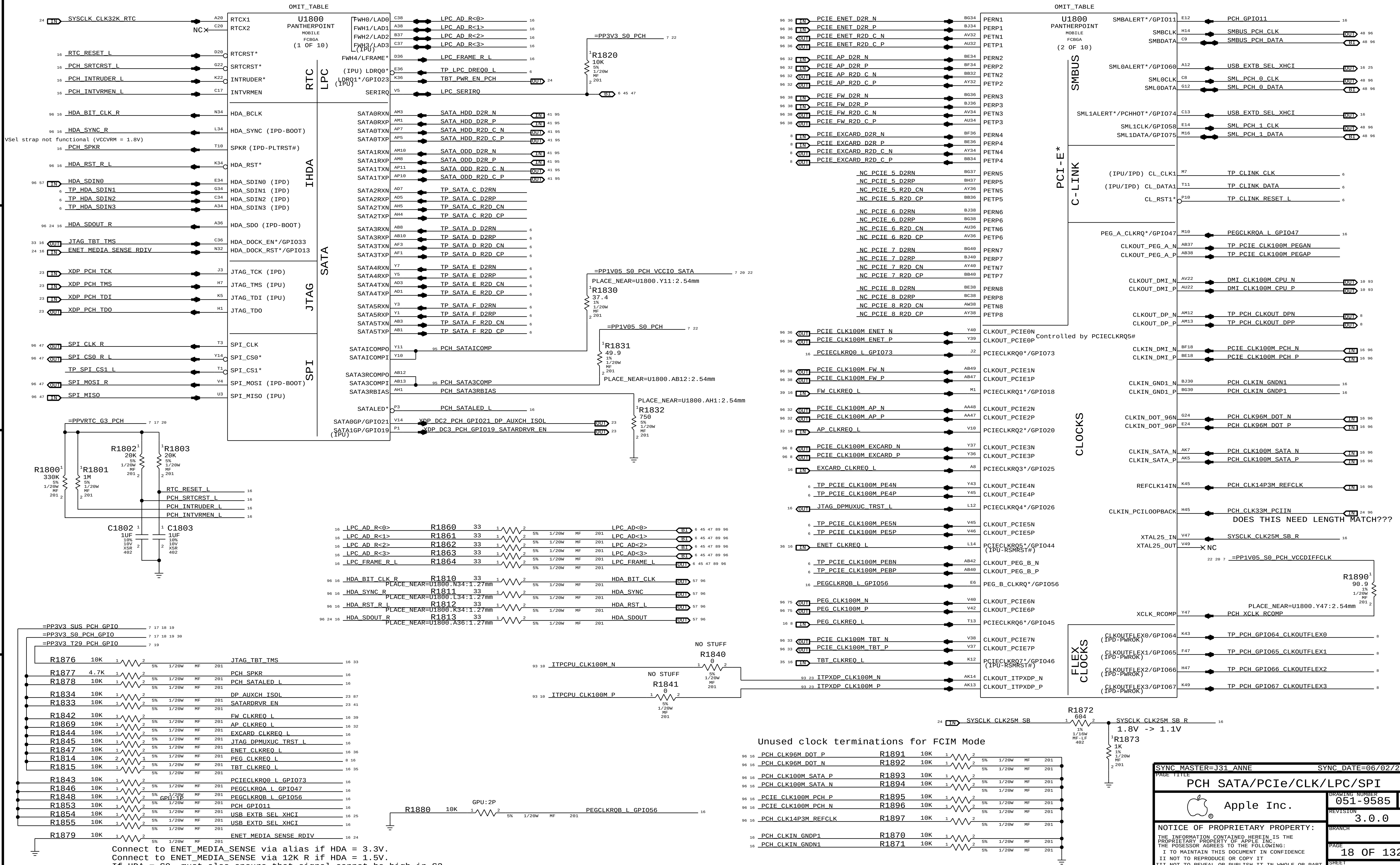
Apple Inc.

DRAWING NUMBER: 051-9585 SIZE: D

REVISION: 3.0.0

BRANCH

NOTICE OF PROPRIETARY PROPERTY:



SYNC_MASTER=J31_ANNE SYNC_DATE=06/02/2011

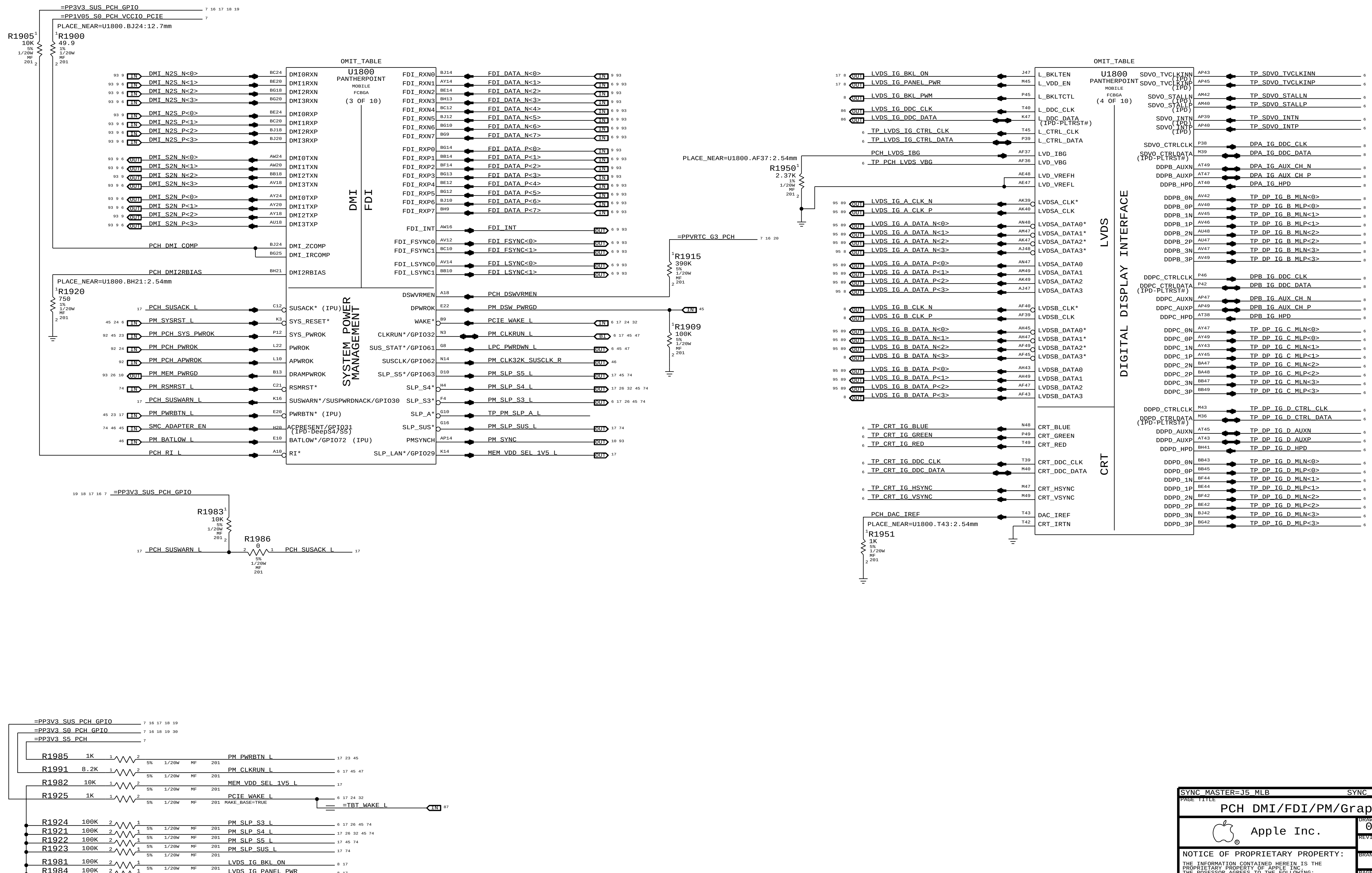
PAGE TITLE
PCH SATA/PCIe/CLK/LPC/SPI

Apple Inc.

DRAWING NUMBER 051-9585 SIZE D

REVISION 3.0.0

BRANCH



SYNC_MASTER=J5_MLB SYNC_DATE=05/26/2011

PAGE TITLE

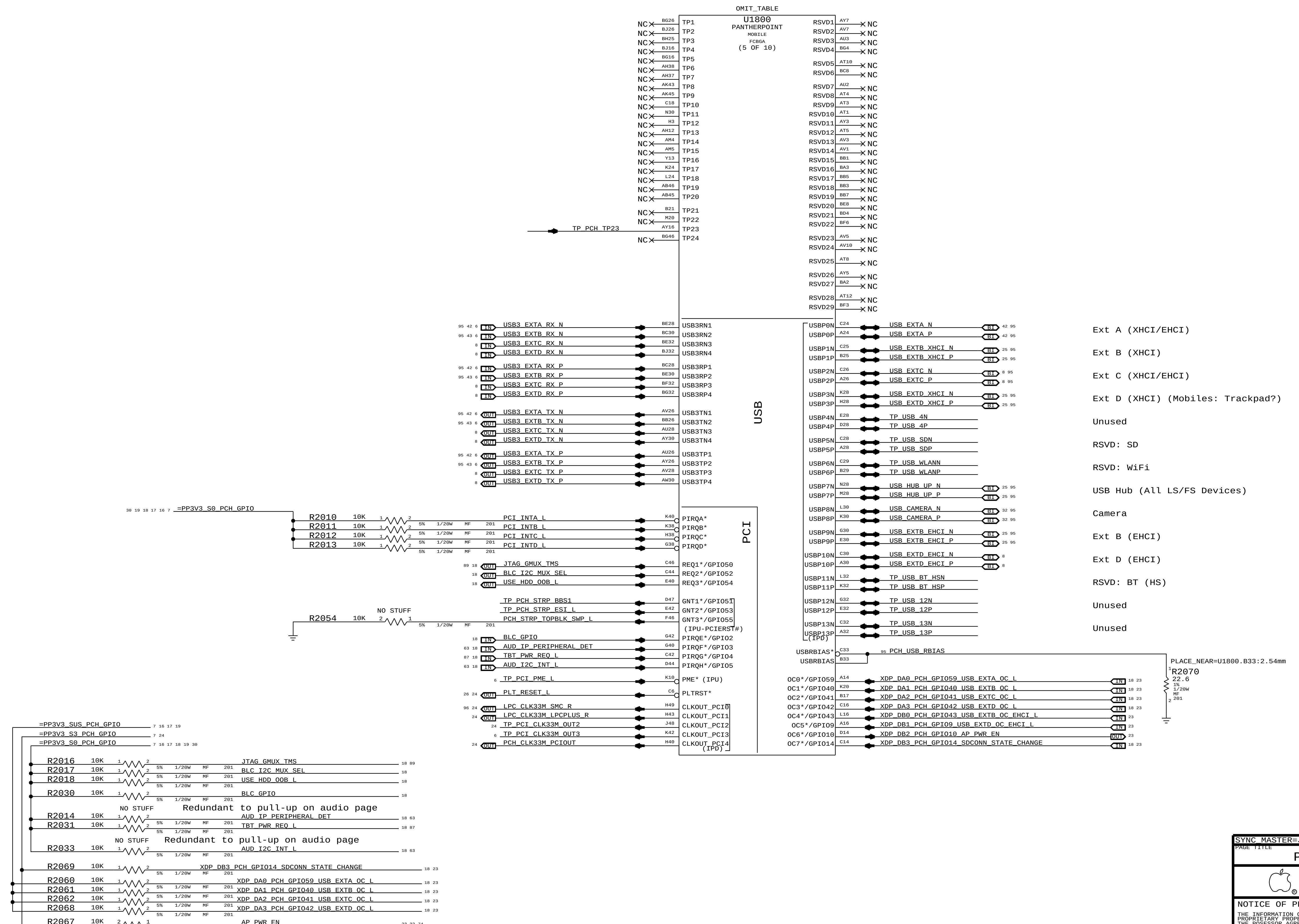
PCH DMI/FDI/PM/Graphics

Apple Inc.

DRAWING NUMBER 051-9585 SIZE D

REVISION 3.0.0

BRANCH



SYNC_MASTER=J31_ANNE SYNC_DATE=06/02/2011

PAGE TITLE

PCH PCI/USB/TP/RSVD

Apple Inc.

DRAWING NUMBER 051-9585 SIZE D

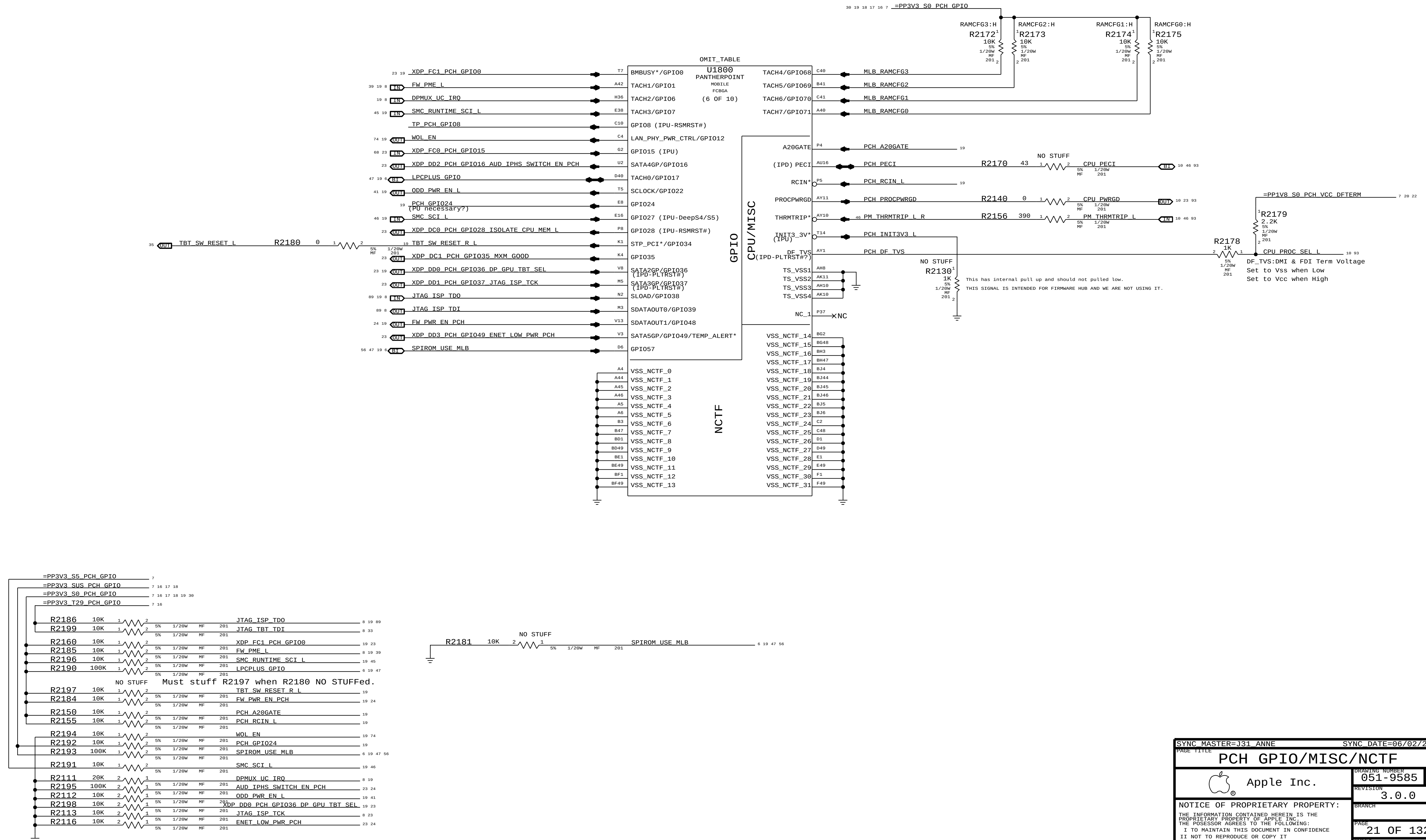
REVISION 3.0.0

BRANCH

BOM GROUP	BOM OPTIONS
RAMCFG_SLOT	RAMCFG3:H,RAMCFG2:H,RAMCFG1:H,RAMCFG0:H

Systems with no chip-down memory should pull all 4 RAMCFG GPIOs high.

Systems with chip-down memory should add pull-downs on another page and set straps per software.



SYNC_MASTER=J31_ANNE SYNC_DATE=06/02/2011

PAGE TITLE

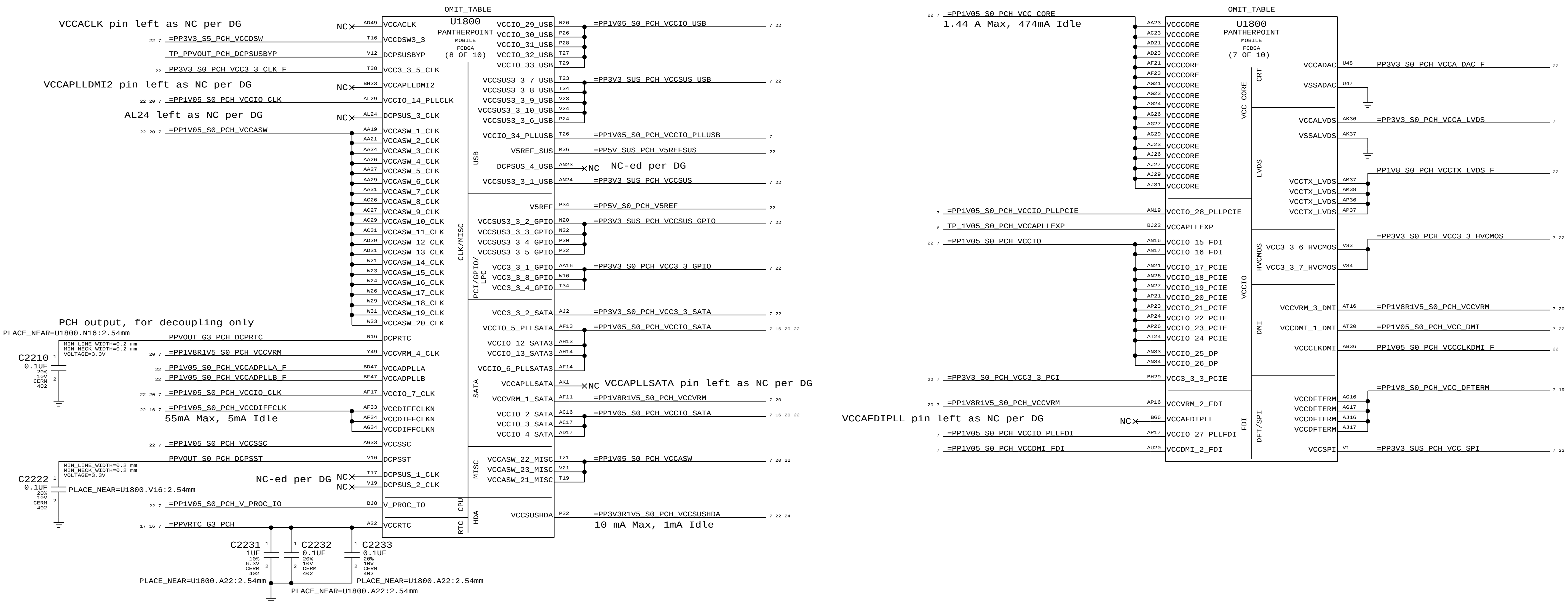
PCH GPIO/MISC/NCTF

Apple Inc.

DRAWING NUMBER 051-9585 SIZE D

REVISION 3.0.0

BRANCH



SYNC_MASTER=J5_MLB SYNC_DATE=03/21/2011

PAGE TITLE

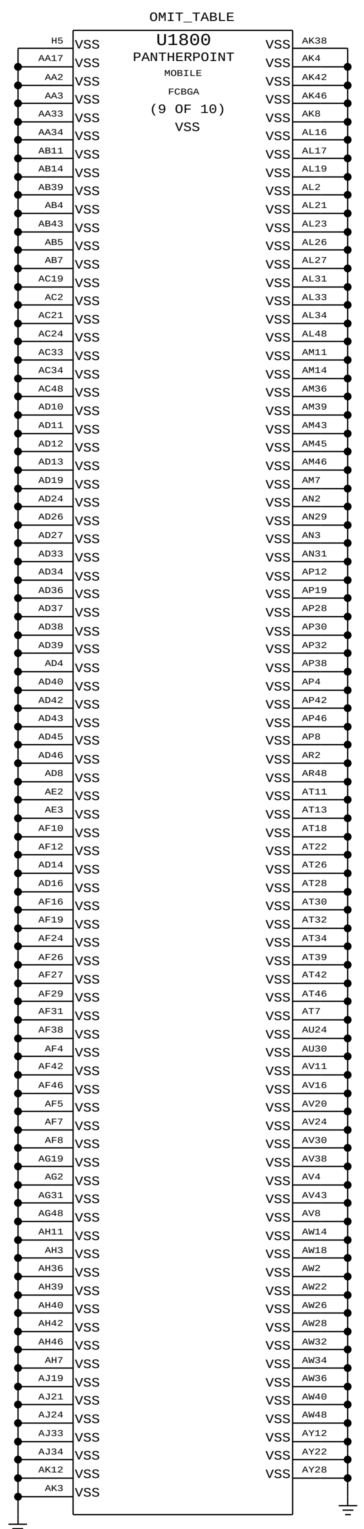
PCH POWER

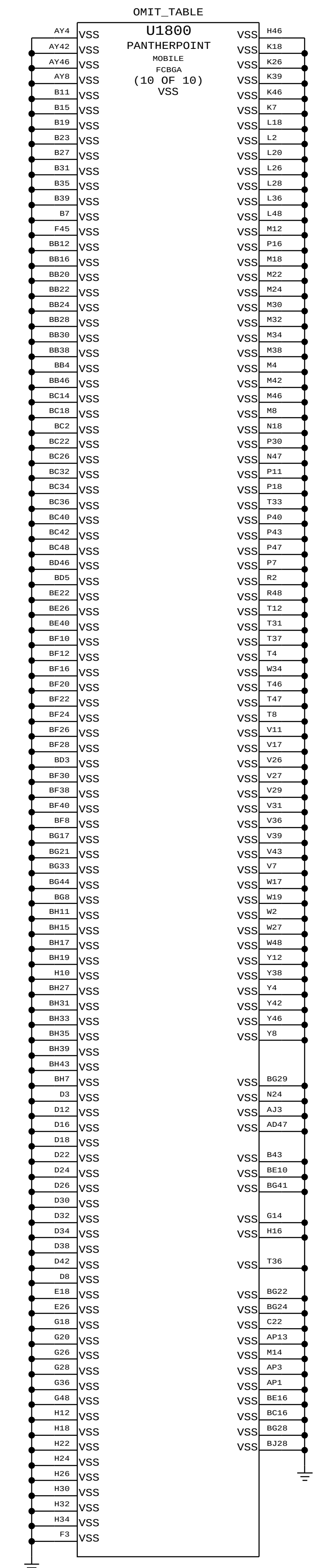
DRAWING NUMBER 051-9585 SIZE D

Apple Inc.

REVISION 3.0.0

BRANCH





SYNC_MASTER=J5_MLB SYNC_DATE=03/21/2011

PAGE TITLE

PCH GROUNDS

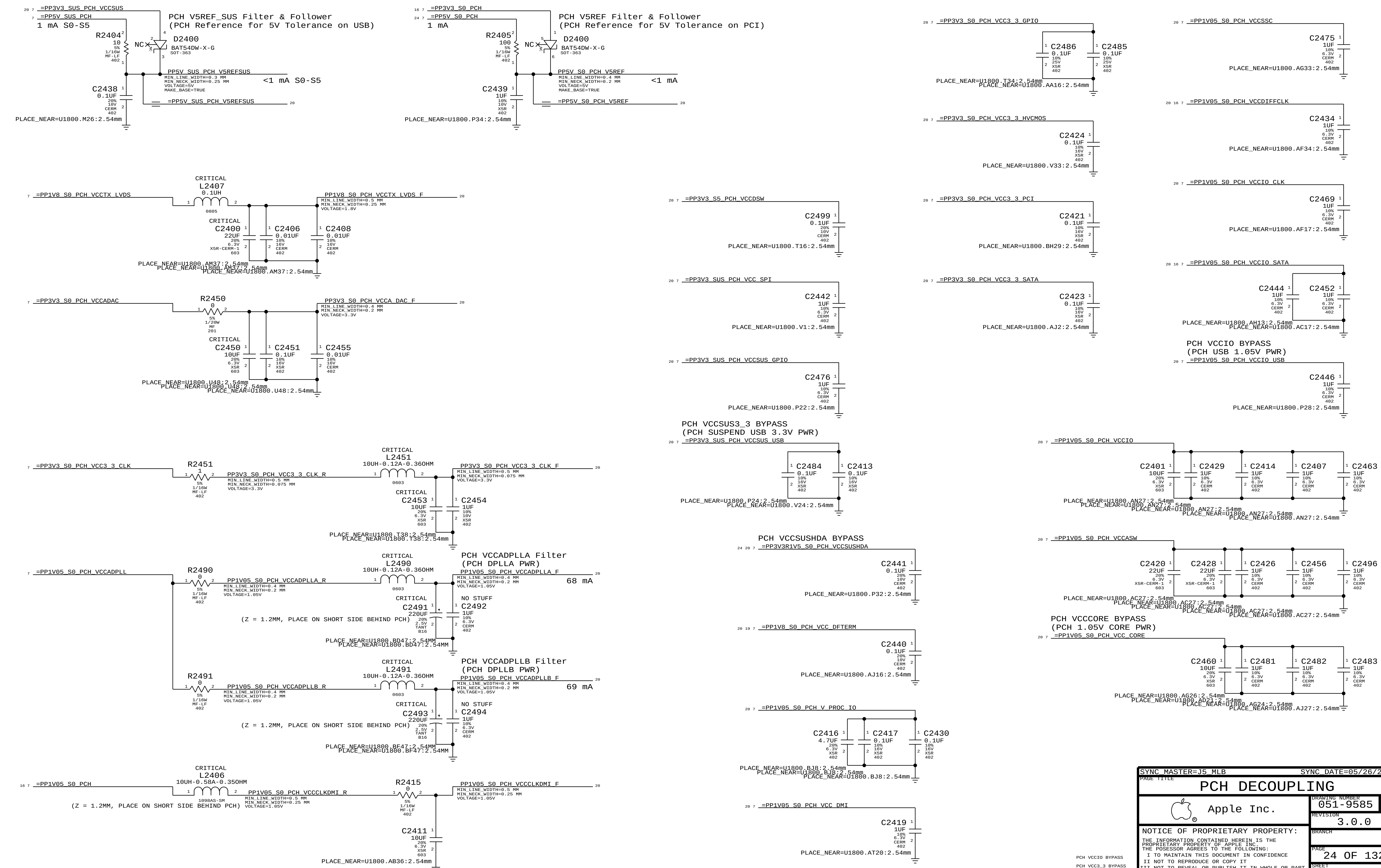
Apple Inc.

DRAWING NUMBER 051-9585 SIZE D

REVISION 3.0.0

BRANCH

NOTICE OF PROPRIETARY PROPERTY:



SYNC_MASTER=J5_MLB SYNC_DATE=05/26/2011

PAGE TITLE

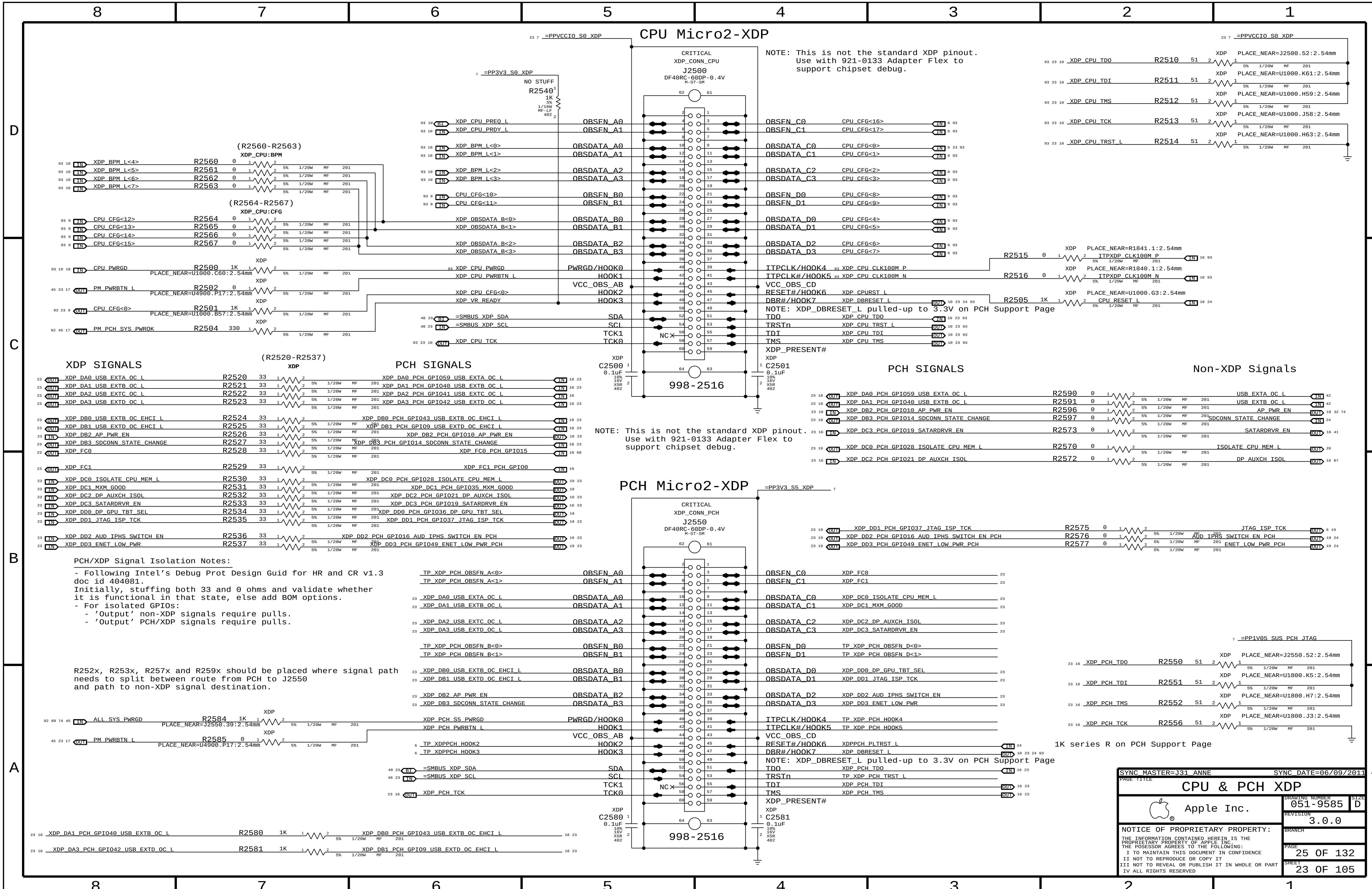
PCH DECOUPLING

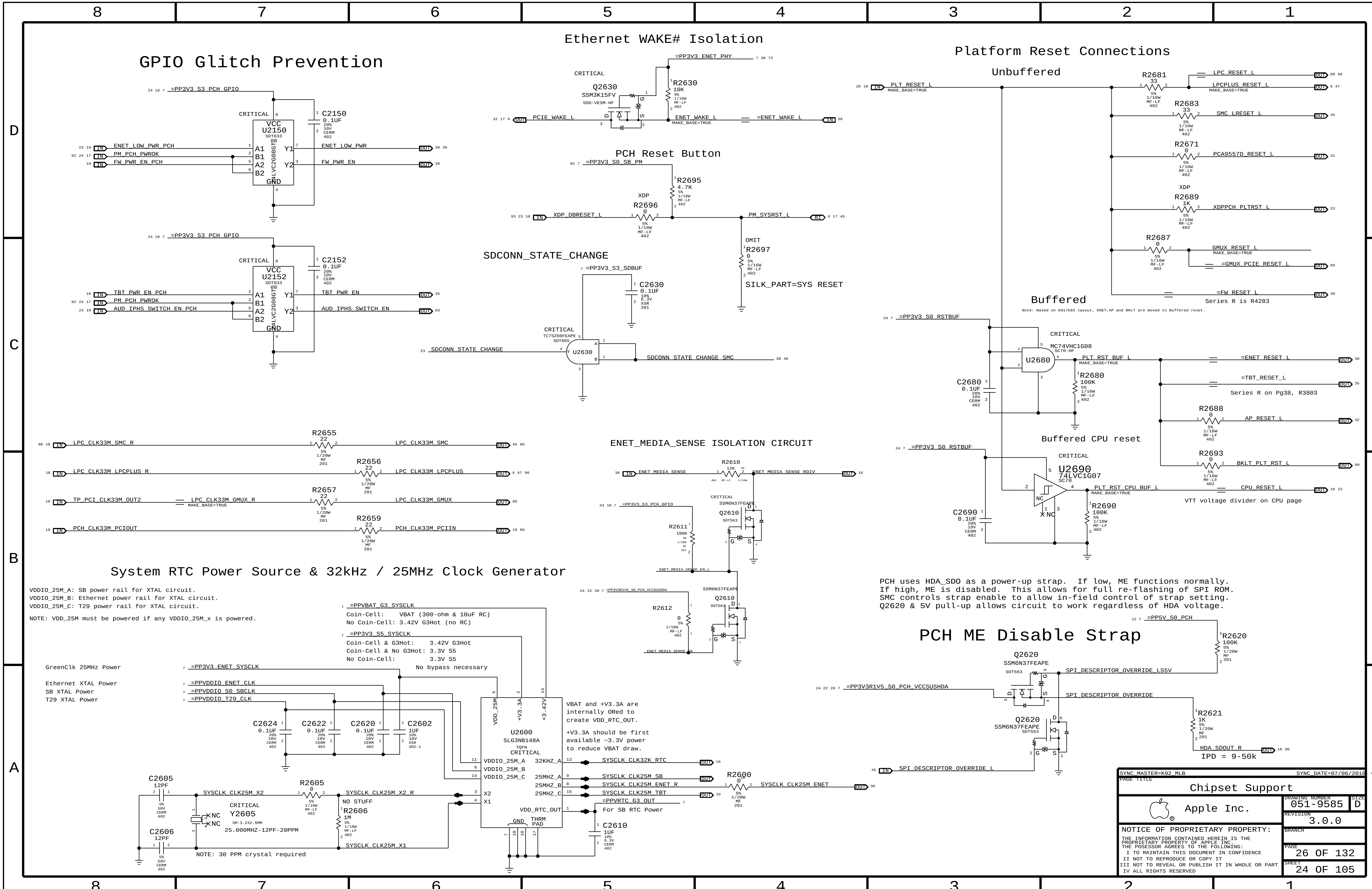
Apple Inc.

DRAWING NUMBER 051-9585 SIZE D

REVISION 3.0.0

BRANCH


CPU Micro2-XDP
CRITICAL
XDP_CONN_CPU
J2500
DF40RC-60DP-0.4V
NOTE: This is not the standard XDP pinout.
Use with 921-0133 Adapter Flex to
support chipset debug.
NOTE: XDP_DBRESET_L pulled-up to 3.3V on PCH Support Page
998-2516
XDP SIGNALS
PCH SIGNALS
Non-XDP Signals
(R2560-R2563)
XDP_CPU:BPM
(R2564-R2567)
XDP_CPU:CFG
(R2520-R2537)
PCH Micro2-XDP
CRITICAL
XDP_CONN_PCH
J2550
DF40RC-60DP-0.4V
PCH/XDP Signal Isolation Notes:
- Following Intel's Debug Prot Design Guid for HR and CR v1.3
doc id 404081.
Initially, stuffing both 33 and 0 ohms and validate whether
it is functional in that state, else add BOM options.
- For isolated GPIOs:
- 'Output' non-XDP signals require pulls.
- 'Output' PCH/XDP signals require pulls.
R252x, R253x, R257x and R259x should be placed where signal path
needs to split between route from PCH to J2550
and path to non-XDP signal destination.
1K series R on PCH Support Page
SYNC_MASTER=J31_ANNE
SYNC_DATE=06/09/2011
PAGE TITLE
CPU & PCH XDP
Apple Inc.
DRAWING NUMBER
051-9585
SIZE
D
REVISION
3.0.0
BRANCH
PAGE
25 OF 132
SHEET
23 OF 105
NOTICE OF PROPRIETARY PROPERTY:
THE INFORMATION CONTAINED HEREIN IS THE
PROPRIETARY PROPERTY OF APPLE INC.
THE POSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED


GPIO Glitch Prevention
Ethernet WAKE# Isolation
Platform Reset Connections
Unbuffered
Buffered
Note: Based on K91/K92 layout, ENET,AP and BKLT are moved to Buffered reset.
Buffered CPU reset
Series R is R4283
Series R on Pg38, R3803
VTT voltage divider on CPU page
PCH Reset Button
SILK_PART=SYS RESET
SDCONN_STATE_CHANGE
ENET_MEDIA_SENSE ISOLATION CIRCUIT
System RTC Power Source & 32kHz / 25MHz Clock Generator
VDDIO_25M_A: SB power rail for XTAL circuit.
VDDIO_25M_B: Ethernet power rail for XTAL circuit.
VDDIO_25M_C: T29 power rail for XTAL circuit.
NOTE: VDD_25M must be powered if any VDDIO_25M_x is powered.
=PPVBAT_G3_SYSCLK
Coin-Cell: VBAT (300-ohm & 10uF RC)
No Coin-Cell: 3.42V G3Hot (no RC)
=PP3V3_S5_SYSCLK
Coin-Cell & G3Hot: 3.42V G3Hot
Coin-Cell & No G3Hot: 3.3V S5
No Coin-Cell: 3.3V S5
No bypass necessary
GreenClk 25MHz Power
Ethernet XTAL Power
SB XTAL Power
T29 XTAL Power
VBAT and +V3.3A are internally ORed to create VDD_RTC_OUT.
+V3.3A should be first available ~3.3V power to reduce VBAT draw.
For SB RTC Power
NOTE: 30 PPM crystal required
PCH uses HDA_SDO as a power-up strap. If low, ME functions normally.
If high, ME is disabled. This allows for full re-flashing of SPI ROM.
SMC controls strap enable to allow in-field control of strap setting.
Q2620 & 5V pull-up allows circuit to work regardless of HDA voltage.
PCH ME Disable Strap
IPD = 9-50k
SYNC_MASTER=K92_MLB
SYNC_DATE=07/06/2010
PAGE TITLE
Chipset Support
Apple Inc.
DRAWING NUMBER 051-9585
SIZE D
REVISION 3.0.0
BRANCH
PAGE 26 OF 132
SHEET 24 OF 105
NOTICE OF PROPRIETARY PROPERTY:
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC.
THE POSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

USB MUX FOR LS/FS INTERNAL DEVICES

BOM GROUP	BOM OPTIONS
HUB_ALLREM	HUB_NONREM1_0,HUB_NONREM0_0
HUB_1NONREM	HUB_NONREM1_0,HUB_NONREM0_1
HUB_2NONREM	HUB_NONREM1_1,HUB_NONREM0_0
HUB_3NONREM	HUB_NONREM1_1,HUB_NONREM0_1

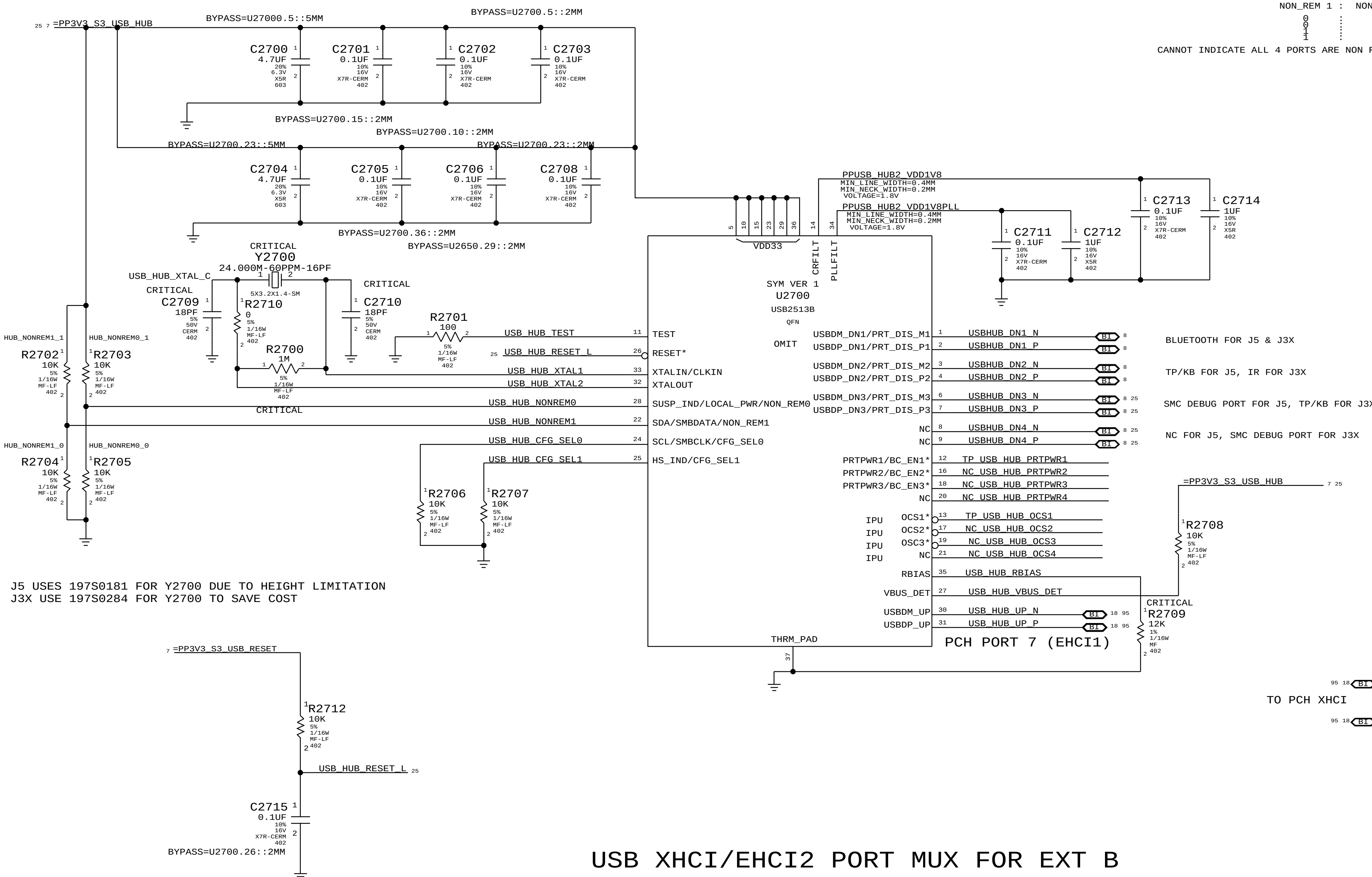
NON_REM 1	NON_REM 0	STRAP PIN CFG
0	0	ALL PORTS ARE REMOVABLE
0	1	PORT 1 IS NON REMOVABLE
1	0	PORT 1&2 ARE NON REMOVABLE
1	1	PORT 1&2&3 ARE NON REMOVABLE

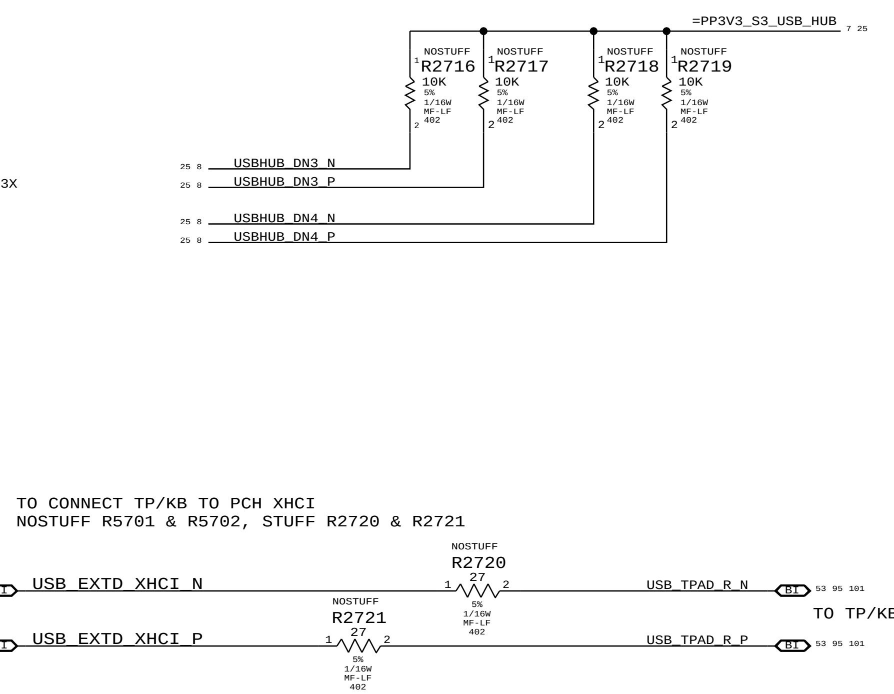
CANNOT INDICATE ALL 4 PORTS ARE NON REMOVABLE ON USB2514B VIA STARPPING, PROGRAM NON_REMOVABLE DEVICE REGISTER 09H

BOM TABLE

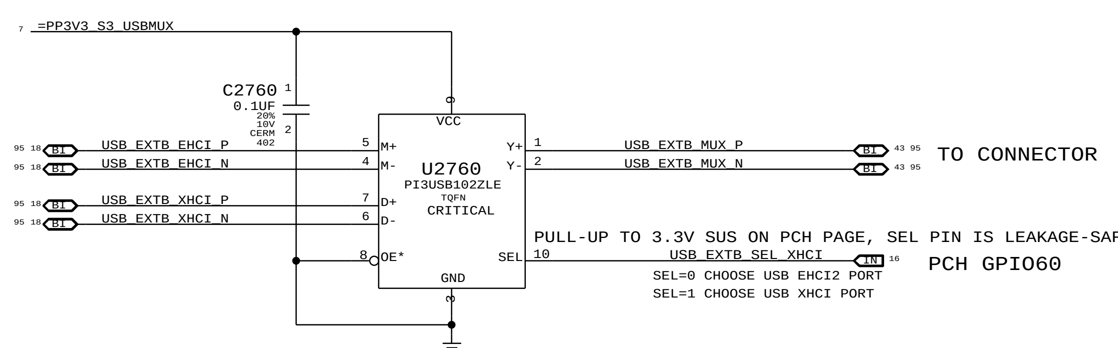
PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
338S0824	1	USB HUB 2514B	U2700	CRITICAL	USBHUB2514B
338S0923	1	USB HUB 2513B	U2700	CRITICAL	USBHUB2513B
338S0983	1	USB HUB 2512B	U2700	CRITICAL	USBHUB2512B

J5 ENGINEERING: USE USB2513B PRODUCTION: USE USB2512B
J3X ENGINEERING: USE USB2514B PRODUCTION: USE USB2513B





USB XHCI/EHCI2 PORT MUX FOR EXT B



SYNC_MASTER=J31_LINDA SYNC_DATE=09/16/2011

PAGE TITLE: USB HUB & MUX

Apple Inc.

DRAWING NUMBER: 051-9585 SIZE: D

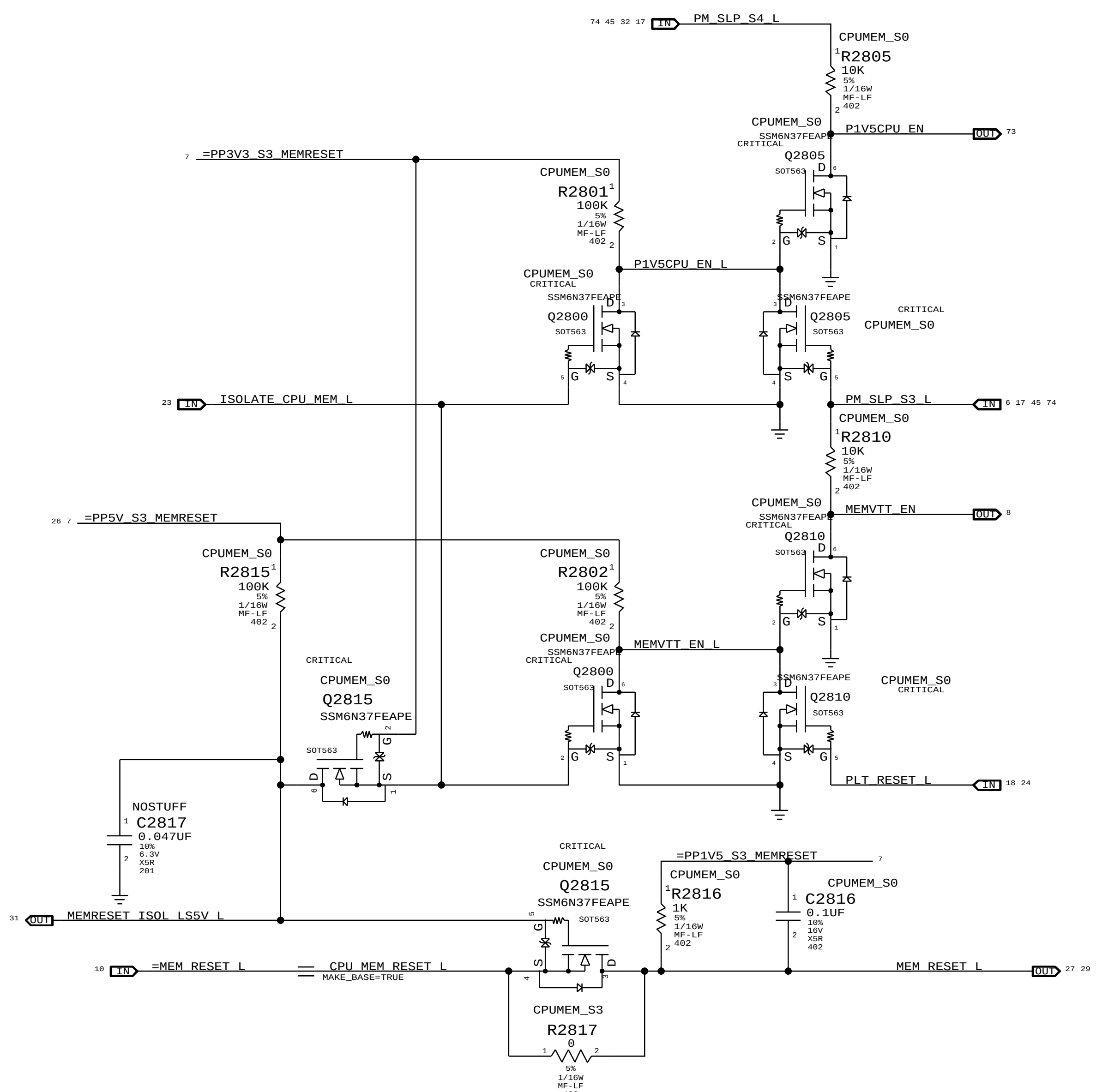
REVISION: 3.0.0

BRANCH

The circuit below handles CPU and VTT power during S0->S3->S0 transitions, as well
as isolating the CPU's SM_DRAMRST# output from the SO-DIMMs when necessary.

ISOLATE_CPU_MEM_L GPIO state during S3<->S0 transitions determines behavior of signals.
WHEN HIGH: CPU 1.5V remains powered in S3, VTT follows S0 rails, MEM_RESET_L not isolated.
WHEN LOW: CPU 1.5V follows S0 rails, VTT ensures clean CKE transition, MEM_RESET_L isolated.

P1V5CPU_EN = (ISOLATE_CPU_MEM_L + PM_SLP_S3_L) * PM_SLP_S4_L
MEMVTT_EN = (ISOLATE_CPU_MEM_L + PLT_RST_L) * PM_SLP_S3_L
MEM_RESET_L = !ISOLATE_CPU_MEM_L + CPU_MEM_RESET_L

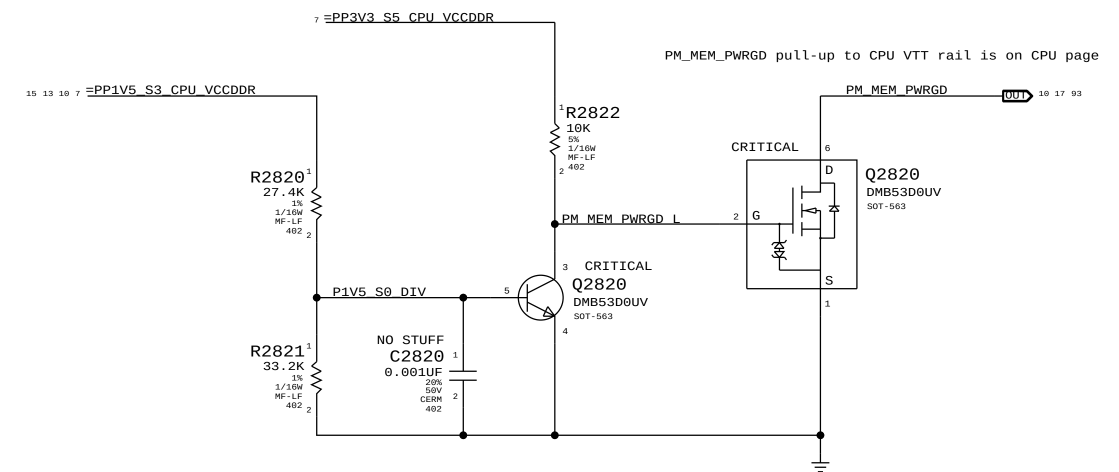


Step	ISOLATE_CPU_MEM_L	PLT_RESET_L	PM_SLP_S3_L	PM_SLP_S4_L	CPU_MEM_RESET_L	MEM_RESET_L	MEMVTT_EN	P1V5CPU_EN
S0 0	1	1	1	1	1	CPU_MEM_RESET_L	1	1
1	0	1	1	1	1	1	1	1
to 2	0	0	1	1	1	1	0	1
3	0	0	0	1	X	1	0	0
S3 4	0	0	1	1	X	1	0	1
to 5	0	1	1	1	0 (*)	1	1	1
6	0	1	1	1	1	1	1	1
S0 7	1	1	1	1	1	CPU_MEM_RESET_L	1	1

(*) CPU_MEM_RESET_L asserts due to loss of PM_MEM_PWRGD, must wait for software to clear before deasserting ISOLATE_CPU_MEM_L GPIO.

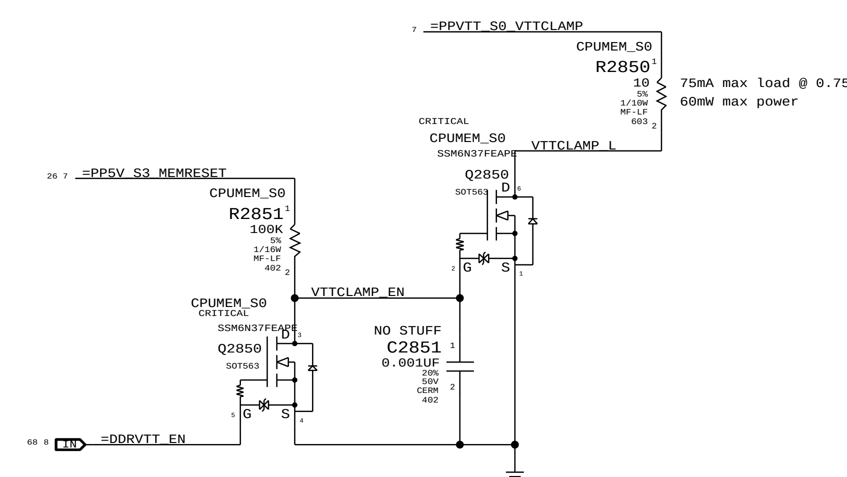
NOTE: In the event of a S3->S5 transition ISOLATE_CPU_MEM_L will still be asserted on next S5->S0 transition. Rails will power-up as if from S3, but MEM_RESET_L will not properly assert. Software must deassert ISOLATE_CPU_MEM_L and then generate a valid reset cycle on CPU_MEM_RESET_L.

1V5 S0 "PGOOD" for CPU



MEMVTT Clamp

Ensures CKE signals are held low in S3



SYNC_MASTER=K18_MLB SYNC_DATE=04/27/2010

PAGE TITLE: CPU Memory S3 Support

Apple Inc.

DRAWING NUMBER 051-9585 SIZE D

REVISION 3.0.0

BRANCH

Page Notes

Power aliases required by this page:

- =PP1V5_S0_MEM_A
- =PP1V5_S3_MEM_A
- =PP0V75_S0_MEM_VTT_A
- =PPSPD_S0_MEM_A (2.5 - 3.3V)

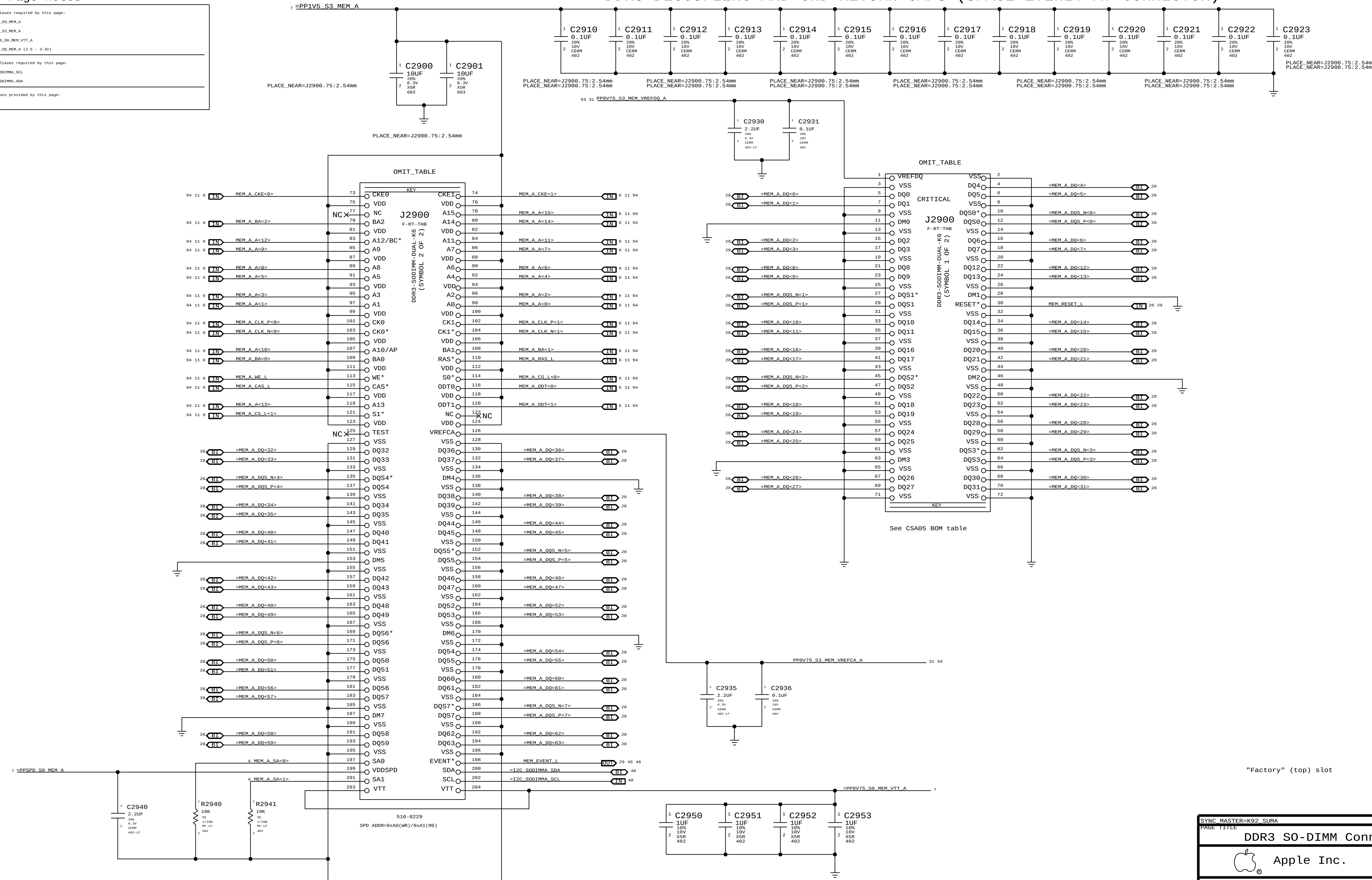
Signal aliases required by this page:

- =I2C_SODIMMA_SCL
- =I2C_SODIMMA_SDA

BOM options provided by this page:

(NONE)

DDR3 DECOUPLING AND GND RETURN CAPS (SPACE EVENLY AT CONNECTOR)



SYNC_MASTER=K92_SUMA SYNC_DATE=06/23/2010

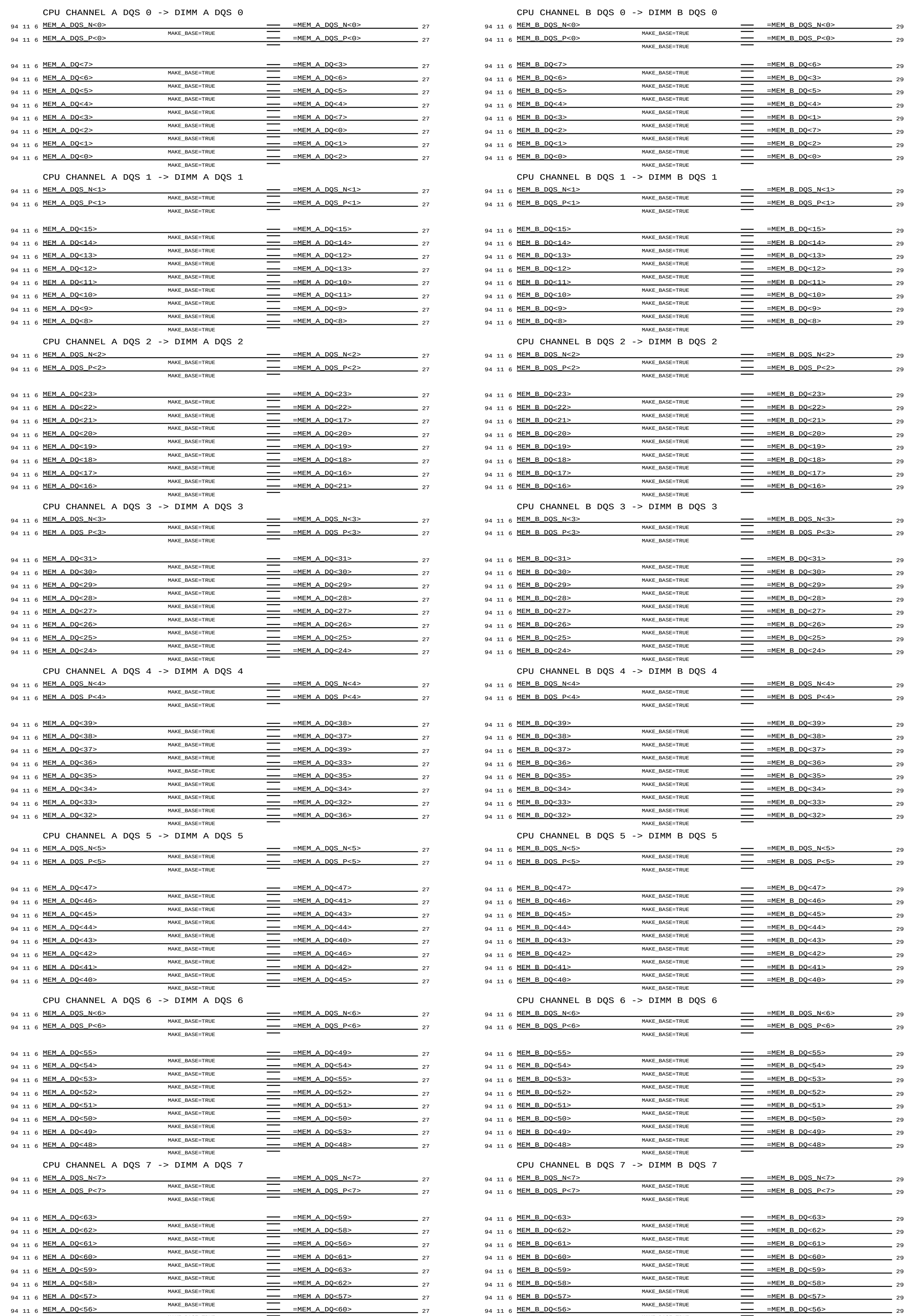
PAGE TITLE: DDR3 SO-DIMM Connector A

Apple Inc.

DRAWING NUMBER: 051-9585 SIZE: D

REVISION: 3.0.0

BRANCH



SYNC_MASTER=K92_SUMA SYNC_DATE=05/10/2010

PAGE TITLE

DDR3 Byte/Bit Swaps

Apple Inc.

DRAWING NUMBER 051-9585 SIZE D

REVISION 3.0.0

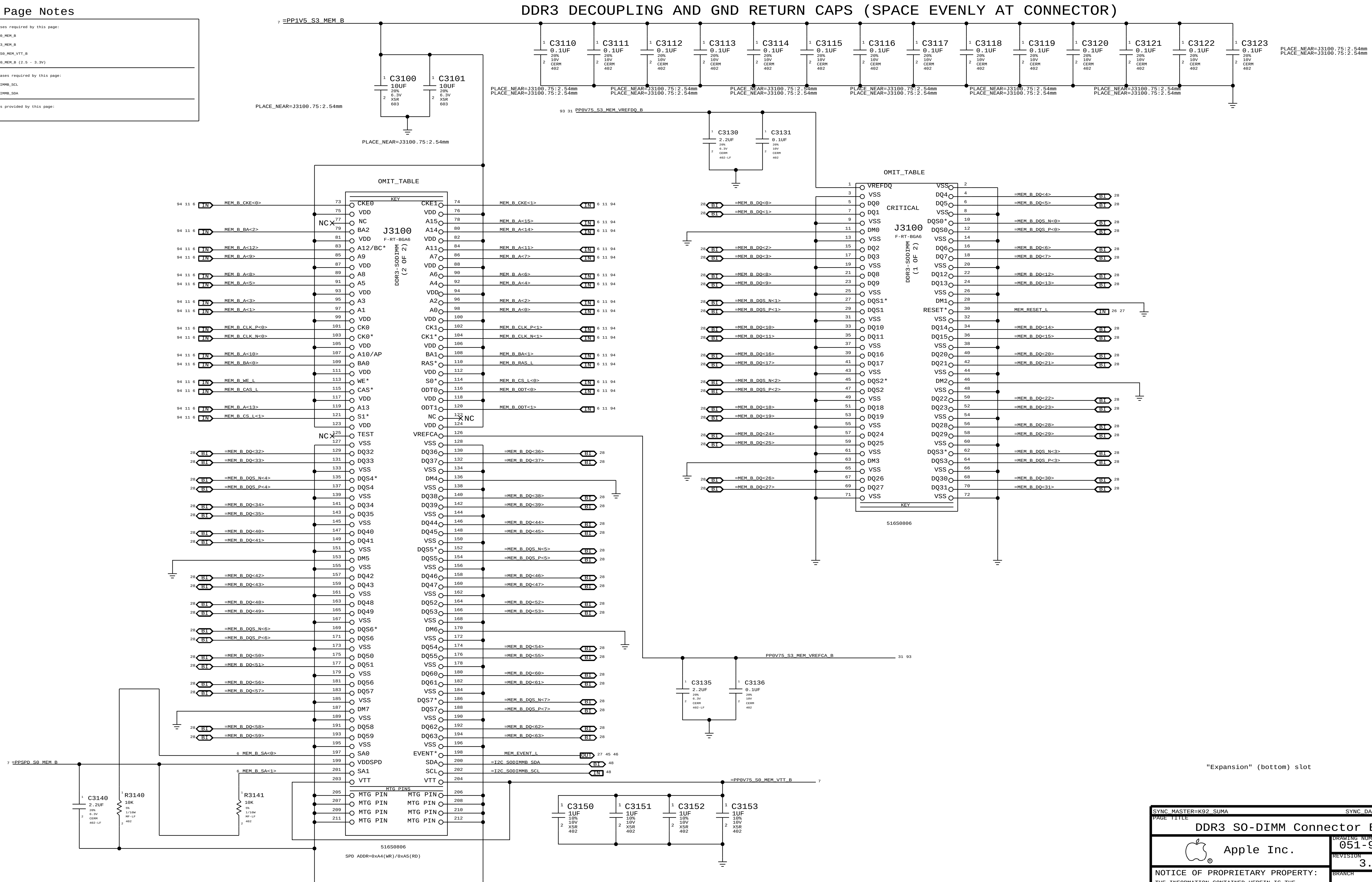
BRANCH

Page Notes

Power aliases required by this page:
- =PP1V5_S0_MEM_B
- =PP1V5_S3_MEM_B
- =PP0V75_S0_MEM_VTT_B
- =PPSPD_S0_MEM_B (2.5 - 3.3V)

Signal aliases required by this page:
- =I2C_SODIMMB_SCL
- =I2C_SODIMMB_SDA

BOM options provided by this page:
(NONE)



SYNC_MASTER=K92_SUMA SYNC_DATE=06/23/2010

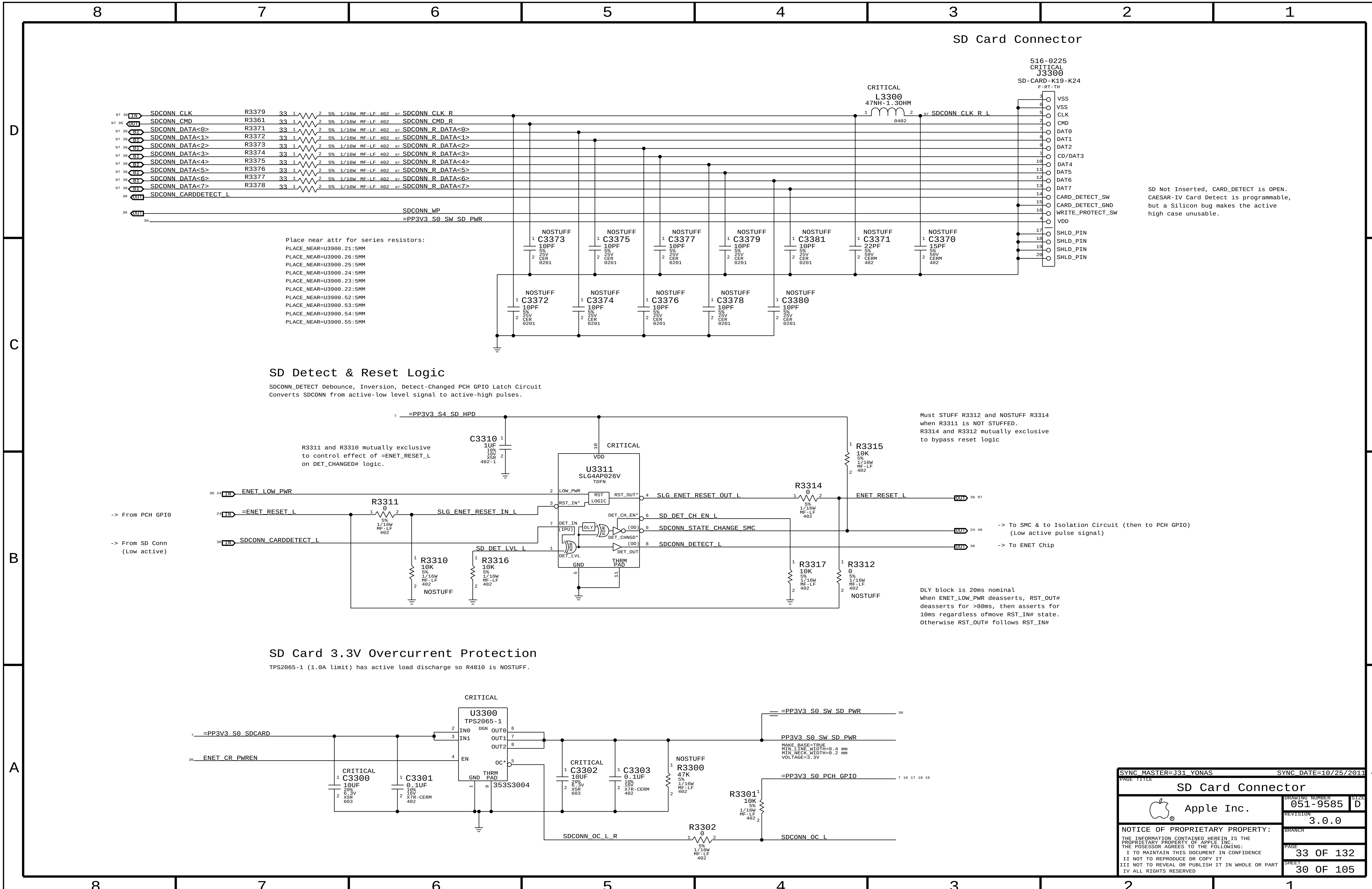
PAGE TITLE
DDR3 SO-DIMM Connector B

Apple Inc.

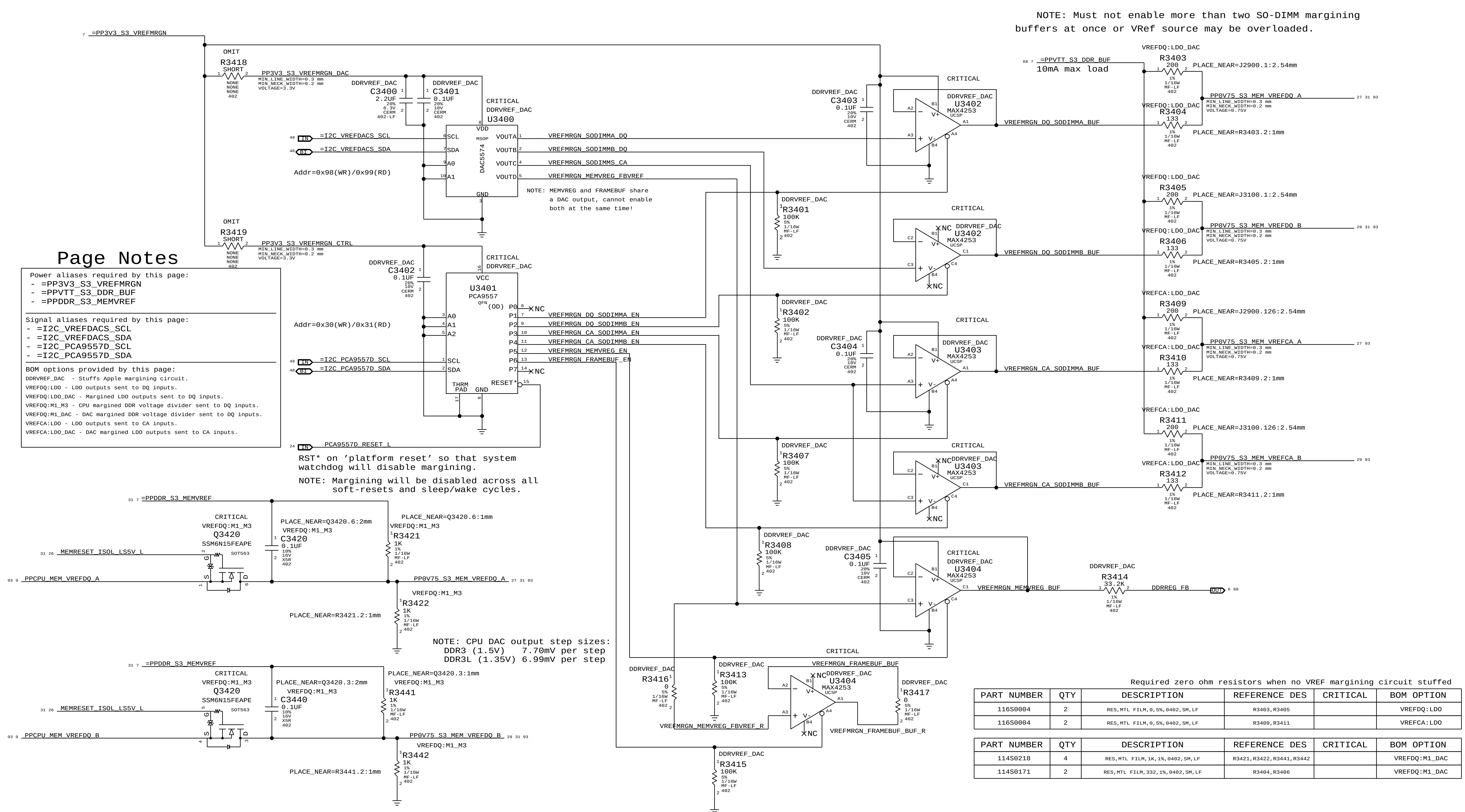
DRAWING NUMBER 051-9585 SIZE D

REVISION 3.0.0

BRANCH


SD Card Connector
516-0225
CRITICAL
J3300
SD-CARD-K19-K24
F-RT-TH
CRITICAL
L3300
47NH-1.3OHM
0402
SDCONN_CLK
SDCONN_CMD
SDCONN_DATA<0>
SDCONN_DATA<1>
SDCONN_DATA<2>
SDCONN_DATA<3>
SDCONN_DATA<4>
SDCONN_DATA<5>
SDCONN_DATA<6>
SDCONN_DATA<7>
SDCONN_CARDDETECT_L
R3379
R3361
R3371
R3372
R3373
R3374
R3375
R3376
R3377
R3378
5% 1/16W MF-LF 402
SDCONN_CLK_R
SDCONN_CMD_R
SDCONN_R_DATA<0>
SDCONN_R_DATA<1>
SDCONN_R_DATA<2>
SDCONN_R_DATA<3>
SDCONN_R_DATA<4>
SDCONN_R_DATA<5>
SDCONN_R_DATA<6>
SDCONN_R_DATA<7>
SDCONN_CLK_R_L
SDCONN_WP
=PP3V3_S0_SW_SD_PWR
VSS
VSS
CLK
CMD
DAT0
DAT1
DAT2
CD/DAT3
DAT4
DAT5
DAT6
DAT7
CARD_DETECT_SW
CARD_DETECT_GND
WRITE_PROTECT_SW
VDD
SHLD_PIN
SHLD_PIN
SHLD_PIN
SHLD_PIN
SD Not Inserted, CARD_DETECT is OPEN.
CAESAR-IV Card Detect is programmable,
but a Silicon bug makes the active
high case unusable.
Place near attr for series resistors:
PLACE_NEAR=U3900.21:5MM
PLACE_NEAR=U3900.26:5MM
PLACE_NEAR=U3900.25:5MM
PLACE_NEAR=U3900.24:5MM
PLACE_NEAR=U3900.23:5MM
PLACE_NEAR=U3900.22:5MM
PLACE_NEAR=U3900.52:5MM
PLACE_NEAR=U3900.53:5MM
PLACE_NEAR=U3900.54:5MM
PLACE_NEAR=U3900.55:5MM
NOSTUFF C3373 10PF 5% 25V CER 0201
NOSTUFF C3375 10PF 5% 25V CER 0201
NOSTUFF C3377 10PF 5% 25V CER 0201
NOSTUFF C3379 10PF 5% 25V CER 0201
NOSTUFF C3381 10PF 5% 25V CER 0201
NOSTUFF C3371 22PF 5% 50V CERM 402
NOSTUFF C3370 15PF 5% 50V CERM 402
NOSTUFF C3372 10PF 5% 25V CER 0201
NOSTUFF C3374 10PF 5% 25V CER 0201
NOSTUFF C3376 10PF 5% 25V CER 0201
NOSTUFF C3378 10PF 5% 25V CER 0201
NOSTUFF C3380 10PF 5% 25V CER 0201
SD Detect & Reset Logic
SDCONN_DETECT Debounce, Inversion, Detect-Changed PCH GPIO Latch Circuit
Converts SDCONN from active-low level signal to active-high pulses.
=PP3V3_S4_SD_HPD
C3310 1UF 10% 10V X5R 402-1
CRITICAL
U3311
SLG4AP026V
TDFN
VDD
LOW_PWR
RST_IN*
RST LOGIC
RST_OUT*
DET_IN
(IPU)
DLY
DET_CH_EN*
DET_CHNGD*
DET_OUT
DET_LVL
GND
THRM PAD
R3311 and R3310 mutually exclusive
to control effect of =ENET_RESET_L
on DET_CHANGED# logic.
ENET_LOW_PWR
=ENET_RESET_L
SDCONN_CARDDETECT_L
-> From PCH GPIO
-> From SD Conn
(Low active)
R3311 0 5% 1/16W MF-LF 402
SLG_ENET_RESET_IN_L
SD_DET_LVL_L
R3310 10K 5% 1/16W MF-LF 402 NOSTUFF
R3316 10K 5% 1/16W MF-LF 402
SLG_ENET_RESET_OUT_L
SD_DET_CH_EN_L
SDCONN_STATE_CHANGE_SMC
SDCONN_DETECT_L
R3314 0 5% 1/16W MF-LF 402
R3315 10K 5% 1/16W MF-LF 402
R3317 10K 5% 1/16W MF-LF 402
R3312 0 5% 1/16W MF-LF 402 NOSTUFF
ENET_RESET_L
-> To SMC & to Isolation Circuit (then to PCH GPIO)
(Low active pulse signal)
-> To ENET Chip
Must STUFF R3312 and NOSTUFF R3314
when R3311 is NOT STUFFED.
R3314 and R3312 mutually exclusive
to bypass reset logic
DLY block is 20ms nominal
When ENET_LOW_PWR deasserts, RST_OUT#
deasserts for >80ms, then asserts for
10ms regardless ofmove RST_IN# state.
Otherwise RST_OUT# follows RST_IN#
SD Card 3.3V Overcurrent Protection
TPS2065-1 (1.0A limit) has active load discharge so R4810 is NOSTUFF.
CRITICAL
U3300
TPS2065-1
DGN
IN0
IN1
EN
OUT0
OUT1
OUT2
OC*
GND
THRM PAD
353S3004
=PP3V3_S0_SDCARD
ENET_CR_PWREN
CRITICAL C3300 10UF 20% 6.3V X5R 603
C3301 0.1UF 10% 16V X7R-CERM 402
CRITICAL C3302 10UF 20% 6.3V X5R 603
C3303 0.1UF 10% 16V X7R-CERM 402
NOSTUFF R3300 47K 5% 1/16W MF-LF 402
=PP3V3_S0_SW_SD_PWR
PP3V3_S0_SW_SD_PWR
MAKE_BASE=TRUE
MIN_LINE_WIDTH=0.4 mm
MIN_NECK_WIDTH=0.2 mm
VOLTAGE=3.3V
=PP3V3_S0_PCH_GPIO
R3301 10K 5% 1/16W MF-LF 402
R3302 0 5% 1/16W MF-LF 402
SDCONN_OC_L_R
SDCONN_OC_L
SYNC_MASTER=J31_YONAS
SYNC_DATE=10/25/2011
PAGE TITLE
SD Card Connector
Apple Inc.
DRAWING NUMBER
051-9585
SIZE
D
REVISION
3.0.0
BRANCH
NOTICE OF PROPRIETARY PROPERTY:
THE INFORMATION CONTAINED HEREIN IS THE
PROPRIETARY PROPERTY OF APPLE INC.
THE POSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED
PAGE
33 OF 132
SHEET
30 OF 105

8	7	6	5	4	3	2	1



Page Notes

Power aliases required by this page:
- =PP3V3_S3_VREFMRGN
- =PPVTT_S3_DDR_BUF
- =PPDDR_S3_MEMVREF

Signal aliases required by this page:
- =I2C_VREFDACS_SCL
- =I2C_VREFDACS_SDA
- =I2C_PCA9557D_SCL
- =I2C_PCA9557D_SDA

BOM options provided by this page:
- DDRVREF_DAC - Stuffs Apple margining circuit.
- VREFDQ:LDO - LDO outputs sent to DQ inputs.
- VREFDQ:LDO_DAC - Margined LDO outputs sent to DQ inputs.
- VREFDQ:M1_M3 - CPU margined DDR voltage divider sent to DQ inputs.
- VREFDQ:M1_DAC - DAC margined DDR voltage divider sent to DQ inputs.
- VREFCA:LDO - LDO outputs sent to CA inputs.
- VREFCA:LDO_DAC - DAC margined LDO outputs sent to CA inputs.

Required zero ohm resistors when no VREF margining circuit stuffed

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
116S0004	2	RES,MTL FILM,0,5%,0402,SM,LF	R3403,R3405		VREFDQ:LDO
116S0004	2	RES,MTL FILM,0,5%,0402,SM,LF	R3409,R3411		VREFCA:LDO

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
114S0218	4	RES,MTL FILM,1K,1%,0402,SM,LF	R3421,R3422,R3441,R3442		VREFDQ:M1_DAC
114S0171	2	RES,MTL FILM,332,1%,0402,SM,LF	R3404,R3406		VREFDQ:M1_DAC

	MEM A VREF DQ	MEM B VREF DQ	MEM A VREF CA	MEM B VREF CA	MEM VREG	GPU Frame Buffer (1.8V, 70% VRef)
DAC Channel:	A	B	C	C	D	D
PCA9557D Pin:	1	2	3	4	5	6
Nominal value	0.75V (DAC: 0x3A)				1.5V (DAC: 0x3A)	1.267V (DAC: 0x8B)
Margined target:	0.300V - 1.200V (+/- 450mV)				1.000V - 2.000V (+/- 500mV)	1.056V - 1.442V (+/- 180mV)
DAC range:	0.000V - 1.501V (0x00 - 0x74)				0.000V - 3.000V (0x00 - 0x74)	0.000V - 3.300V (0x00 - 0xFF)
VRef current:	+3.4mA - -3.4mA (- = sourced)				+61uA - -61uA (- = sourced)	+6.0mA - -5.0mA (- = sourced)
DAC step size:	7.69mV / step @ output				8.59mV / step @ output	1.51mV / step @ output

SYNC_MASTER=J31_ANNE SYNC_DATE=06/09/2011

PAGE TITLE: DDR3/FRAMEBUF VREF MARGINING

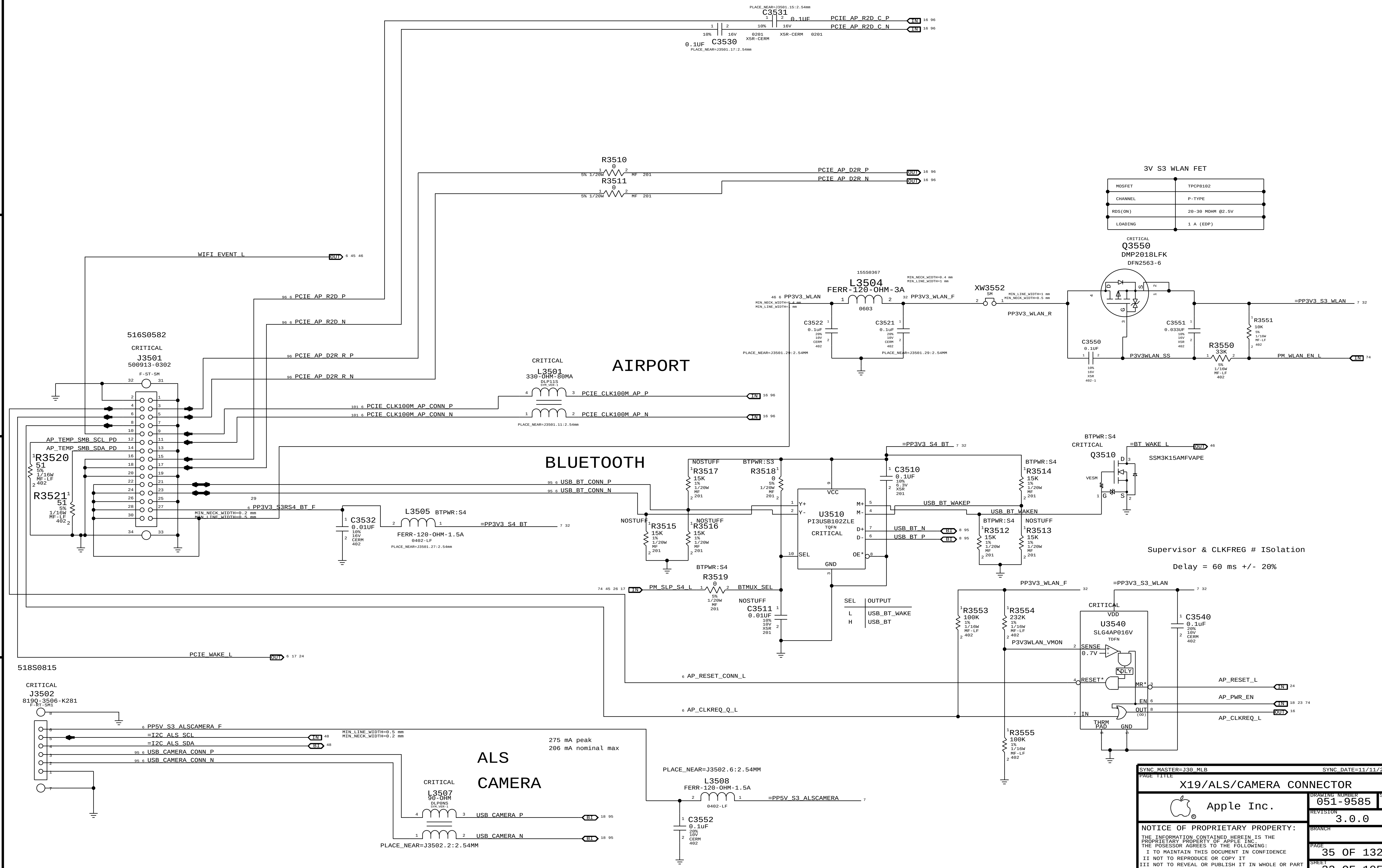
Apple Inc.

DRAWING NUMBER: 051-9585 SIZE: D

REVISION: 3.0.0

BRANCH

8	7	6	5	4	3	2	1



3V S3 WLAN FET

MOSFET	TPCP8102
CHANNEL	P-TYPE
RDS(ON)	20-30 MOHM @2.5V
LOADING	1 A (EDP)

SYNC_MASTER=J30_MLB SYNC_DATE=11/11/2011

PAGE TITLE

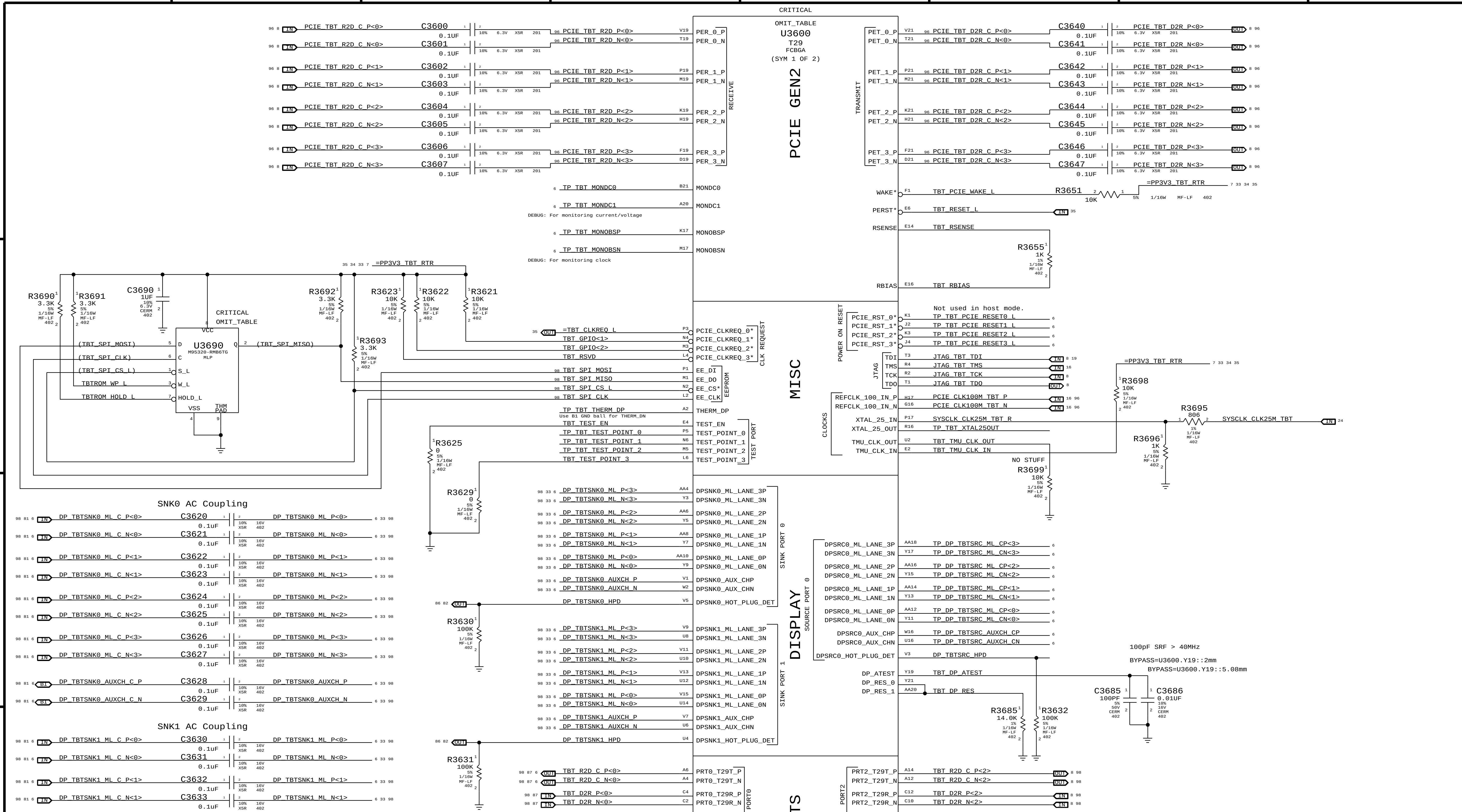
X19/ALS/CAMERA CONNECTOR

Apple Inc.

DRAWING NUMBER 051-9585 SIZE D

REVISION 3.0.0

BRANCH



SYNC_MASTER=T29_REF SYNC_DATE=06/14/2011

PAGE TITLE

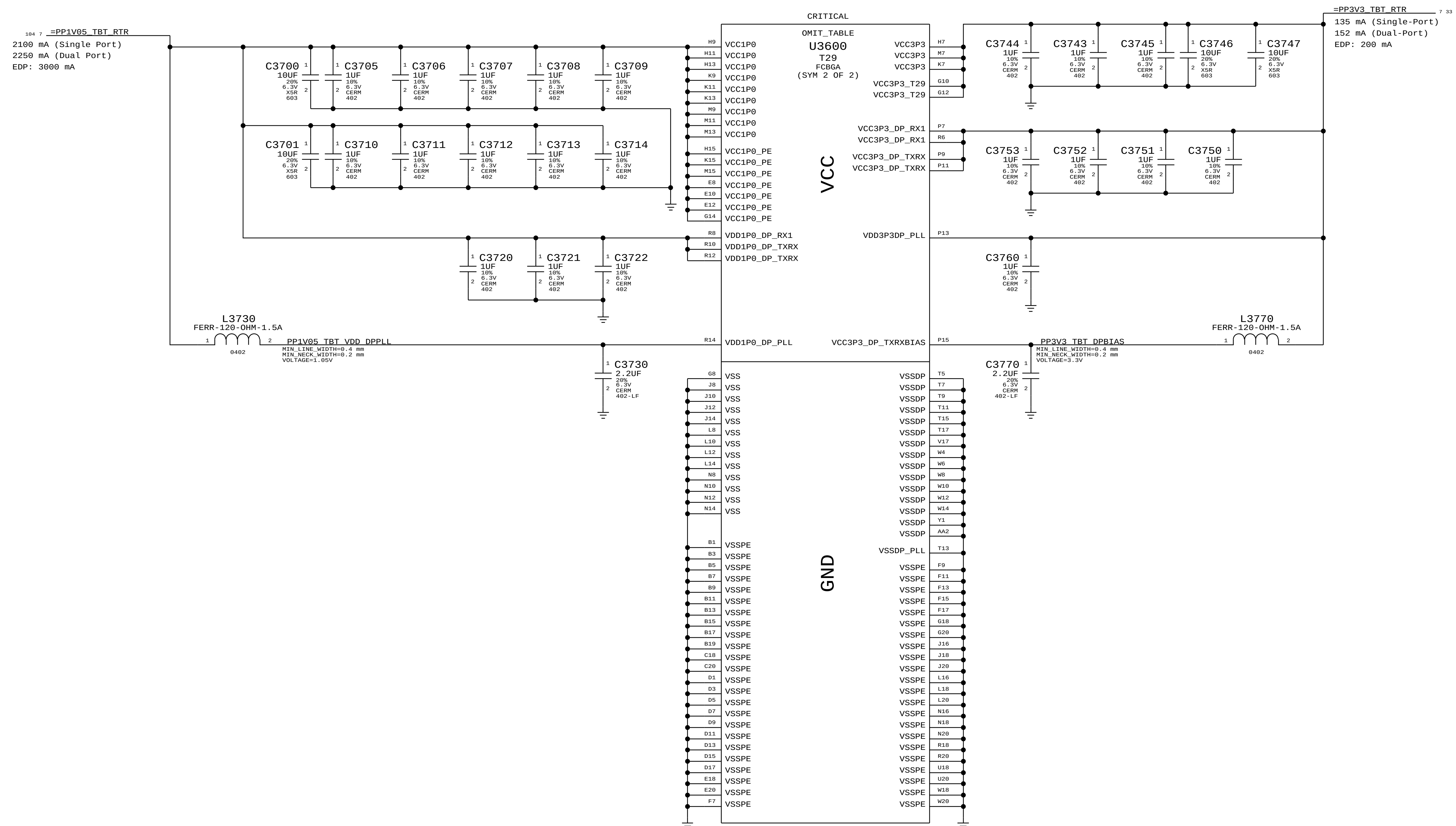
Thunderbolt Host (1 of 2)

Apple Inc.

DRAWING NUMBER 051-9585 SIZE D

REVISION 3.0.0

BRANCH



Current numbers from Vendor slide (<REDACTED> power measure 1.ppt), emailed 6/21/2010, TDP @ 90C.

SYNC_MASTER=T29_REF SYNC_DATE=06/14/2011

PAGE TITLE

Thunderbolt Host (2 of 2)

Apple Inc.

DRAWING NUMBER 051-9585 SIZE D

REVISION 3.0.0

BRANCH

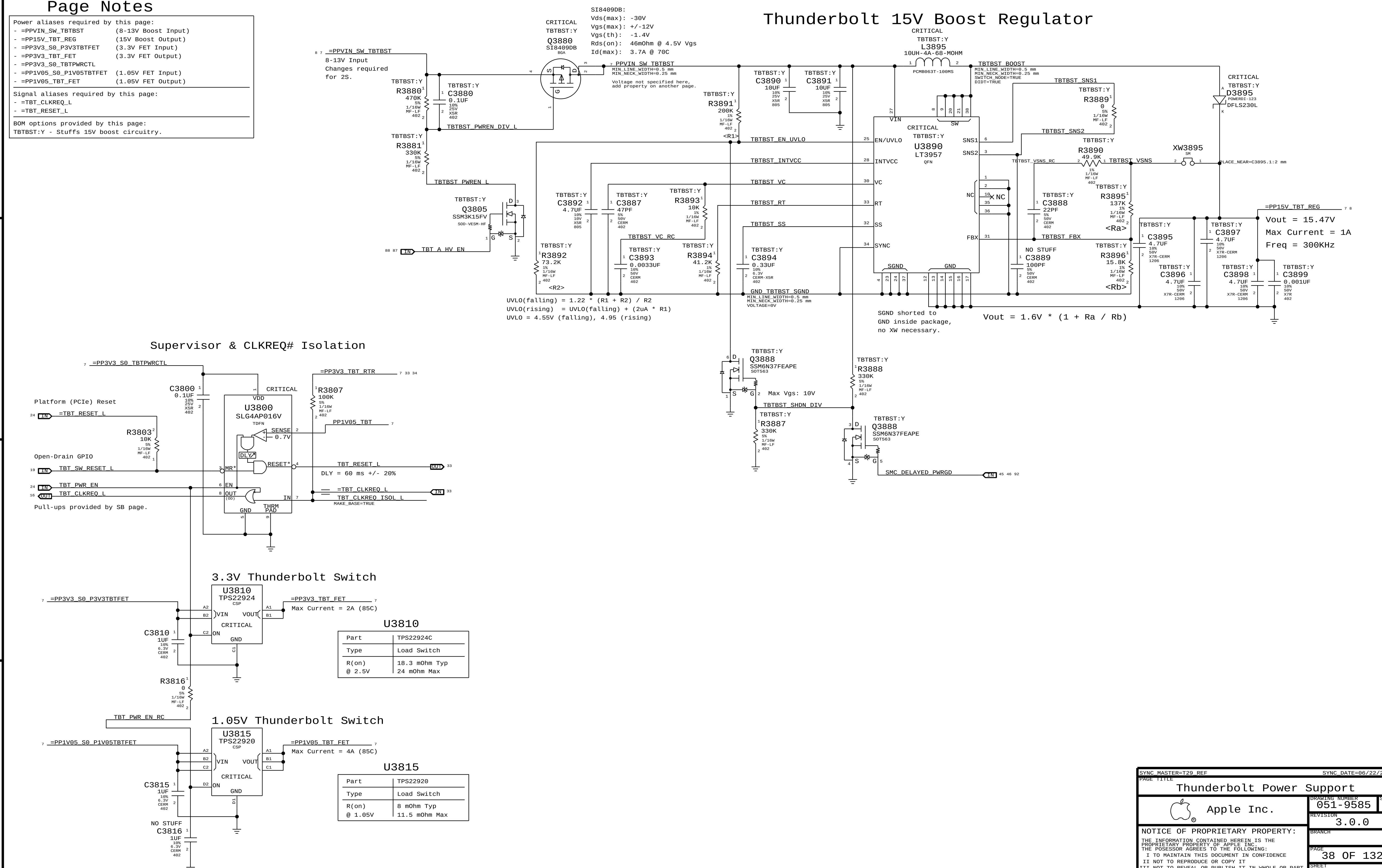
Page Notes

Power aliases required by this page:
- =PPVIN_SW_TBTBST (8-13V Boost Input)
- =PP15V_TBT_REG (15V Boost Output)
- =PP3V3_S0_P3V3TBTFET (3.3V FET Input)
- =PP3V3_TBT_FET (3.3V FET Output)
- =PP3V3_S0_TBTPWRCTL
- =PP1V05_S0_P1V05TBTFET (1.05V FET Input)
- =PP1V05_TBT_FET (1.05V FET Output)

Signal aliases required by this page:
- =TBT_CLKREQ_L
- =TBT_RESET_L

BOM options provided by this page:
TBTBST:Y - Stuffs 15V boost circuitry.

Thunderbolt 15V Boost Regulator



U3810

Part	TPS22924C
Type	Load Switch
R(on) @ 2.5V	18.3 mOhm Typ 24 mOhm Max

U3815

Part	TPS22920
Type	Load Switch
R(on) @ 1.05V	8 mOhm Typ 11.5 mOhm Max

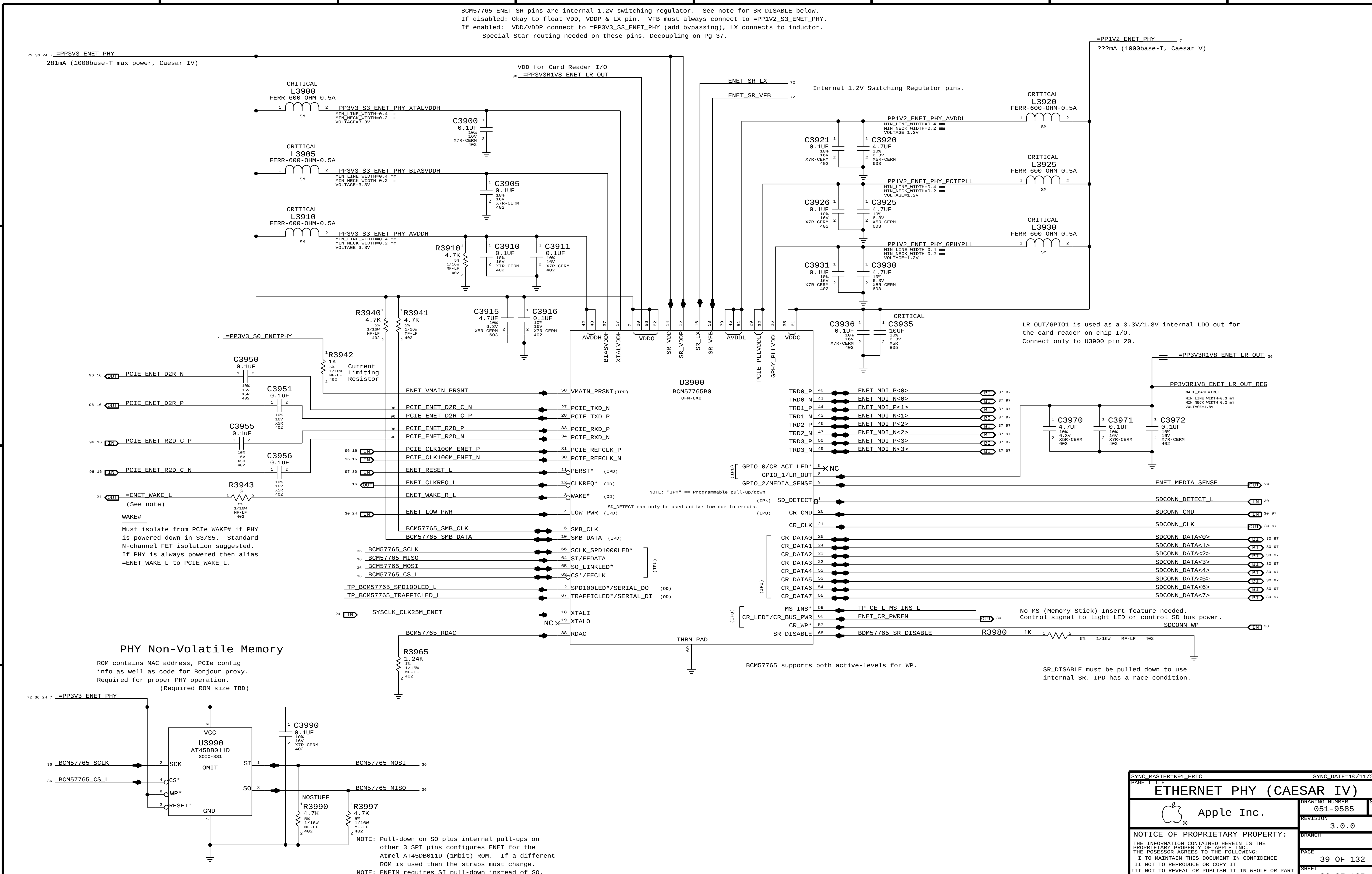
SYNC_MASTER=T29_REF SYNC_DATE=06/22/2011

PAGE TITLE: Thunderbolt Power Support

Apple Inc.

DRAWING NUMBER 051-9585 SIZE D

REVISION 3.0.0



SYNC_MASTER=K91_ERIC SYNC_DATE=10/11/2010

PAGE TITLE

ETHERNET PHY (CAESAR IV)

Apple Inc.

DRAWING NUMBER 051-9585 SIZE D

REVISION 3.0.0

BRANCH

NOTICE OF PROPRIETARY PROPERTY:

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC.
THE POSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

PAGE 39 OF 132

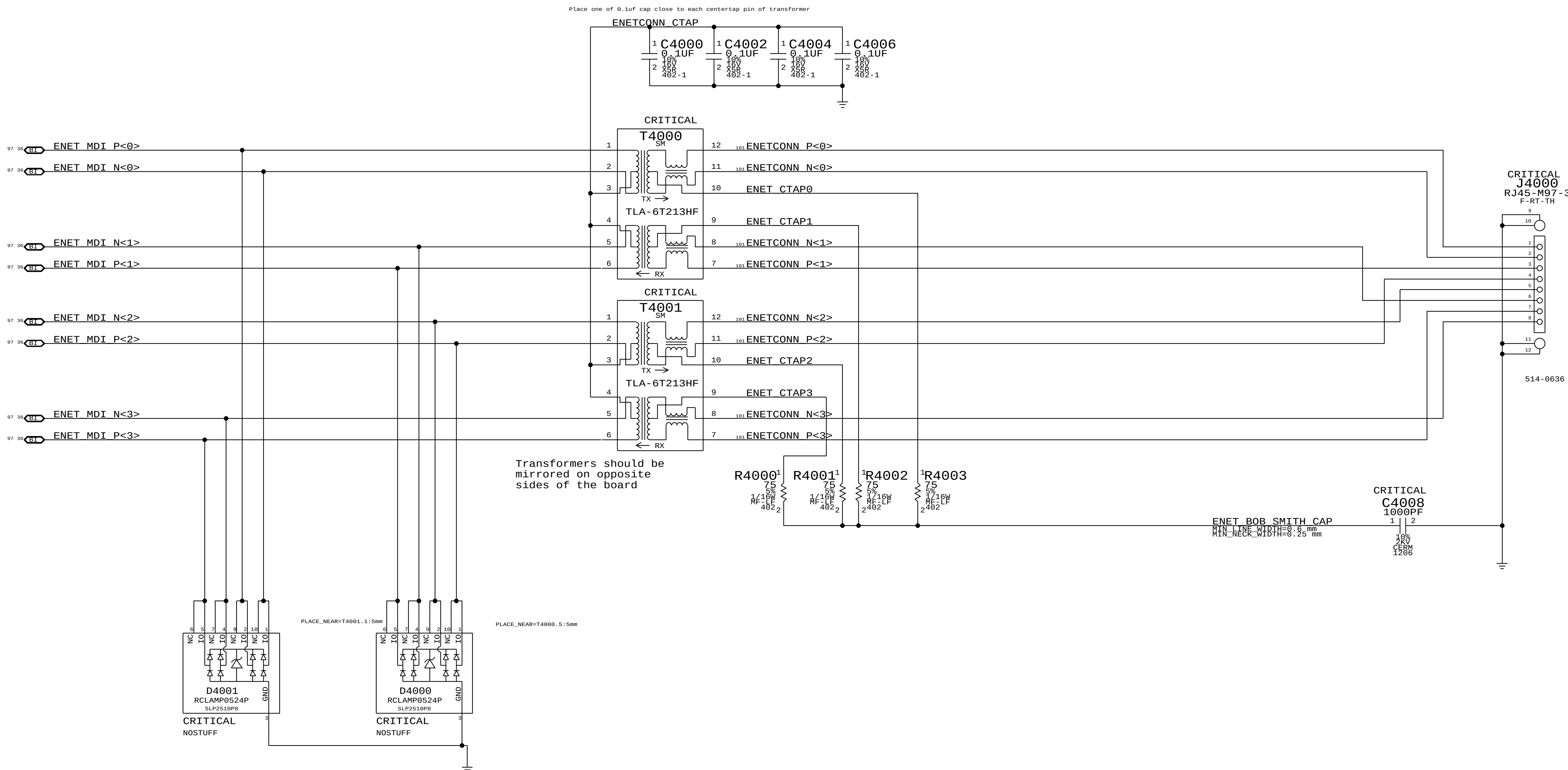
SHEET 36 OF 105

Page Notes

Power aliases required by this page:
(NONE)

Signal aliases required by this page:
(NONE)

BOM options provided by this page:
(NONE)



SYNC_MASTER=K91_TRINHNI SYNC_DATE=05/26/2010

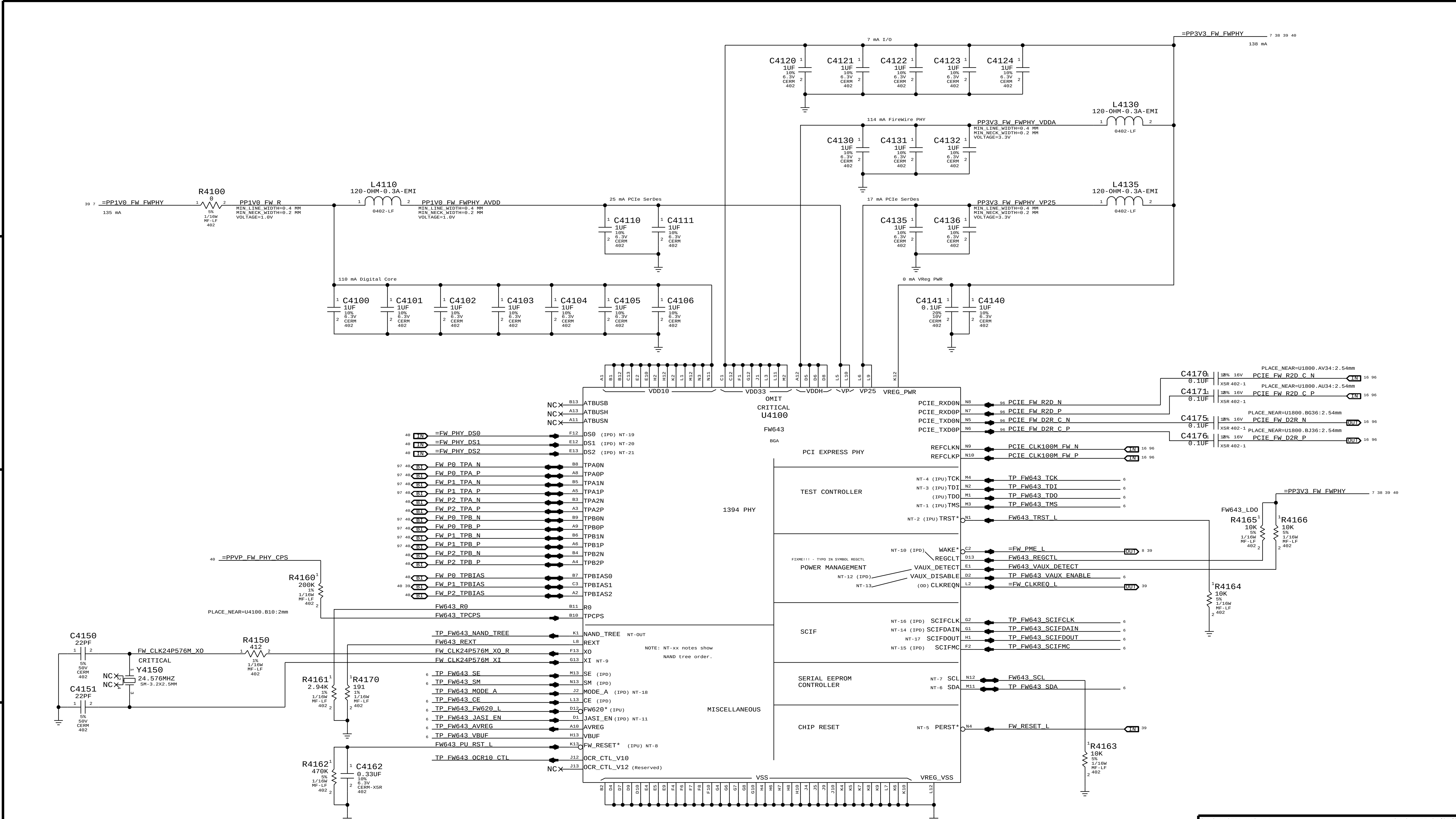
PAGE TITLE
Ethernet Connector

Apple Inc.

DRAWING NUMBER 051-9585 SIZE D

REVISION 3.0.0

BRANCH



SYNC_MASTER=K18_MLB SYNC_DATE=04/27/2010

PAGE TITLE

FireWire LLC/PHY (FW643)

Apple Inc.

DRAWING NUMBER	SIZE
051-9585	D

REVISION 3.0.0

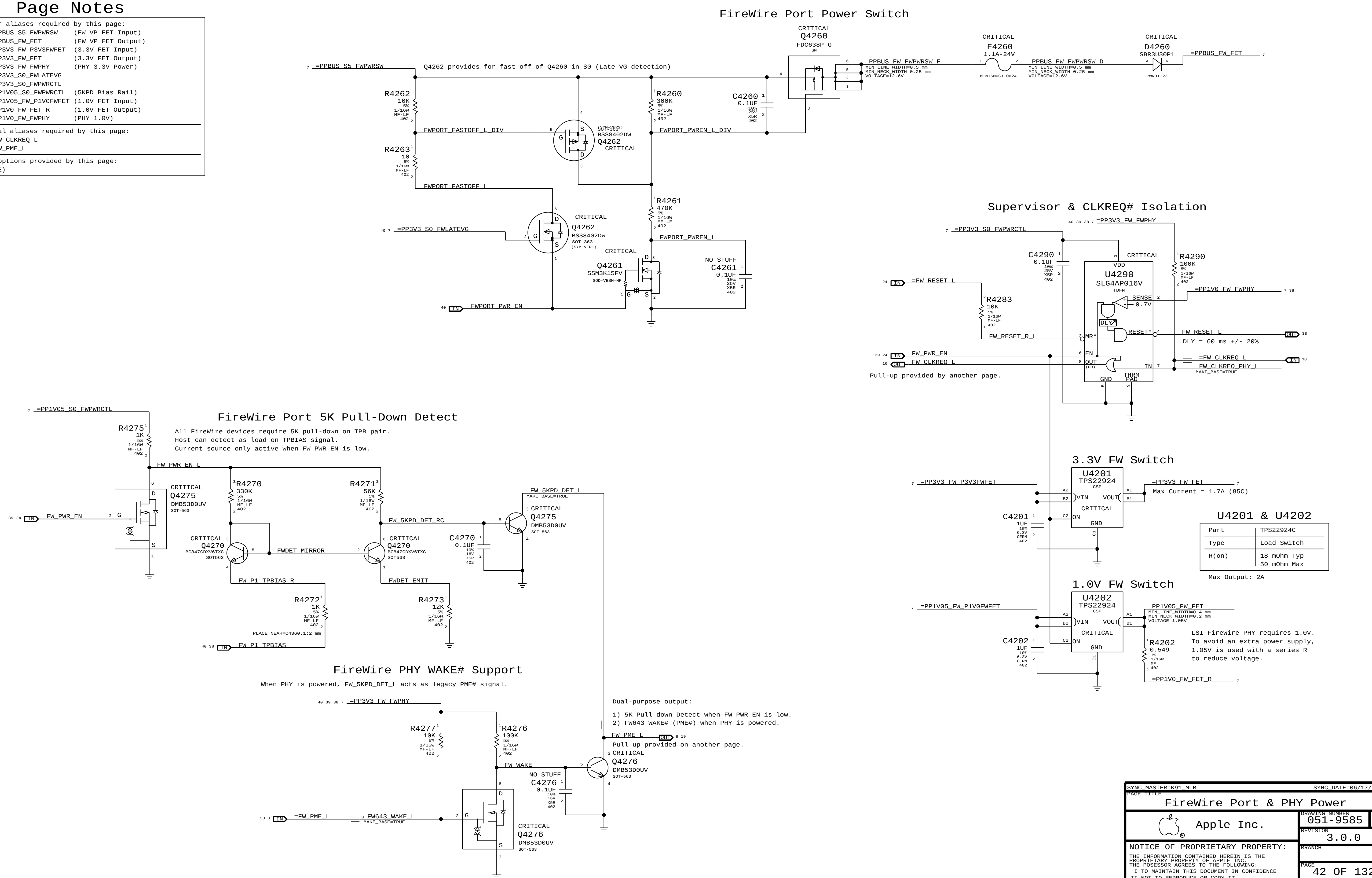
BRANCH

Page Notes

Power aliases required by this page:
- =PPBUS_S5_FWPWRSW (FW VP FET Input)
- =PPBUS_FW_FET (FW VP FET Output)
- =PP3V3_FW_P3V3FWFET (3.3V FET Input)
- =PP3V3_FW_FET (3.3V FET Output)
- =PP3V3_FW_FWPHY (PHY 3.3V Power)
- =PP3V3_S0_FWLATEVG
- =PP3V3_S0_FWPWRCTL
- =PP1V05_S0_FWPWRCTL (5KPD Bias Rail)
- =PP1V05_FW_P1V0FWFET (1.0V FET Input)
- =PP1V0_FW_FET_R (1.0V FET Output)
- =PP1V0_FW_FWPHY (PHY 1.0V)

Signal aliases required by this page:
- =FW_CLKREQ_L
- =FW_PME_L

BOM options provided by this page:
(NONE)



SYNC_MASTER=K91_MLB SYNC_DATE=06/17/2011

PAGE TITLE: FireWire Port & PHY Power

Apple Inc.

DRAWING NUMBER: 051-9585 SIZE: D

REVISION: 3.0.0

BRANCH

Page Notes

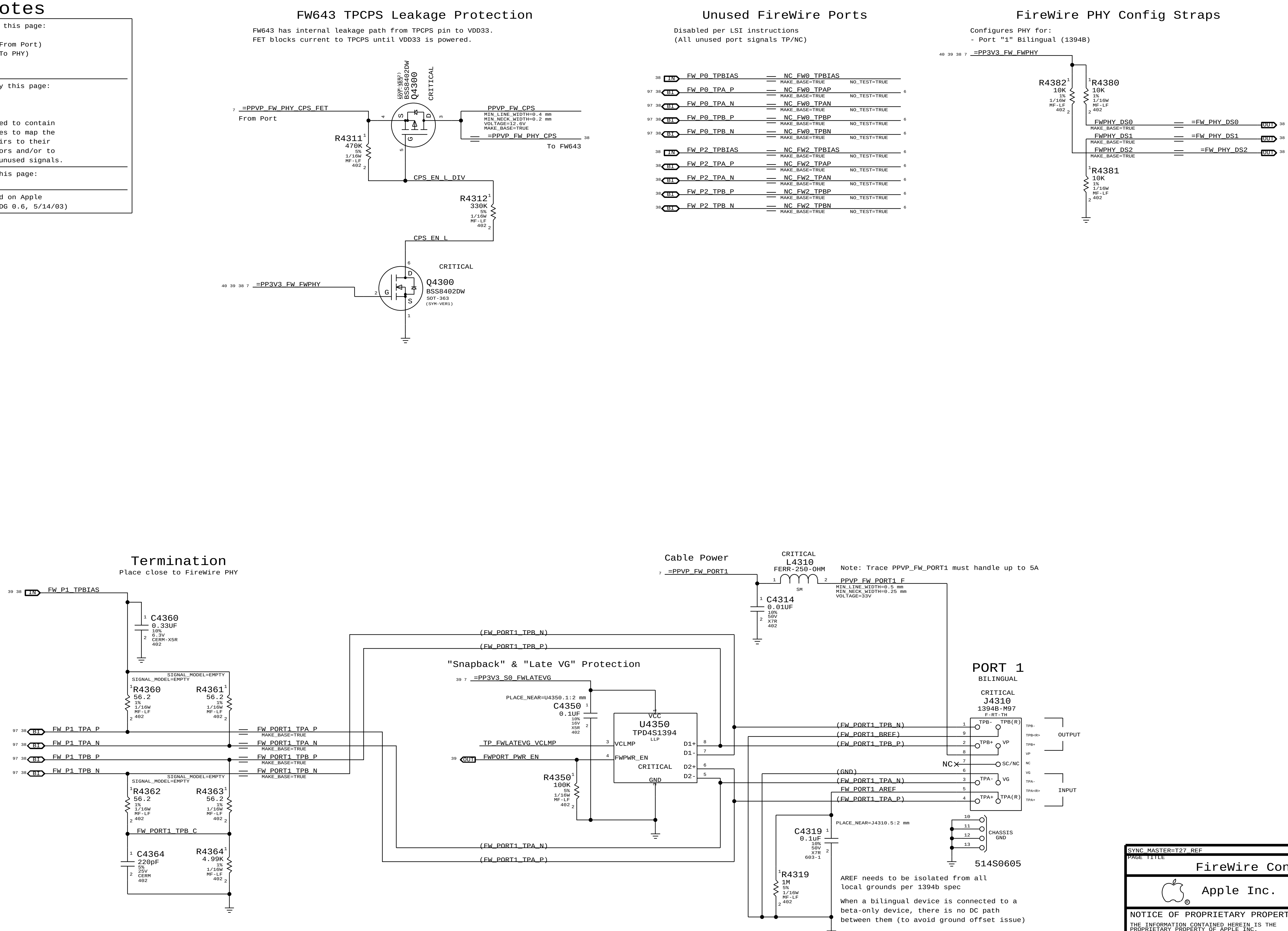
Power aliases required by this page:
- =PPVP_FW_PORT1
- =PPVP_FW_PHY_CPS_FET (From Port)
- =PPVP_FW_PHY_CPS (To PHY)
- =PP3V3_FW_FWPHY
- =PP3V3_S0_FWLATEVG

Signal aliases required by this page:
- =FW_PHY_DS0
- =FW_PHY_DS1
- =FW_PHY_DS2

NOTE: This page is expected to contain the necessary aliases to map the FireWire TPA/TPB pairs to their appropriate connectors and/or to properly terminate unused signals.

BOM options provided by this page:
(NONE)

1394b implementation based on Apple FireWire Design Guide (FWDG 0.6, 5/14/03)



SYNC_MASTER=T27_REF SYNC_DATE=06/10/2010

PAGE TITLE: FireWire Connector

Apple Inc.

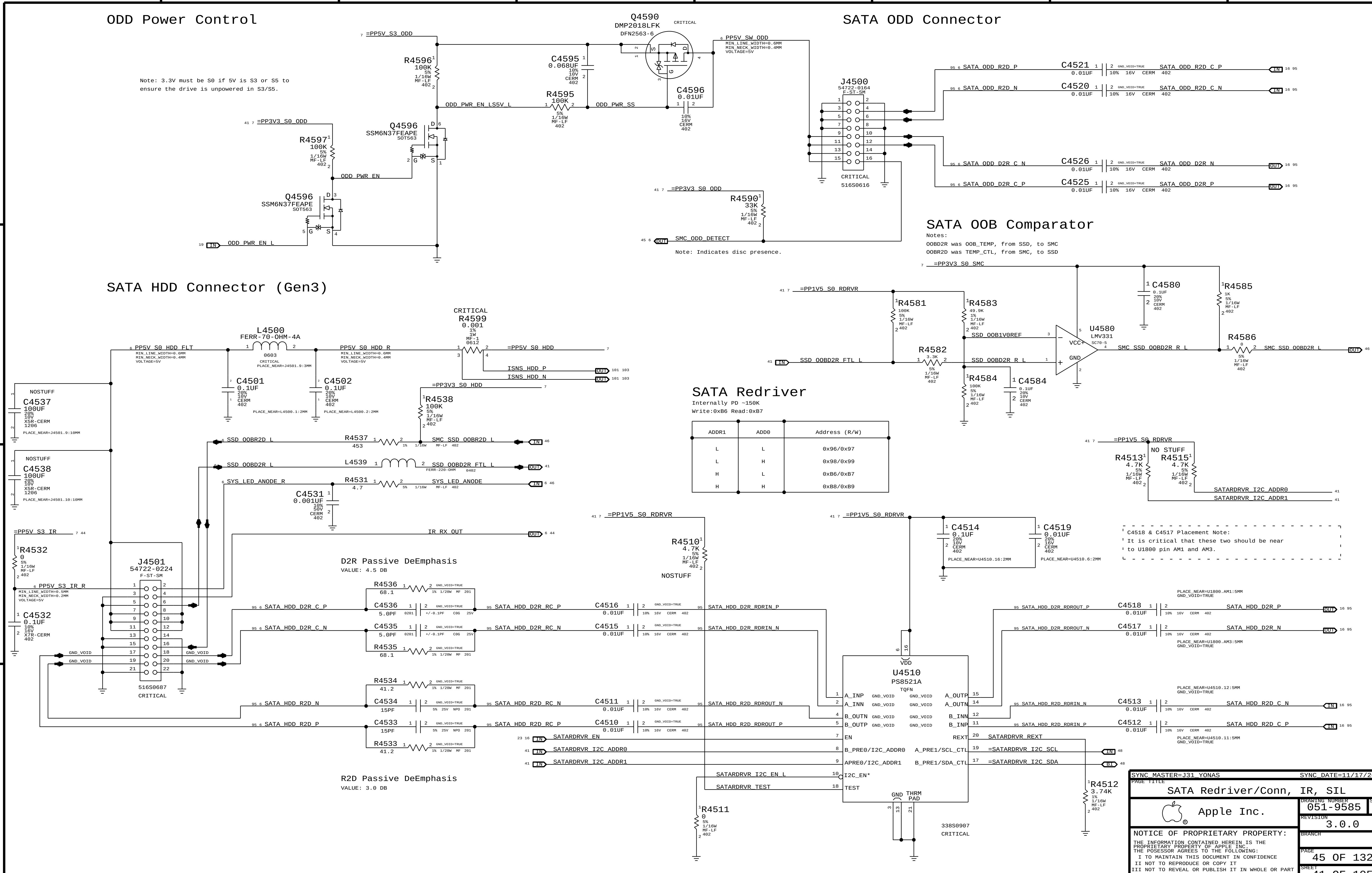
DRAWING NUMBER: 051-9585 SIZE: D

REVISION: 3.0.0

BRANCH

PAGE: 43 OF 132

SHEET: 40 OF 105



ADDR1	ADD0	Address (R/W)
L	L	0x96/0x97
L	H	0x98/0x99
H	L	0xB6/0xB7
H	H	0xB8/0xB9

SYNC_MASTER=J31_YONAS SYNC_DATE=11/17/2011

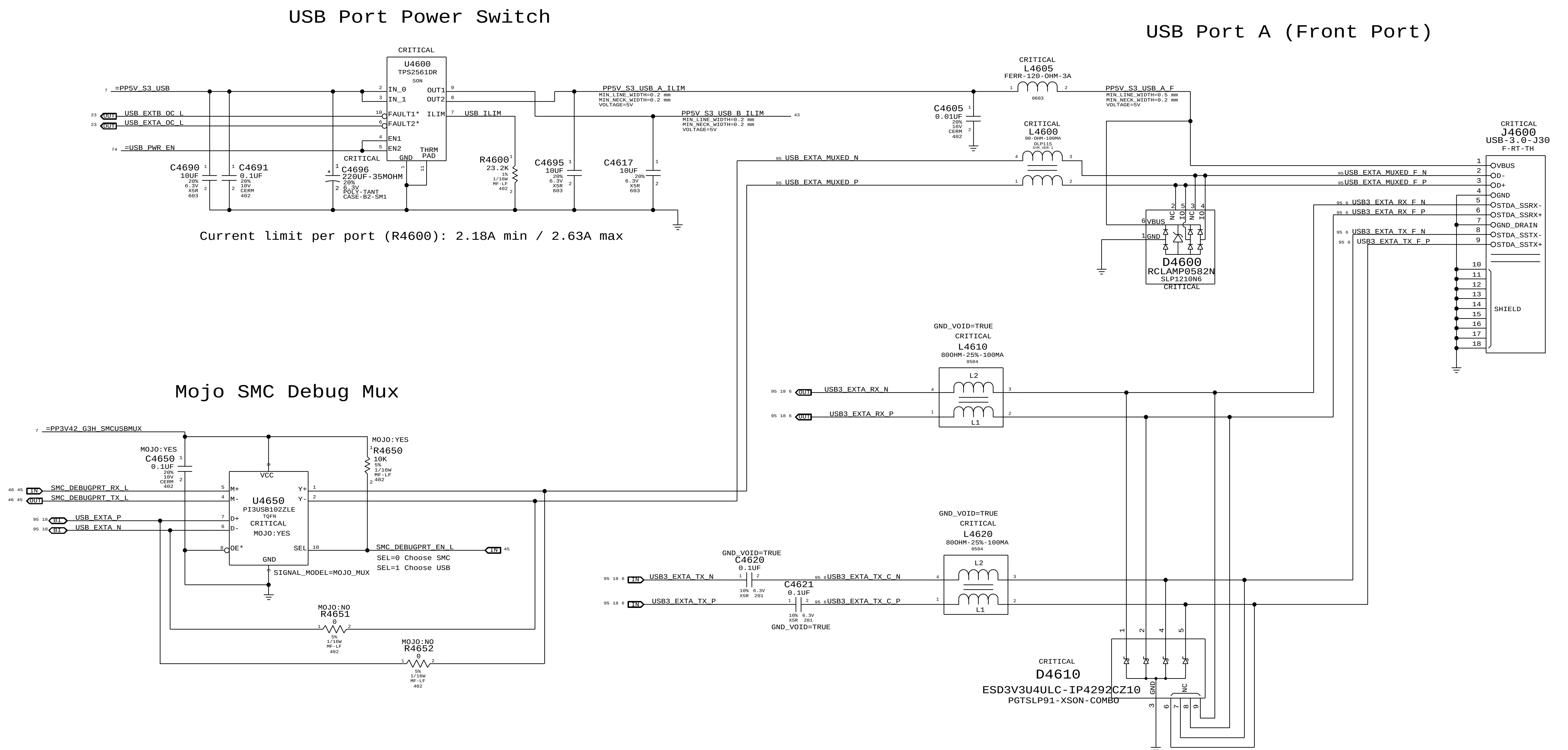
PAGE TITLE: SATA Redriver/Conn, IR, SIL

Apple Inc.

DRAWING NUMBER: 051-9585 SIZE: D

REVISION: 3.0.0

BRANCH



SYNC_MASTER=J31_LINDA SYNC_DATE=09/21/2011

PAGE TITLE

External A USB3 Connector

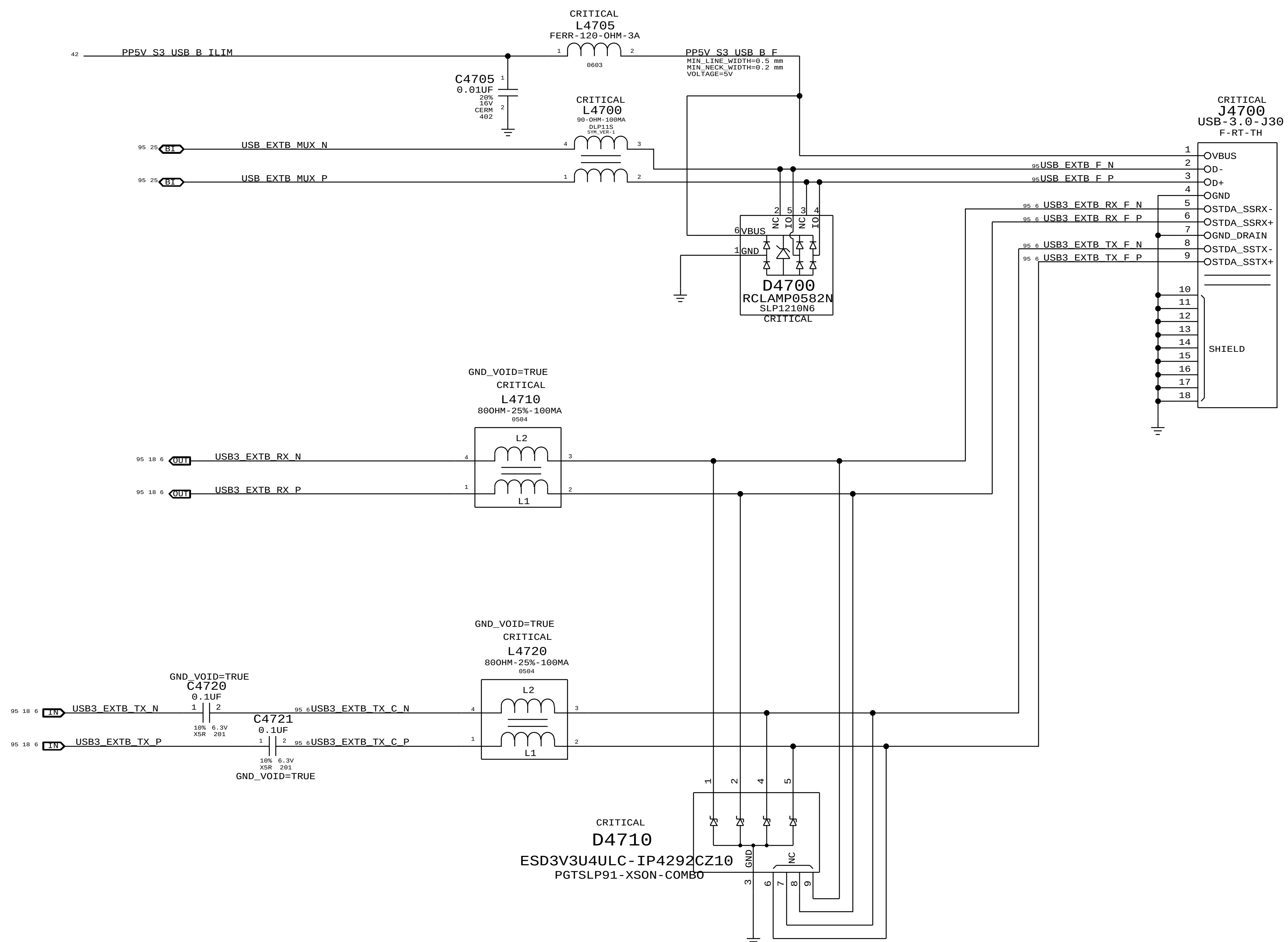
Apple Inc.

DRAWING NUMBER 051-9585 SIZE D

REVISION 3.0.0

BRANCH

USB Port B (Back Port)



NOTE: Swapped pin4 and 5, pin6 and 7 for layout.

SYNC_MASTER=J30_MLB SYNC_DATE=08/04/2011

PAGE TITLE: External B USB3 Connector

Apple Inc.

DRAWING NUMBER: 051-9585 SIZE: D

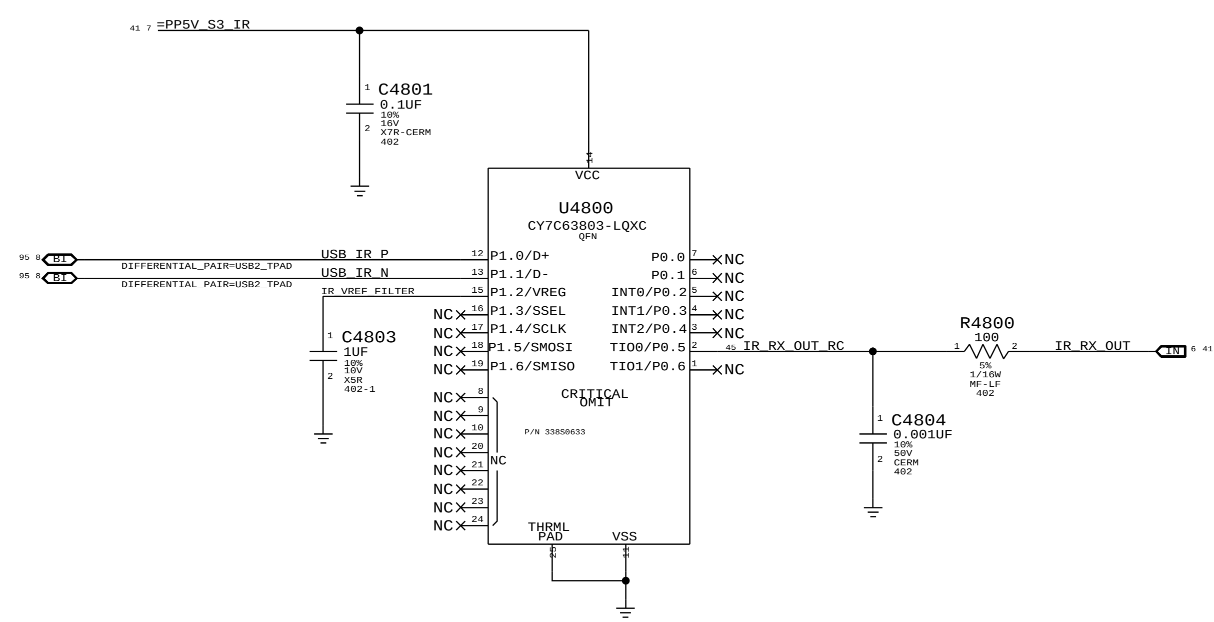
REVISION: 3.0.0

BRANCH

PAGE: 47 OF 132

SHEET: 43 OF 105

IR SUPPORT



SYNC_MASTER=K18_MLB SYNC_DATE=04/27/2010

PAGE TITLE

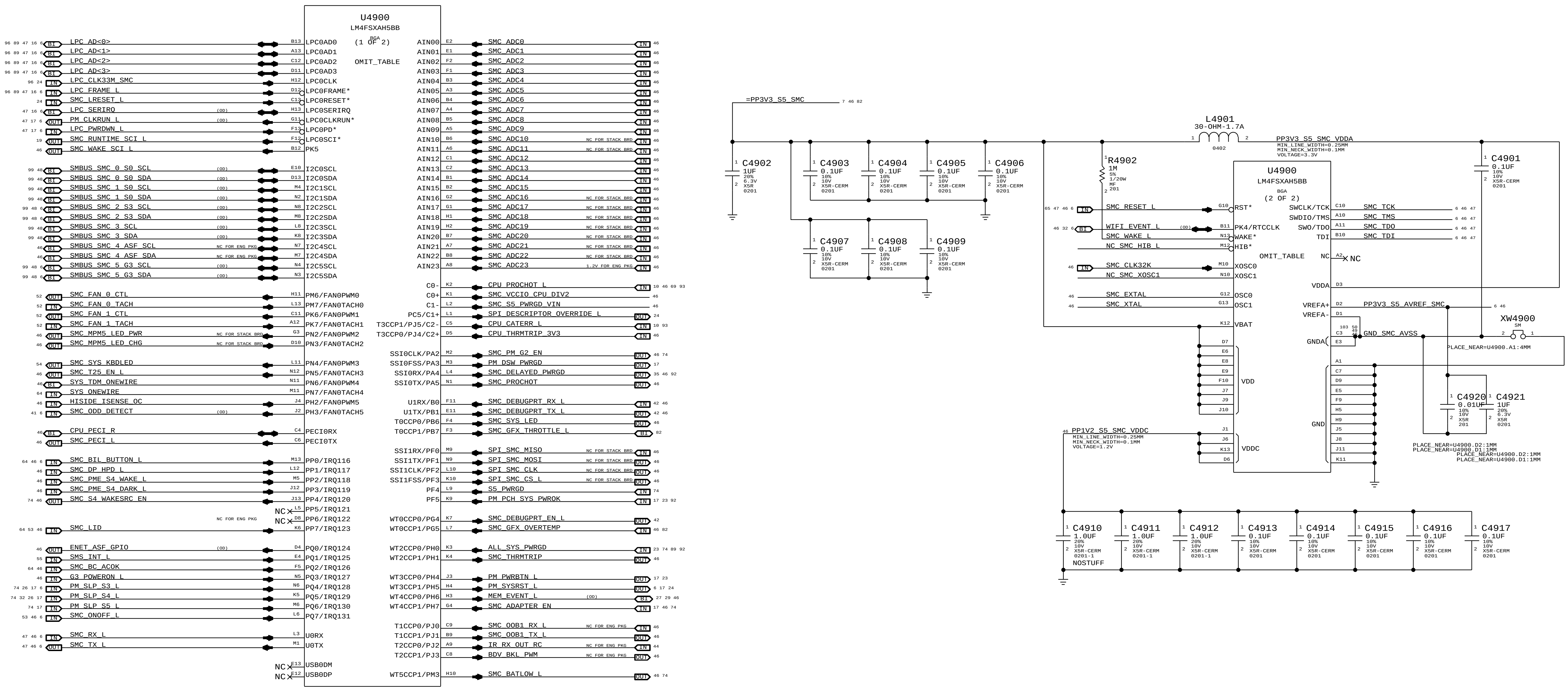
Front Flex Support

Apple Inc.

DRAWING NUMBER 051-9585 SIZE D

REVISION 3.0.0

BRANCH



NOTE:
SMS Interrupt can be active high or low, rename net accordingly.
If SMS interrupt is not used, pull up to SMC rail.

NOTE:
Unused pins have "SMC_Pxx" names. Unused
pins designed as outputs can be left floating,
those designated as inputs require pull-ups.

SYNC_MASTER=J31_YONAS SYNC_DATE=12/19/2011

PAGE TITLE

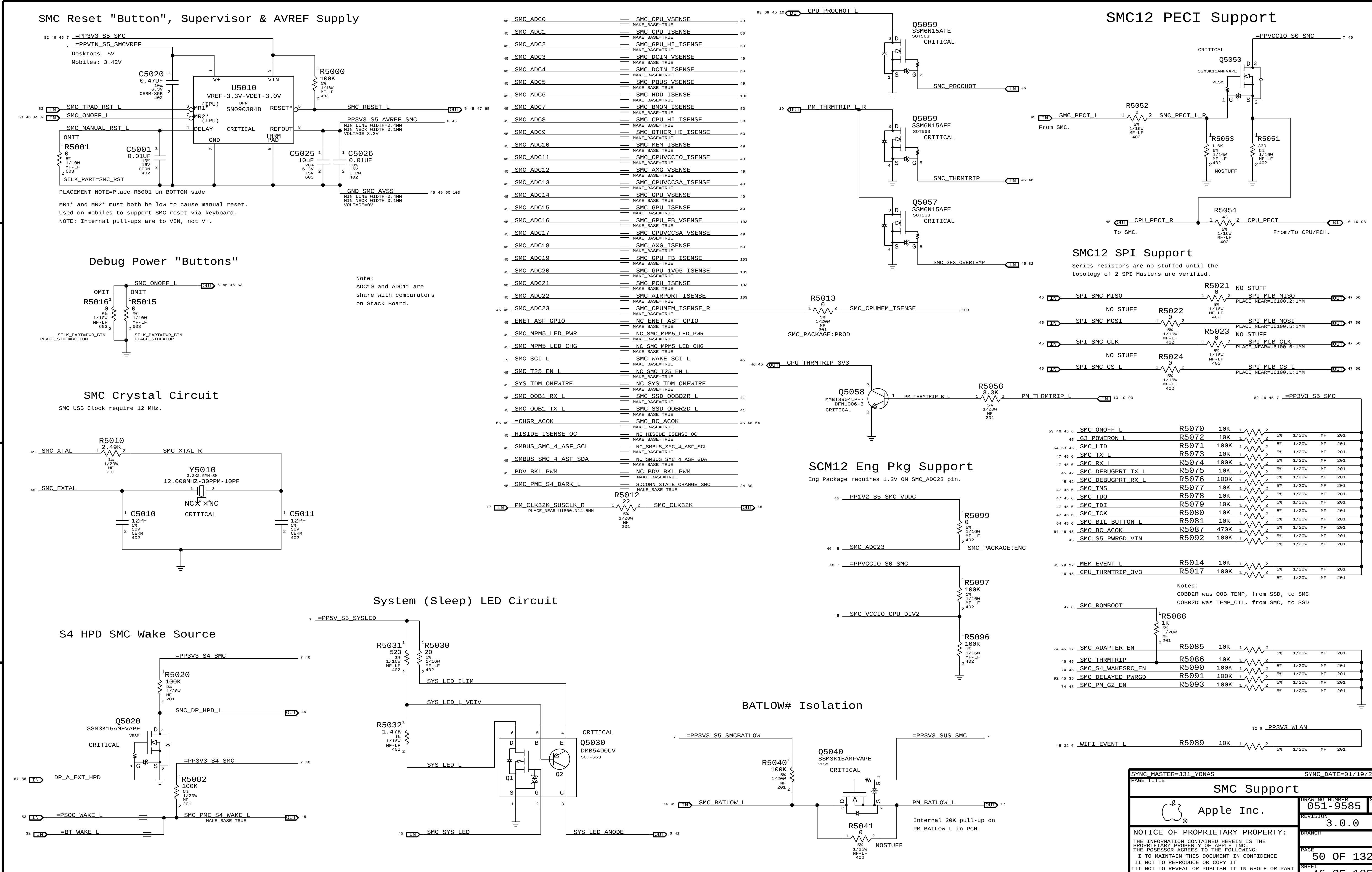
SMC

Apple Inc.

DRAWING NUMBER 051-9585 SIZE D

REVISION 3.0.0

BRANCH



SYNC_MASTER=J31_YONAS SYNC_DATE=01/19/2012

PAGE TITLE

SMC Support

Apple Inc.

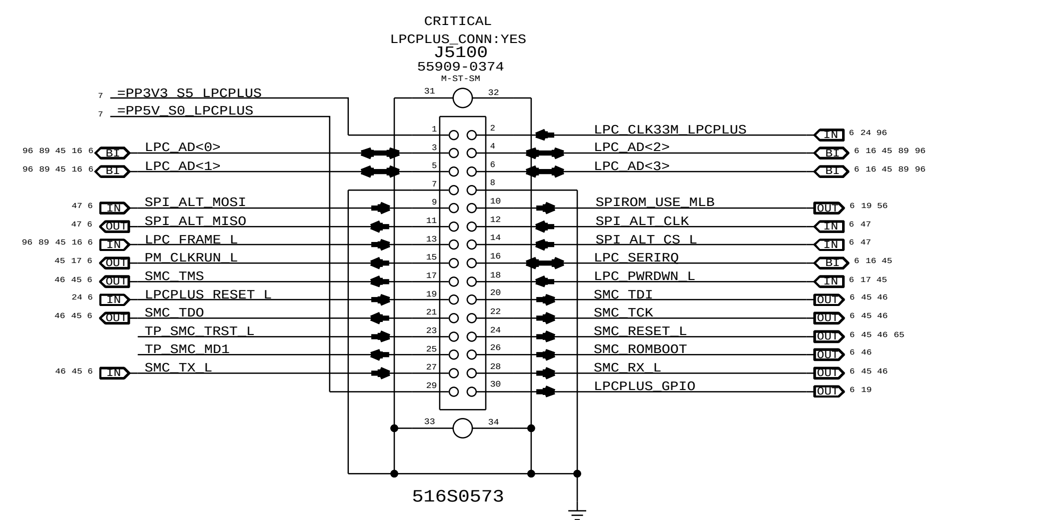
DRAWING NUMBER 051-9585 SIZE D

REVISION 3.0.0

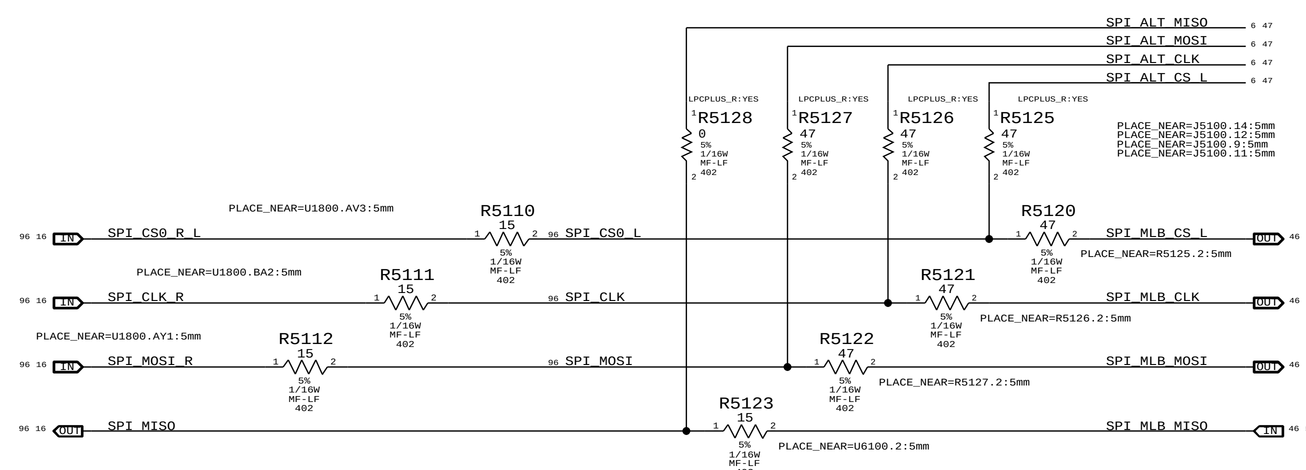
BRANCH

NOTICE OF PROPRIETARY PROPERTY:

LPC+SPI Connector



SPI Bus Series Termination



SYNC_MASTER=J5_MLB SYNC_DATE=05/26/2011

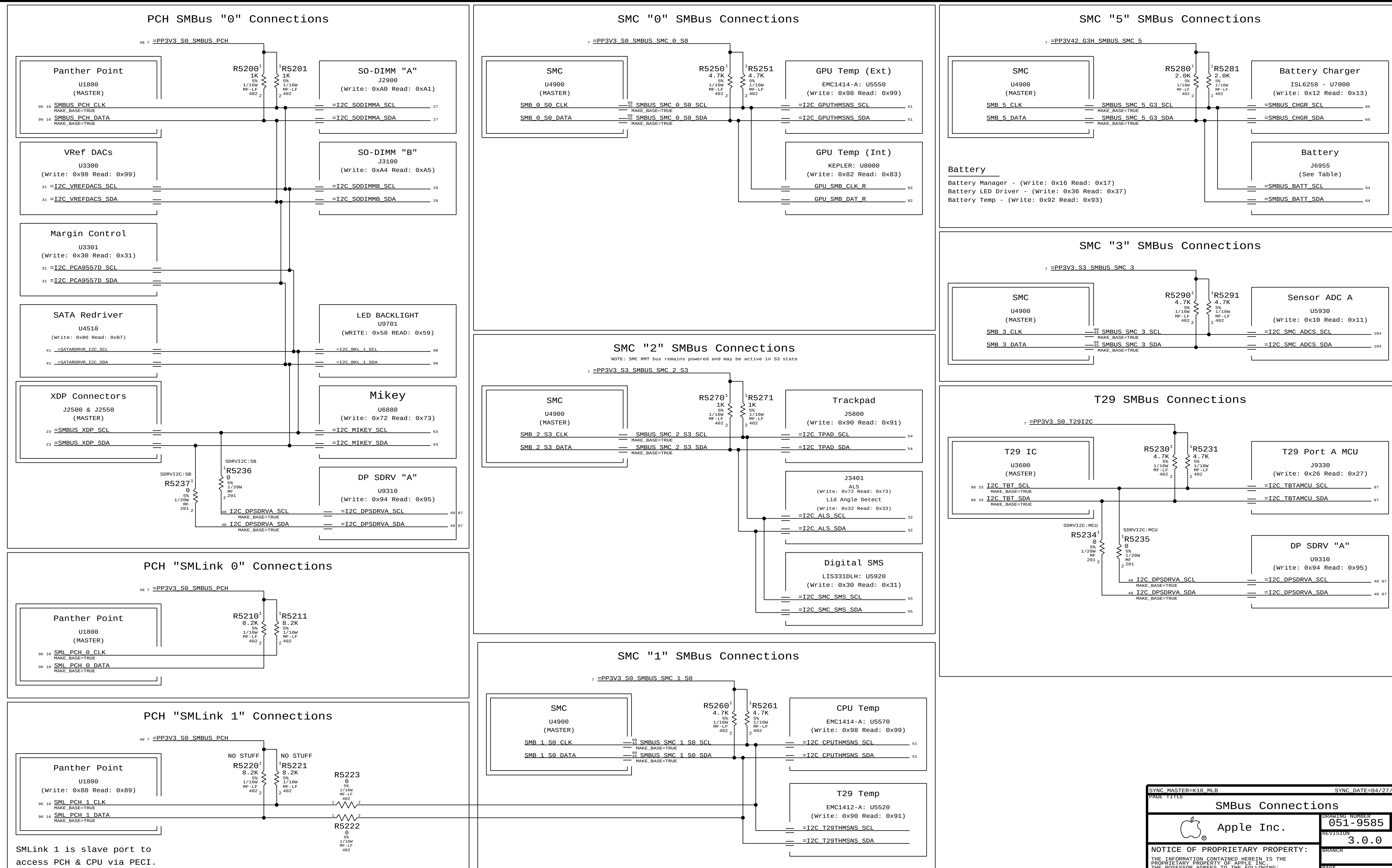
PAGE TITLE: LPC+SPI Debug Connector

Apple Inc.

DRAWING NUMBER: 051-9585 SIZE: D

REVISION: 3.0.0

BRANCH



SYNC_MASTER=K18_MLB SYNC_DATE=04/27/2010

PAGE TITLE

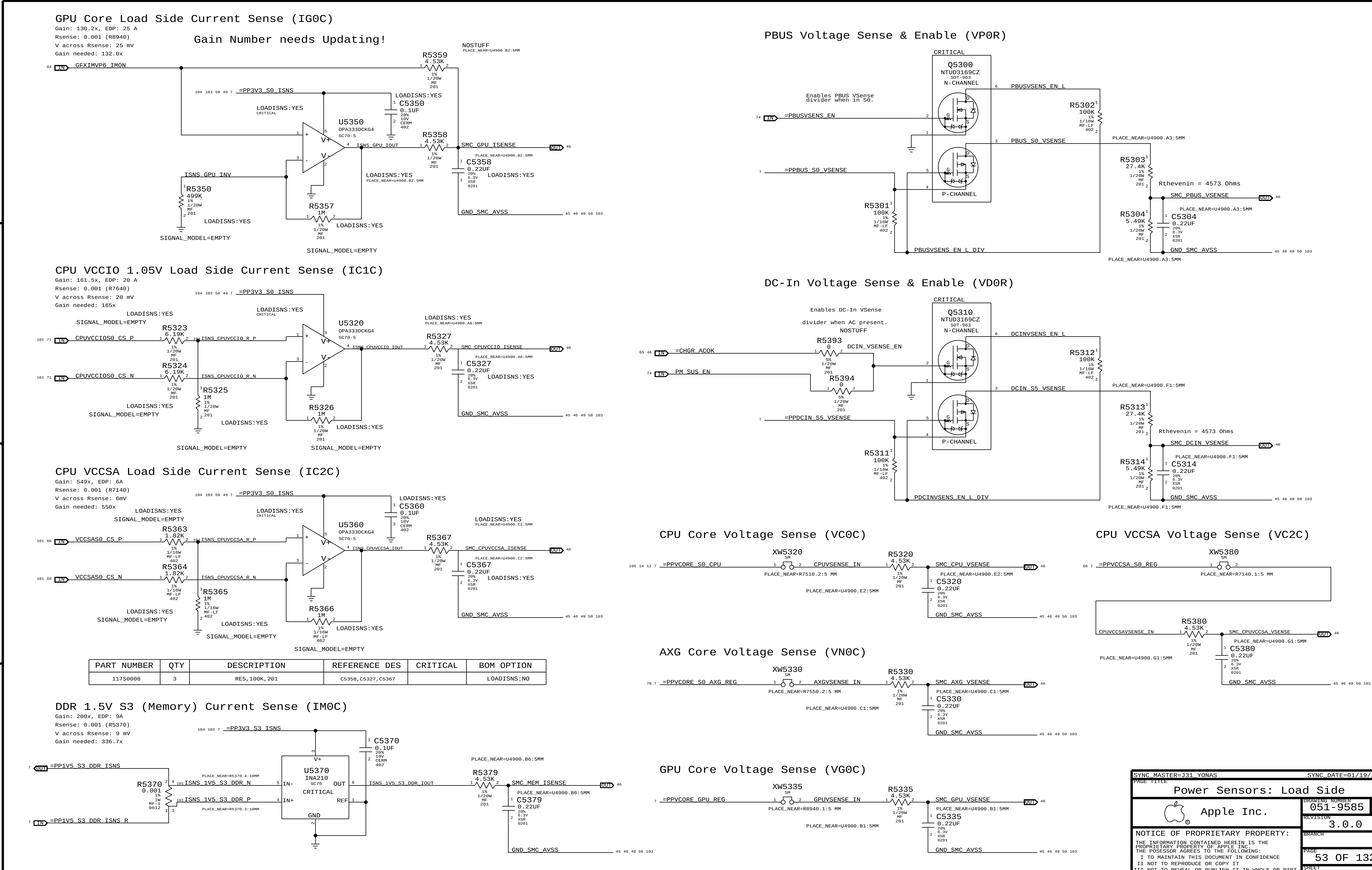
SMBus Connections

Apple Inc.

DRAWING NUMBER 051-9585 SIZE D

REVISION 3.0.0

BRANCH



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
117S0008	3	RES,100K,201	C5358,C5327,C5367		LOADISNS:NO

SYNC_MASTER=J31_YONAS SYNC_DATE=01/19/2012

PAGE TITLE: Power Sensors: Load Side

Apple Inc.

DRAWING NUMBER: 051-9585 SIZE: D

REVISION: 3.0.0

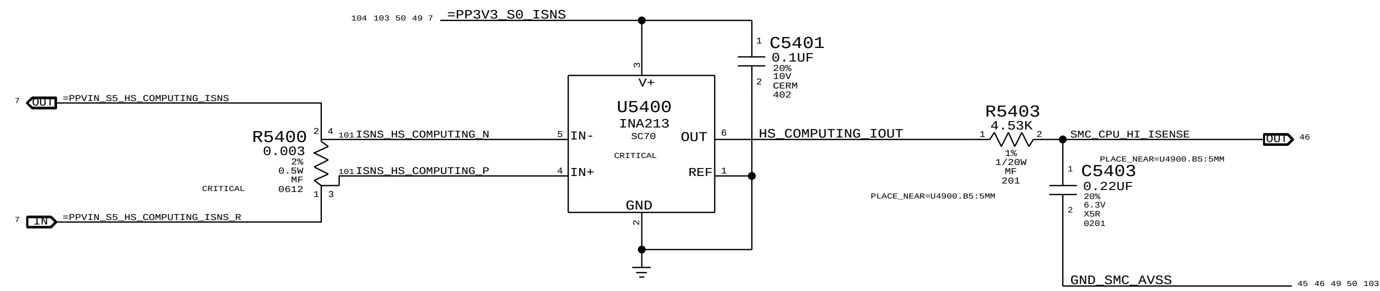
BRANCH

PAGE: 53 OF 132

SHEET: 49 OF 105

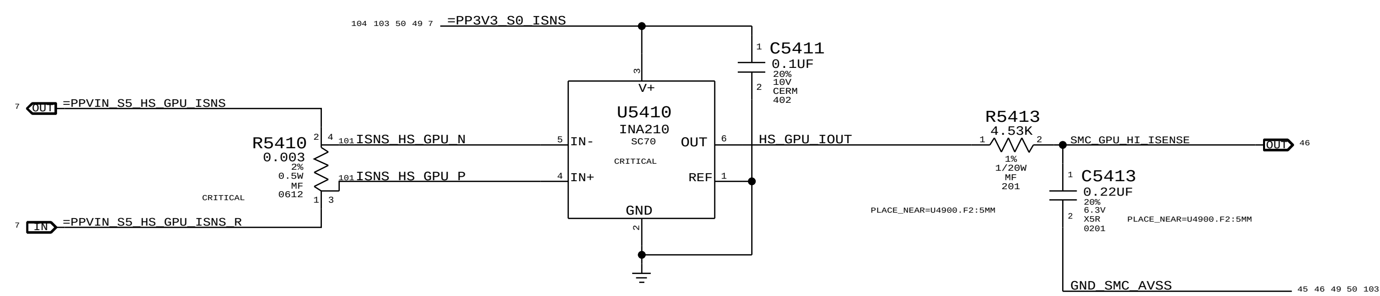
CPU High Side Current Sense (IC0R)

Gain: 50x, EDP: 22.8 A
Rsense: 0.003 (R5400)
V across Rsense: 68.4 mV
Gain needed: 48.25x



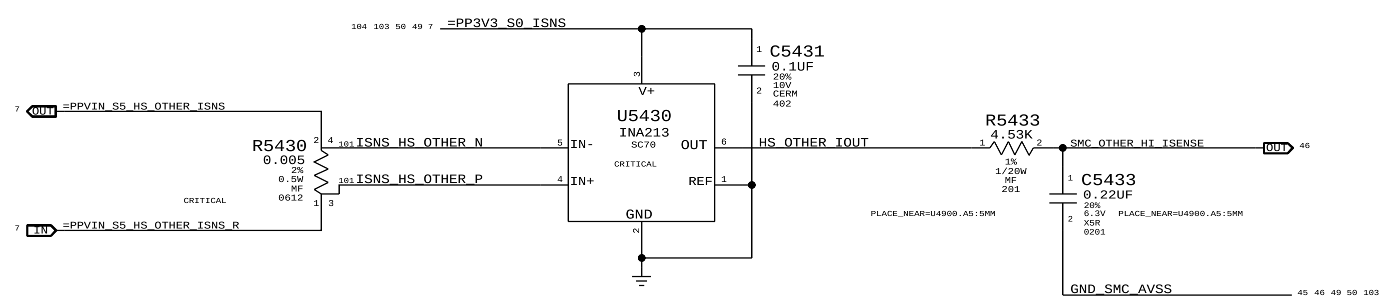
GPU High Side Current Sense (IG0R)

Gain: 200x, EDP: 5.2 A (Kepler)
Rsense: 0.003 (R5410)
V across Rsense: 15.6 mV
Gain needed: 211.54x (Kepler)



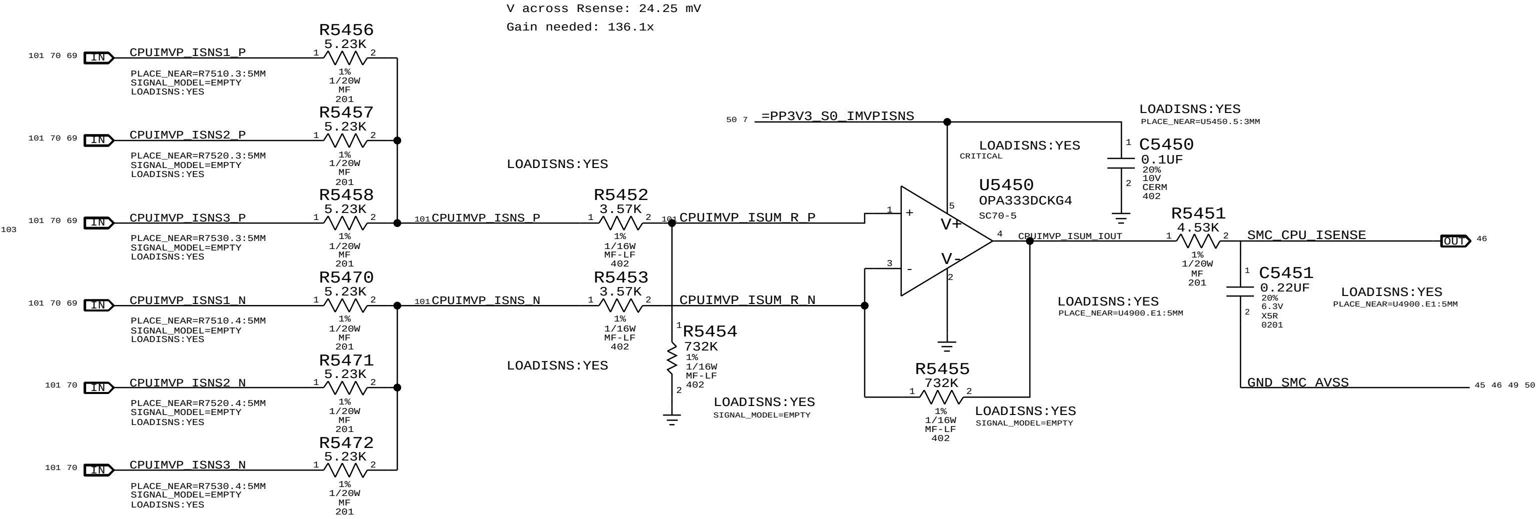
OTHER High Side Current Sense (IO0R)

Gain: 50x, EDP: 10.3 A
Rsense: 0.005 (R5440)
V across Rsense: 51.5 mV
Gain needed: 64.1x



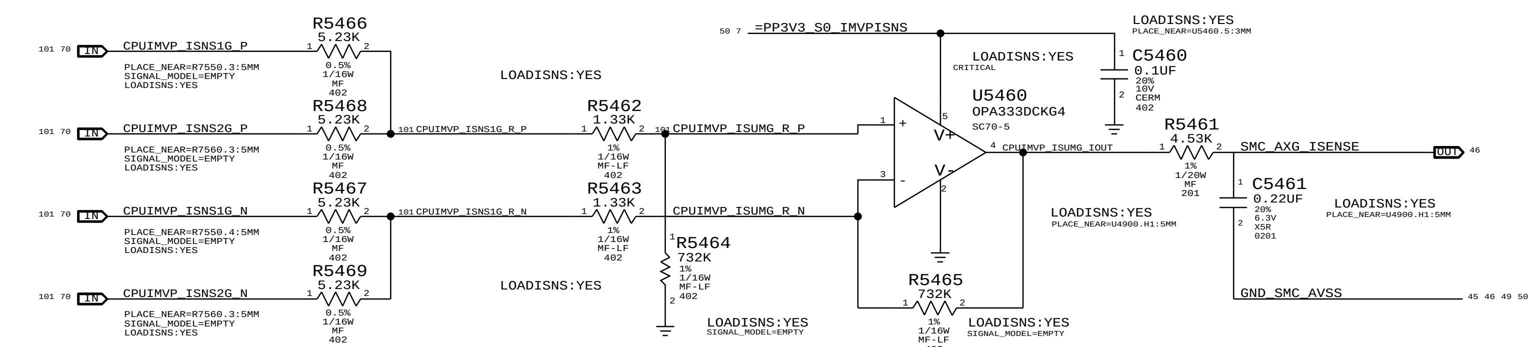
CPU Core Load Side Current Sense (IC0C)

Gain: 136.1x, EDP: 97 A
Rsense: 3x of 0.00075 (R7510, R7520. R7530), Rsum: 0.00025.
V across Rsense: 24.25 mV
Gain needed: 136.1x



AXG Core Load Side Current Sense (IN0C)

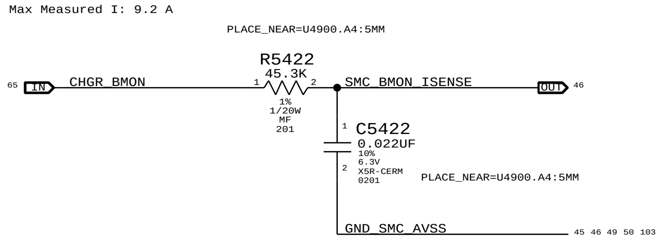
Gain: 185.5x, EDP: 46 A
Rsense: 2x of 0.00075 (R7550, R7560), Rsum: 0.000375.
V across Rsense: 17.25 mV
Gain needed: 191.3x



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
117S0008	2	RES,100K,201	C5451,C5461		LOADISNS:NO

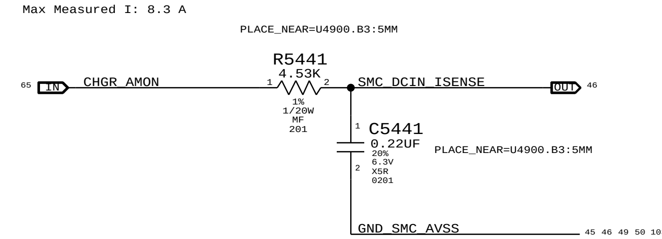
Charger (BMON Prod) Current Sense (IPBR)

Charger Gain: 36x
Rsense: 0.010 (R7050)
Max Measured I: 9.2 A



DC-In (AMON) Current Sense (ID0R)

Charger Gain: 20x
Rsense: 0.020 (R7020)
Max Measured I: 8.3 A



SYNC_MASTER=J31_YONAS SYNC_DATE=10/25/2011

PAGE TITLE

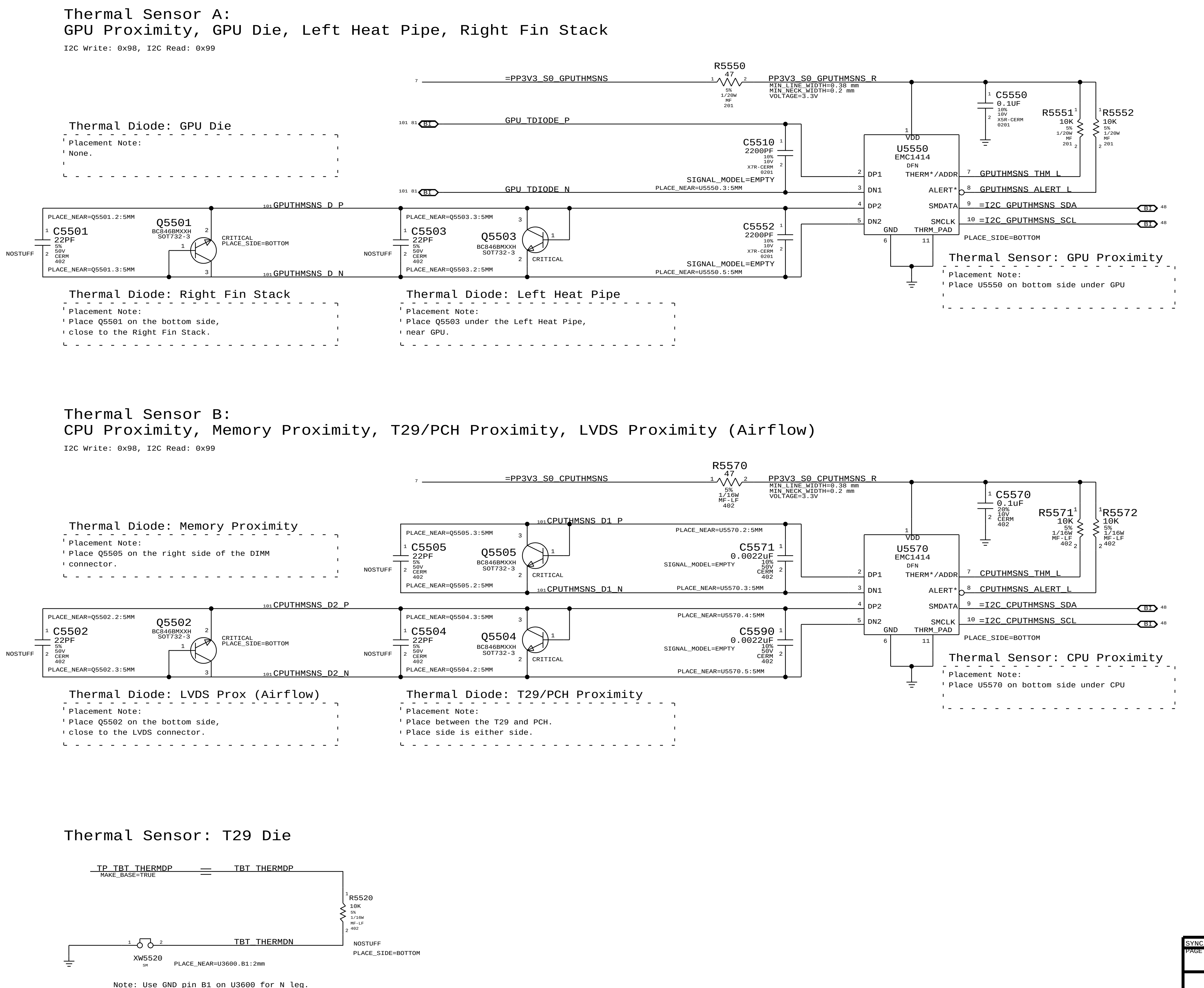
Power Sensors: High Side, CPU, AXG

Apple Inc.

DRAWING NUMBER 051-9585 SIZE D

REVISION 3.0.0

BRANCH



SYNC_MASTER=J31_YONAS SYNC_DATE=09/08/2011

PAGE TITLE: Thermal Sensors

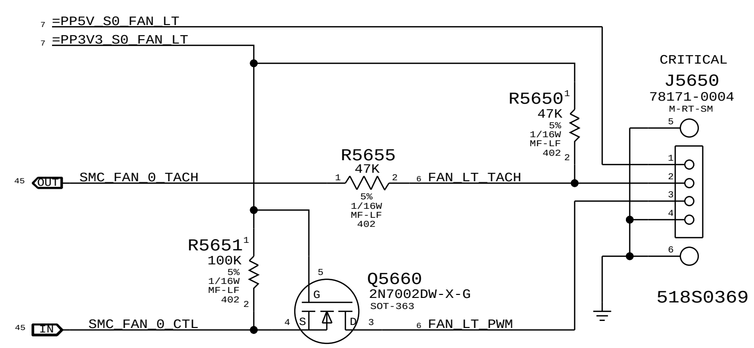
Apple Inc.

DRAWING NUMBER: 051-9585 SIZE: D

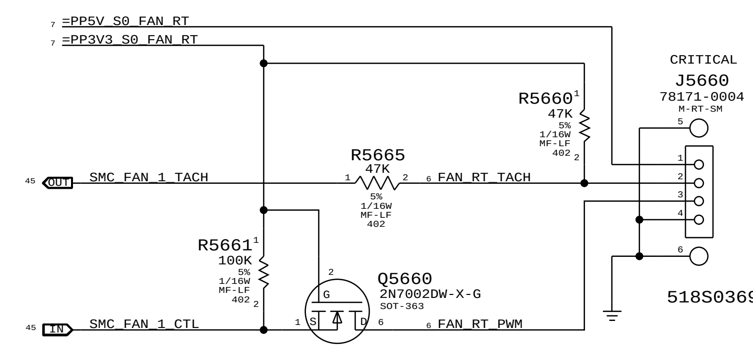
REVISION: 3.0.0

BRANCH

Left Fan



Right Fan



SYNC_MASTER=K18_MLB SYNC_DATE=04/27/2010

PAGE TITLE

Fan Connectors

Apple Inc.

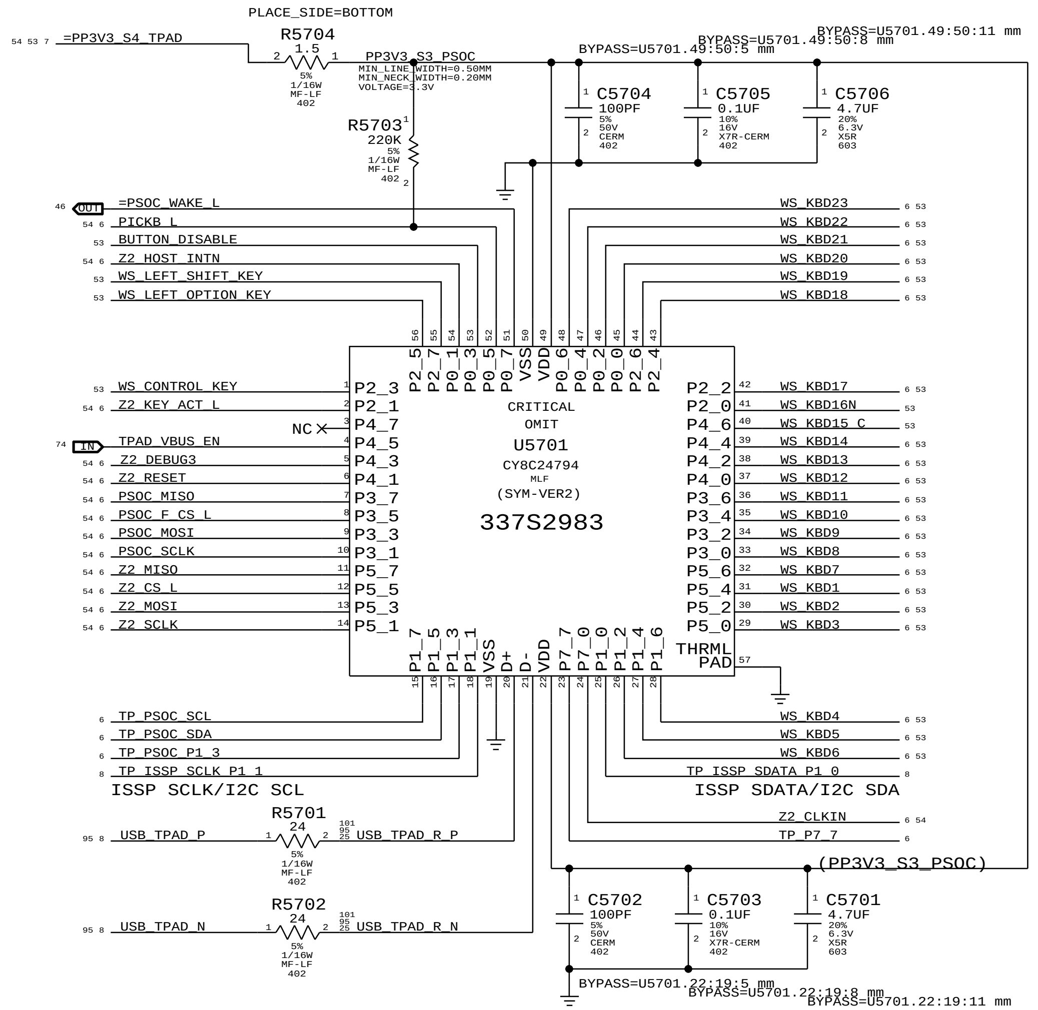
DRAWING NUMBER 051-9585 SIZE D

REVISION 3.0.0

BRANCH

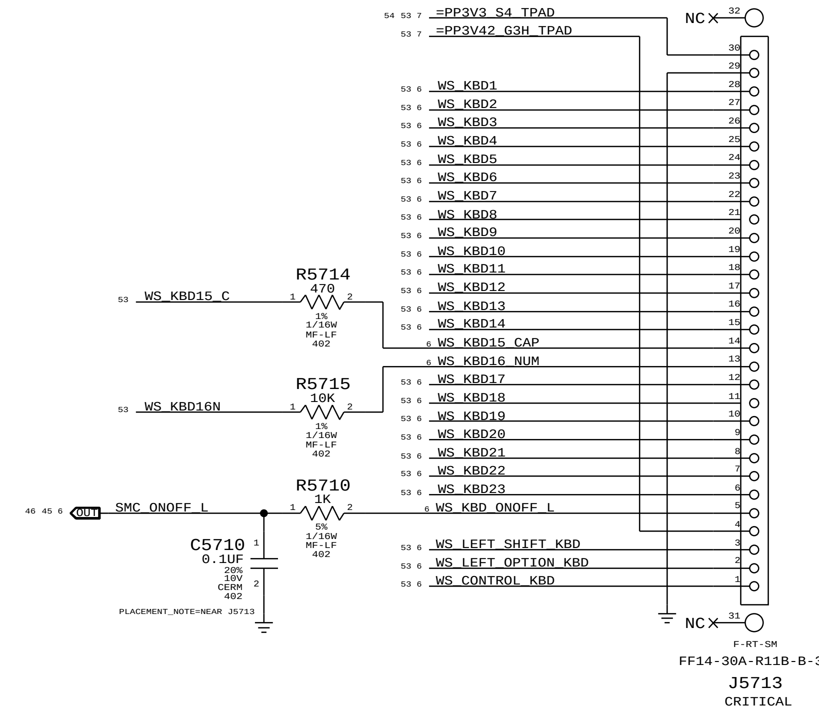
PSOC USB CONTROLLER

- USB INTERFACES TO MLB
- SPI HOST TO Z2
- TRACKPAD PICK BUTTONS
- KEYBOARD SCANNER



IC	PIN NAME	CURRENT	R_SNS	V_SNS	POWER
TMP102	V+	10UA 80UA	2.55 KOHM	0.0255 V 0.204 V	0.255E-6 W 16.32E-6 W
3V3 LDO	VDD VOUT	60MA (MAX) 60MA (MAX)	10 OHM 0.2 OHM	0.6 V 0.012 V	36E-3 W 0.72E-3 W
PSOC	VDD	8MA (TYP) 14MA (MAX)	1.5 OHM	0.012 V 0.021 V	96E-6 W 294E-6 W
18V BOOSTER	VIN	4MA (MAX)	4.7 OHM	0.0188 V	75.2E-6 W

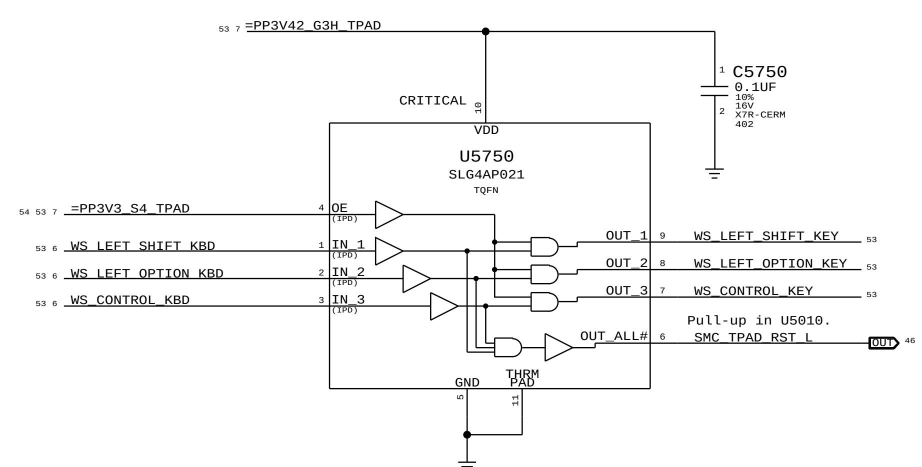
Keyboard Connector



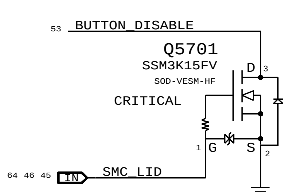
SMC Manual Reset & Isolation

Left shift, option & control keys combined with power button cause SMC RESET# assertion.

Keys ANDed with MSP power to isolate when MSP is not powered. No IPD on OE input pin PP3V3_S4 (symbol error).



TPAD Buttons Disable



PLACE THESE COMPONENTS CLOSE TO J5800
THIS ASSUMES THERE'S A PP3V42_G3H PULL UP ON MLB

THE TPAD BUTTONS WILL BE DISABLE
WHEN THE LID IS CLOSED
LID OPEN => SMC_LID_LC ~ 3.42V
LID CLOSE => SMC_LID_LC < 0.50V

SYNC_MASTER=J30_MLB SYNC_DATE=06/10/2011

PAGE TITLE: WELLSPRING 1

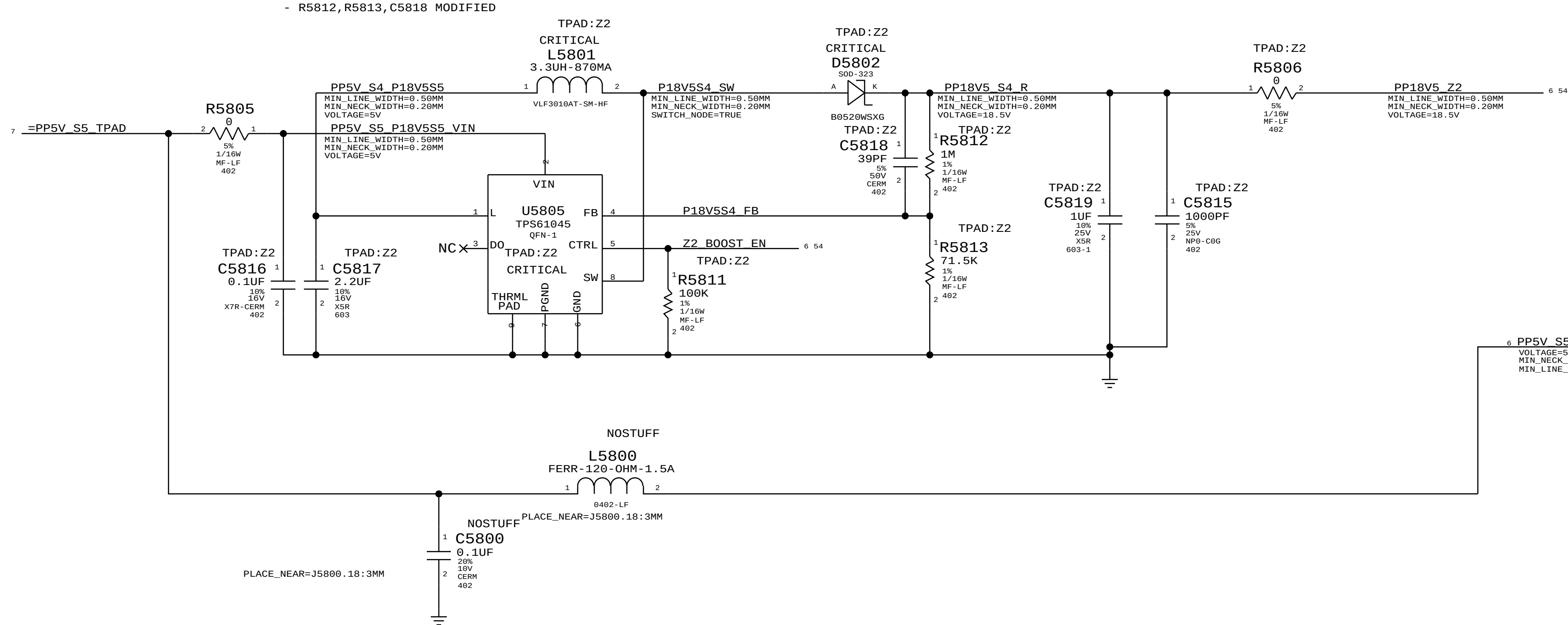
Apple Inc.

DRAWING NUMBER: 051-9585 SIZE: D

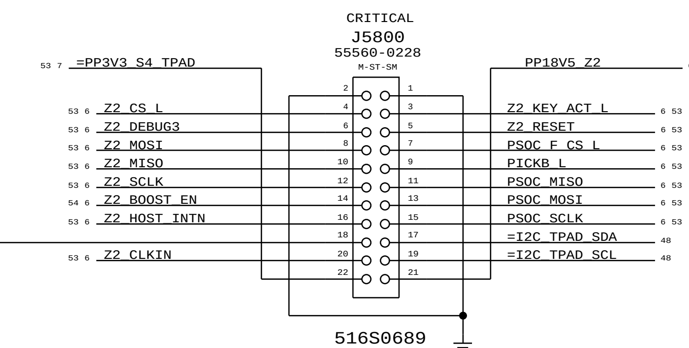
REVISION: 3.0.0

BOOSTER +18.5VDC FOR SENSORS

BOOSTER DESIGN CONSIDERATION:
- POWER CONSUMPTION
- DROOP LINE REGULATION
- RIPPLE TO MEET ERS
- 100-300 KHZ CLEAN SPECTRUM
- STARTUP TIME LESS THAN 2MS
- R5812,R5813,C5818 MODIFIED

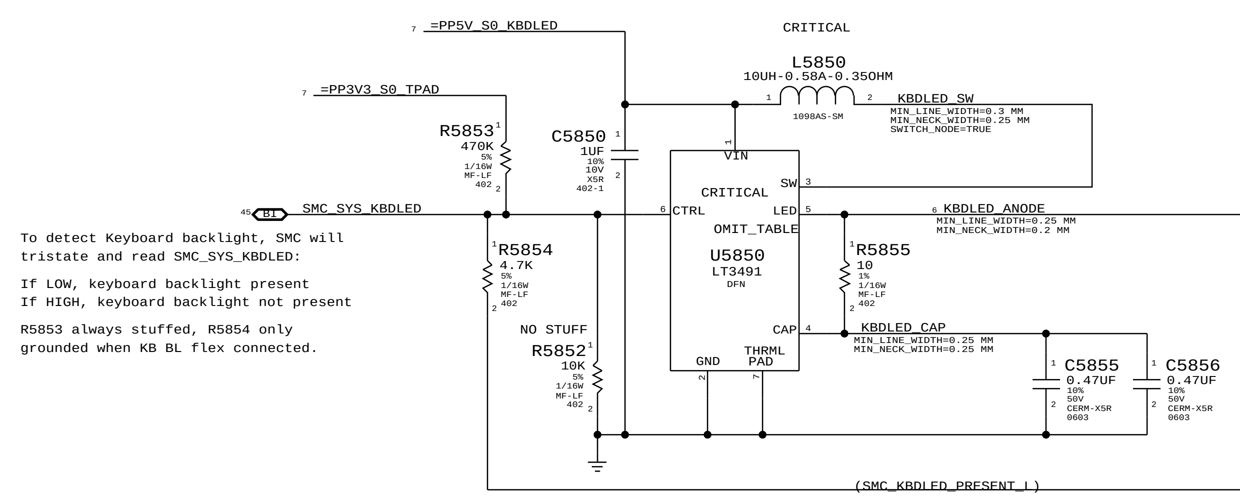


IPD Flex Connector



PIN 21 IS NC ON CUMULUS FLEX
PIN 18 IS NC ON Z2 FLEX

Keyboard Backlight Driver & Detection

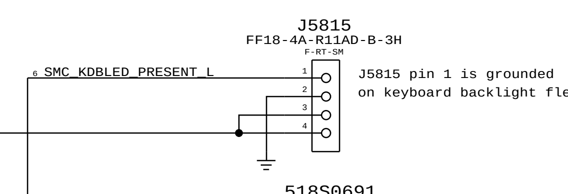


To detect Keyboard backlight, SMC will tristate and read SMC_SYS_KBDLED:

If LOW, keyboard backlight present
If HIGH, keyboard backlight not present

R5853 always stuffed, R5854 only grounded when KB BL flex connected.

Keyboard Backlight Connector



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
353S3085	1	IC,LT3491,1-STRING LED DRIVER,2X2DFN-6	U5850	CRITICAL	

SYNC_MASTER=J31_LINDA SYNC_DATE=07/01/2011

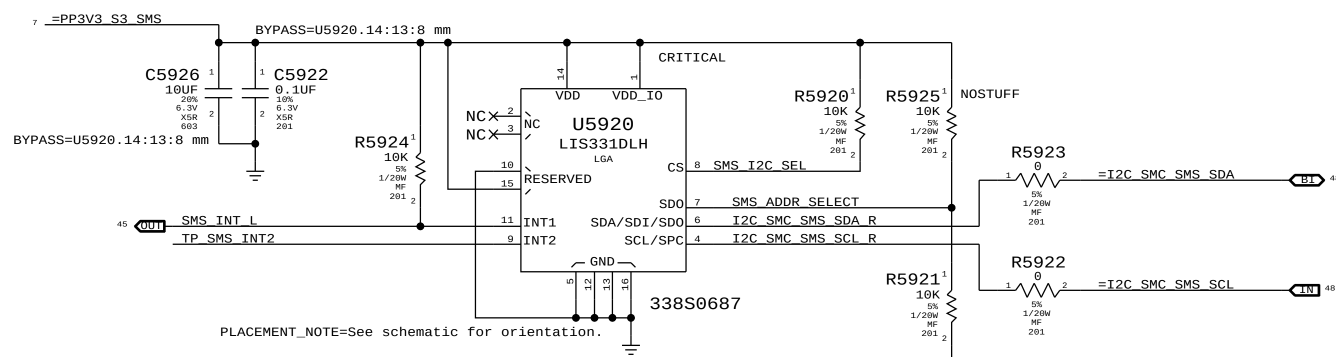
PAGE TITLE: WELLSPRING 2

Apple Inc.

DRAWING NUMBER: 051-9585 SIZE: D

REVISION: 3.0.0

BRANCH

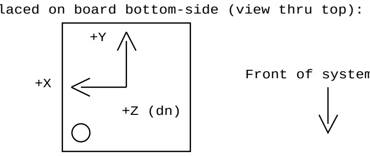


PLACEMENT_NOTE=See schematic for orientation.

SMS_ADDR_SELECT=0 Addr: 0x30(Wr)/0x31(Rd)
SMS_ADDR_SELECT=1 Addr: 0x32(Wr)/0x33(Rd)

NOTE: SDA and SCL have internal pull-ups to VDD_IO.

Desired orientation when
placed on board bottom-side (view thru top):



Circle indicates pin 1 location when placed
in correct orientation

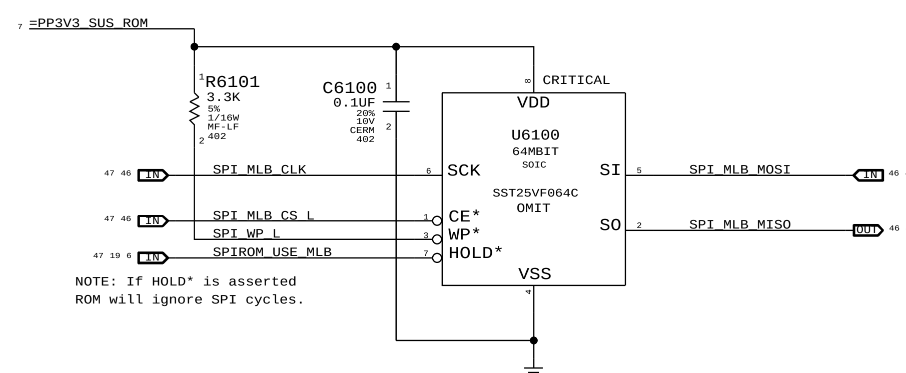
SYNC_MASTER=J31_YONAS
SYNC_DATE=08/11/2011

PAGE TITLE
Digital Accelerometer

Apple Inc.

DRAWING NUMBER 051-9585
SIZE D
REVISION 3.0.0
BRANCH

NOTICE OF PROPRIETARY PROPERTY:



SYNC_MASTER=K91_BEN SYNC_DATE=06/08/2010

PAGE TITLE

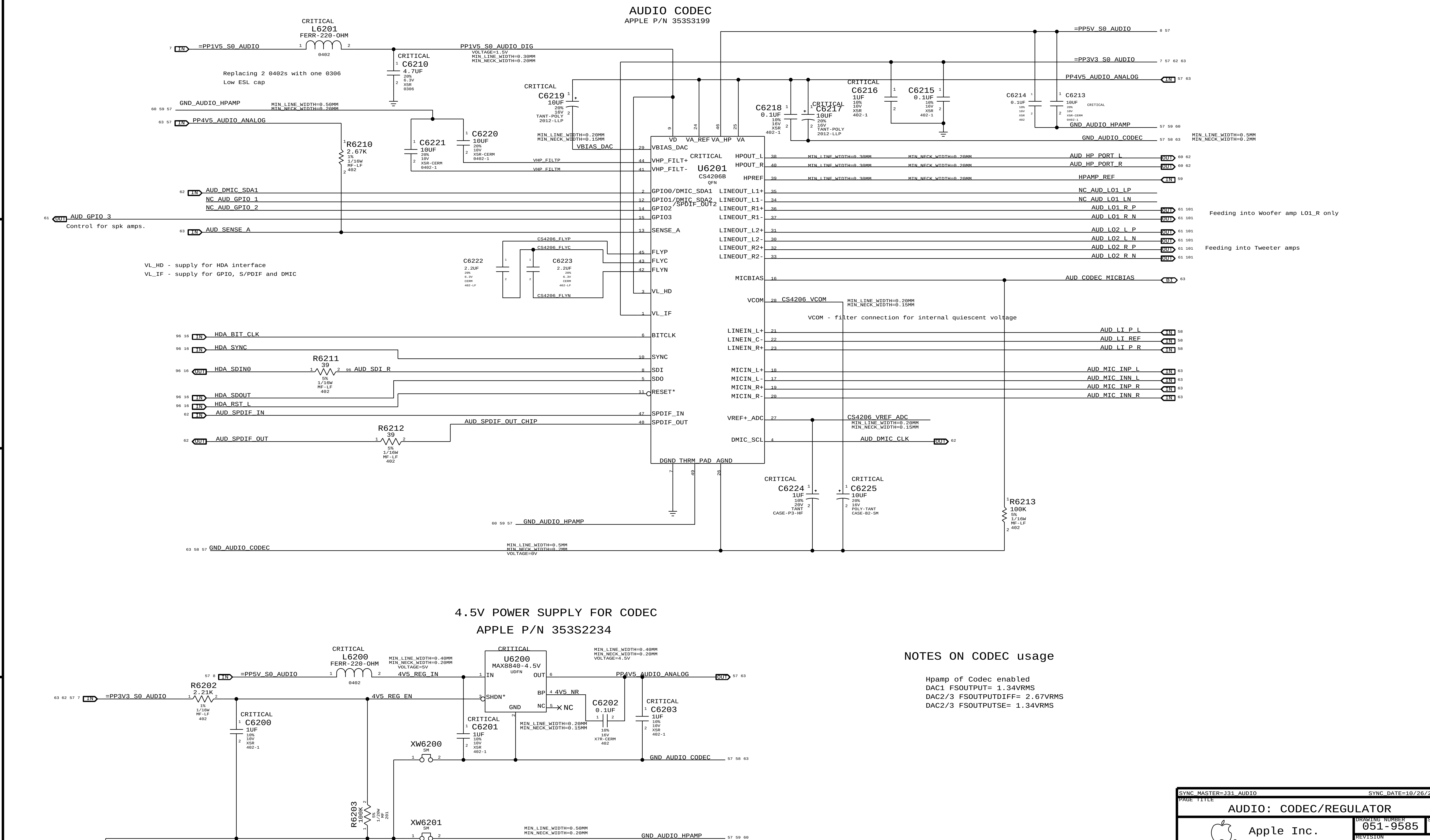
SPI ROM

Apple Inc.

DRAWING NUMBER 051-9585 SIZE D

REVISION 3.0.0

BRANCH



NOTES ON CODEC usage

Hpamp of Codec enabled
DAC1 FSOUTPUT= 1.34VRMS
DAC2/3 FSOUTPUTDIFF= 2.67VRMS
DAC2/3 FSOUTPUTSE= 1.34VRMS

SYNC_MASTER=J31_AUDIO SYNC_DATE=10/26/2011

PAGE TITLE: AUDIO: CODEC/REGULATOR

Apple Inc.

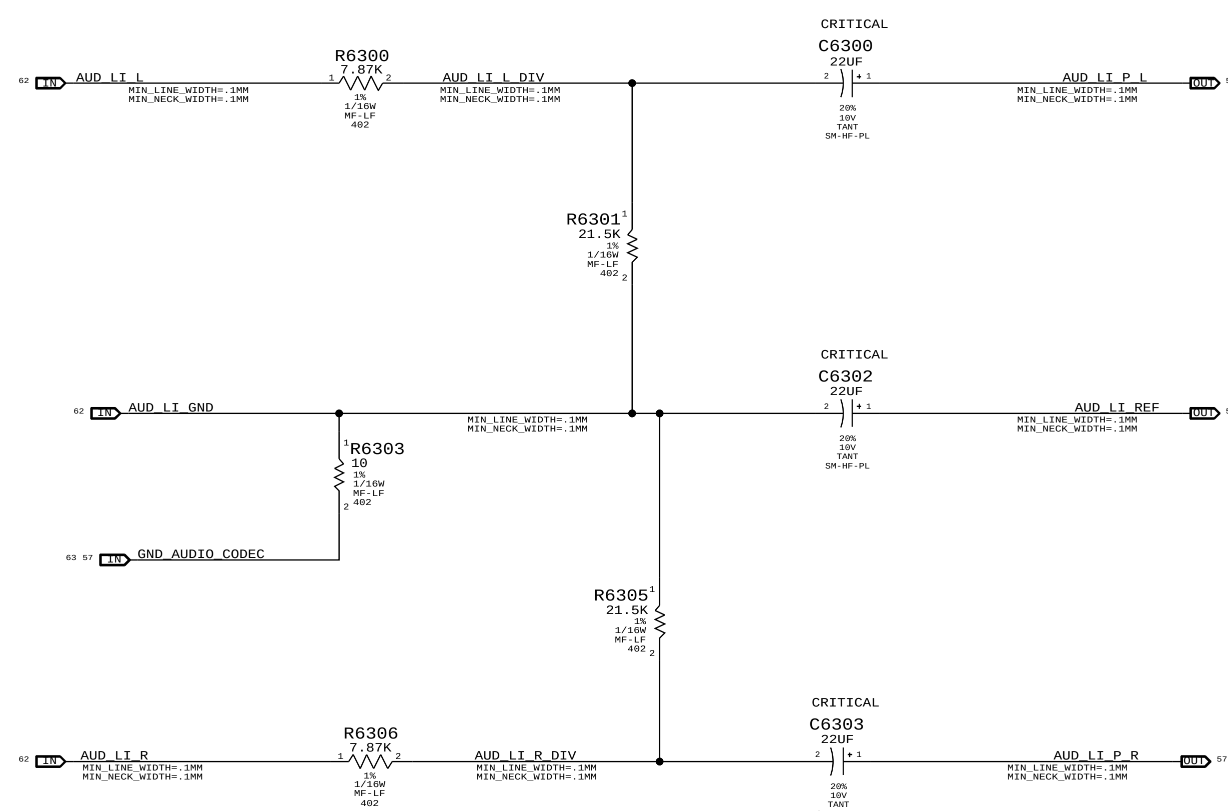
DRAWING NUMBER: 051-9585 SIZE: D

REVISION: 3.0.0

BRANCH

LINE INPUT VOLTAGE DIVIDER

CODEC RIN = 20K OHMS
NET RIN = 18K OHMS
FC = 0.36 HZ
VIN = 2VRMS, CODEC VIN = 1.14 VRMS

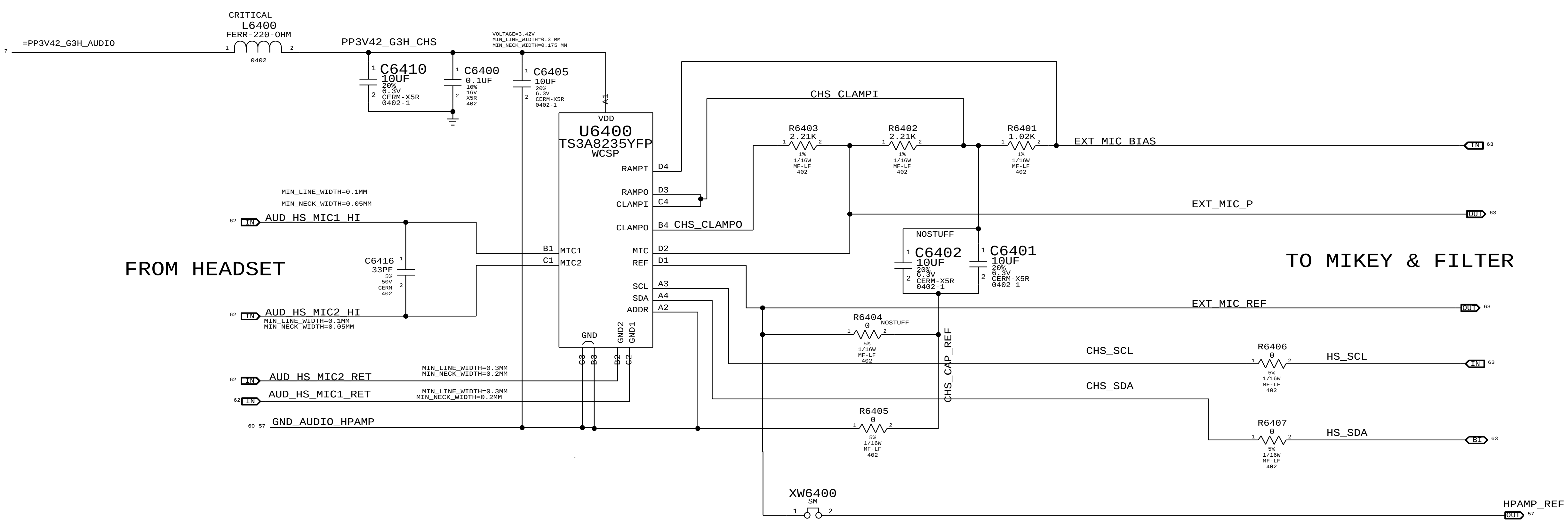


SYNC_MASTER=J31_AUDIO		SYNC_DATE=10/26/2011
PAGE TITLE: AUDIO: LINE INPUT FILTER		
Apple Inc.	DRAWING NUMBER 051-9585	SIZE D
	REVISION 3.0.0	
NOTICE OF PROPRIETARY PROPERTY:	BRANCH	
	PAGE 63 OF 132	
	SHEET 58 OF 105	

EXTERNAL (HEADSET) MIC INPUT CIRCUITRY

APN: 353S3066 as of JUly 2011

U6400 should get VDD from battery. Should be powered all the time.



I2C ADDRESSES: CHS uses SMBus 0 connections

CHS	U6400	READ	0111	0111	0x77
CHS	U6400	WRITE	0111	0110	0x76

SYNC_MASTER=J31_AUDIO SYNC_DATE=10/26/2011

PAGE TITLE
AUDIO: DETECT/MIC BIAS

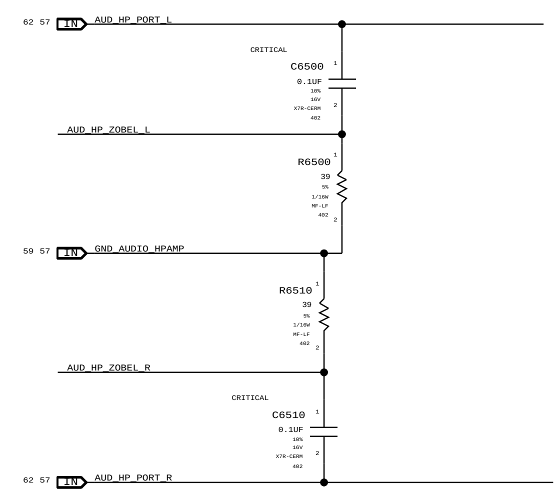
Apple Inc.

DRAWING NUMBER 051-9585 SIZE D

REVISION 3.0.0

BRANCH

ZOBEL NETWORK & 1ST ORDER DAC FILTER PLACEHOLDER



SYNC_MASTER=J31_AUDIO SYNC_DATE=10/26/2011

3X MONO SPEAKER AMPLIFIERS (SSM2375)

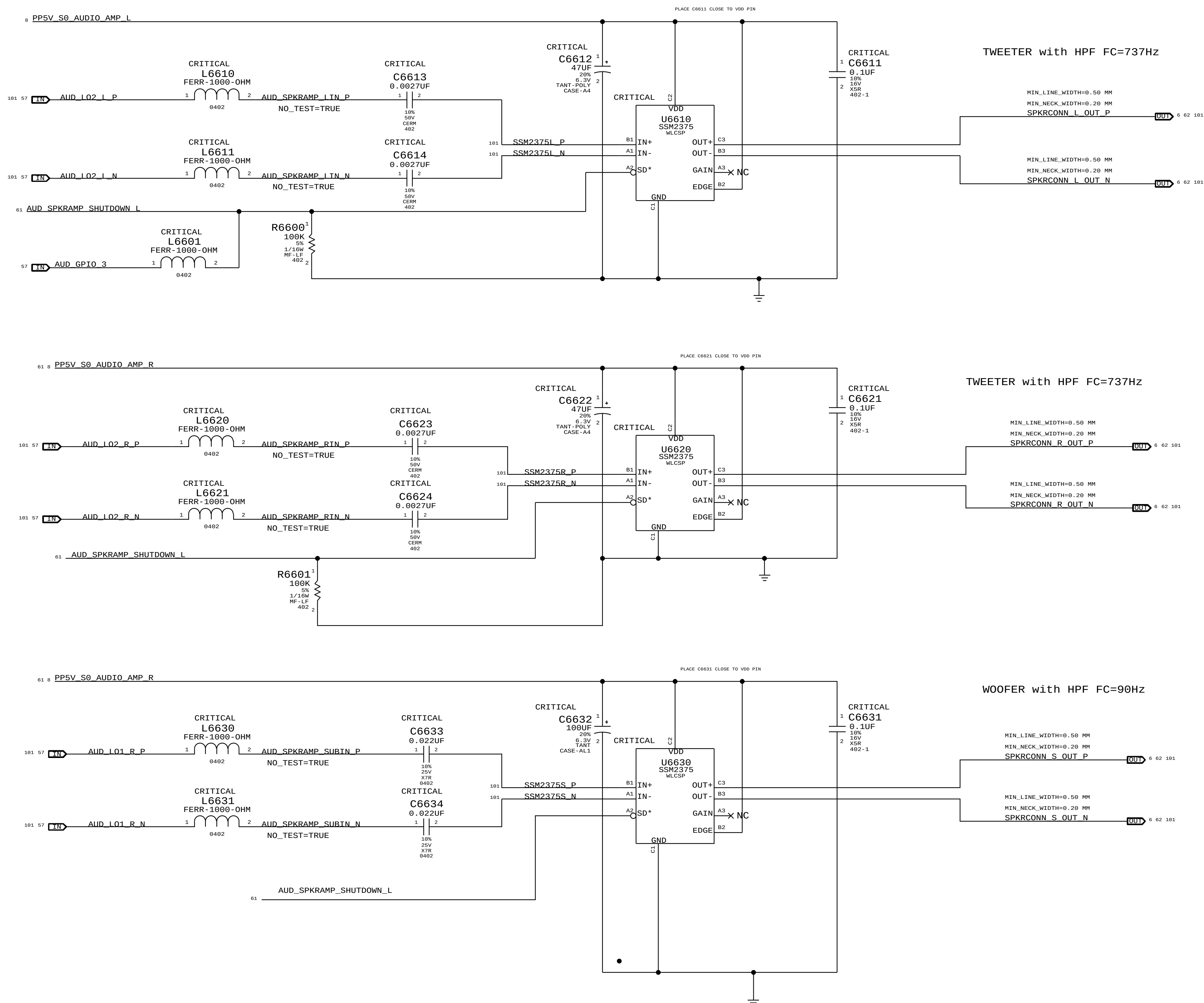
APN: 353S2958 as of July 2011

GAIN = +3 DB   Rin=80k irrespective of gain

1ST ORDER FC (L&R) = ~737 HZ

1ST ORDER FC (SUB) = ~90 HZ

Gain Pin	Gain dB
Connect to VDD	6
Connect to VDD through 47k	12
Not connected	3
Connect to GND through 47k	9
Connect to GND	0



SYNC_MASTER=J31_AUDIO   SYNC_DATE=10/26/2011

PAGE TITLE

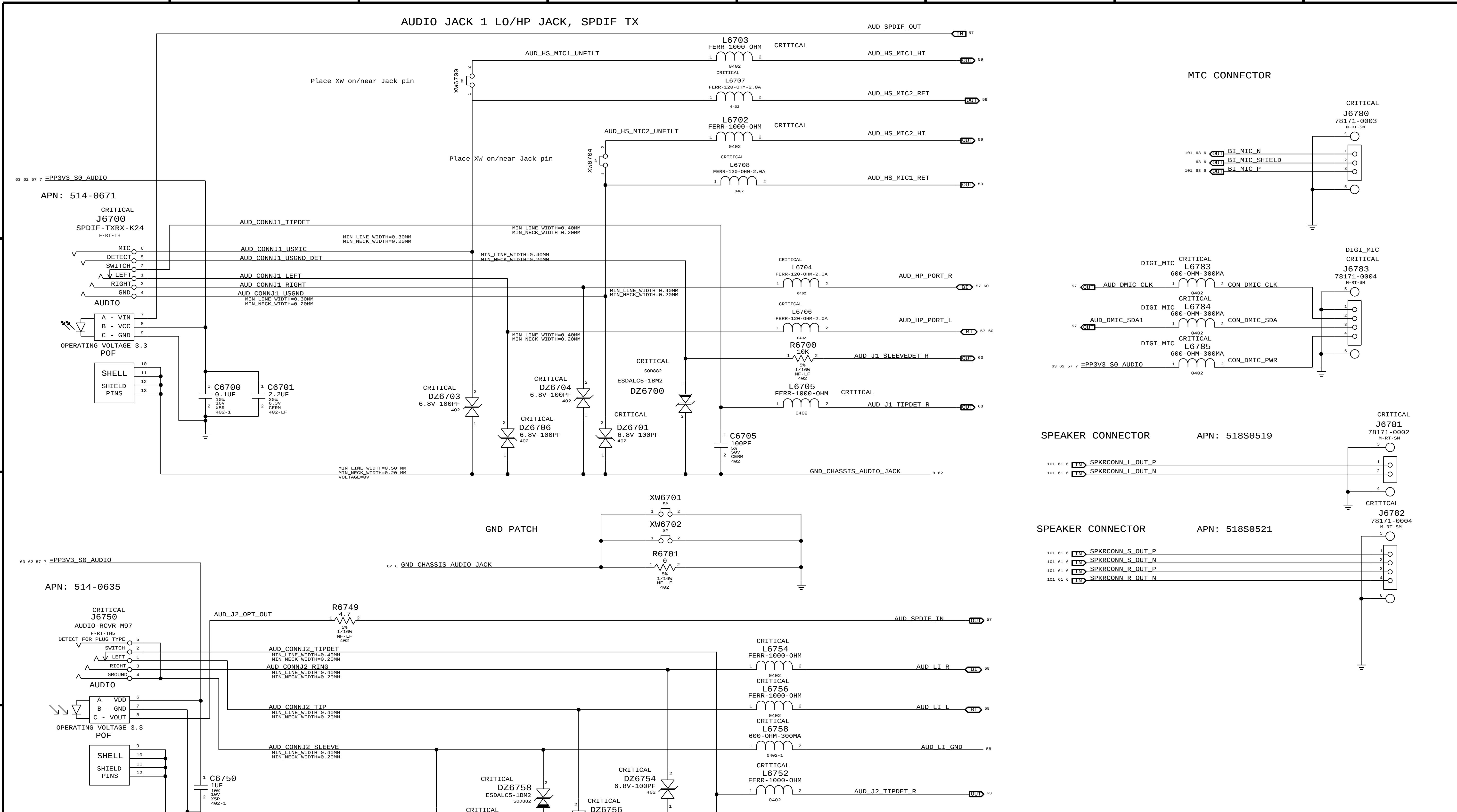
AUDIO: SPEAKER AMP

Apple Inc.

DRAWING NUMBER 051-9585   SIZE D

REVISION 3.0.0

BRANCH



SYNC_MASTER=J31_AUDIO SYNC_DATE=10/26/2011

PAGE TITLE

AUDIO: JACKS

Apple Inc.

DRAWING NUMBER 051-9585 SIZE D

REVISION 3.0.0

BRANCH

NOTICE OF PROPRIETARY PROPERTY:

CODEC OUTPUT SIGNAL PATHS

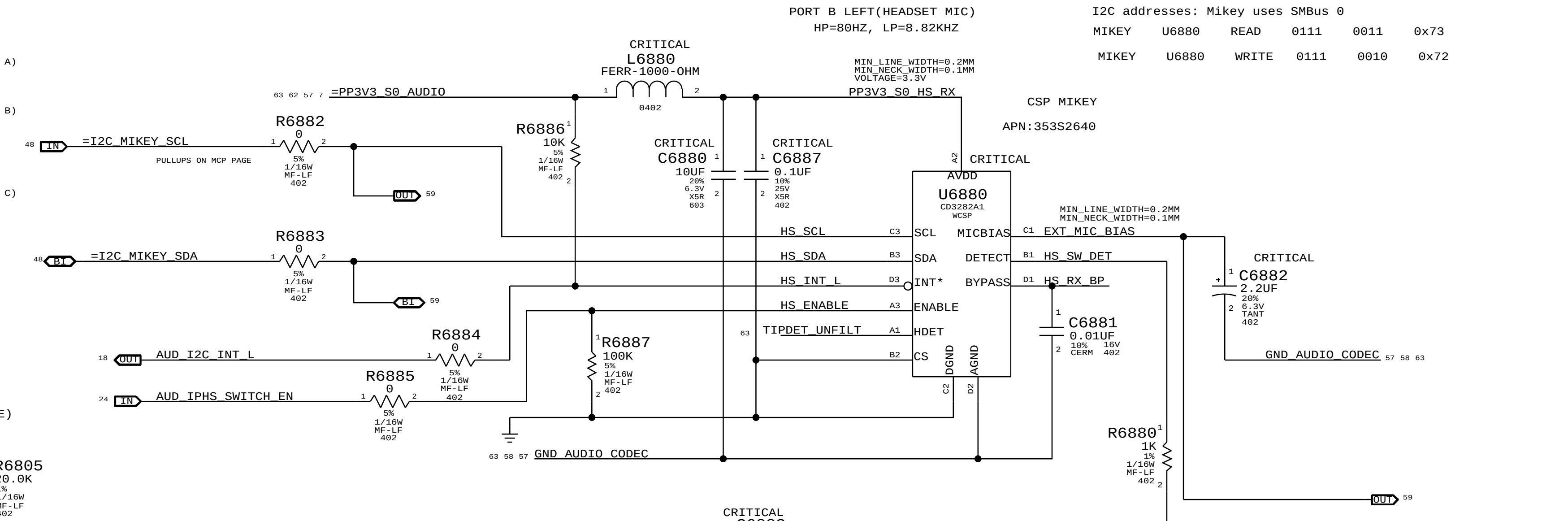
FUNCTION	VOLUME	CONVERTER	PIN COMPLEX	MUTE CONTROL	DET ASSIGNMENT
HP/LINE OUT	0X02 (2)	0X02 (2)	0X09 (9,A)	NA	0X09 (Jack Detect A)
SATELLITES	0X04 (4)	0X04 (4)	0X0B (11)	GPIO_3	N/A
SUB	0X03 (3)	0X03 (03)	0X0A (10)	GPIO_3	N/A
SPDIF OUT	N/A	0X08 (8)	0X10 (16)	N/A	0X0C (Jack detect B)

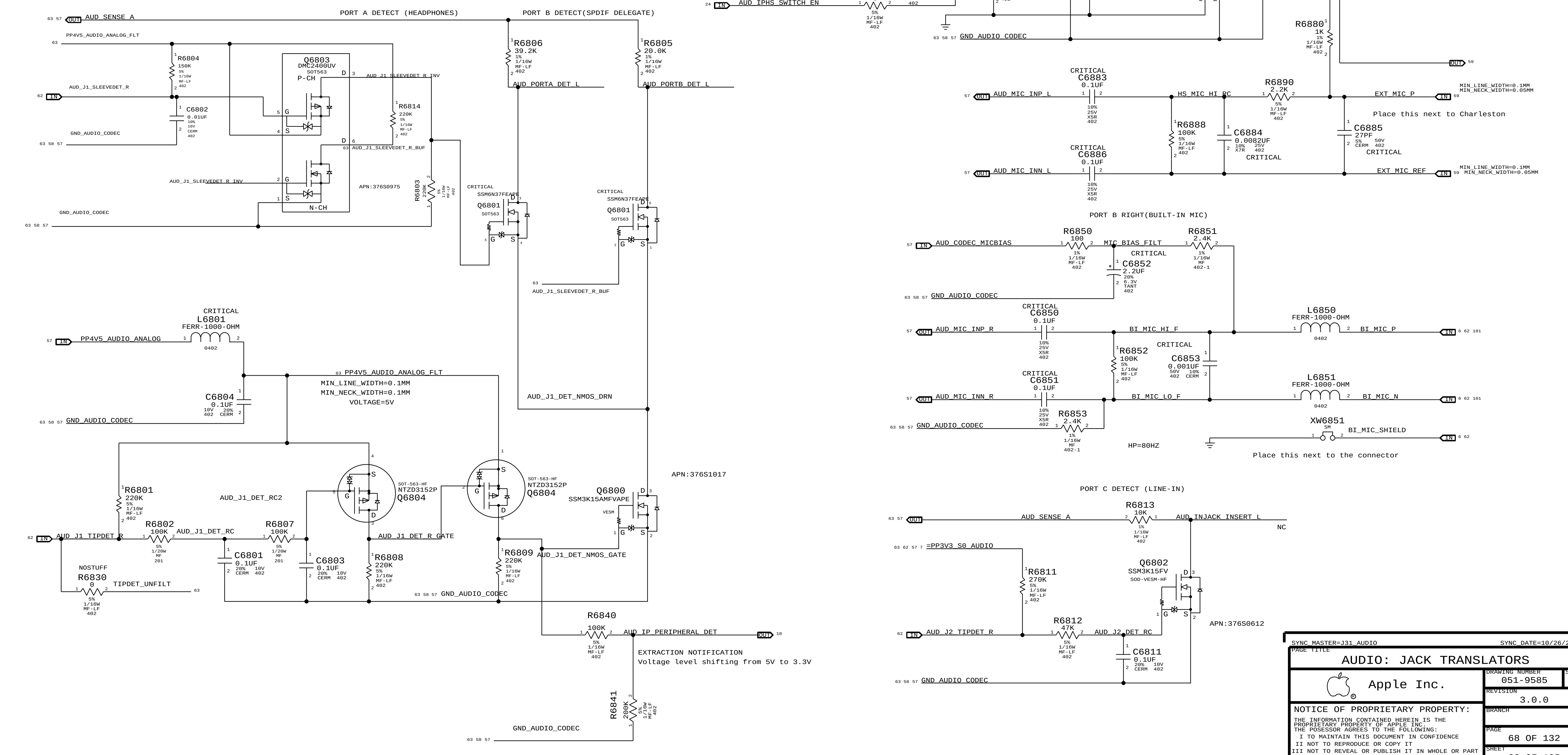
CODEC INPUT SIGNAL PATHS

FUNCTION	CONVERTER	PIN COMPLEX	VREF	DET ASSIGNMENT
LINE IN	0X05 (5)	0X0C (12,C)	N/A	0X0C (Jack detect C)
SPDIF IN	0X07 (7)	0X0F (15)	N/A	N/A
BUILT-IN MIC	0X06 (6)	0X0D (13)	N/A	N/A
HEADSET MIC	0X06 (6)	0X0D (13,V22,B,LEFT)	MIKEY	MIKEY

SYSTEM INT AND GPIO LINES

FUNCTION	INT	GPIO
MIKEY ENABLE		SATA4GP/GPIO 16
MIKEY INTERRUPT	PIRQ H	GPIO 5
PERIPHERAL DETECT	PIRQ F	GPIO 3





SYNC_MASTER=J31_AUDIO SYNC_DATE=10/26/2011

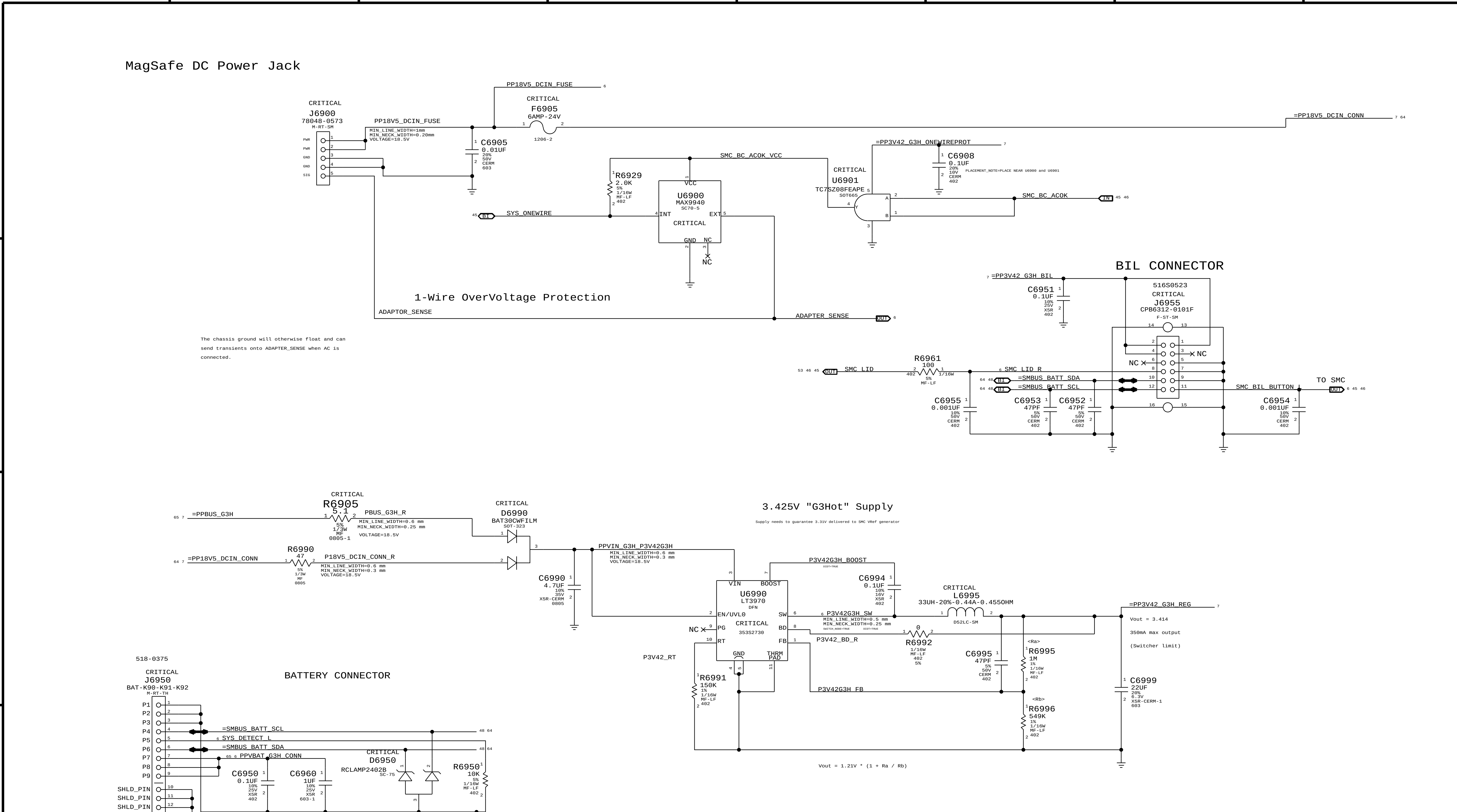
PAGE TITLE: AUDIO: JACK TRANSLATORS

Apple Inc.

DRAWING NUMBER: 051-9585 SIZE: D

REVISION: 3.0.0

BRANCH



SYNC_MASTER=J31_JACK SYNC_DATE=09/02/2011

PAGE TITLE

DC-In & Battery Connectors

Apple Inc.

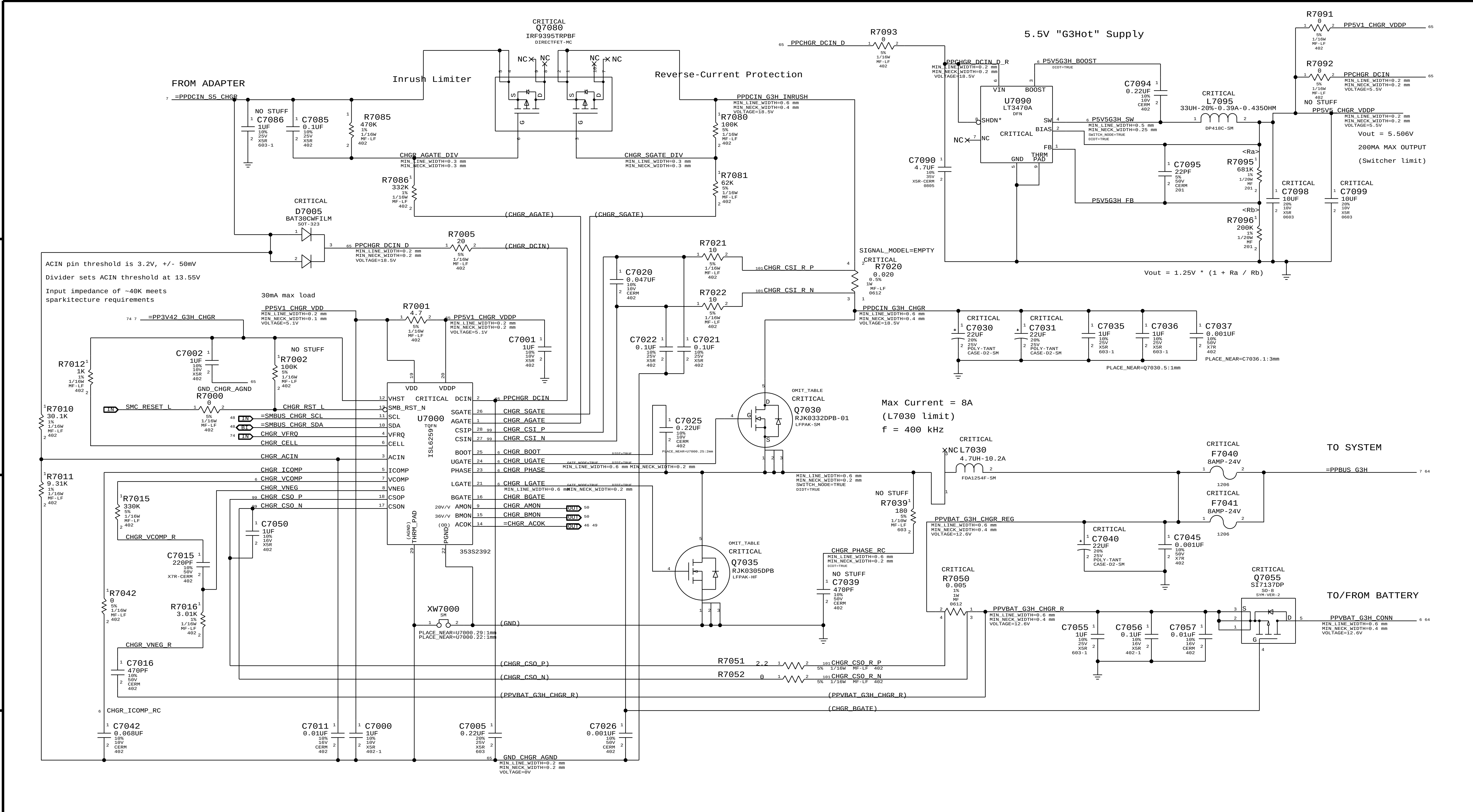
DRAWING NUMBER 051-9585 SIZE D

REVISION 3.0.0

BRANCH

PAGE 69 OF 132

SHEET 64 OF 105



SYNC_MASTER=J31_JACK SYNC_DATE=11/14/2011

PAGE TITLE

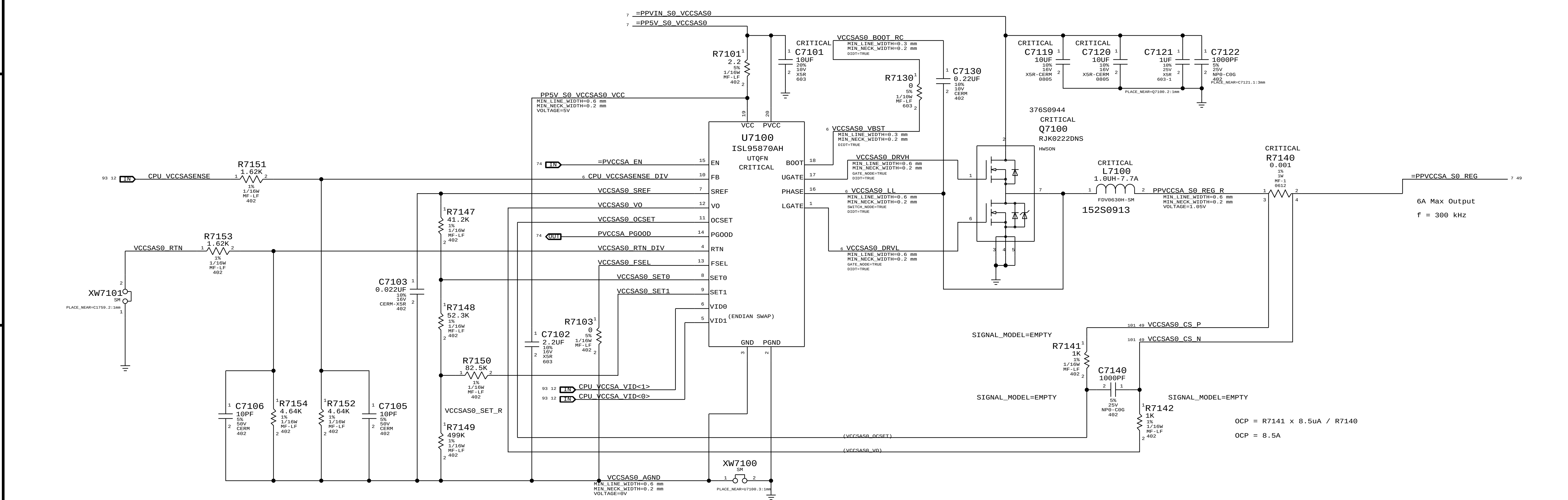
PBus Supply & Battery Charger

Apple Inc.

DRAWING NUMBER 051-9585 SIZE D

REVISION 3.0.0

BRANCH



INTEL TABLE:

VID1	VID0	Voltage
0	0	0.9V
1	0	0.8V
0	1	0.725V
1	1	0.675V

SYNC_MASTER=J31_JACK SYNC_DATE=09/14/2011

PAGE TITLE

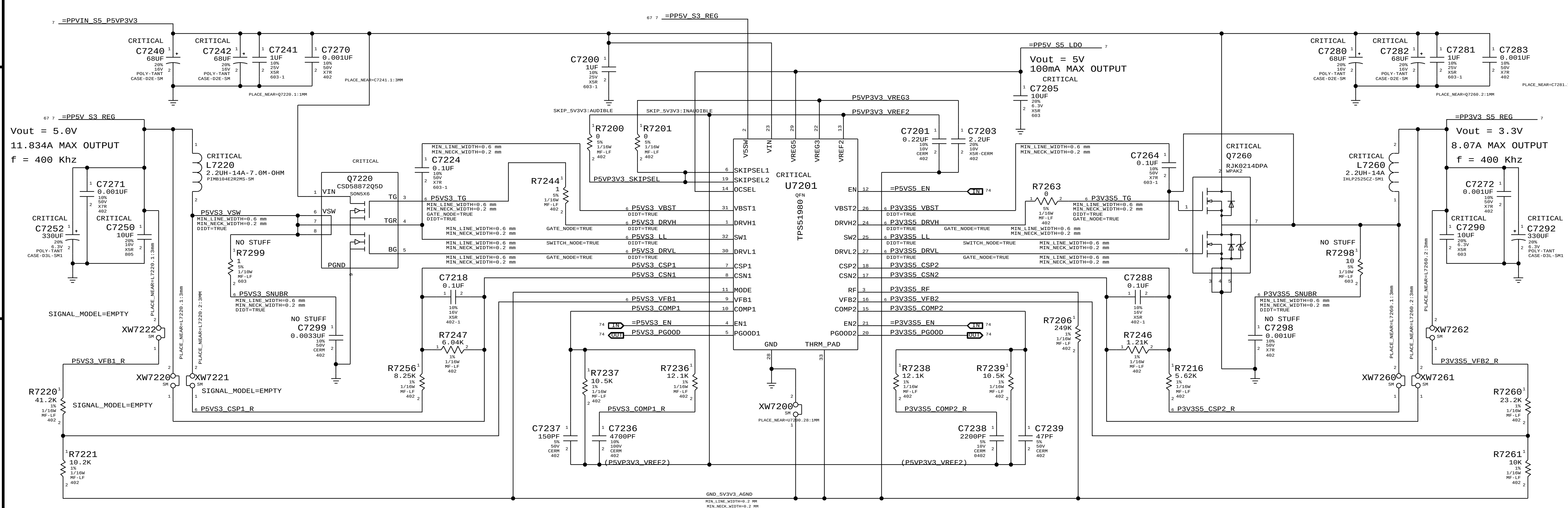
System Agent Supply

Apple Inc.

DRAWING NUMBER 051-9585 SIZE D

REVISION 3.0.0

BRANCH



SYNC_MASTER=J31_JACK SYNC_DATE=11/09/2011

PAGE TITLE

5V / 3.3V Power Supply

Apple Inc.

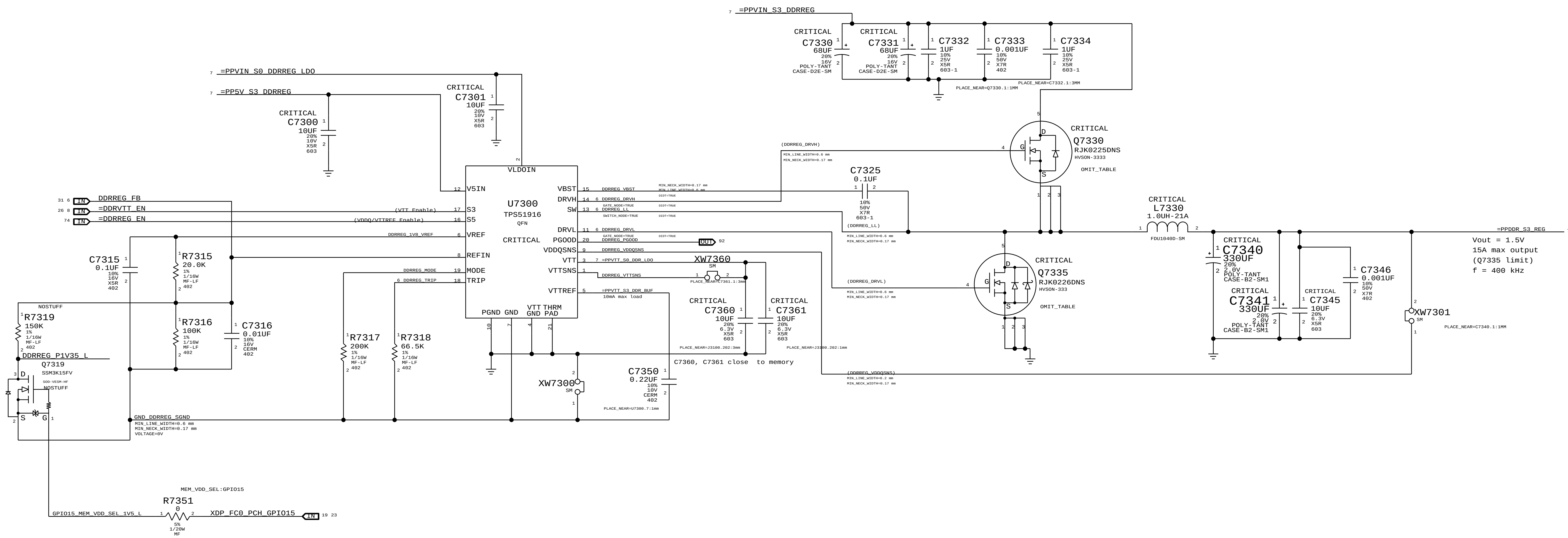
DRAWING NUMBER 051-9585 SIZE D

REVISION 3.0.0

BRANCH

PAGE 72 OF 132

SHEET 67 OF 105



SYNC_MASTER=J31_JACK

SYNC_DATE=07/07/2011

PAGE TITLE

1.5V DDR3 Supply

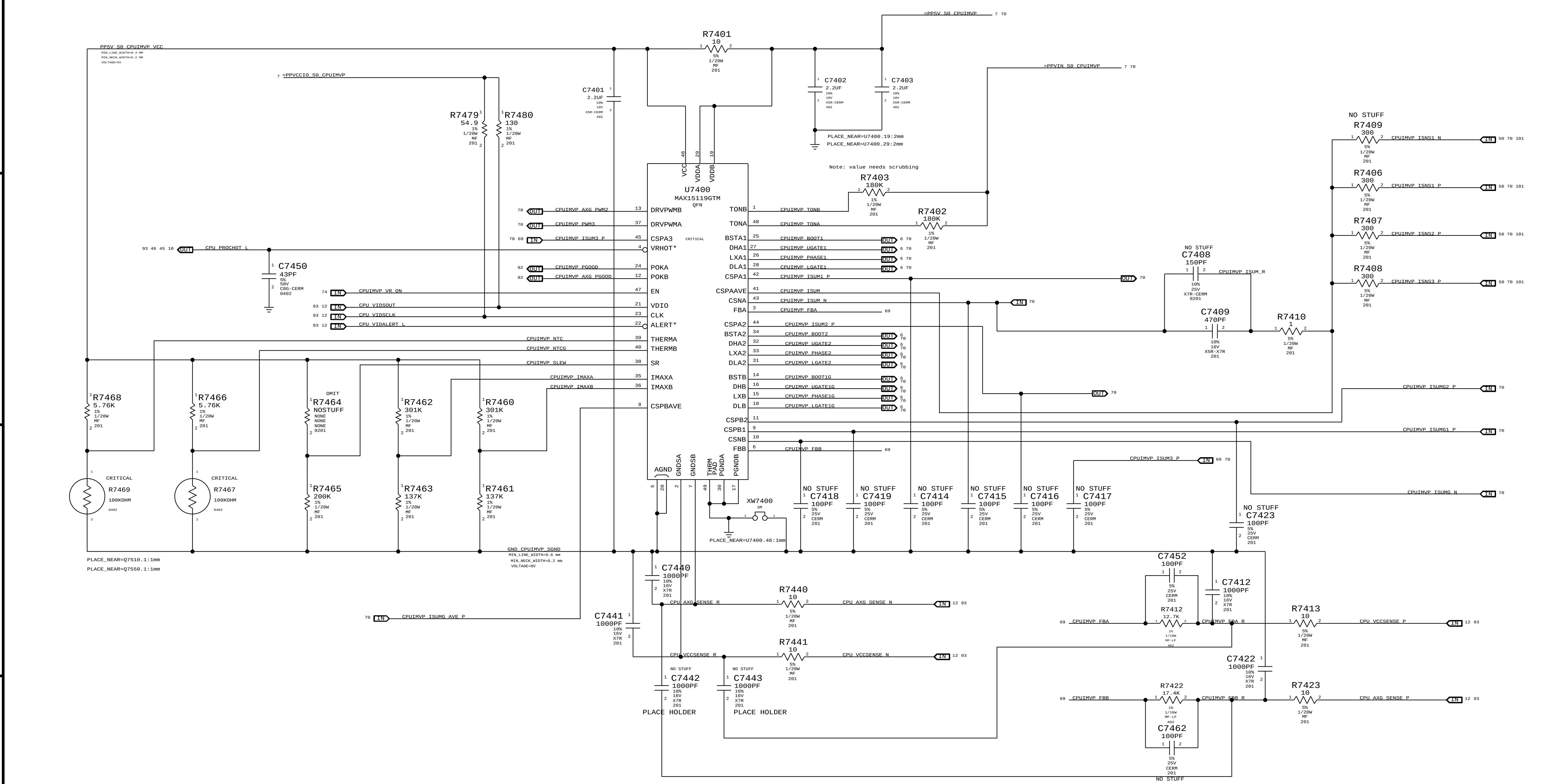
Apple Inc.

DRAWING NUMBER 051-9585

SIZE D

REVISION 3.0.0

BRANCH



SYNC_MASTER=J31_JACK SYNC_DATE=11/11/2011

PAGE TITLE

CPU IMVP7 & AXG VCore Regulator

Apple Inc.

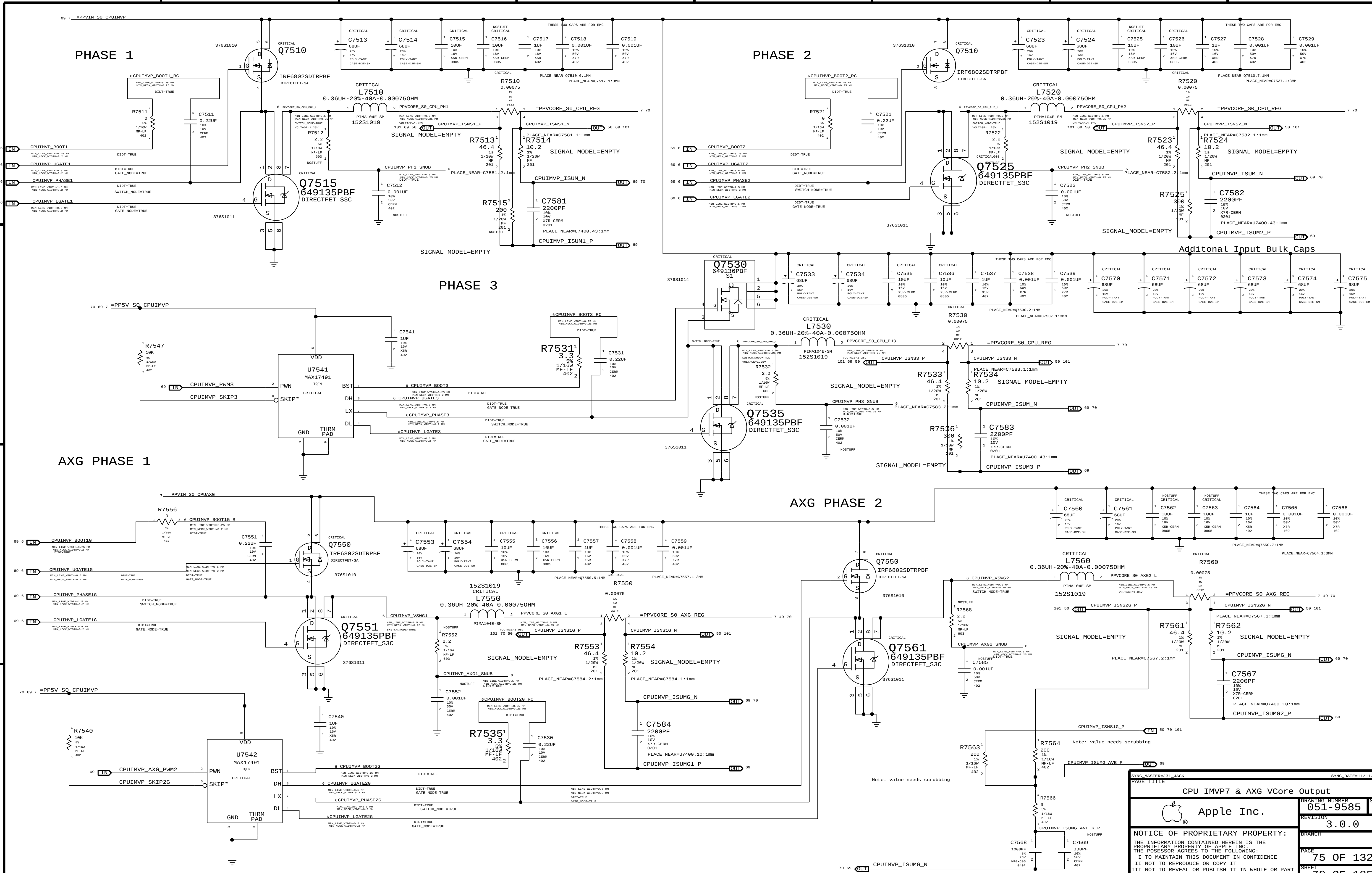
DRAWING NUMBER	SIZE
051-9585	D

REVISION 3.0.0

BRANCH

PAGE 74 OF 132

SHEET 69 OF 105



SYNC_MASTER=J31_JACK SYNC_DATE=11/11/2011

PAGE TITLE

CPU IMVP7 & AXG VCore Output

Apple Inc.

DRAWING NUMBER 051-9585 SIZE D

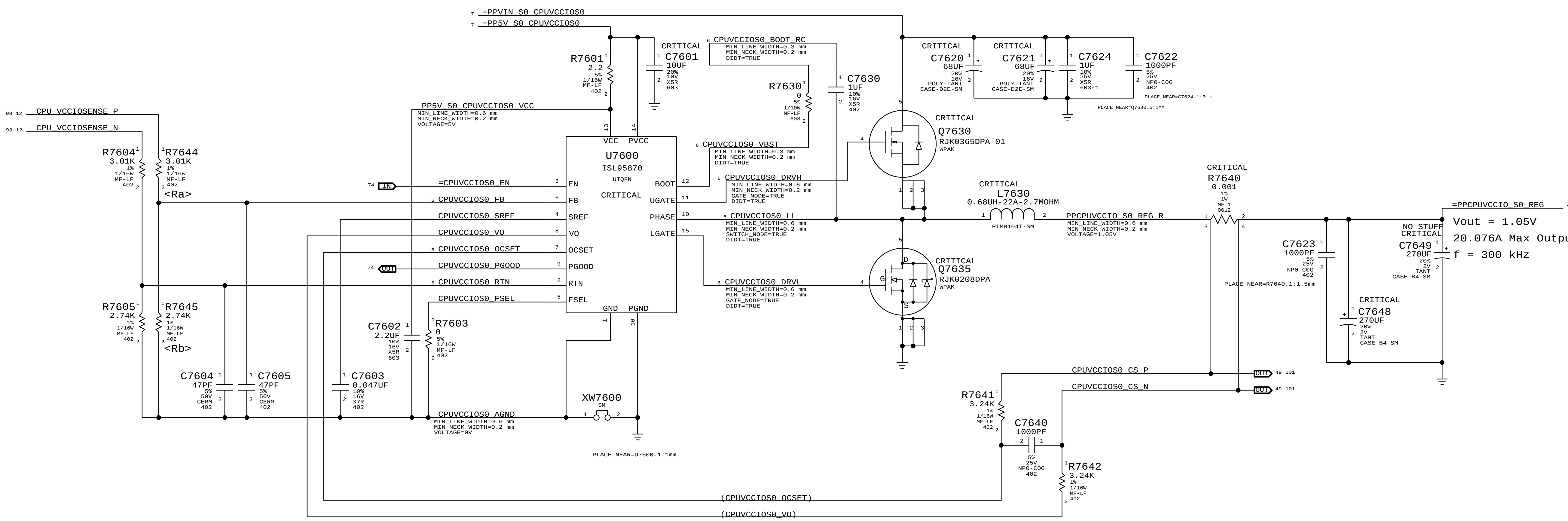
REVISION 3.0.0

BRANCH

PAGE 75 OF 132

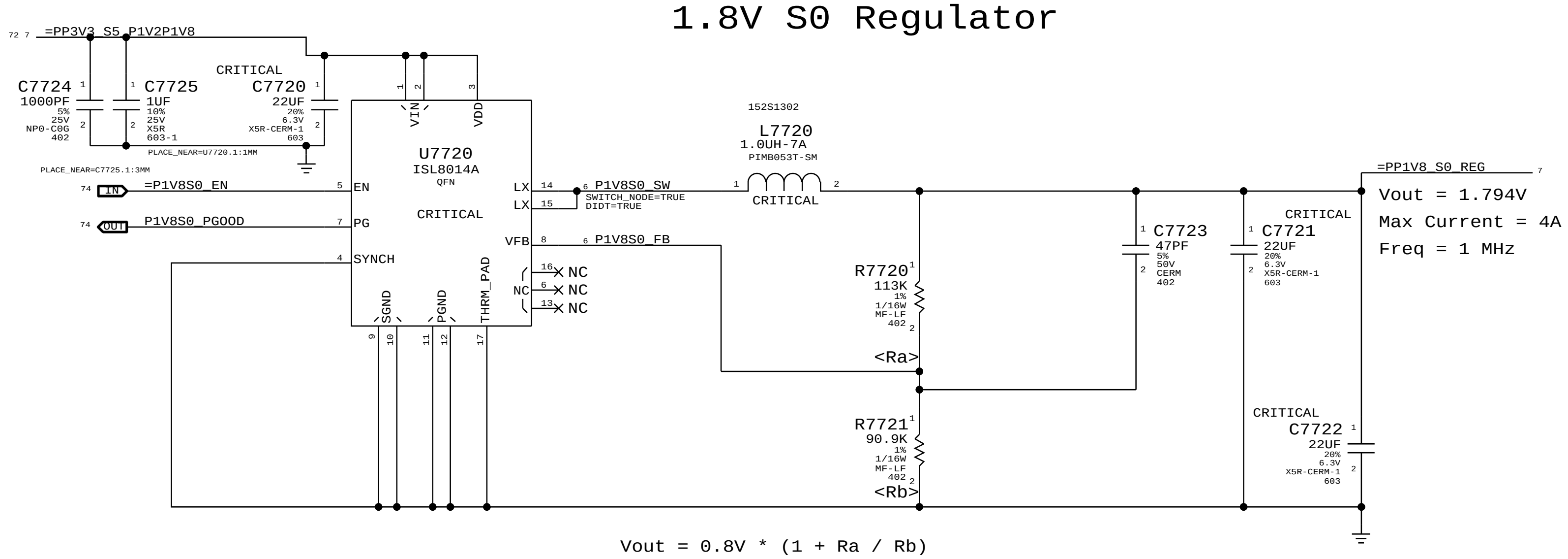
SHEET 70 OF 105

CPU VCCIO REGULATOR



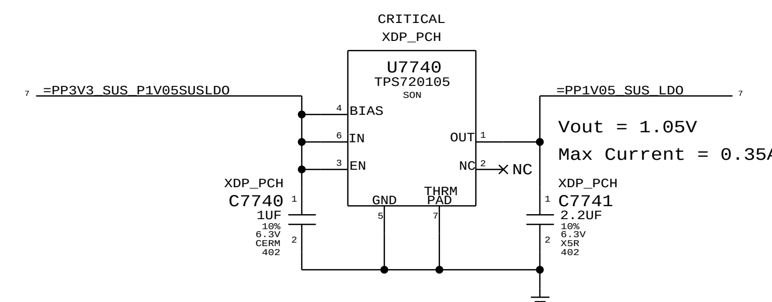
OCP = R7641 x 8.5uA / R7640
OCP = 27.54A
Vout = 0.5V * (1 + Ra / Rb)

1.8V S0 Regulator

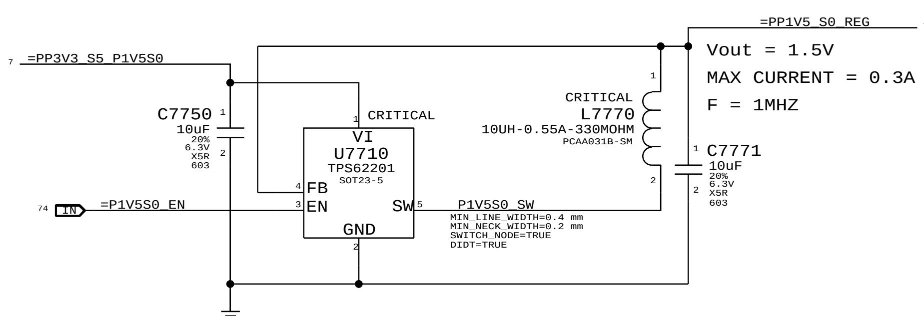


1.05V SUS LDO

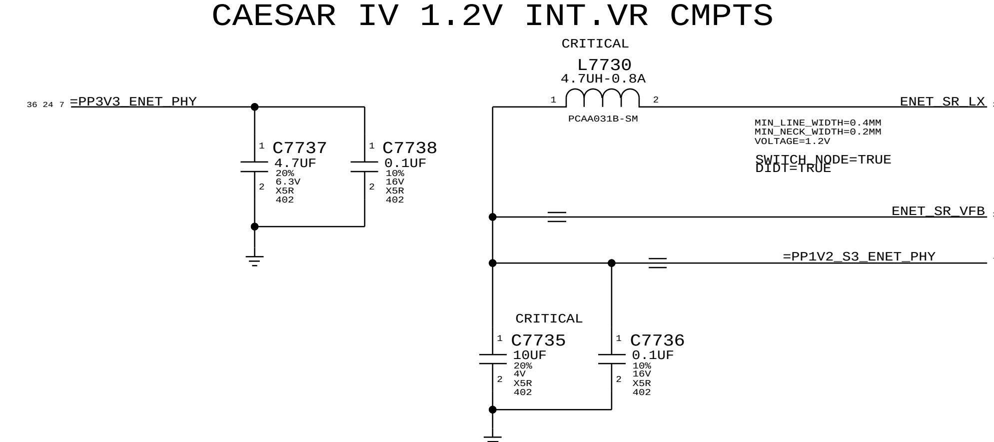
Cougar Point-M requires JTAG pull-ups to be powered at 1.05V in Sus.
Pull-ups (3) must be 51 ohms to support XDP (not required in production).
70mA is required to support pull-ups. Alternative is strong voltage
dividers (200/100) to 3.3V Sus, which burns 100mW in all S-states.



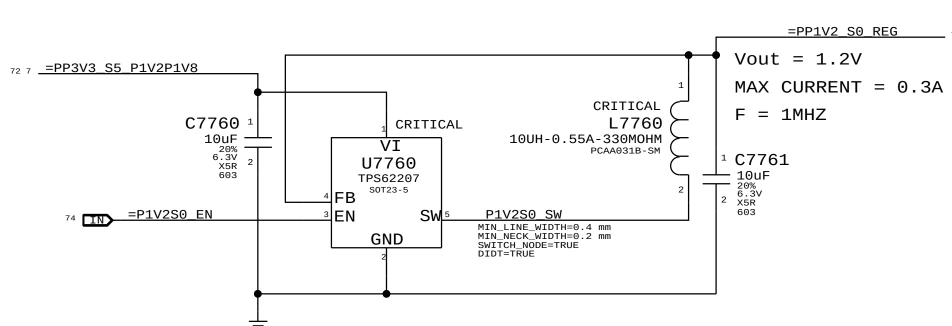
1.5V S0 Regulator



CAESAR IV 1.2V INT.VR CMPTS



1.2V S0 (GMUX) Regulator



SYNC_MASTER=J31_JACK SYNC_DATE=06/10/2011

PAGE TITLE: Misc Power Supplies

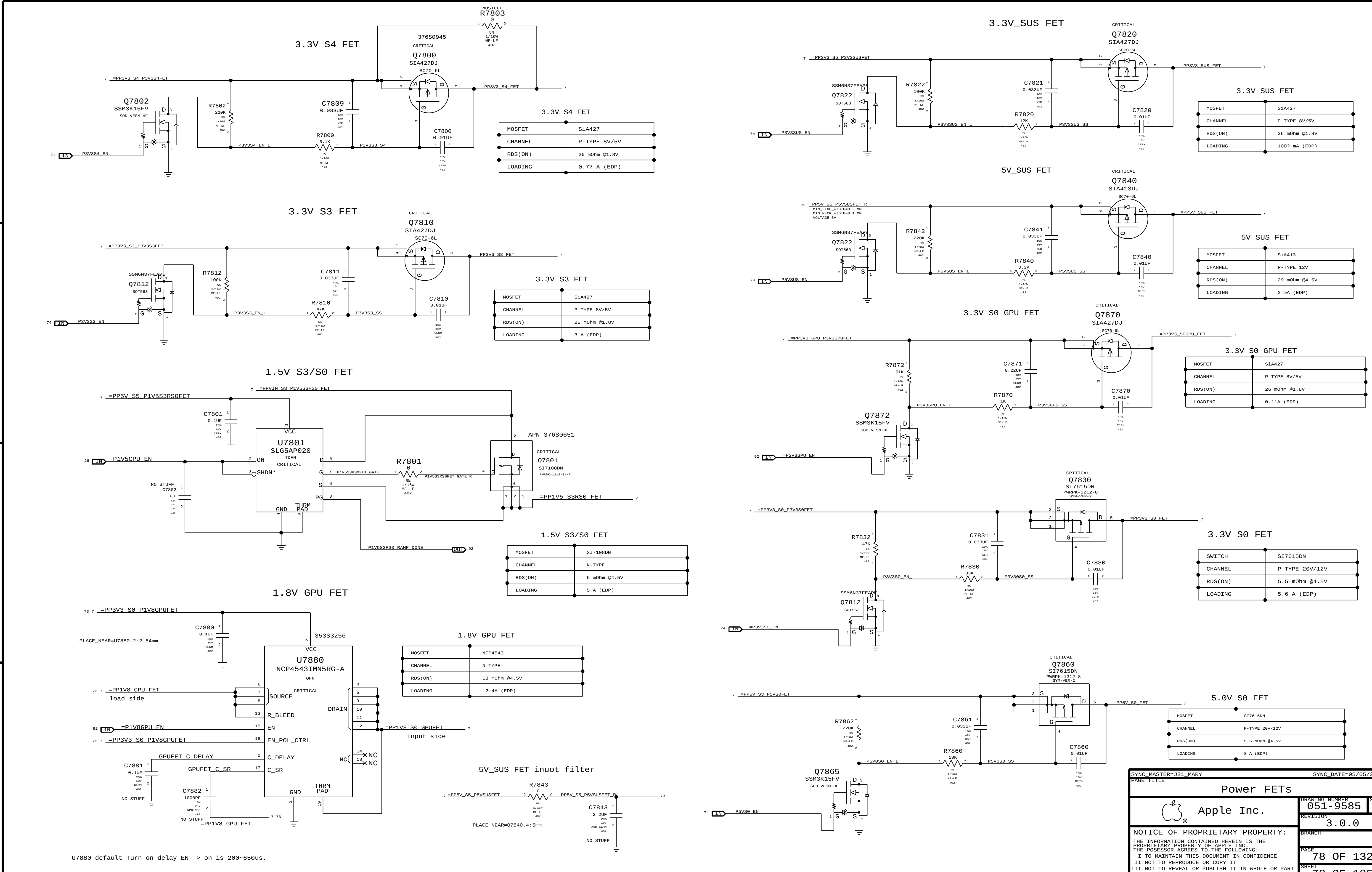
Apple Inc.

DRAWING NUMBER: 051-9585 SIZE: D

REVISION: 3.0.0

BRANCH

NOTICE OF PROPRIETARY PROPERTY:



3.3V S4 FET

MOSFET	SiA427
CHANNEL	P-TYPE 8V/5V
RDS(ON)	26 mOhm @1.8V
LOADING	0.7? A (EDP)

3.3V S3 FET

MOSFET	SiA427
CHANNEL	P-TYPE 8V/5V
RDS(ON)	26 mOhm @1.8V
LOADING	3 A (EDP)

1.5V S3/S0 FET

MOSFET	SI7108DN
CHANNEL	N-TYPE
RDS(ON)	6 mOhm @4.5V
LOADING	5 A (EDP)

1.8V GPU FET

MOSFET	NCP4543
CHANNEL	N-TYPE
RDS(ON)	18 mOhm @4.5V
LOADING	2.4A (EDP)

3.3V SUS FET

MOSFET	SiA427
CHANNEL	P-TYPE 8V/5V
RDS(ON)	26 mOhm @1.8V
LOADING	100? mA (EDP)

5V SUS FET

MOSFET	SiA413
CHANNEL	P-TYPE 12V
RDS(ON)	29 mOhm @4.5V
LOADING	2 mA (EDP)

3.3V S0 GPU FET

MOSFET	SiA427
CHANNEL	P-TYPE 8V/5V
RDS(ON)	26 mOhm @1.8V
LOADING	0.11A (EDP)

3.3V S0 FET

SWITCH	SI7615DN
CHANNEL	P-TYPE 20V/12V
RDS(ON)	5.5 mOhm @4.5V
LOADING	5.6 A (EDP)

5.0V S0 FET

MOSFET	SI7615DN
CHANNEL	P-TYPE 20V/12V
RDS(ON)	5.5 MOHM @4.5V
LOADING	8 A (EDP)

SYNC_MASTER=J31_MARY SYNC_DATE=05/05/2011

PAGE TITLE

Power FETs

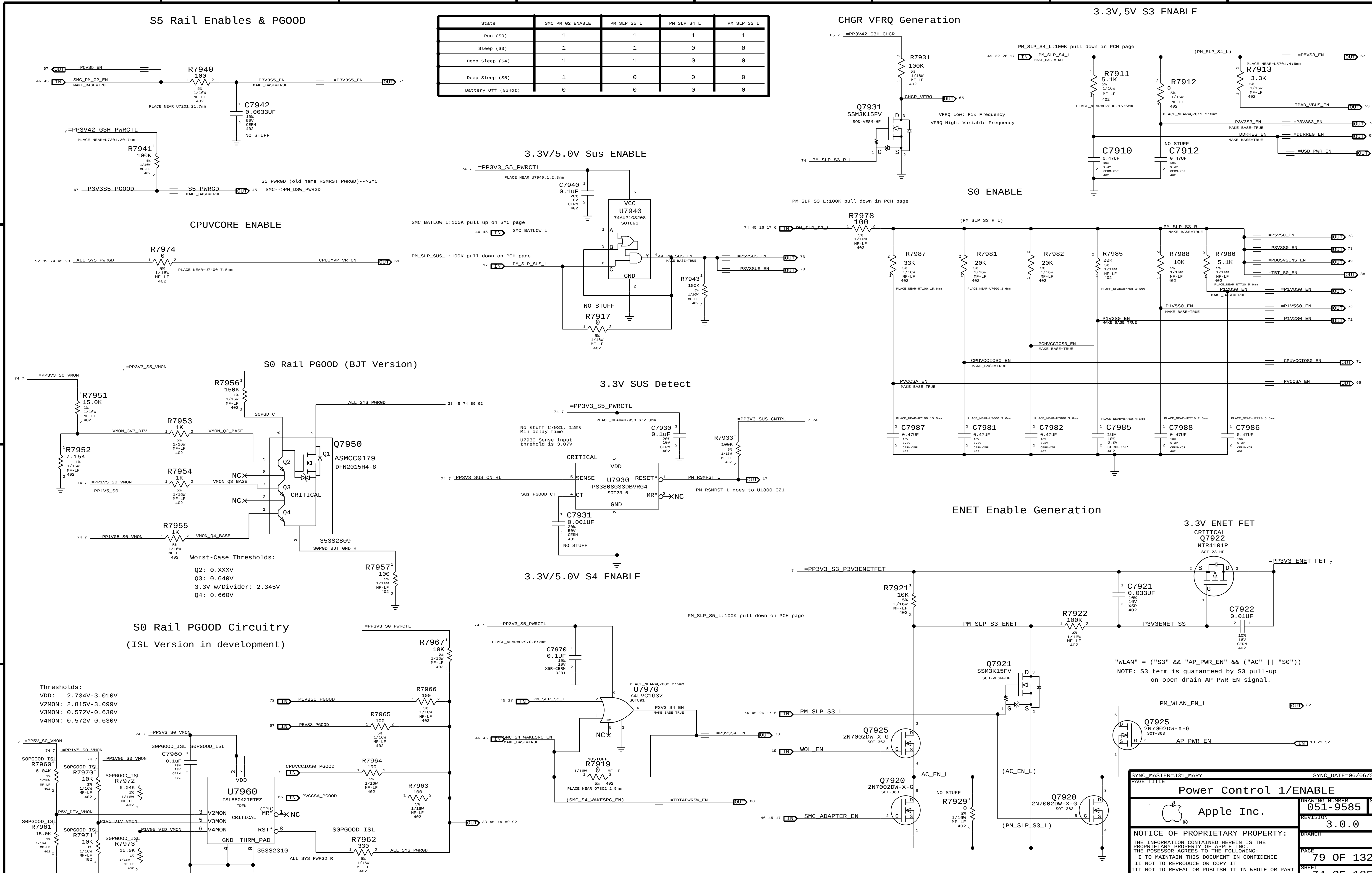
Apple Inc.

DRAWING NUMBER 051-9585 SIZE D

REVISION 3.0.0

BRANCH

State	SMC_PM_G2_ENABLE	PM_SLP_S5_L	PM_SLP_S4_L	PM_SLP_S3_L
Run (S0)	1	1	1	1
Sleep (S3)	1	1	1	0
Deep Sleep (S4)	1	1	0	0
Deep Sleep (S5)	1	0	0	0
Battery Off (G3Hot)	0	0	0	0



SYNC_MASTER=J31_MARY SYNC_DATE=06/06/2011

PAGE TITLE

Power Control 1/ENABLE

Apple Inc.

DRAWING NUMBER 051-9585 SIZE D

REVISION 3.0.0

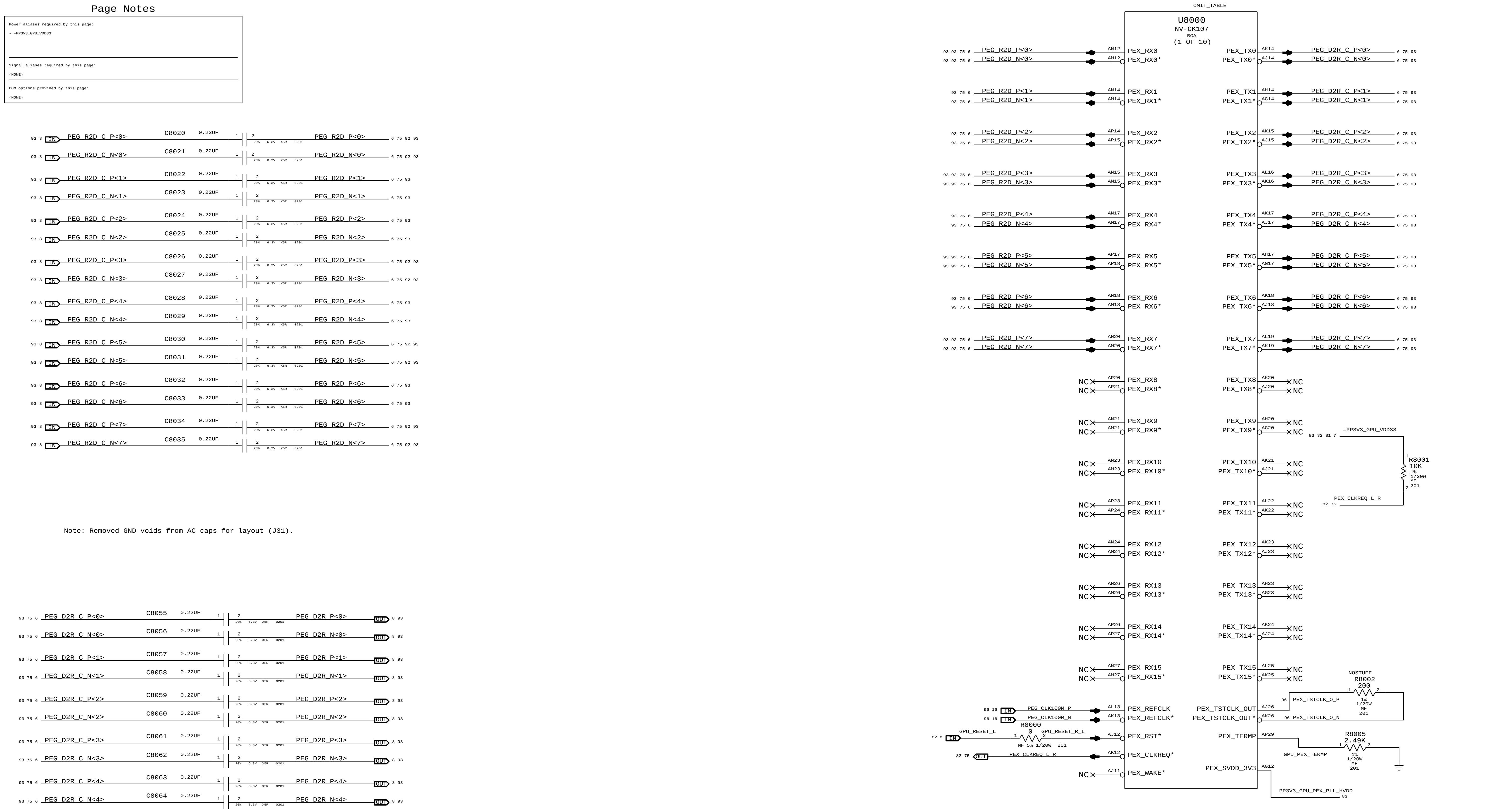
BRANCH

Page Notes

Power aliases required by this page:
- =PP3V3_GPU_VDD33

Signal aliases required by this page:
(NONE)

BOM options provided by this page:
(NONE)



Note: Removed GND voids from AC caps for layout (J31).

SYNC_MASTER=J31_SREE SYNC_DATE=10/25/2011

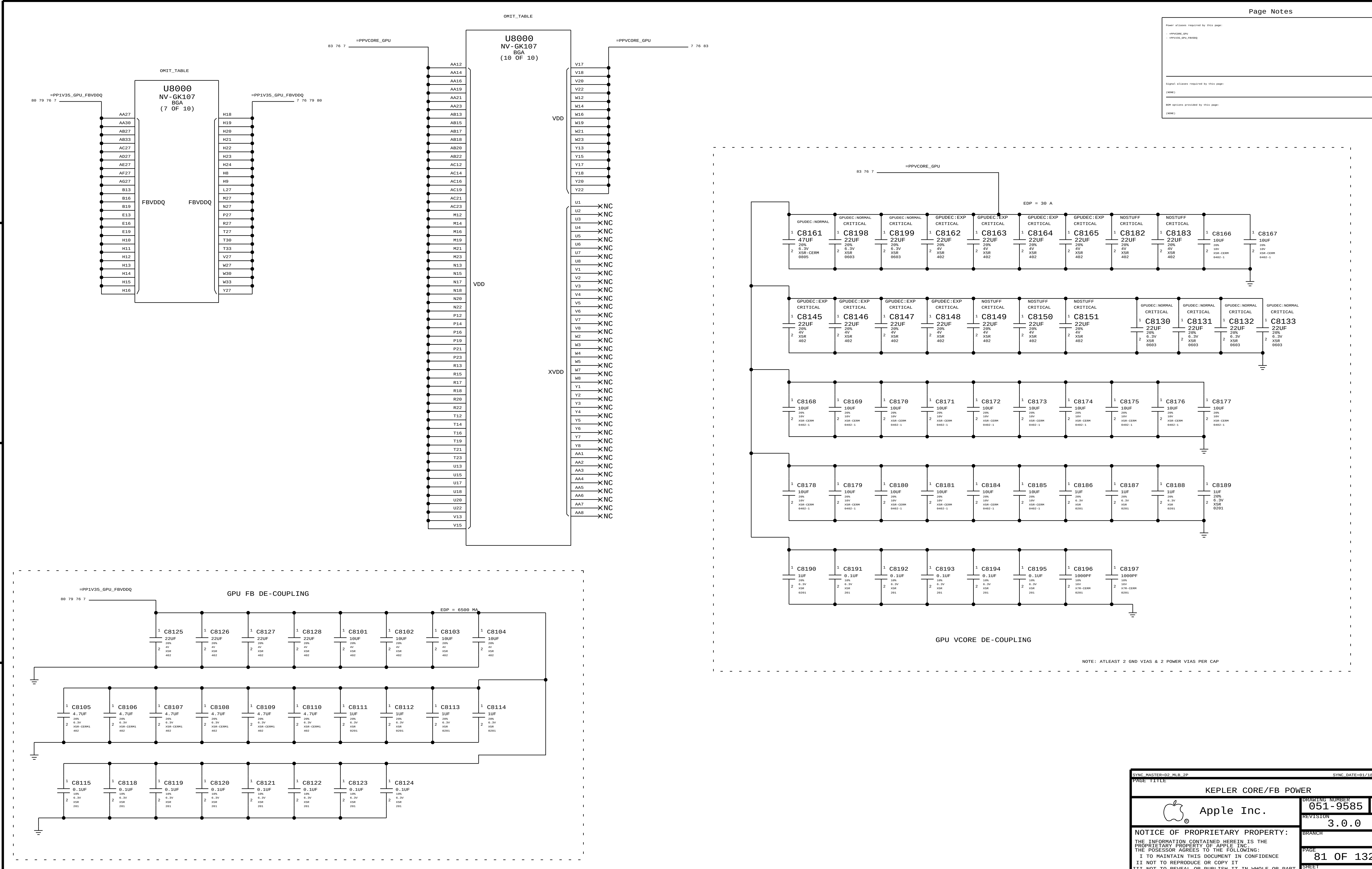
PAGE TITLE
KEPLER PCI-E

Apple Inc.

DRAWING NUMBER 051-9585 SIZE D

REVISION 3.0.0

BRANCH



SYNC_MASTER=D2_MLB_2P SYNC_DATE=01/18/2012

PAGE TITLE

KEPLER CORE/FB POWER

Apple Inc.

DRAWING NUMBER 051-9585 SIZE D

REVISION 3.0.0

BRANCH

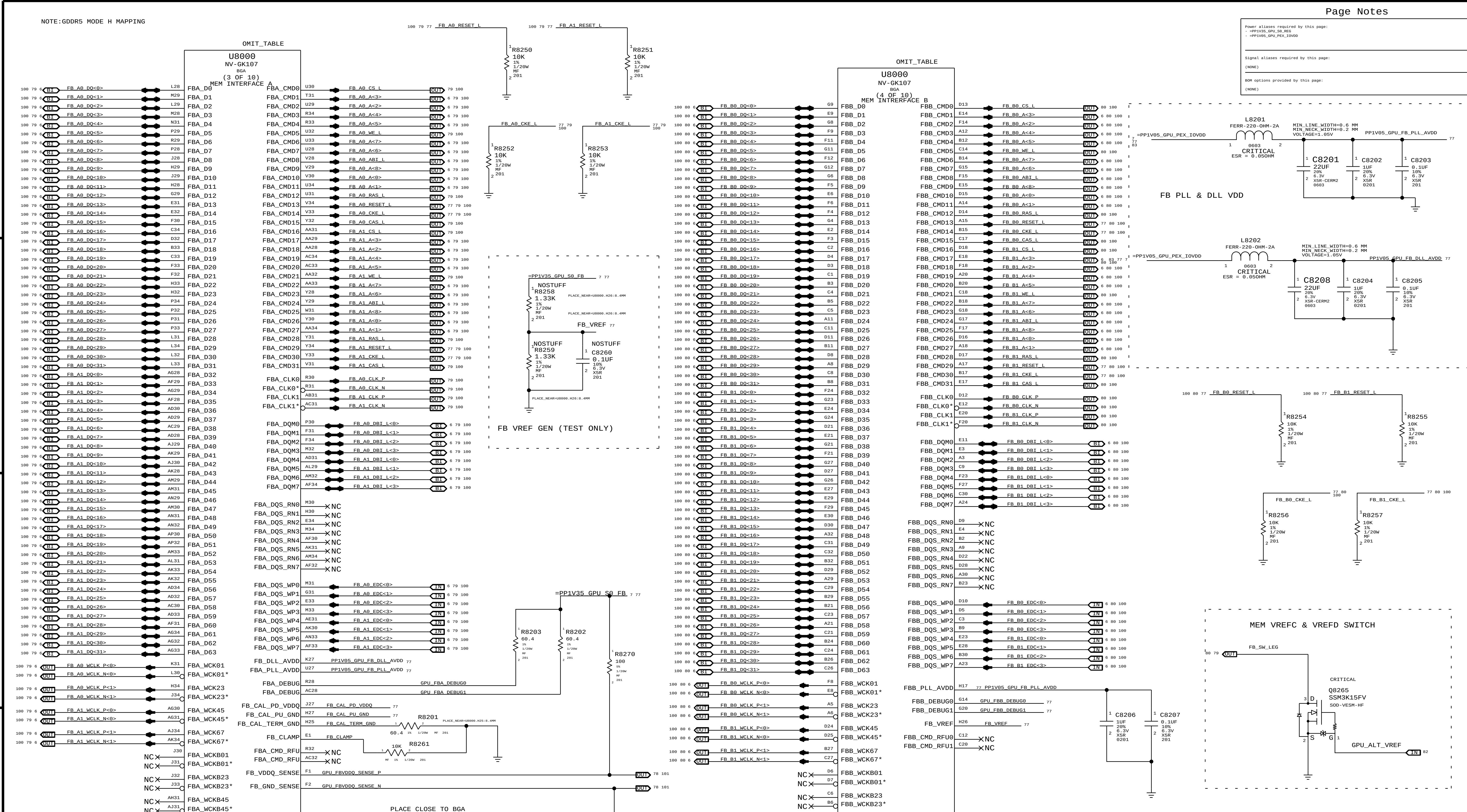
PAGE 81 OF 132

SHEET 76 OF 105

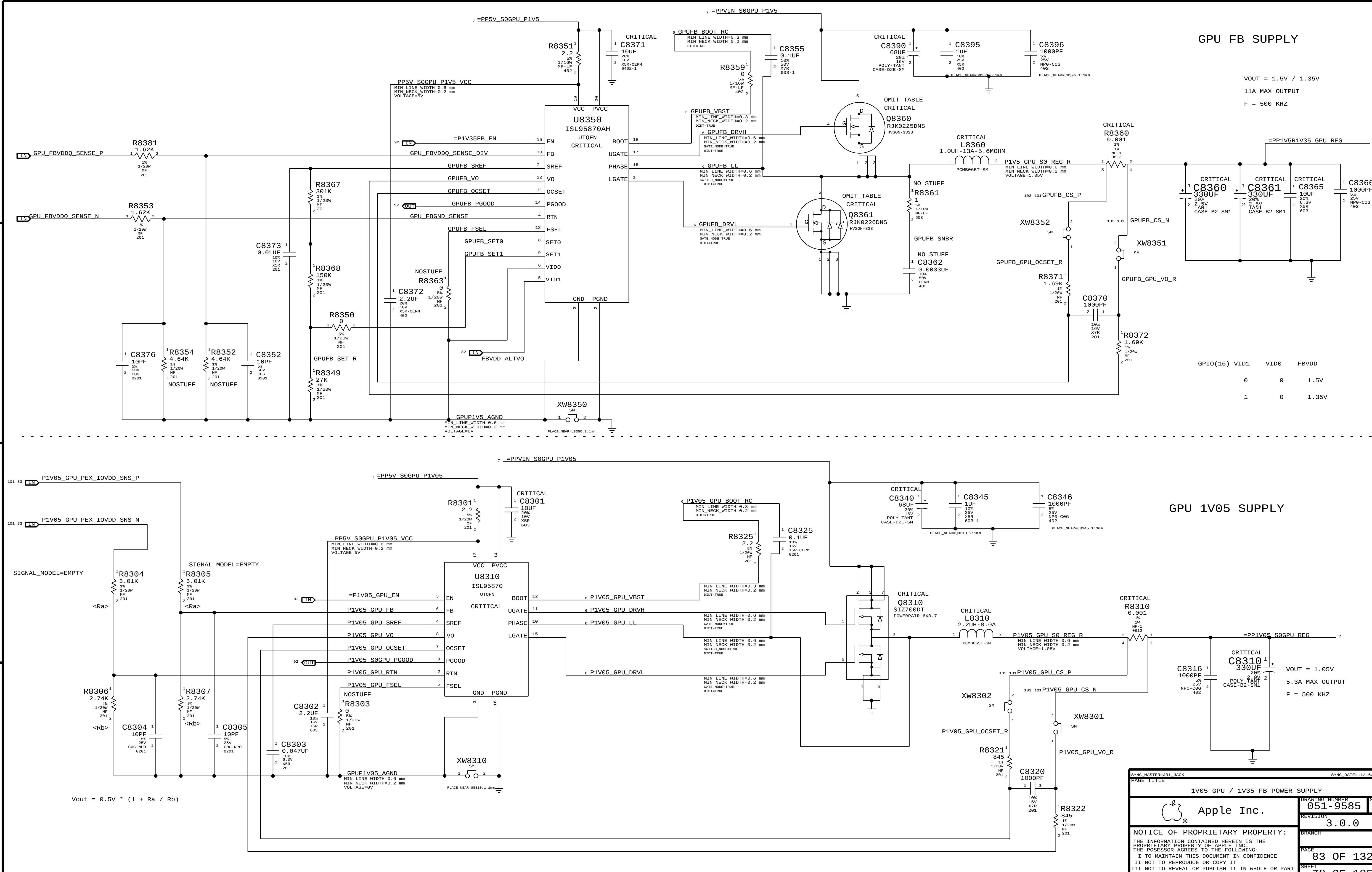
NOTICE OF PROPRIETARY PROPERTY:

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSESSOR AGREES TO THE FOLLOWING:

I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART



SYNC_MASTER=J31_SREE
SYNC_DATE=10/25/2011
PAGE TITLE
KEPLER FRAME BUFFER I/F
DRAWING NUMBER 051-9585
SIZE D
Apple Inc.
REVISION 3.0.0
BRANCH
PAGE 82 OF 132
SHEET 77 OF 105



SYNC_MASTER=J31_JACK SYNC_DATE=11/16/2011

PAGE TITLE

1V05 GPU / 1V35 FB POWER SUPPLY

Apple Inc.

DRAWING NUMBER 051-9585 SIZE D

REVISION 3.0.0

BRANCH

Page Notes

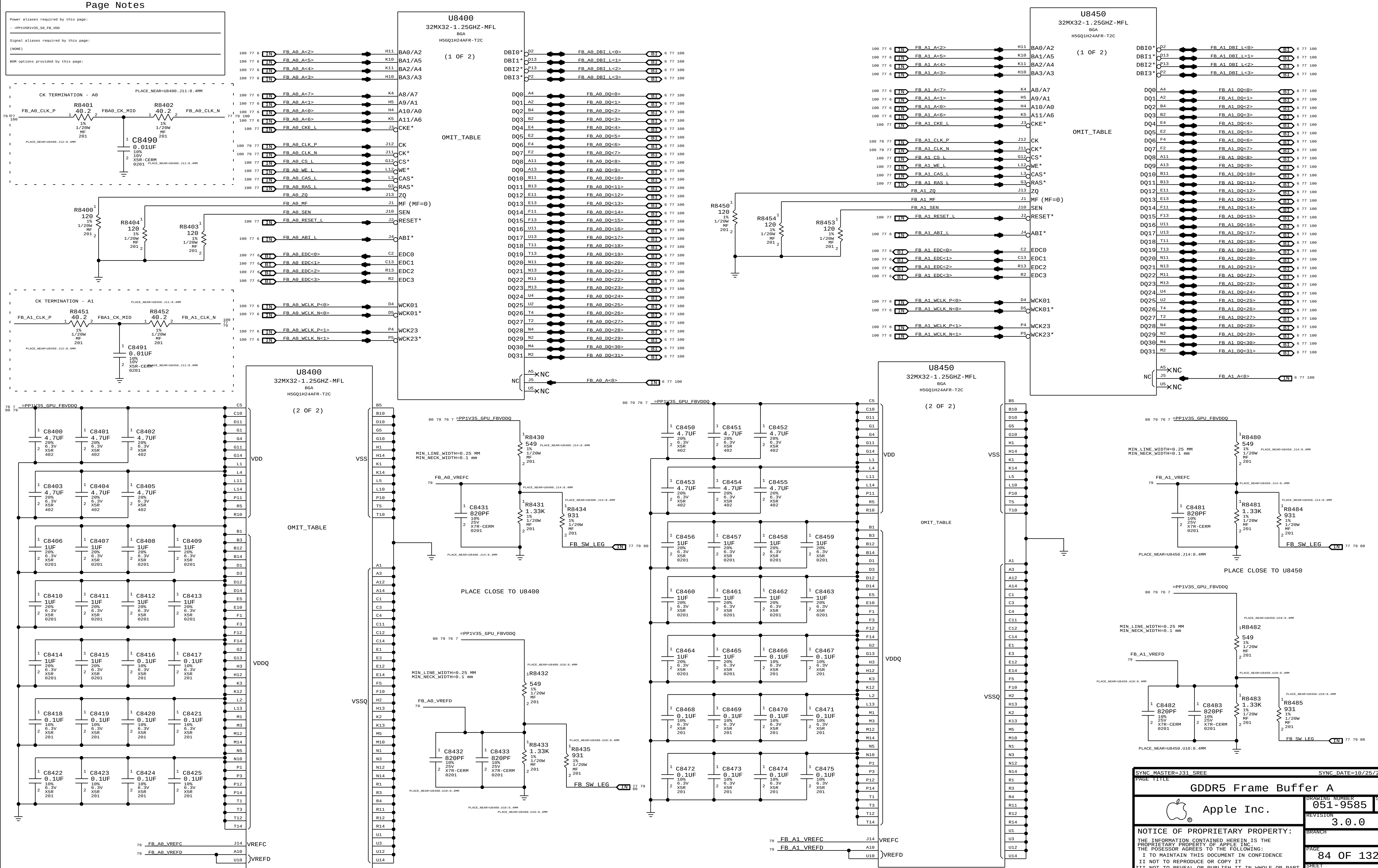
Power aliases required by this page:

- =PP1V5R1V35_S0_FB_VDD

Signal aliases required by this page:

(NONE)

BOM options provided by this page:



SYNC_MASTER=J31_SREE SYNC_DATE=10/25/2011

PAGE TITLE

GDDR5 Frame Buffer A

DRAWING NUMBER 051-9585 SIZE D

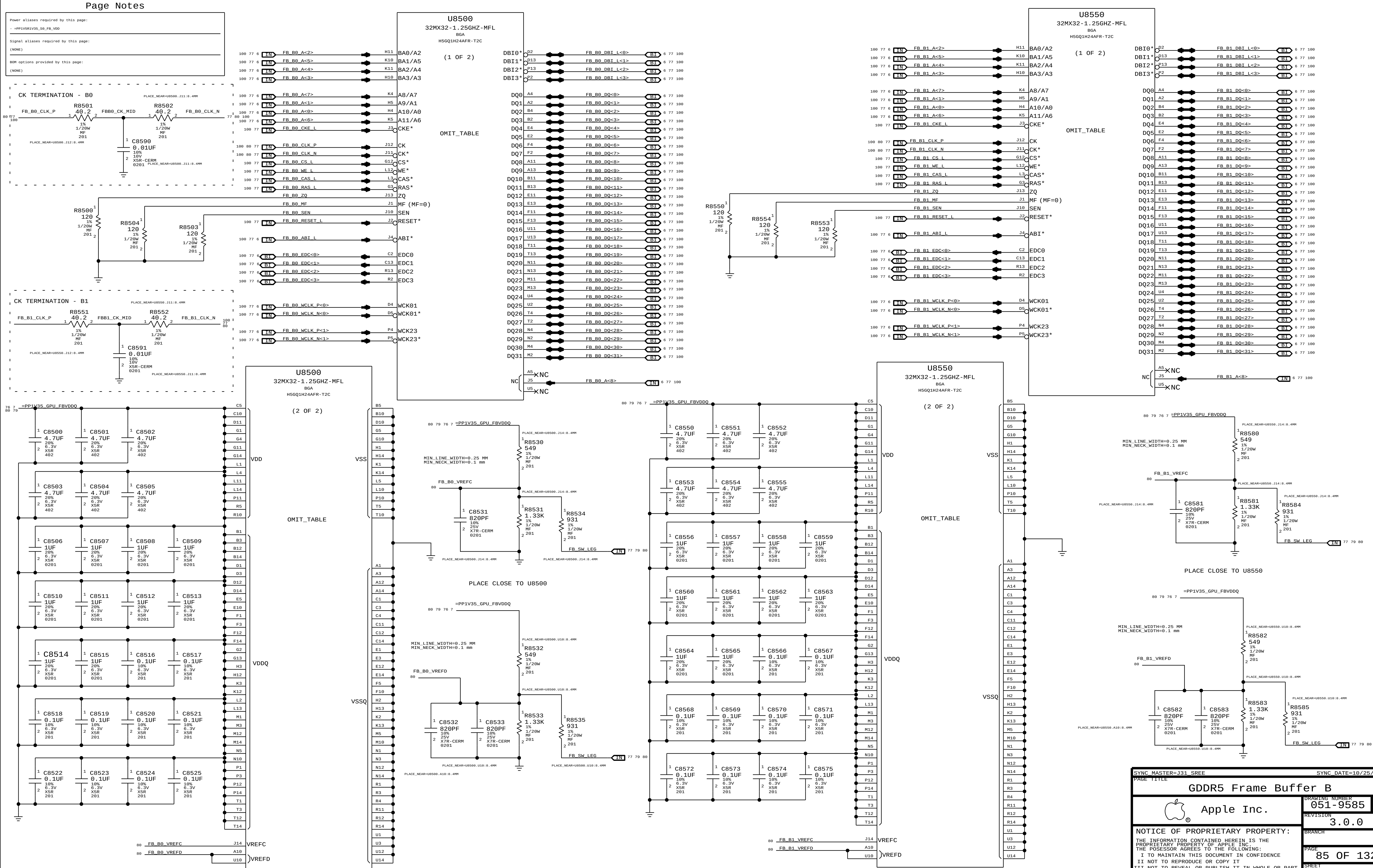
Apple Inc.

REVISION 3.0.0

BRANCH

PAGE 84 OF 132

SHEET 79 OF 105



SYNC_MASTER=J31_SREE SYNC_DATE=10/25/2011

PAGE TITLE

GDDR5 Frame Buffer B

Apple Inc.

DRAWING NUMBER 051-9585 SIZE D

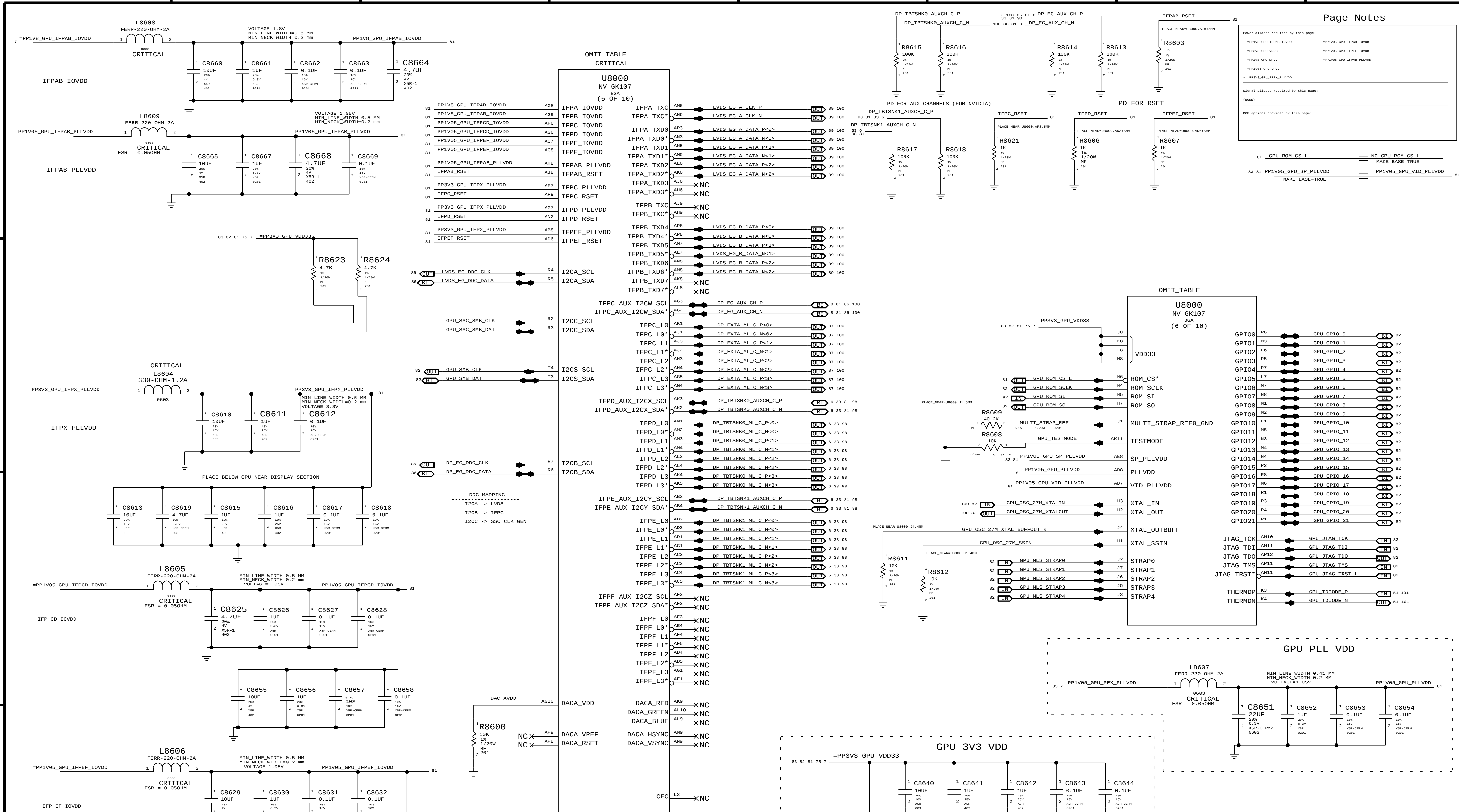
REVISION 3.0.0

BRANCH

NOTICE OF PROPRIETARY PROPERTY:

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC.
THE POSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

PAGE 85 OF 132



SYNC_MASTER=J31_SREE SYNC_DATE=10/25/2011

PAGE TITLE

KEPLER LVDS/DP/GPIO

Apple Inc.

DRAWING NUMBER 051-9585 SIZE D

REVISION 3.0.0

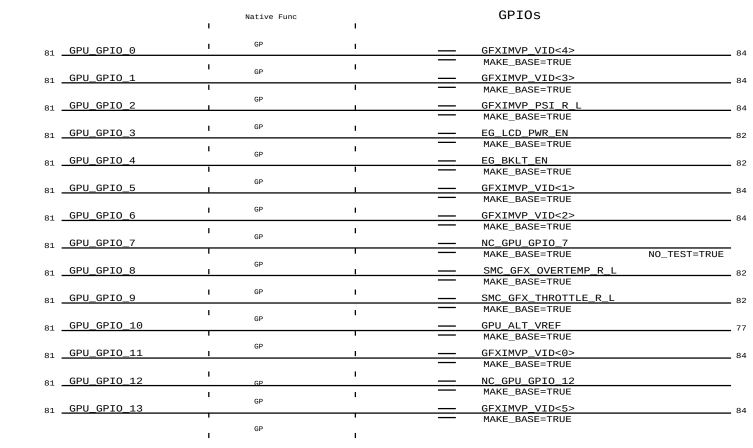
BRANCH

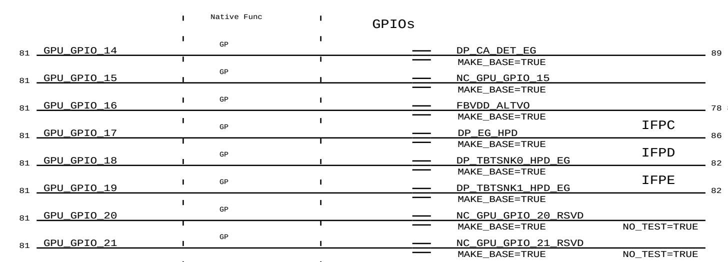
NOTICE OF PROPRIETARY PROPERTY:

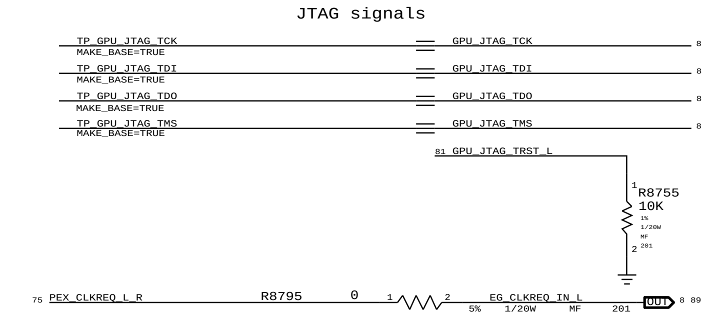
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC.
THE POSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

PAGE 86 OF 132

SHEET 81 OF 105





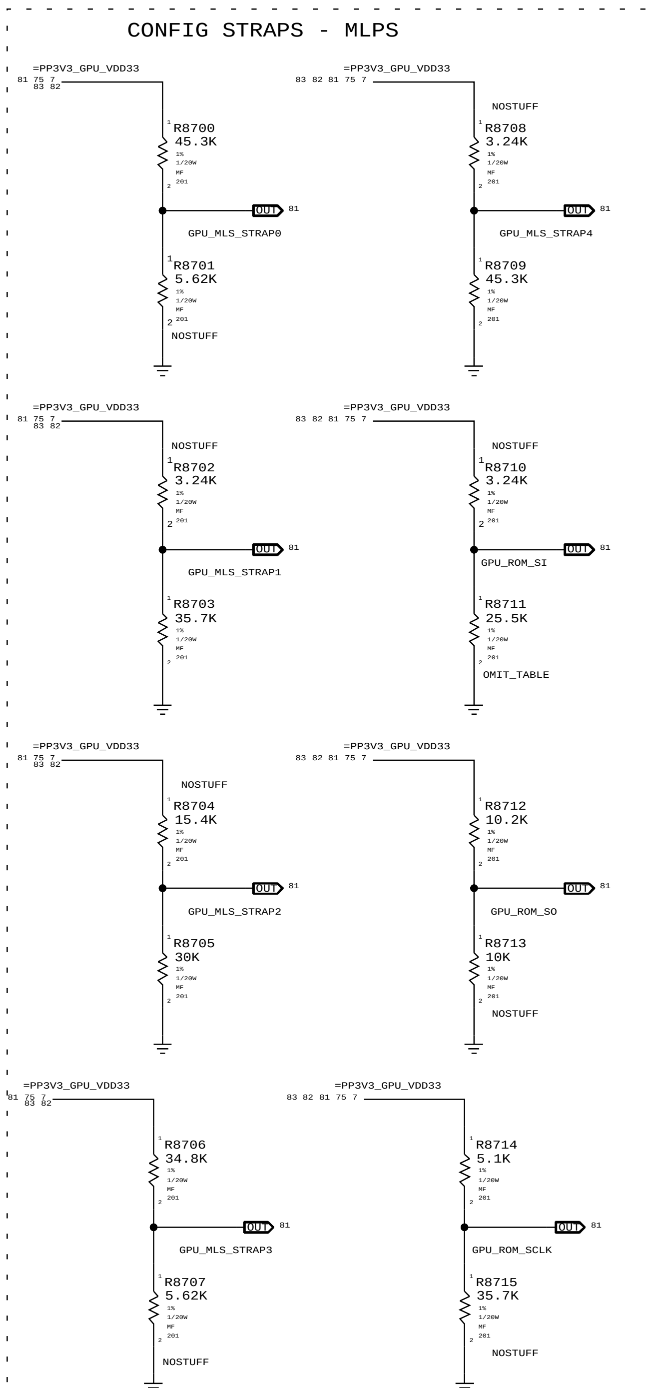


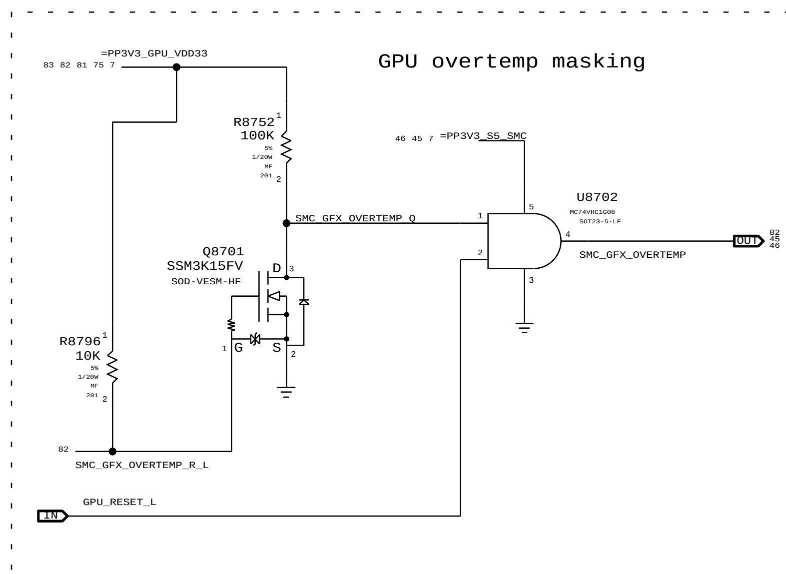
STRAP NOTES:

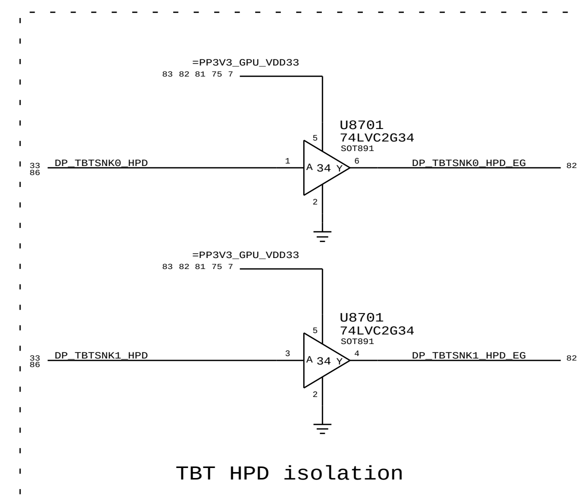
CURRENTLY STUFFED FOR GK107-GTX (R8705)

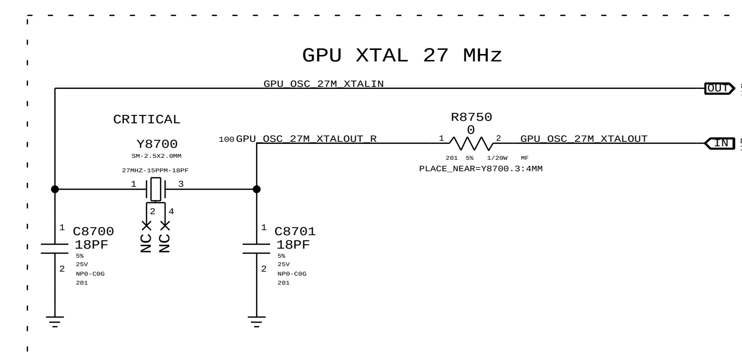
STUFF R8711 = 5KOHM FOR HYNIX 1GB - M die
STUFF R8711 = 10KOHM FOR SAMSUNG 1GB
STUFF R8711 = 15KOHM FOR HYNIX 512MB
STUFF R8711 = 20KOHM FOR SAMSUNG 512MB
STUFF R8711 = 24.9KOHM FOR HYNIX 1GB - A die

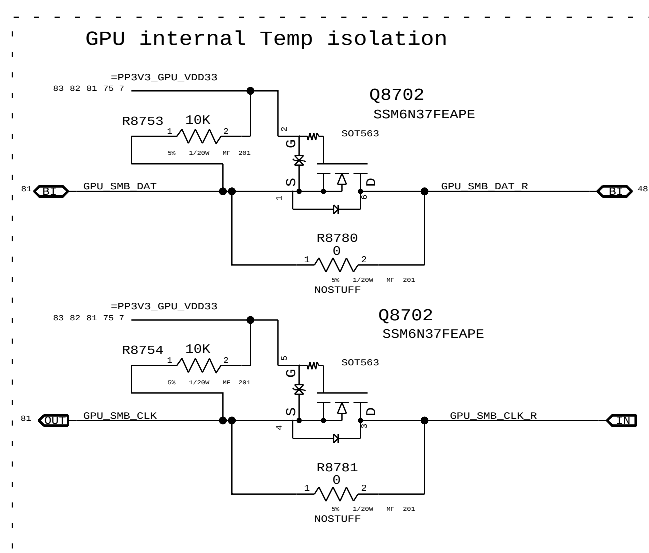
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION	Strap values
118S0019	1	RES, 10.2KOHM, 0201	R8711	CRITICAL	FB_1G_SAMSUNG	0x01
118S0414	1	RES, 5.1KOHM, 0201	R8711	CRITICAL	FB_1G_HYNIX_M_DIE	0x00
118S0105	1	RES,15KOHM, 0201	R8711	CRITICAL	FB_512_HYNIX	0x02
118S0175	1	RES, 20KOHM, 0201	R8711	CRITICAL	FB_512_SAMSUNG	0x03
118S0230	1	RES, 24.9KOHM, 0201	R8711	CRITICAL	FB_1G_HYNIX_A_DIE	0x04

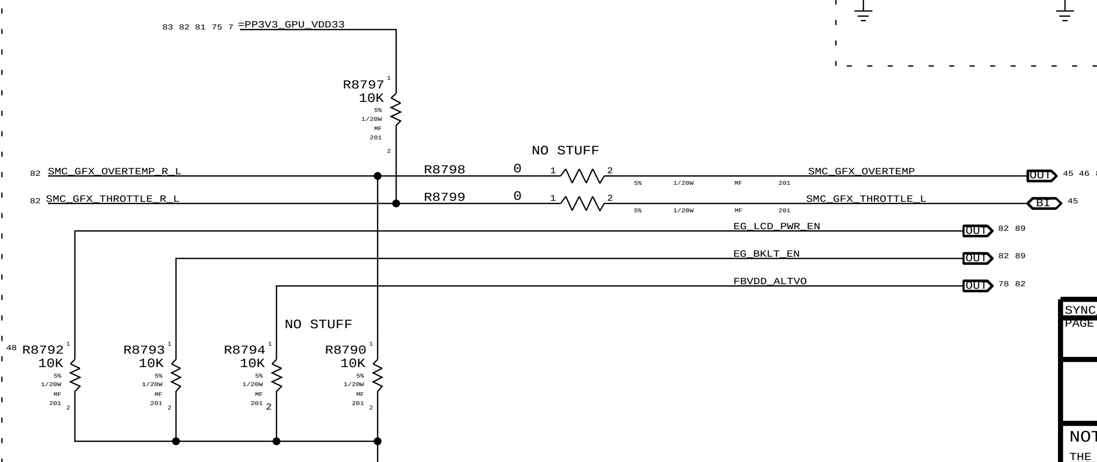












SYNC_MASTER=J31_SREE SYNC_DATE=11/16/2011

PAGE TITLE

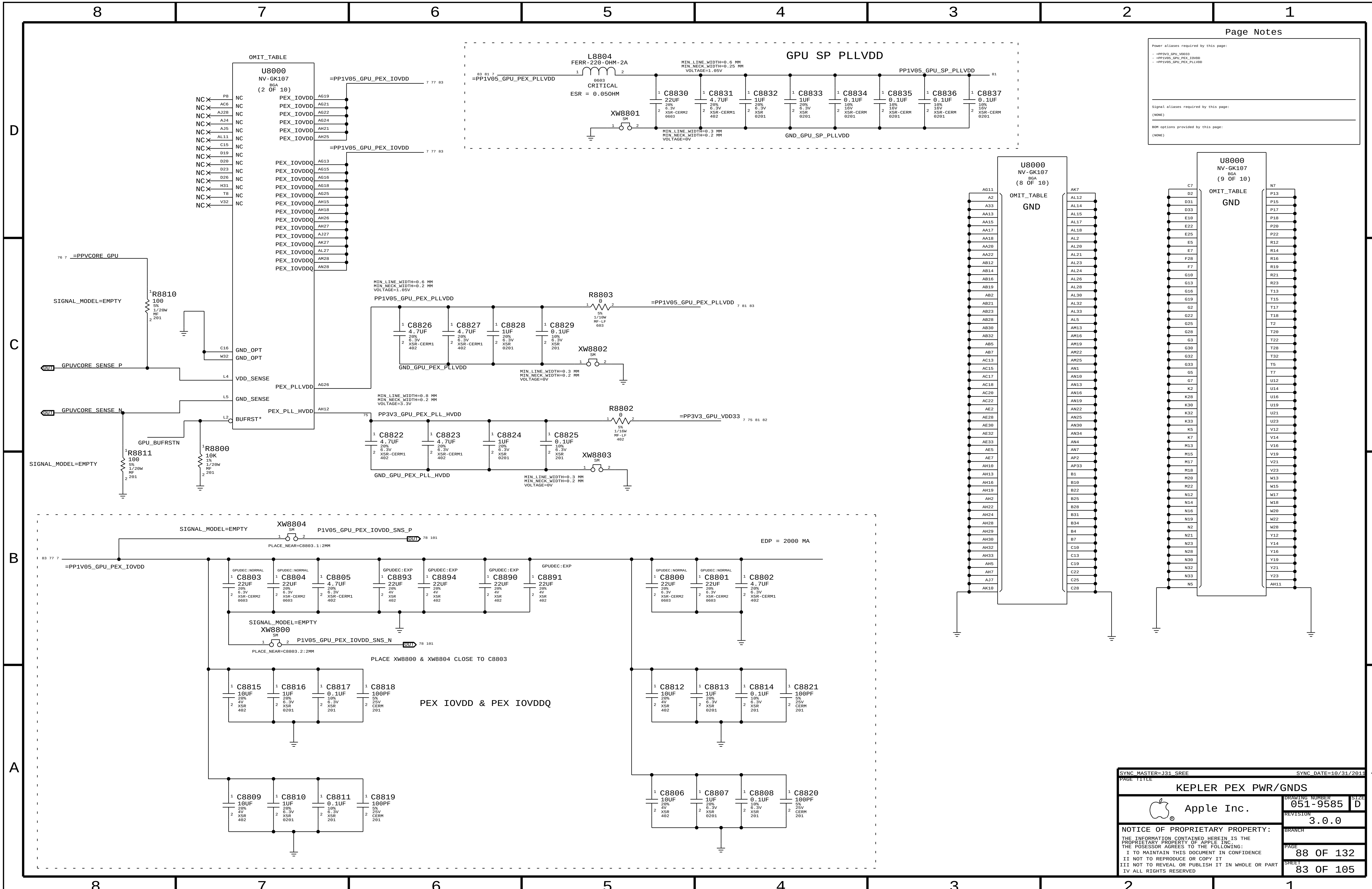
KEPLER GPIOS,CLK & STRAPS

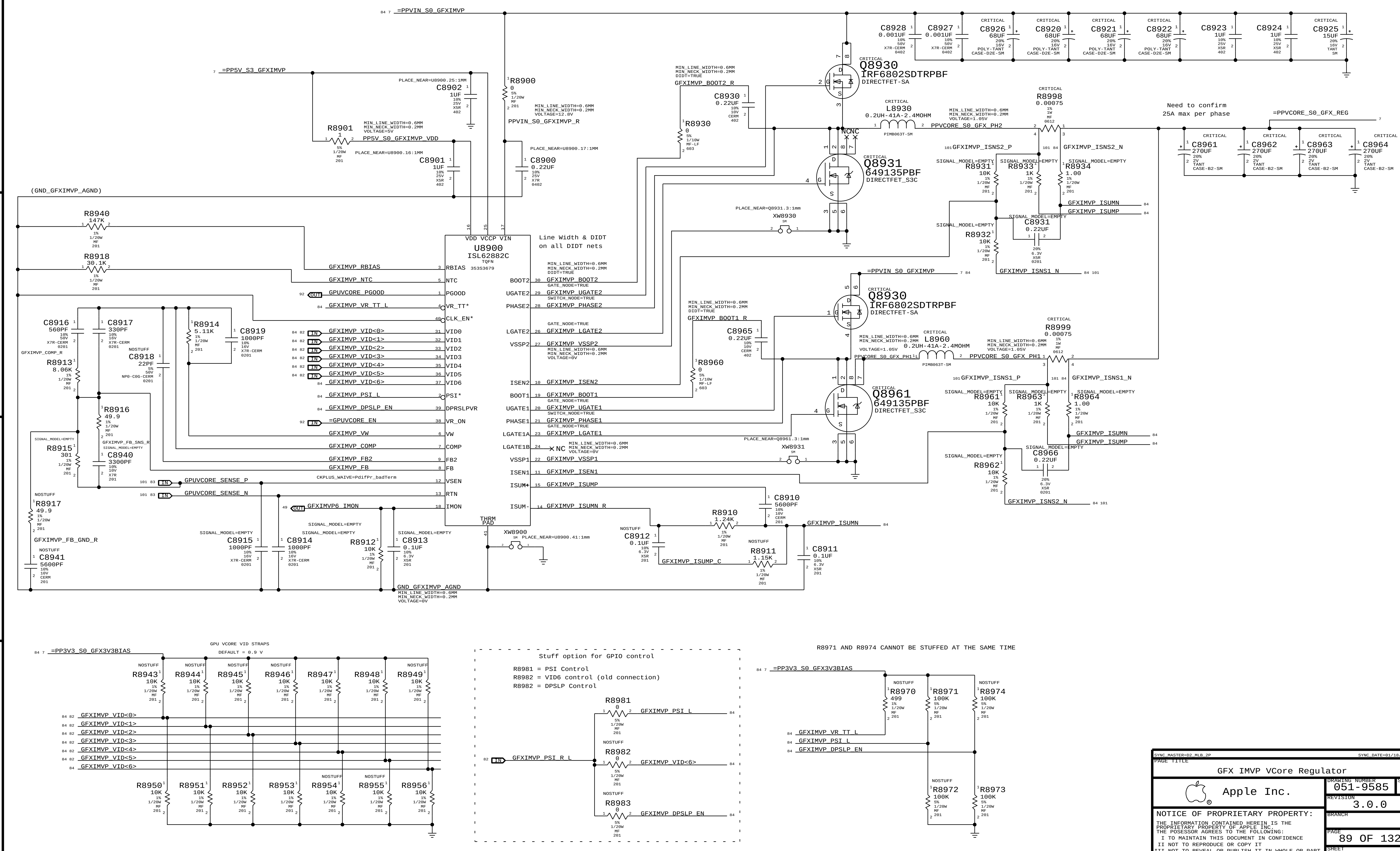
Apple Inc.

DRAWING NUMBER 051-9585 SIZE D

REVISION 3.0.0

BRANCH


GPU SP PLLVDD
PEX IOVDD & PEX IOVDDQ
Page Notes
Power aliases required by this page:
- =PP3V3_GPU_VDD33
- =PP1V05_GPU_PEX_IOVDD
- =PP1V05_GPU_PEX_PLLVDD
Signal aliases required by this page:
(NONE)
BOM options provided by this page:
(NONE)
U8000
NV-GK107
BGA
(2 OF 10)
(8 OF 10)
(9 OF 10)
OMIT_TABLE
GND
=PP1V05_GPU_PEX_IOVDD
=PP1V05_GPU_PEX_PLLVDD
PP1V05_GPU_SP_PLLVDD
GND_GPU_SP_PLLVDD
PP1V05_GPU_PEX_PLLVDD
GND_GPU_PEX_PLLVDD
PP3V3_GPU_PEX_PLL_HVDD
GND_GPU_PEX_PLL_HVDD
=PP3V3_GPU_VDD33
=PPVCORE_GPU
GPUVCORE_SENSE_P
GPUVCORE_SENSE_N
GPU_BUFRSTN
SIGNAL_MODEL=EMPTY
P1V05_GPU_PEX_IOVDD_SNS_P
P1V05_GPU_PEX_IOVDD_SNS_N
PLACE XW8800 & XW8804 CLOSE TO C8803
EDP = 2000 MA
SYNC_MASTER=J31_SREE
SYNC_DATE=10/31/2011
PAGE TITLE
KEPLER PEX PWR/GNDS
Apple Inc.
DRAWING NUMBER 051-9585
SIZE D
REVISION 3.0.0
BRANCH
PAGE 88 OF 132
SHEET 83 OF 105
NOTICE OF PROPRIETARY PROPERTY:
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC.
THE POSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED



SYNC_MASTER=D2_MLB_2P SYNC_DATE=01/18/2012

PAGE TITLE

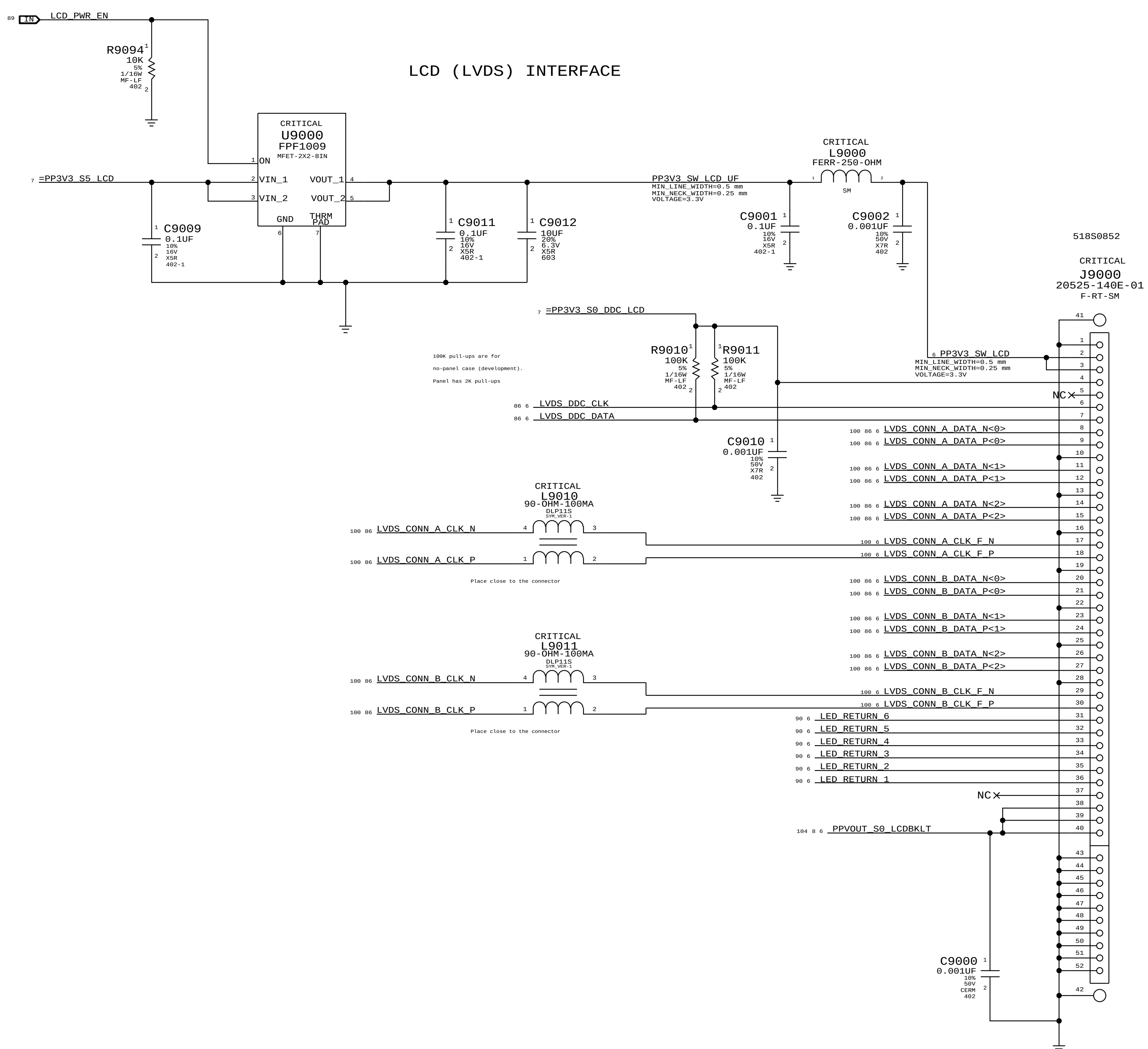
GFX IMVP VCore Regulator

Apple Inc.

DRAWING NUMBER 051-9585 SIZE D

REVISION 3.0.0

BRANCH



SYNC_MASTER=K18_MLB SYNC_DATE=04/27/2010

PAGE TITLE

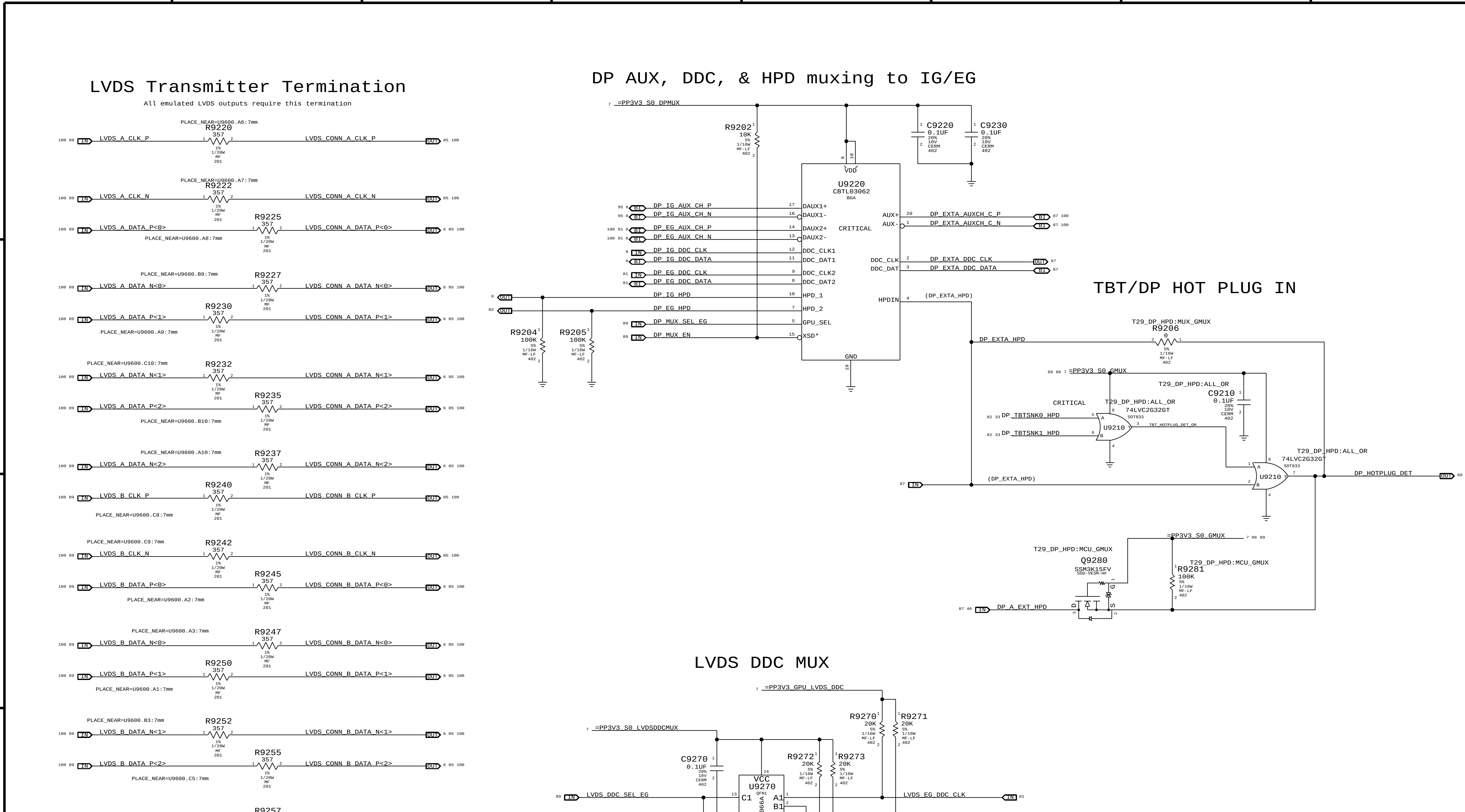
LVDS Display Connector

Apple Inc.

DRAWING NUMBER 051-9585 SIZE D

REVISION 3.0.0

BRANCH



SYNC_MASTER=K92_MLB SYNC_DATE=11/21/2010

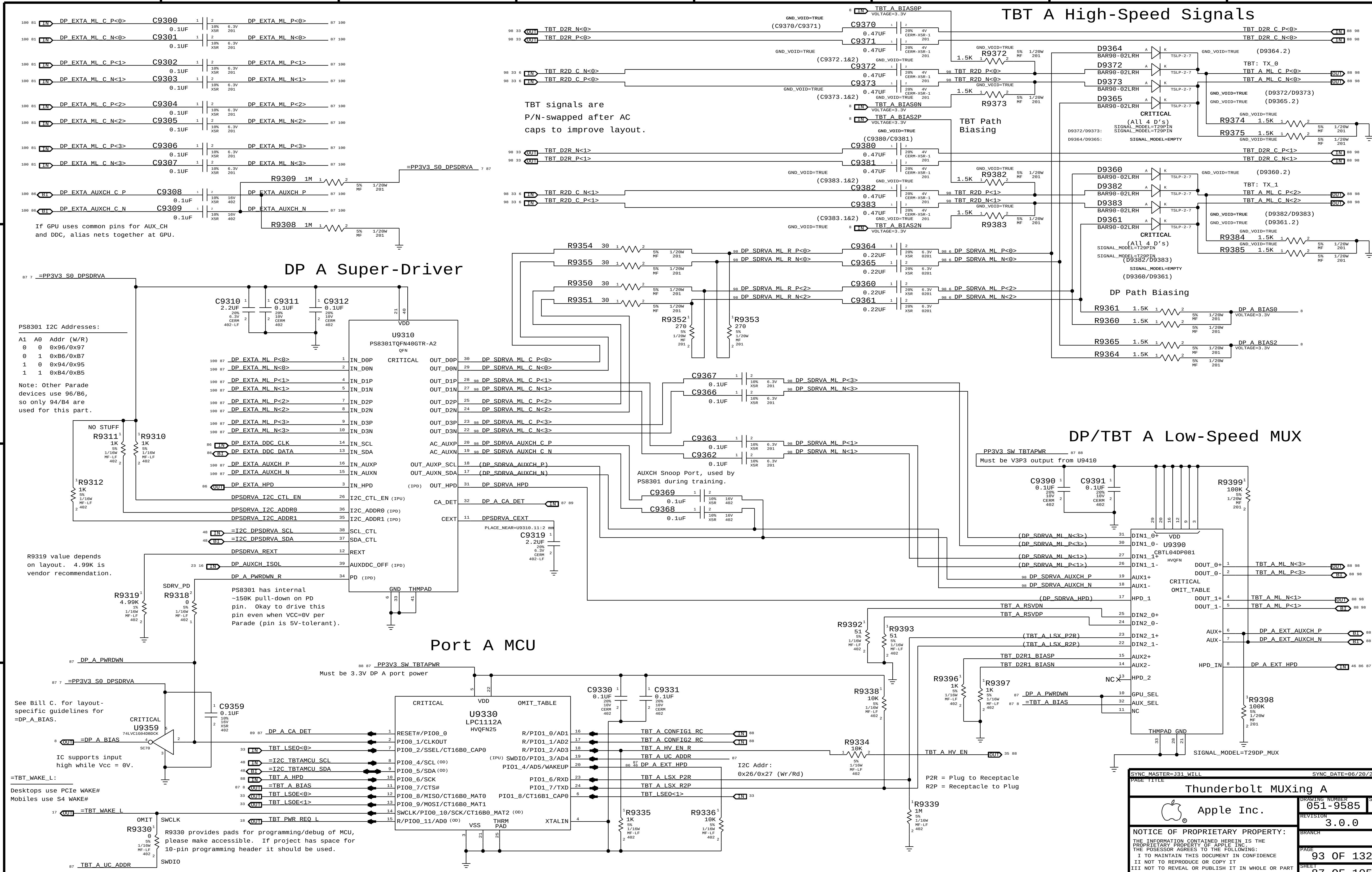
PAGE TITLE: Muxed Graphics Support

Apple Inc.

DRAWING NUMBER: 051-9585 SIZE: D

REVISION: 3.0.0

BRANCH



SYNC_MASTER=J31_WILL SYNC_DATE=06/20/2011

PAGE TITLE

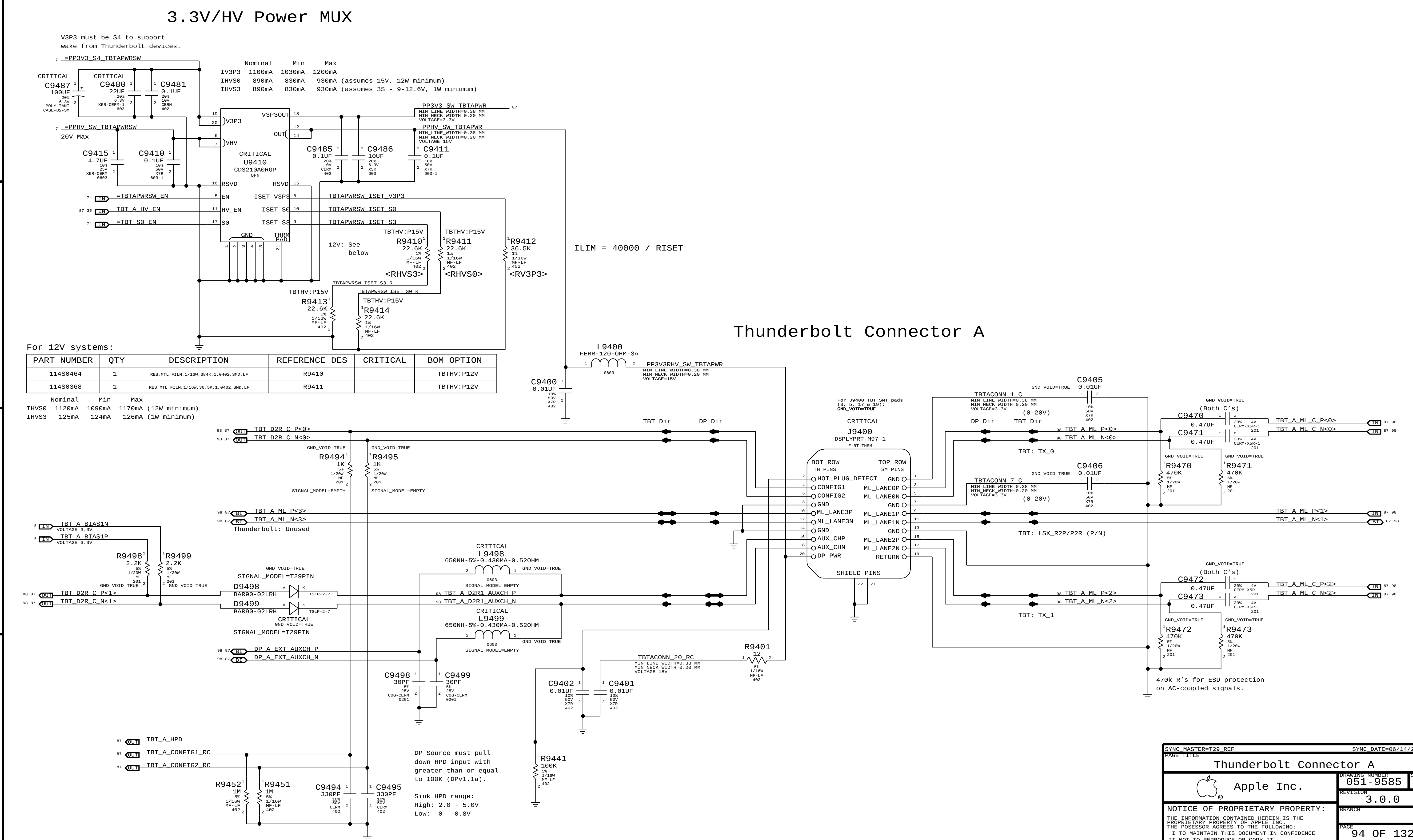
Thunderbolt MUXing A

Apple Inc.

DRAWING NUMBER 051-9585 SIZE D

REVISION 3.0.0

3.3V/HV Power MUX



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
114S0464	1	RES,MTL FILM,1/16W,384K,1,0402,SMD,LF	R9410		TBTHV:P12V
114S0368	1	RES,MTL FILM,1/16W,36.5K,1,0402,SMD,LF	R9411		TBTHV:P12V

SYNC_MASTER=T29_REF SYNC_DATE=06/14/2011

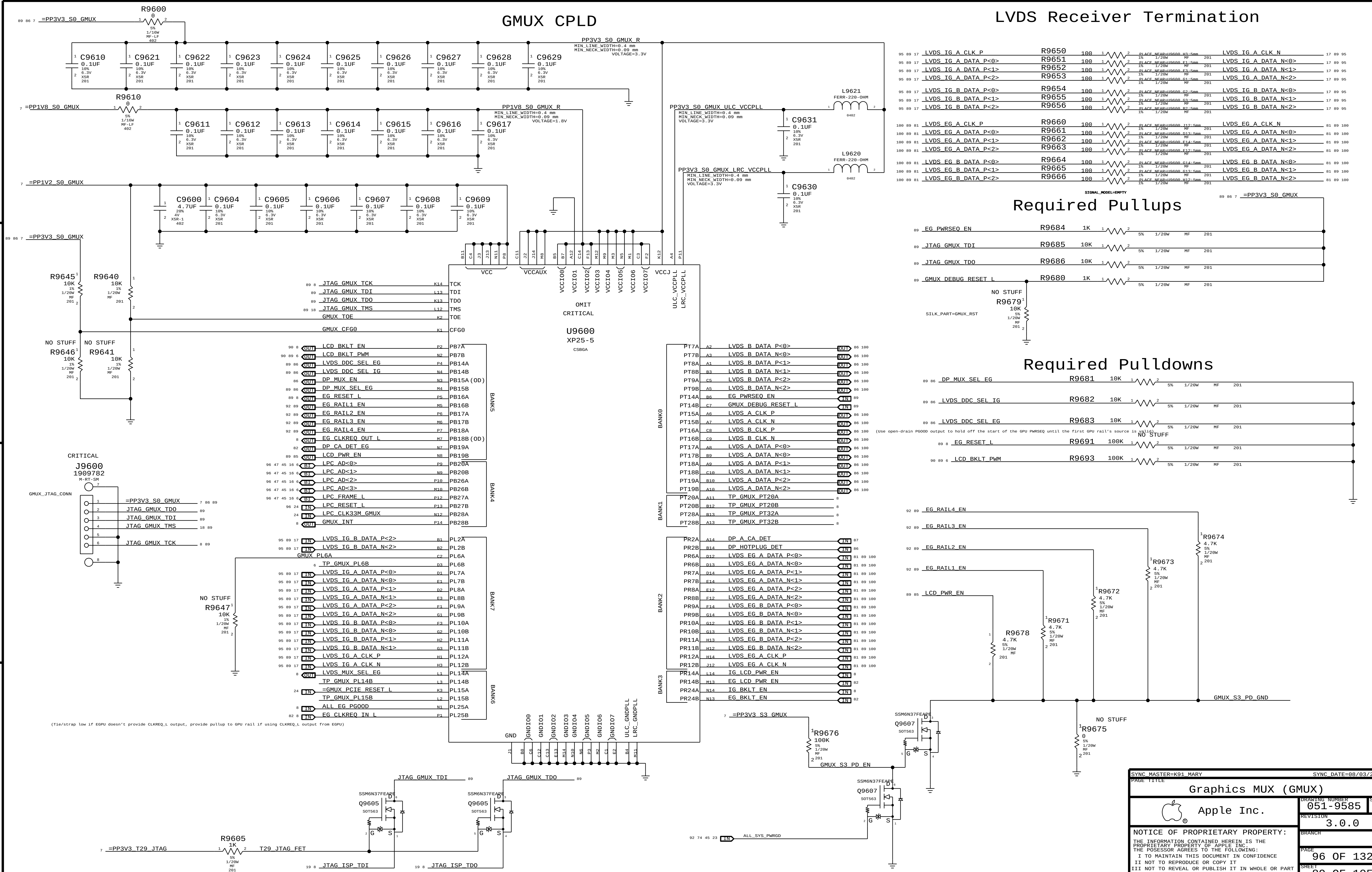
PAGE TITLE: Thunderbolt Connector A

Apple Inc.

DRAWING NUMBER: 051-9585 SIZE: D

REVISION: 3.0.0

BRANCH



SYNC_MASTER=K91_MARY SYNC_DATE=08/03/2010

PAGE TITLE

Graphics MUX (GMUX)

Apple Inc.

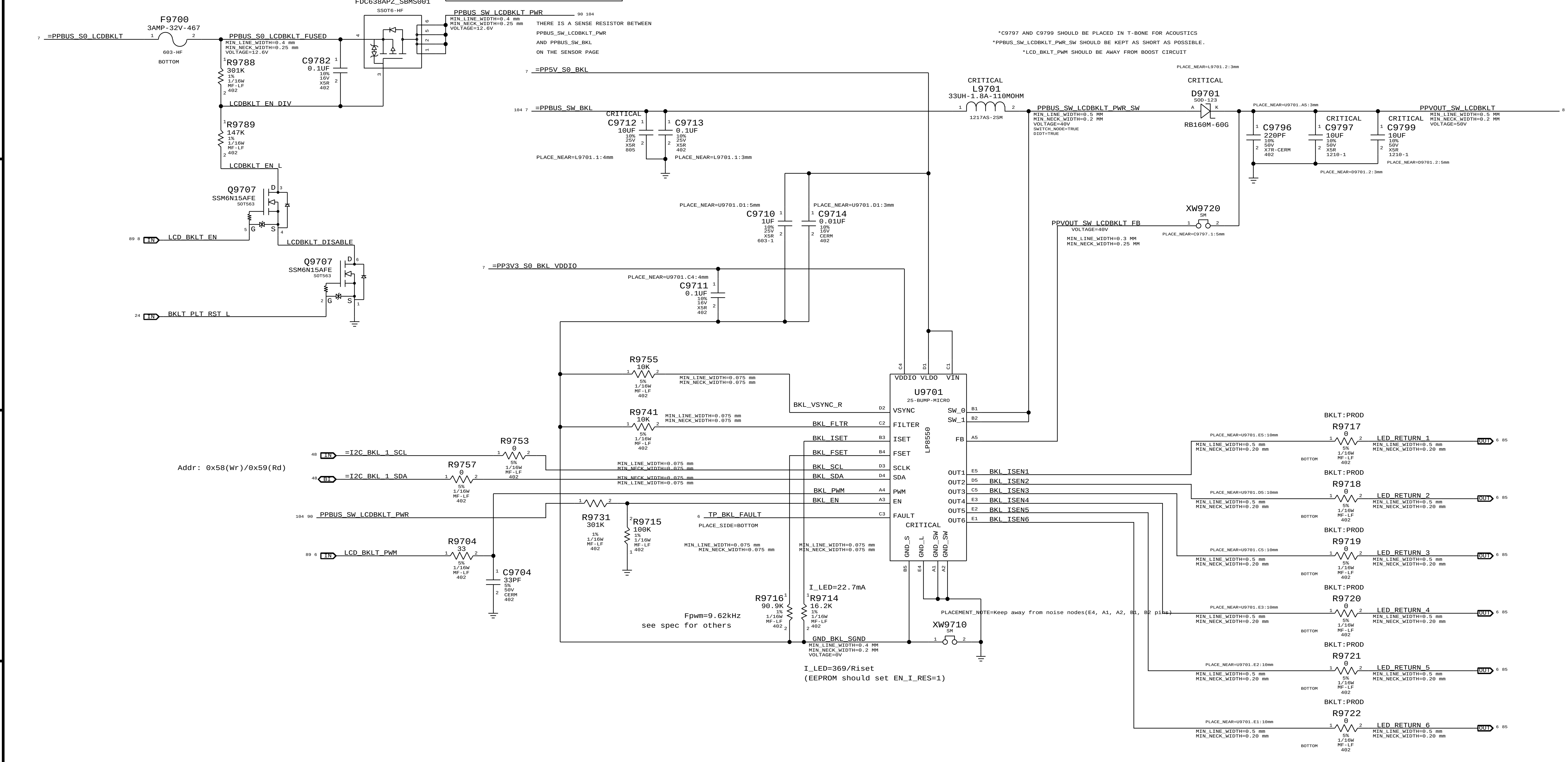
DRAWING NUMBER 051-9585 SIZE D

REVISION 3.0.0

BRANCH

PPBUS S0 LCDBKLT FET

MOSFET	FDC638APZ
CHANNEL	P-TYPE
RDS(ON)	43 mOhm @4.5V
LOADING	0.715 A (EDP)



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
103S0198	3	RES,THIN FLIM,1/16W,10.2 OHM,0.1,0402,SM	R9717,R9718,R9719		BKLT:ENG
103S0198	3	RES,THIN FLIM,1/16W,10.2 OHM,0.1,0402,SM	R9720,R9721,R9722		BKLT:ENG

10.2 ohm resistors for current measurement on LED strings.

SYNC_MASTER=J31_KIRAN SYNC_DATE=03/21/2011

PAGE TITLE

LCD Backlight Driver

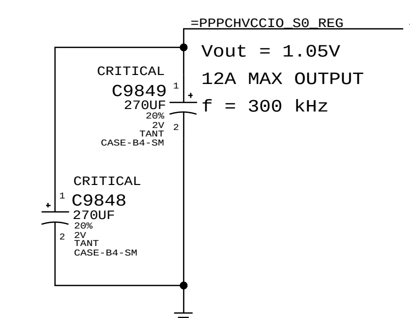
Apple Inc.

DRAWING NUMBER 051-9585 SIZE D

REVISION 3.0.0

BRANCH

PCH VCCIO (1.05V S0) REGULATOR



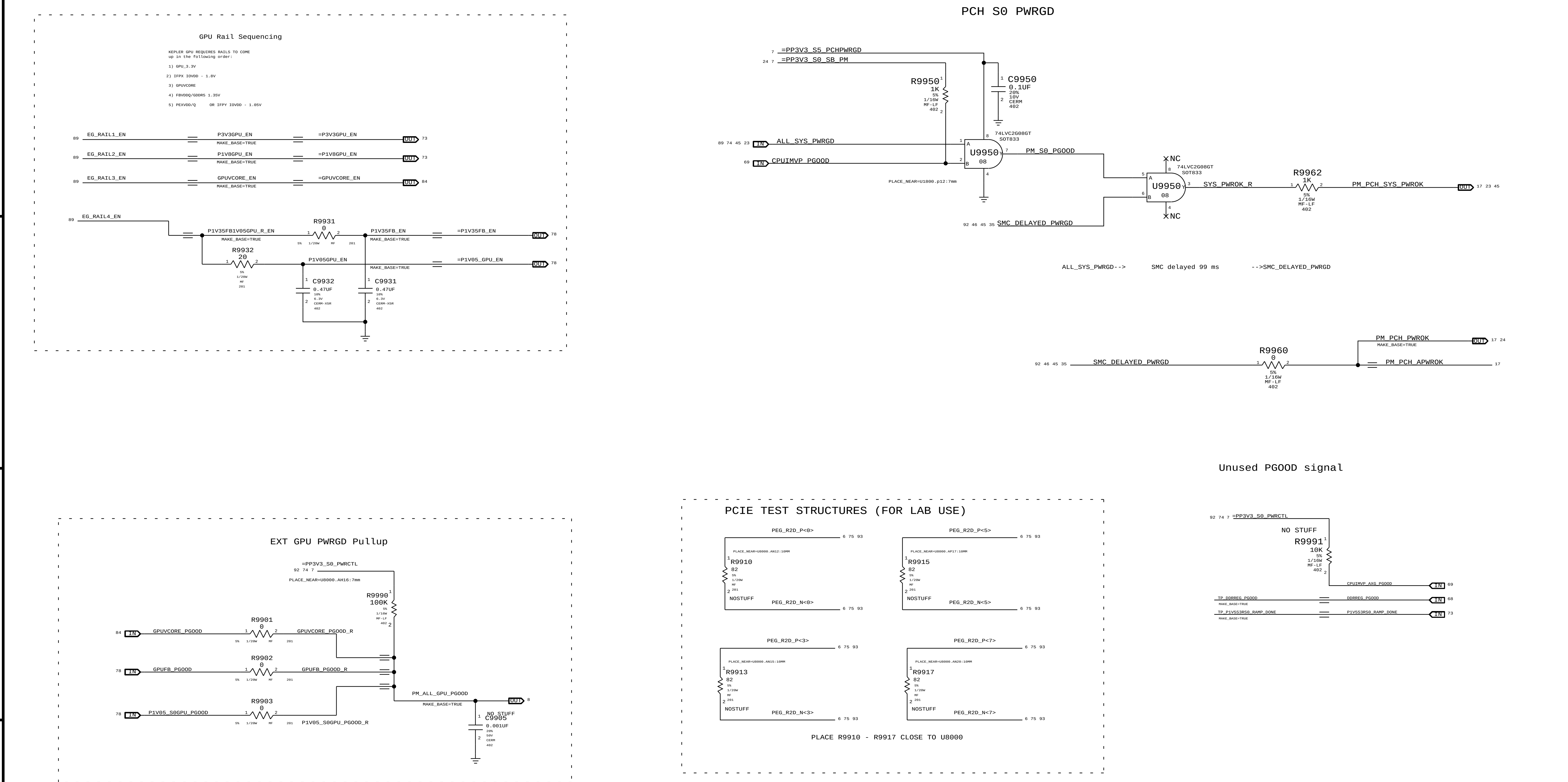
SYNC_MASTER=J31_JACK SYNC_DATE=09/16/2011

PAGE TITLE

PCH VCCIO (1.05V) POWER SUPPLY

Apple Inc.

DRAWING NUMBER	051-9585
SIZE	D
REVISION	3.0.0
BRANCH	
PAGE	98 OF 132
SHEET	91 OF 105



SYNC_MASTER=J31_SREE SYNC_DATE=09/19/2011

PAGE TITLE

Power Sequencing EG/PCH S0

Apple Inc.

DRAWING NUMBER 051-9585 SIZE D

REVISION 3.0.0

BRANCH

CPU Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CPU_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
CPU_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CPU_27P4S	*	=27P4_OHM_SE	=27P4_OHM_SE	=27P4_OHM_SE	=27P4_OHM_SE	7 MIL	7 MIL

NOTE: 7 mil gap is for VCCSense pair, which Intel says to route with 7 mil spacing without specifying a target impedance.

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CPU_AGTL	*	=STANDARD	?
CPU_8MIL	*	8 MIL	?
CPU_COMP	*	20 MIL	?
CPU_ITP	*	=2:1_SPACING	?
CPU_VCCSENSE	*	25 MIL	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CPU_AGTL	TOP,BOTTOM	=2x_DIELECTRIC	?
CPU_VID	*	0.457 MM	?
CPU_VREF	*	12 MIL	?

Most CPU signals with impedance requirements are 50-ohm single-ended.
Some signals require 27.4-ohm single-ended impedance.

SOURCE: IVB PLATFORM DG , Tables 205-207

PCI-Express

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCIE_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
CLK_PCIE_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCIE	*	15 MIL	?
CLK_PCIE	*	20 MIL	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCIE	TOP,BOTTOM	15 MIL	?

PEG

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PEG_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PEG_RXRX	*	=3X_DIELECTRIC	?
PEG_TXTX	*	=3X_DIELECTRIC	?
PEG_TXRX	*	=10X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
PEG_D2R	PEG_D2R	*	PEG_RXRX
PEG_R2D	PEG_R2D	*	PEG_TXTX
PEG_D2R	PEG_R2D	*	PEG_TXRX

CPU Net Properties

	ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING		
	DMI_S2N	PCIE_85D	PCIE	DMI_S2N_P<3:0>	6 9 17
	DMI_S2N	PCIE_85D	PCIE	DMI_S2N_N<3:0>	6 9 17
	DMI_N2S	PCIE_85D	PCIE	DMI_N2S_P<3:0>	6 9 17
	DMI_N2S	PCIE_85D	PCIE	DMI_N2S_N<3:0>	6 9 17
	FDI_DATA	PCIE_85D	PCIE	FDI_DATA_P<7:0>	6 9 17
	FDI_DATA	PCIE_85D	PCIE	FDI_DATA_N<7:0>	6 9 17
	FDI_FSYNC	CPU_50S	CPU_AGTL	FDI_FSYNC<1..0>	6 9 17
	FDI_LSYNC	CPU_50S	CPU_AGTL	FDI_LSYNC<1..0>	6 9 17
	FDI_INT	CPU_50S	CPU_AGTL	FDI_INT	6 9 17
I125	DMI_CLK100M	CLK_PCIE_90D	CLK_PCIE	DMI_CLK100M_CPU_P	10 16
I126	DMI_CLK100M	CLK_PCIE_90D	CLK_PCIE	DMI_CLK100M_CPU_N	10 16
I127		DP_85D	DISPLAYPORT	DP_INT_IG_ML_P<3:0>	8 9
I128		DP_85D	DISPLAYPORT	DP_INT_IG_ML_N<3:0>	8 9
I129	DP_INT_AUX	DP_85D	DISPLAYPORT	DP_INT_IG_AUX_P	8 9
I131	DP_INT_AUX	DP_85D	DISPLAYPORT	DP_INT_IG_AUX_N	8 9
I132	CPU_EDP_COMP	CPU_27P4S	CPU_COMP	CPU_EDP_COMP	9
I130	CPU_PEG_COMP	CPU_27P4S	CPU_COMP	CPU_PEG_COMP	9
I133	CPU_CFG	CPU_50S	CPU_ITP	CPU_CFG<17..0>	9 23
	XDP_CLK_CPU	CLK_PCIE_90D	CLK_PCIE	ITPCPU_CLK100M_P	10 16
	XDP_CLK_CPU	CLK_PCIE_90D	CLK_PCIE	ITPCPU_CLK100M_N	10 16
	XDP_CLK_PCH	CLK_PCIE_90D	CLK_PCIE	ITPXDP_CLK100M_P	16 23
	XDP_CLK_PCH	CLK_PCIE_90D	CLK_PCIE	ITPXDP_CLK100M_N	16 23
I138	DPLL_REF_CLK120M	CLK_PCIE_90D	CLK_PCIE	DPLL_REF_CLKP	8
I139	DPLL_REF_CLK120M	CLK_PCIE_90D	CLK_PCIE	DPLL_REF_CLKN	8
	XDP_TDI	CPU_50S	CPU_ITP	XDP_CPU_TDI	10 23
	XDP_TDO	CPU_50S	CPU_ITP	XDP_CPU_TDO	10 23
	XDP_TMS	CPU_50S	CPU_ITP	XDP_CPU_TMS	10 23
	XDP_TCK	CPU_50S	CPU_ITP	XDP_CPU_TCK	10 23
	XDP_TRST_L	CPU_50S	CPU_ITP	XDP_CPU_TRST_L	10 23
	XDP_BPM	CPU_50S	CPU_ITP	XDP_BPM_L<3..0>	10 23
	XDP_BPM_L	CPU_50S	CPU_ITP	XDP_BPM_L<7..4>	10 23
I134	XDP_BDRESET_L	CPU_50S	CPU_ITP	XDP_DBRESET_L	10 23 24
I135	XDP_PRDY_L	CPU_50S	CPU_ITP	XDP_CPU_PRDY_L	10 23
I136	XDP_PREQ_L	CPU_50S	CPU_ITP	XDP_CPU_PREQ_L	10 23
	CPU_CATERR_L	CPU_50S	CPU_AGTL	CPU_CATERR_L	10 45
I115	CPU_PROC_SEL_L	CPU_50S	CPU_AGTL	CPU_PROC_SEL_L	10 19
	CPU_PECI	CPU_50S	CPU_VID	CPU_PECI	10 19 46
	CPU_PROCHOT_L	CPU_50S	CPU_AGTL	CPU_PROCHOT_L	10 45 46 69
	XDP_CPU_PWRGD	CPU_50S	CPU_ITP	XDP_CPU_PWRGD	23
	PM_THRMTRIP_L	CPU_50S	CPU_8MIL	PM_THRMTRIP_L	10 19 46
	PM_SYNC	CPU_50S	CPU_AGTL	PM_SYNC	10 17
	PM_MEM_PWRGD	CPU_50S	CPU_AGTL	PM_MEM_PWRGD	10 17 26
I150	CPU_PWRGD	CPU_50S	CPU_AGTL	CPU_PWRGD	10 19 23
	CPU_SM_RCOMP	CPU_27P4S	CPU_COMP	CPU_SM_RCOMP<2..0>	10
		CPU_50S	CPU_VID	CPU_VIDSOUT	12 69
		CPU_50S	CPU_VID	CPU_VIDSCLK	12 69
		CPU_50S	CPU_VID	CPU_VIDALERT_L	12 69
		CPU_55S	CPU_VID	CPU_VCCSA_VID<1..0>	12 66
	SENSE_DIFFPAIR	SENSE_1TO1_55S	SENSE	CPU_VCCSENSE_P	12 69
	SENSE_DIFFPAIR	SENSE_1TO1_55S	SENSE	CPU_VCCSENSE_N	12 69
	SENSE_DIFFPAIR	SENSE_1TO1_55S	SENSE	CPU_VCCIOSENSE_P	12 71
	SENSE_DIFFPAIR	SENSE_1TO1_55S	SENSE	CPU_VCCIOSENSE_N	12 71
	SENSE_DIFFPAIR	SENSE_1TO1_55S	SENSE	CPU_AXG_SENSE_P	12 69
	SENSE_DIFFPAIR	SENSE_1TO1_55S	SENSE	CPU_AXG_SENSE_N	12 69
I120	SENSE_DIFFPAIR	SENSE_1TO1_55S	SENSE	CPU_VCC_VALSENSE_P	12
I121	SENSE_DIFFPAIR	SENSE_1TO1_55S	SENSE	CPU_VCC_VALSENSE_N	12
I122	SENSE_DIFFPAIR	SENSE_1TO1_55S	SENSE	CPU_AXG_VALSENSE_P	12
I123	SENSE_DIFFPAIR	SENSE_1TO1_55S	SENSE	CPU_AXG_VALSENSE_N	12
I137	CPU_VCCSASENSE	CPU_50S	CPU_AGTL	CPU_VCCSASENSE	12 66
I140	CPU_MEM_VREF		CPU_VREF	PPCPU_MEM_VREFDQ_A	9 31
I141	CPU_MEM_VREF		CPU_VREF	PPCPU_MEM_VREFDQ_B	9 31
I144	CPU_MEM_VREF		CPU_VREF	PP0V75_S3_MEM_VREFDQ_A	27 31
I145	CPU_MEM_VREF		CPU_VREF	PP0V75_S3_MEM_VREFDQ_B	29 31
I146	CPU_MEM_VREF		CPU_VREF	PP0V75_S3_MEM_VREFCA_A	27 31
I147	CPU_MEM_VREF		CPU_VREF	PP0V75_S3_MEM_VREFCA_B	29 31
I148	XDP_CLK_ITP	CLK_PCIE_90D	CLK_PCIE	XDP_CPU_CLK100M_P	23
I149	XDP_CLK_ITP	CLK_PCIE_90D	CLK_PCIE	XDP_CPU_CLK100M_N	23
		PEG_80D	PEG_R2D	PEG_R2D_P<7..0>	6 75 92
		PEG_80D	PEG_R2D	PEG_R2D_N<7..0>	6 75 92
	PEG_R2D	PEG_80D	PEG_R2D	PEG_R2D_C_P<7..0>	8 75
		PEG_80D	PEG_R2D	PEG_R2D_C_N<7..0>	8 75
	PEG_D2R	PEG_80D	PEG_D2R	PEG_D2R_P<7..0>	8 75
		PEG_80D	PEG_D2R	PEG_D2R_N<7..0>	8 75
		PEG_80D	PEG_D2R	PEG_D2R_C_P<7..0>	6 75
		PEG_80D	PEG_D2R	PEG_D2R_C_N<7..0>	6 75

SYNC_MASTER=K92_MLB SYNC_DATE=08/09/2010

PAGE TITLE

CPU Constraints

Apple Inc.

DRAWING NUMBER 051-9585 SIZE D

REVISION 3.0.0

BRANCH

8	7	6	5	4	3	2	1

Memory Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_37S	*	=37_OHM_SE	=37_OHM_SE	=37_OHM_SE	=37_OHM_SE	=STANDARD	=STANDARD
MEM_40S	*	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=STANDARD	=STANDARD
MEM_72D	*	=72_OHM_DIFF	=72_OHM_DIFF	=72_OHM_DIFF	=72_OHM_DIFF	=72_OHM_DIFF	=72_OHM_DIFF
MEM_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
MEM_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MEM_CLK2MEM	*	=4:1_SPACING	?
MEM_CTRL2CTRL	*	=3:1_SPACING	?
MEM_CTRL2MEM	*	=2.5:1_SPACING	?
MEM_CMD2CMD	*	=1.5:1_SPACING	?
MEM_CMD2MEM	*	=3:1_SPACING	?
MEM_DATA2DATA	*	=1.5:1_SPACING	?
MEM_DATA2MEM	*	=3:1_SPACING	?
MEM_DQS2MEM	*	=3:1_SPACING	?
MEM_2OTHER	*	25 MILS	?

Memory Bus Spacing Group Assignments

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CLK	MEM_*	*	MEM_CLK2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CMD	MEM_*	*	MEM_CMD2MEM
MEM_CMD	MEM_CMD	*	MEM_CMD2CMD

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CTRL	MEM_*	*	MEM_CTRL2MEM
MEM_CTRL	MEM_CTRL	*	MEM_CTRL2CTRL

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DATA	MEM_*	*	MEM_DATA2MEM
MEM_DATA	MEM_DATA	*	MEM_DATA2DATA

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DQS	MEM_*	*	MEM_DQS2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_*	*	*	MEM_2OTHER

DDR3:
DQ/DM signals should be matched within 0.508mm of associated DQS pair.
DQS intra-pair matching should be within 0.127mm, no inter-pair matching requirement.
DQS to clock matching should be within [CLK-12.7mm] and [CLK+25.4mm].
CLK intra-pair matching should be within 0.127mm, inter-pair matching should be within 0.508mm.
CONTROL signals should be matched within [CLK-12.7mm] to [CLK+0.0mm] of CLK pairs.
A/BA/CMD signals should be matched within [CLK-12.7mm] to [CLK+12.7mm] of CLK pairs.
DQ/DQS/A/BA/cmd signal spacing is 4x dielectric, CLK is 5x dielectric.
Maximum length of any signal from die pad to SODIMM pad is 139.7mm, from procesor ball to SODIMM pad is 114.3mm.

SOURCE: Calpella SFF Platform DG, Rev 1.5 (#407364), Section 2.2

Memory Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING		
MEM_A_CLK	MEM_72D	MEM_CLK	MEM_A_CLK_P<5..0>	6 11 27
MEM_A_CLK	MEM_72D	MEM_CLK	MEM_A_CLK_N<5..0>	6 11 27
MEM_A_CNTL	MEM_37S	MEM_CTRL	MEM_A_CKE<3..0>	6 11 27
MEM_A_CNTL	MEM_37S	MEM_CTRL	MEM_A_CS_L<3..0>	6 11 27
MEM_A_CNTL	MEM_37S	MEM_CTRL	MEM_A_ODT<3..0>	6 11 27
MEM_A_CMD	MEM_40S	MEM_CMD	MEM_A_A<15..0>	6 11 27
MEM_A_CMD	MEM_40S	MEM_CMD	MEM_A_BA<2..0>	6 11 27
MEM_A_CMD	MEM_40S	MEM_CMD	MEM_A_RAS_L	6 11 27
MEM_A_CMD	MEM_40S	MEM_CMD	MEM_A_CAS_L	6 11 27
MEM_A_CMD	MEM_40S	MEM_CMD	MEM_A_WE_L	6 11 27
MEM_A_DQ_BYTE0	MEM_50S	MEM_DATA	MEM_A_DQ<7..0>	6 11 28
MEM_A_DQ_BYTE1	MEM_50S	MEM_DATA	MEM_A_DQ<15..8>	6 11 28
MEM_A_DQ_BYTE2	MEM_50S	MEM_DATA	MEM_A_DQ<23..16>	6 11 28
MEM_A_DQ_BYTE3	MEM_50S	MEM_DATA	MEM_A_DQ<31..24>	6 11 28
MEM_A_DQ_BYTE4	MEM_50S	MEM_DATA	MEM_A_DQ<39..32>	6 11 28
MEM_A_DQ_BYTE5	MEM_50S	MEM_DATA	MEM_A_DQ<47..40>	6 11 28
MEM_A_DQ_BYTE6	MEM_50S	MEM_DATA	MEM_A_DQ<55..48>	6 11 28
MEM_A_DQ_BYTE7	MEM_50S	MEM_DATA	MEM_A_DQ<63..56>	6 11 28
MEM_A_DQS0	MEM_85D	MEM_DQS	MEM_A_DQS_P<0>	6 11 28
MEM_A_DQS0	MEM_85D	MEM_DQS	MEM_A_DQS_N<0>	6 11 28
MEM_A_DQS1	MEM_85D	MEM_DQS	MEM_A_DQS_P<1>	6 11 28
MEM_A_DQS1	MEM_85D	MEM_DQS	MEM_A_DQS_N<1>	6 11 28
MEM_A_DQS2	MEM_85D	MEM_DQS	MEM_A_DQS_P<2>	6 11 28
MEM_A_DQS2	MEM_85D	MEM_DQS	MEM_A_DQS_N<2>	6 11 28
MEM_A_DQS3	MEM_85D	MEM_DQS	MEM_A_DQS_P<3>	6 11 28
MEM_A_DQS3	MEM_85D	MEM_DQS	MEM_A_DQS_N<3>	6 11 28
MEM_A_DQS4	MEM_85D	MEM_DQS	MEM_A_DQS_P<4>	6 11 28
MEM_A_DQS4	MEM_85D	MEM_DQS	MEM_A_DQS_N<4>	6 11 28
MEM_A_DQS5	MEM_85D	MEM_DQS	MEM_A_DQS_P<5>	6 11 28
MEM_A_DQS5	MEM_85D	MEM_DQS	MEM_A_DQS_N<5>	6 11 28
MEM_A_DQS6	MEM_85D	MEM_DQS	MEM_A_DQS_P<6>	6 11 28
MEM_A_DQS6	MEM_85D	MEM_DQS	MEM_A_DQS_N<6>	6 11 28
MEM_A_DQS7	MEM_85D	MEM_DQS	MEM_A_DQS_P<7>	6 11 28
MEM_A_DQS7	MEM_85D	MEM_DQS	MEM_A_DQS_N<7>	6 11 28
MEM_B_CLK	MEM_72D	MEM_CLK	MEM_B_CLK_P<5..0>	6 11 29
MEM_B_CLK	MEM_72D	MEM_CLK	MEM_B_CLK_N<5..0>	6 11 29
MEM_B_CNTL	MEM_37S	MEM_CTRL	MEM_B_CKE<3..0>	6 11 29
MEM_B_CNTL	MEM_37S	MEM_CTRL	MEM_B_CS_L<3..0>	6 11 29
MEM_B_CNTL	MEM_37S	MEM_CTRL	MEM_B_ODT<3..0>	6 11 29
MEM_B_CMD	MEM_40S	MEM_CMD	MEM_B_A<15..0>	6 11 29
MEM_B_CMD	MEM_40S	MEM_CMD	MEM_B_BA<2..0>	6 11 29
MEM_B_CMD	MEM_40S	MEM_CMD	MEM_B_RAS_L	6 11 29
MEM_B_CMD	MEM_40S	MEM_CMD	MEM_B_CAS_L	6 11 29
MEM_B_CMD	MEM_40S	MEM_CMD	MEM_B_WE_L	6 11 29
MEM_B_DQ_BYTE0	MEM_50S	MEM_DATA	MEM_B_DQ<7..0>	6 11 28
MEM_B_DQ_BYTE1	MEM_50S	MEM_DATA	MEM_B_DQ<15..8>	6 11 28
MEM_B_DQ_BYTE2	MEM_50S	MEM_DATA	MEM_B_DQ<23..16>	6 11 28
MEM_B_DQ_BYTE3	MEM_50S	MEM_DATA	MEM_B_DQ<31..24>	6 11 28
MEM_B_DQ_BYTE4	MEM_50S	MEM_DATA	MEM_B_DQ<39..32>	6 11 28
MEM_B_DQ_BYTE5	MEM_50S	MEM_DATA	MEM_B_DQ<47..40>	6 11 28
MEM_B_DQ_BYTE6	MEM_50S	MEM_DATA	MEM_B_DQ<55..48>	6 11 28
MEM_B_DQ_BYTE7	MEM_50S	MEM_DATA	MEM_B_DQ<63..56>	6 11 28
MEM_B_DQS0	MEM_85D	MEM_DQS	MEM_B_DQS_P<0>	6 11 28
MEM_B_DQS0	MEM_85D	MEM_DQS	MEM_B_DQS_N<0>	6 11 28
MEM_B_DQS1	MEM_85D	MEM_DQS	MEM_B_DQS_P<1>	6 11 28
MEM_B_DQS1	MEM_85D	MEM_DQS	MEM_B_DQS_N<1>	6 11 28
MEM_B_DQS2	MEM_85D	MEM_DQS	MEM_B_DQS_P<2>	6 11 28
MEM_B_DQS2	MEM_85D	MEM_DQS	MEM_B_DQS_N<2>	6 11 28
MEM_B_DQS3	MEM_85D	MEM_DQS	MEM_B_DQS_P<3>	6 11 28
MEM_B_DQS3	MEM_85D	MEM_DQS	MEM_B_DQS_N<3>	6 11 28
MEM_B_DQS4	MEM_85D	MEM_DQS	MEM_B_DQS_P<4>	6 11 28
MEM_B_DQS4	MEM_85D	MEM_DQS	MEM_B_DQS_N<4>	6 11 28
MEM_B_DQS5	MEM_85D	MEM_DQS	MEM_B_DQS_P<5>	6 11 28
MEM_B_DQS5	MEM_85D	MEM_DQS	MEM_B_DQS_N<5>	6 11 28
MEM_B_DQS6	MEM_85D	MEM_DQS	MEM_B_DQS_P<6>	6 11 28
MEM_B_DQS6	MEM_85D	MEM_DQS	MEM_B_DQS_N<6>	6 11 28
MEM_B_DQS7	MEM_85D	MEM_DQS	MEM_B_DQS_P<7>	6 11 28
MEM_B_DQS7	MEM_85D	MEM_DQS	MEM_B_DQS_N<7>	6 11 28

8 7 6 5 4 3 2 1

Digital Video Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DP_85D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF
LVDS_85D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DISPLAYPORT	ISL3,ISL4,ISL9,ISL10	=4:1_SPACING	?
LVDS	ISL3,ISL4,ISL9,ISL10	=4:1_SPACING	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DISPLAYPORT	TOP,BOTTOM	=4:1_SPACING	?
LVDS	TOP,BOTTOM	=4:1_SPACING	?

SOURCE: HR PLATFORM DESIGN GUIDE, TABLES 191,193

SATA Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SATA_90D_ALT	*	=90_OHM_DIFF_ALT	=90_OHM_DIFF_ALT	=90_OHM_DIFF_ALT	=90_OHM_DIFF_ALT	=90_OHM_DIFF_ALT	=90_OHM_DIFF_ALT
SATA_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF
SATA_37SE	*	=37_OHM_SE	=37_OHM_SE	=37_OHM_SE	=37_OHM_SE	=37_OHM_SE	=37_OHM_SE
SATA_50SE	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SATA	*	=5:1_SPACING	?
SATA_ICOMP	*	15 MIL	?

SOURCE: HR PLATFORM DESIGN GUIDE, TABLES 191,193

USB 2.0 Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCH_USB_RBIAS	*	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
USB_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB	*	=4:1_SPACING	?
USB_RBIAS	*	15 MIL	?

SOURCE: CR SFF PLATFORM DESIGN GUIDE V0.7, TABLE 4-211, 1X1+

USB 3.0 Interface Constraints

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB3	*	=5:1_SPACING	?

PCH Net Properties

	ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	
	DP_AUX_CH	DP_85D	DISPLAYPORT	DP_IG_AUX_CH_P	8 86
	DP_AUX_CH	DP_85D	DISPLAYPORT	DP_IG_AUX_CH_N	8 86
	LVDS_IG_A_CLK	LVDS_85D	LVDS	LVDS_IG_A_CLK_P	17 89
	LVDS_IG_A_CLK	LVDS_85D	LVDS	LVDS_IG_A_CLK_N	17 89
	LVDS_IG_A_DATA	LVDS_85D	LVDS	LVDS_IG_A_DATA_P<2..0>	17 89
	LVDS_IG_A_DATA	LVDS_85D	LVDS	LVDS_IG_A_DATA_N<2..0>	17 89
	LVDS_IG_A_DATA3	LVDS_85D	LVDS	LVDS_IG_A_DATA_P<3>	8 17
	LVDS_IG_A_DATA3	LVDS_85D	LVDS	LVDS_IG_A_DATA_N<3>	8 17
	LVDS_IG_B_DATA	LVDS_85D	LVDS	LVDS_IG_B_DATA_P<2..0>	17 89
	LVDS_IG_B_DATA	LVDS_85D	LVDS	LVDS_IG_B_DATA_N<2..0>	17 89
		SATA_90D_ALT	SATA	SATA_HDD_R2D_C_P	16 41
		SATA_90D_ALT	SATA	SATA_HDD_R2D_C_N	16 41
		SATA_90D_ALT	SATA	SATA_HDD_R2D_RC_P	41
		SATA_90D_ALT	SATA	SATA_HDD_R2D_RC_N	41
I270	SATA_HDD_R2D	SATA_90D_ALT	SATA	SATA_HDD_R2D_P	6 41
I271		SATA_90D_ALT	SATA	SATA_HDD_R2D_N	6 41
	SATA_HDD_D2R	SATA_90D_ALT	SATA	SATA_HDD_D2R_P	16 41
		SATA_90D_ALT	SATA	SATA_HDD_D2R_N	16 41
		SATA_90D_ALT	SATA	SATA_HDD_D2R_C_P	6 41
		SATA_90D_ALT	SATA	SATA_HDD_D2R_C_N	6 41
I272		SATA_90D_ALT	SATA	SATA_HDD_D2R_RC_P	41
I273		SATA_90D_ALT	SATA	SATA_HDD_D2R_RC_N	41
I242		SATA_90D_ALT	SATA	SATA_HDD_R2D_RDROUT_P	41
I243		SATA_90D_ALT	SATA	SATA_HDD_R2D_RDROUT_N	41
I246		SATA_90D_ALT	SATA	SATA_HDD_D2R_RDRIN_P	41
I247		SATA_90D_ALT	SATA	SATA_HDD_D2R_RDRIN_N	41
I248		SATA_90D_ALT	SATA	SATA_HDD_D2R_RDROUT_P	41
I249		SATA_90D_ALT	SATA	SATA_HDD_D2R_RDROUT_N	41
I250		SATA_90D_ALT	SATA	SATA_HDD_R2D_RDRIN_P	41
I251		SATA_90D_ALT	SATA	SATA_HDD_R2D_RDRIN_N	41
	SATA_ODD_R2D	SATA_90D	SATA	SATA_ODD_R2D_C_P	16 41
		SATA_90D	SATA	SATA_ODD_R2D_C_N	16 41
	SATA_ODD_R2D	SATA_90D	SATA	SATA_ODD_R2D_P	6 41
		SATA_90D	SATA	SATA_ODD_R2D_N	6 41
	SATA_ODD_D2R	SATA_90D	SATA	SATA_ODD_D2R_P	16 41
		SATA_90D	SATA	SATA_ODD_D2R_N	16 41
	SATA_ODD_D2R	SATA_90D	SATA	SATA_ODD_D2R_C_P	6 41
		SATA_90D	SATA	SATA_ODD_D2R_C_N	6 41
I213	PCH_SATA3_ICOMP	SATA_50SE	SATA_ICOMP	PCH_SATA3COMP	16
	PCH_SATA_ICOMP	SATA_37SE	SATA_ICOMP	PCH_SATAICOMP	16
	USB_EXTA	USB_85D	USB	USB_EXTA_P	18 42
		USB_85D	USB	USB_EXTA_N	18 42
	USB_EXTB_MUX	USB_85D	USB	USB_EXTB_MUX_P	25 43
	USB_EXTB_MUX	USB_85D	USB	USB_EXTB_MUX_N	25 43
	USB_EXTC	USB_85D	USB	USB_EXTC_P	8 18
		USB_85D	USB	USB_EXTC_N	8 18
	USB_CAMERA	USB_85D	USB	USB_CAMERA_CONN_P	6 32
		USB_85D	USB	USB_CAMERA_CONN_N	6 32
I228	USB_CAMERA	USB_85D	USB	USB_CAMERA_P	18 32
I229		USB_85D	USB	USB_CAMERA_N	18 32
	USB_BT	USB_85D	USB	USB_BT_P	8 32
		USB_85D	USB	USB_BT_N	8 32
I278	USB_BT	USB_85D	USB	USB_BT_CONN_P	6 32
I279		USB_85D	USB	USB_BT_CONN_N	6 32
	USB_TPAD	USB_85D	USB	USB_TPAD_P	8 53
		USB_85D	USB	USB_TPAD_N	8 53
I268	USB_TPAD	USB_85D	USB	USB_TPAD_R_P	25 53 101
I269		USB_85D	USB	USB_TPAD_R_N	25 53 101
I275		USB_85D	USB	USB_EXTD_XHCI_P	18 25
I274		USB_85D	USB	USB_EXTD_XHCI_N	18 25
I276		USB_85D	USB	USB_HUB_UP_P	18 25
I277		USB_85D	USB	USB_HUB_UP_N	18 25
	USB_IR	USB_85D	USB	USB_IR_P	8 44
		USB_85D	USB	USB_IR_N	8 44
	PCH_USB_RBIAS	PCH_USB_RBIAS	USB_RBIAS	PCH_USB_RBIAS	18
I221	USB3_EXT_TX	USB_85D	USB3	USB3_EXTA_TX_P	6 18 42
I220		USB_85D	USB3	USB3_EXTA_TX_N	6 18 42
I222	USB3_EXT_RX	USB_85D	USB3	USB3_EXTA_RX_P	6 18 42
I223		USB_85D	USB3	USB3_EXTA_RX_N	6 18 42
I256	USB3_EXT_TX	USB_85D	USB3	USB3_EXTA_TX_F_P	6 42
I258		USB_85D	USB3	USB3_EXTA_TX_F_N	6 42
I257	USB3_EXT_RX	USB_85D	USB3	USB3_EXTA_RX_F_P	6 42
I259		USB_85D	USB3	USB3_EXTA_RX_F_N	6 42
I260	USB3_EXT_TX	USB_85D	USB3	USB3_EXTA_TX_C_P	6 42
I261		USB_85D	USB3	USB3_EXTA_TX_C_N	6 42
I266	USB3_EXT_TX	USB_85D	USB3	USB3_EXTB_TX_C_P	6 43
I267		USB_85D	USB3	USB3_EXTB_TX_C_N	6 43
I264	USB3_EXT_TX	USB_85D	USB3	USB3_EXTB_TX_F_P	6 43
I265		USB_85D	USB3	USB3_EXTB_TX_F_N	6 43
I262	USB3_EXT_RX	USB_85D	USB3	USB3_EXTB_RX_F_P	6 43
I263		USB_85D	USB3	USB3_EXTB_RX_F_N	6 43
I224	USB3_EXT_TX	USB_85D	USB3	USB3_EXTB_TX_P	6 18 43
I225		USB_85D	USB3	USB3_EXTB_TX_N	6 18 43
I226	USB3_EXT_RX	USB_85D	USB3	USB3_EXTB_RX_P	6 18 43
I227		USB_85D	USB3	USB3_EXTB_RX_N	6 18 43
I230		USB_85D	USB	USB_EXTB_EHCI_P	18 25
I231		USB_85D	USB	USB_EXTB_EHCI_N	18 25
I232		USB_85D	USB	USB_EXTB_XHCI_P	18 25
I233		USB_85D	USB	USB_EXTB_XHCI_N	18 25
I234	USB_EXTB_MUX	USB_85D	USB	USB_EXTB_F_P	43
I235	USB_EXTB_MUX	USB_85D	USB	USB_EXTB_F_N	43
I236	USB_SMC	USB_85D	USB	USB_SMC_N	8
I237	USB_SMC	USB_85D	USB	USB_SMC_P	8
I253	USB_EXTA	USB_85D	USB	USB_EXTA_MUXED_P	42
I252		USB_85D	USB	USB_EXTA_MUXED_N	42
I254	USB_EXTA	USB_85D	USB	USB_EXTA_MUXED_F_P	42
I255		USB_85D	USB	USB_EXTA_MUXED_F_N	42

D C B A

SYNC_MASTER=K92_MLB SYNC_DATE=08/09/2010

PAGE TITLE: PCH Constraints 1

Apple Inc.

DRAWING NUMBER: 051-9585 SIZE: D

REVISION: 3.0.0

BRANCH

8 7 6 5 4 3 2 1

LPC Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
LPC_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
CLK_LPC_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
LPC	*	6 MIL	?
CLK_LPC	*	8 MIL	?

SMBus Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SMB_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SMB	*	=2x_DIELECTRIC	?

HD Audio Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
HDA_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
HDA	*	=2x_DIELECTRIC	?

SIO Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_SLOW_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_SLOW	*	8 MIL	?

SPI Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SPI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SPI	*	8 MIL	?

PCH Net Properties

NET_TYPE

	ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING		
		LPC_50S	LPC	LPC_AD<3..0>	6 16 45 47 89
	LPC_FRAME_L	LPC_50S	LPC	LPC_FRAME_L	6 16 45 47 89
	LPC_RESET_L	LPC_50S	LPC	LPC_RESET_L	24 89
	PCH_LPC_CLK0	CLK_LPC_50S	CLK_LPC	LPC_CLK33M_SMC_R	18 24
		CLK_LPC_50S	CLK_LPC	LPC_CLK33M_SMC	24 45
		CLK_LPC_50S	CLK_LPC	LPC_CLK33M_LPCPLUS	6 24 47
	SMBUS_PCH_CLK	SMB_50S	SMB	SMBUS_PCH_CLK	16 48
	SMBUS_PCH_DATA	SMB_50S	SMB	SMBUS_PCH_DATA	16 48
	SMBUS_PCH_0_CLK	SMB_50S	SMB	SML_PCH_0_CLK	16 48
	SMBUS_PCH_0_DATA	SMB_50S	SMB	SML_PCH_0_DATA	16 48
	SMBUS_PCH_1_CLK	SMB_50S	SMB	SML_PCH_1_CLK	16 48
	SMBUS_PCH_1_DATA	SMB_50S	SMB	SML_PCH_1_DATA	16 48
	HDA_BIT_CLK	HDA_50S	HDA	HDA_BIT_CLK	16 57
		HDA_50S	HDA	HDA_BIT_CLK_R	16
	HDA_SYNC	HDA_50S	HDA	HDA_SYNC	16 57
		HDA_50S	HDA	HDA_SYNC_R	16
	HDA_RST_L	HDA_50S	HDA	HDA_RST_R_L	16
		HDA_50S	HDA	HDA_RST_L	16 57
	HDA_SDIN0	HDA_50S	HDA	HDA_SDIN0	16 57
		HDA_50S	HDA	AUD_SDI_R	57
	HDA_SDOUT	HDA_50S	HDA	HDA_SDOUT	16 57
		HDA_50S	HDA	HDA_SDOUT_R	16 24
	SPI_CLK	SPI_55S	SPI	SPI_CLK_R	16 47
		SPI_55S	SPI	SPI_CLK	47
	SPI_MOSI	SPI_55S	SPI	SPI_MOSI_R	16 47
		SPI_55S	SPI	SPI_MOSI	47
	SPI_MISO	SPI_55S	SPI	SPI_MISO	16 47
	SPI_CS0	SPI_55S	SPI	SPI_CS0_R_L	16 47
		SPI_55S	SPI	SPI_CS0_L	47
		PCIE_85D	PCIE	PCIE_ENET_R2D_P	36
		PCIE_85D	PCIE	PCIE_ENET_R2D_N	36
	PCIE_ENET_R2D	PCIE_85D	PCIE	PCIE_ENET_R2D_C_P	16 36
		PCIE_85D	PCIE	PCIE_ENET_R2D_C_N	16 36
	PCIE_ENET_D2R	PCIE_85D	PCIE	PCIE_ENET_D2R_P	16 36
		PCIE_85D	PCIE	PCIE_ENET_D2R_N	16 36
		PCIE_85D	PCIE	PCIE_ENET_D2R_C_P	36
		PCIE_85D	PCIE	PCIE_ENET_D2R_C_N	36
		PCIE_85D	PCIE	PCIE_AP_R2D_P	6 32
		PCIE_85D	PCIE	PCIE_AP_R2D_N	6 32
	PCIE_AP_R2D	PCIE_85D	PCIE	PCIE_AP_R2D_C_P	16 32
		PCIE_85D	PCIE	PCIE_AP_R2D_C_N	16 32
	PCIE_AP_D2R	PCIE_85D	PCIE	PCIE_AP_D2R_P	16 32
		PCIE_85D	PCIE	PCIE_AP_D2R_N	16 32
I270		PCIE_85D	PCIE	PCIE_AP_D2R_R_P	32
I269		PCIE_85D	PCIE	PCIE_AP_D2R_R_N	32
		PCIE_85D	PCIE	PCIE_FW_R2D_P	38
		PCIE_85D	PCIE	PCIE_FW_R2D_N	38
	PCIE_FW_R2D	PCIE_85D	PCIE	PCIE_FW_R2D_C_P	16 38
		PCIE_85D	PCIE	PCIE_FW_R2D_C_N	16 38
	PCIE_FW_D2R	PCIE_85D	PCIE	PCIE_FW_D2R_P	16 38
		PCIE_85D	PCIE	PCIE_FW_D2R_N	16 38
		PCIE_85D	PCIE	PCIE_FW_D2R_C_P	38
		PCIE_85D	PCIE	PCIE_FW_D2R_C_N	38
I253		CLK_PCIE_90D	CLK_PCIE	PCIE_CLK100M_PCH_P	16
I254		CLK_PCIE_90D	CLK_PCIE	PCIE_CLK100M_PCH_N	16
I262	PCIE_CLK100M_T29_	CLK_PCIE_90D	CLK_PCIE	PCIE_CLK100M_TBT_P	16 33
I261		CLK_PCIE_90D	CLK_PCIE	PCIE_CLK100M_TBT_N	16 33
I255		CLK_PCIE_90D	CLK_PCIE	PCH_CLK96M_DOT_P	16
I257		CLK_PCIE_90D	CLK_PCIE	PCH_CLK96M_DOT_N	16
I256		CLK_PCIE_90D	CLK_PCIE	PCH_CLK100M_SATA_P	16
I259		CLK_PCIE_90D	CLK_PCIE	PCH_CLK100M_SATA_N	16
I258		CPU_50S	CLK_PCIE	PCH_CLK14P3M_REFCLK	16
I260		CPU_50S	CLK_PCIE	PCH_CLK33M_PCIIN	16 24
I277	PCIE_CLK100M	CLK_PCIE_90D	CLK_PCIE	PEX_TSTCLK_O_P	75
I278		CLK_PCIE_90D	CLK_PCIE	PEX_TSTCLK_O_N	75
	PCIE_CLK100M	CLK_PCIE_90D	CLK_PCIE	PEG_CLK100M_P	16 75
		CLK_PCIE_90D	CLK_PCIE	PEG_CLK100M_N	16 75
	PCIE_CLK100M_ENET	CLK_PCIE_90D	CLK_PCIE	PCIE_CLK100M_ENET_P	16 36
		CLK_PCIE_90D	CLK_PCIE	PCIE_CLK100M_ENET_N	16 36
	PCIE_CLK100M_AP	CLK_PCIE_90D	CLK_PCIE	PCIE_CLK100M_AP_P	16 32
		CLK_PCIE_90D	CLK_PCIE	PCIE_CLK100M_AP_N	16 32
	PCIE_CLK100M_FW	CLK_PCIE_90D	CLK_PCIE	PCIE_CLK100M_FW_P	16 38
		CLK_PCIE_90D	CLK_PCIE	PCIE_CLK100M_FW_N	16 38
	PCIE_CLK100M_EXCARD	CLK_PCIE_90D	CLK_PCIE	PCIE_CLK100M_EXCARD_P	8 16
		CLK_PCIE_90D	CLK_PCIE	PCIE_CLK100M_EXCARD_N	8 16
I263	PCIE_T29_R2D	PCIE_85D	PCIE	PCIE_TBT_R2D_C_P<3..0>	8 33
I264	PCIE_T29_R2D	PCIE_85D	PCIE	PCIE_TBT_R2D_C_N<3..0>	8 33
I266	PCIE_T29_D2R	PCIE_85D	PCIE	PCIE_TBT_D2R_P<3..0>	8 33
I268	PCIE_T29_D2R	PCIE_85D	PCIE	PCIE_TBT_D2R_N<3..0>	8 33
I273	PCIE_T29_R2D	PCIE_85D	PCIE	PCIE_TBT_R2D_P<3..0>	33
I274	PCIE_T29_R2D	PCIE_85D	PCIE	PCIE_TBT_R2D_N<3..0>	33
I275	PCIE_T29_D2R	PCIE_85D	PCIE	PCIE_TBT_D2R_C_P<3..0>	33
I276	PCIE_T29_D2R	PCIE_85D	PCIE	PCIE_TBT_D2R_C_N<3..0>	33

SYNC_MASTER=J31_YONAS SYNC_DATE=05/05/2011

PAGE TITLE

PCH Constraints 2

Apple Inc.

DRAWING NUMBER 051-9585 SIZE D

REVISION 3.0.0

BRANCH

CAESAR IV (Ethernet) Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
ENET_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
ENET_3X	*	=3:1_SPACING	?

SOURCE: Broadcom 5764-DS04-RDS Page 38

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
ENET_CR	*	=3X_DIELECTRIC	?

SOURCE: Attila Farkas Email - 8/2/10

CAESAR IV (Ethernet PHY) Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
ENET_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
ENET_MDI	*	0.6 MM	?

SOURCE: Broadcom 5764-DS04-RDS Page 38

Ethernet Net Properties

	ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	Net	
	ENET_RESET_L	ENET_50S	ENET_3X	ENET_RESET_L	30 36
	ENET_MDI	ENET_100D	ENET_MDI	ENET_MDI_P<3..0>	36 37
	ENET_MDI	ENET_100D	ENET_MDI	ENET_MDI_N<3..0>	36 37
I169		ENET_50S	ENET_CR	SDCONN_DATA<7..0>	30 36
I170	CR_DATA_A0	ENET_50S	ENET_CR	SDCONN_CMD	30 36
I171	CR_CLK	ENET_50S	ENET_CR	SDCONN_CLK	30 36
I172		ENET_50S	ENET_CR	SDCONN_CLK_R	30
I173		ENET_50S	ENET_CR	SDCONN_CLK_R_L	30
I174		ENET_50S	ENET_CR	SDCONN_R_DATA<7..0>	30

FireWire Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
FW_110D	*	=110_OHM_DIFF	=110_OHM_DIFF	=110_OHM_DIFF	=110_OHM_DIFF	=110_OHM_DIFF	=110_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
FW_TP	*	=3:1_SPACING	?

FireWire Net Properties

	ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	Net	
I158	FW_P0_TPA	FW_110D	FW_TP	FW_P0_TPA_P	38 40
I159	FW_P0_TPA	FW_110D	FW_TP	FW_P0_TPA_N	38 40
I160	FW_P0_TPB	FW_110D	FW_TP	FW_P0_TPB_P	38 40
I161	FW_P0_TPB	FW_110D	FW_TP	FW_P0_TPB_N	38 40
I162	FW_P1_TPA	FW_110D	FW_TP	FW_P1_TPA_P	38 40
I163	FW_P1_TPA	FW_110D	FW_TP	FW_P1_TPA_N	38 40
I164	FW_P1_TPB	FW_110D	FW_TP	FW_P1_TPB_P	38 40
I165	FW_P1_TPB	FW_110D	FW_TP	FW_P1_TPB_N	38 40

Port 2 Not Used

SYNC_MASTER=K91_ERIC SYNC_DATE=08/03/2010

PAGE TITLE: Ethernet/FW Constraints

Apple Inc.

DRAWING NUMBER: 051-9585 SIZE: D

REVISION: 3.0.0

BRANCH:

DisplayPort Signal Constraints

NOTE: DisplayPort Physical/Spacing Constraints provided by Chipset or GPU page.

Thunderbolt I2C Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
TBT_I2C_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
TBT_I2C	*	=2x_DIELECTRIC	?

Thunderbolt SPI Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
TBT_SPI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
TBT_SPI	*	=2x_DIELECTRIC	?

Thunderbolt/DP Connector Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
TBTDP_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF
TBTDP_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
TBTDP	*	=5x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
TBTDP	TOP,BOTTOM	=7x_DIELECTRIC	?

SOURCE: Bill Cornelius's Thunderbolt Routing Notes

Thunderbolt IC Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
	DP_85D	DISPLAYPORT	DP_TBTSNK0_ML_C_P<3..0>	6 33 81
	DP_85D	DISPLAYPORT	DP_TBTSNK0_ML_C_N<3..0>	6 33 81
DP_TBTSNK0_ML	DP_85D	DISPLAYPORT	DP_TBTSNK0_ML_P<3..0>	6 33
DP_TBTSNK0_ML	DP_85D	DISPLAYPORT	DP_TBTSNK0_ML_N<3..0>	6 33
	DP_85D	DISPLAYPORT	DP_TBTSNK0_AUXCH_C_P	6 33 81
	DP_85D	DISPLAYPORT	DP_TBTSNK0_AUXCH_C_N	6 33 81
DP_TBTSNK0_AUXCH	DP_85D	DISPLAYPORT	DP_TBTSNK0_AUXCH_P	6 33
DP_TBTSNK0_AUXCH	DP_85D	DISPLAYPORT	DP_TBTSNK0_AUXCH_N	6 33
	DP_85D	DISPLAYPORT	DP_TBTSNK1_ML_C_P<3..0>	6 33 81
	DP_85D	DISPLAYPORT	DP_TBTSNK1_ML_C_N<3..0>	6 33 81
DP_TBTSNK1_ML	DP_85D	DISPLAYPORT	DP_TBTSNK1_ML_P<3..0>	6 33
DP_TBTSNK1_ML	DP_85D	DISPLAYPORT	DP_TBTSNK1_ML_N<3..0>	6 33
	DP_85D	DISPLAYPORT	DP_TBTSNK1_AUXCH_C_P	6 33 81
	DP_85D	DISPLAYPORT	DP_TBTSNK1_AUXCH_C_N	6 33 81
DP_TBTSNK1_AUXCH	DP_85D	DISPLAYPORT	DP_TBTSNK1_AUXCH_P	6 33
DP_TBTSNK1_AUXCH	DP_85D	DISPLAYPORT	DP_TBTSNK1_AUXCH_N	6 33
	DP_85D	DISPLAYPORT	DP_TBTSRC_ML_C_P<3..0>	
	DP_85D	DISPLAYPORT	DP_TBTSRC_ML_C_N<3..0>	
	DP_85D	DISPLAYPORT	DP_TBTSRC_AUXCH_C_P	
	DP_85D	DISPLAYPORT	DP_TBTSRC_AUXCH_C_N	
	TBT_I2C_55S	TBT_I2C	I2C_TBT_SCL	33 48
	TBT_I2C_55S	TBT_I2C	I2C_TBT_SDA	33 48
TBT_SPI_CLK	TBT_SPI_55S	TBT_SPI	TBT_SPI_CLK	33
TBT_SPI_MOSI	TBT_SPI_55S	TBT_SPI	TBT_SPI_MOSI	33
TBT_SPI_MISO	TBT_SPI_55S	TBT_SPI	TBT_SPI_MISO	33
TBT_SPI_CS_L	TBT_SPI_55S	TBT_SPI	TBT_SPI_CS_L	33
	TBTDP_80D	TBTDP	TBT_R2D_C_P<3..0>	6 8 33 87
	TBTDP_80D	TBTDP	TBT_R2D_C_N<3..0>	6 8 33 87
	TBTDP_100D	TBTDP	TBT_D2R_P<3..0>	8 33 87
	TBTDP_100D	TBTDP	TBT_D2R_N<3..0>	8 33 87

Only used on hosts supporting Thunderbolt video-in

Thunderbolt/DP Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
TBT_R2D0	TBTDP_80D	TBTDP	TBT_R2D_P<0>	87
TBT_R2D0	TBTDP_80D	TBTDP	TBT_R2D_N<0>	87
TBT_R2D1	TBTDP_80D	TBTDP	TBT_R2D_P<1>	87
TBT_R2D1	TBTDP_80D	TBTDP	TBT_R2D_N<1>	87
TBT_D2R0	TBTDP_100D	TBTDP	TBT_D2R_C_P<0>	87 88
TBT_D2R0	TBTDP_100D	TBTDP	TBT_D2R_C_N<0>	87 88
TBT_D2R1	TBTDP_100D	TBTDP	TBT_D2R_C_P<1>	87 88
TBT_D2R1	TBTDP_100D	TBTDP	TBT_D2R_C_N<1>	87 88
	TBTDP_100D	TBTDP	TBT_A_D2R1_AUXCH_P	88
	TBTDP_100D	TBTDP	TBT_A_D2R1_AUXCH_N	88
	TBTDP_80D	TBTDP	DP_SDRVA_ML_C_P<3..0>	6 87
	TBTDP_80D	TBTDP	DP_SDRVA_ML_C_N<3..0>	6 87
	TBTDP_80D	TBTDP	DP_SDRVA_ML_R_P<3..0>	87
	TBTDP_80D	TBTDP	DP_SDRVA_ML_R_N<3..0>	87
DP_SDRVA_ML_ODD	TBTDP_80D	TBTDP	DP_SDRVA_ML_P<3>	87
DP_SDRVA_ML_ODD	TBTDP_80D	TBTDP	DP_SDRVA_ML_N<3>	87
I276 DP_SDRVA_ML_ODD	TBTDP_80D	TBTDP	DP_SDRVA_ML_P<1>	87
I277 DP_SDRVA_ML_ODD	TBTDP_80D	TBTDP	DP_SDRVA_ML_N<1>	87
I278 DP_SDRVA_ML_EVEN	TBTDP_80D	TBTDP	DP_SDRVA_ML_P<2>	6 87
I279 DP_SDRVA_ML_EVEN	TBTDP_80D	TBTDP	DP_SDRVA_ML_N<2>	6 87
I280 DP_SDRVA_ML_EVEN	TBTDP_80D	TBTDP	DP_SDRVA_ML_P<0>	6 87
I281 DP_SDRVA_ML_EVEN	TBTDP_80D	TBTDP	DP_SDRVA_ML_N<0>	6 87
	TBTDP_80D	TBTDP	DP_SDRVA_AUXCH_P	87
	TBTDP_80D	TBTDP	DP_SDRVA_AUXCH_N	87
	TBTDP_80D	TBTDP	DP_SDRVA_AUXCH_C_P	87
	TBTDP_80D	TBTDP	DP_SDRVA_AUXCH_C_N	87
	TBTDP_80D	TBTDP	TBT_A_ML_P<3..0>	87 88
	TBTDP_80D	TBTDP	TBT_A_ML_N<3..0>	87 88
	TBTDP_80D	TBTDP	TBT_A_ML_C_P<3..0>	87 88
	TBTDP_80D	TBTDP	TBT_A_ML_C_N<3..0>	87 88
	TBTDP_80D	TBTDP	DP_A_EXT_AUXCH_P	87 88
	TBTDP_80D	TBTDP	DP_A_EXT_AUXCH_N	87 88

SYNC_MASTER=T29_REF SYNC_DATE=06/14/2011

PAGE TITLE: Thunderbolt Constraints

Apple Inc.

DRAWING NUMBER: 051-9585 SIZE: D

REVISION: 3.0.0

BRANCH

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
1TO1_DIFFPAIR	*	=STANDARD	=STANDARD	=STANDARD	=STANDARD	0.1 MM	0.1 MM

SMC SMBus Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
SMBUS_SMC_0_S0_SCL	SMB_50S	SMB	SMBUS_SMC_0_S0_SCL	45 48
SMBUS_SMC_0_S0_SDA	SMB_50S	SMB	SMBUS_SMC_0_S0_SDA	45 48
SMBUS_SMC_1_S0_SCL	SMB_50S	SMB	SMBUS_SMC_1_S0_SCL	45 48
SMBUS_SMC_1_S0_SDA	SMB_50S	SMB	SMBUS_SMC_1_S0_SDA	45 48
SMBUS_SMC_2_S3_SCL	SMB_50S	SMB	SMBUS_SMC_2_S3_SCL	6 45 48
SMBUS_SMC_2_S3_SDA	SMB_50S	SMB	SMBUS_SMC_2_S3_SDA	6 45 48
SMBUS_SMC_3_SCL	SMB_50S	SMB	SMBUS_SMC_3_SCL	45 48
SMBUS_SMC_3_SDA	SMB_50S	SMB	SMBUS_SMC_3_SDA	45 48
SMBUS_SMC_5_G3_SCL	SMB_50S	SMB	SMBUS_SMC_5_G3_SCL	6 45 48
SMBUS_SMC_5_G3_SDA	SMB_50S	SMB	SMBUS_SMC_5_G3_SDA	6 45 48

SMBus Charger Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
CHGR_CSI	1TO1_DIFFPAIR		CHGR_CSI_P	65
	1TO1_DIFFPAIR		CHGR_CSI_N	65
CHGR_CSO	1TO1_DIFFPAIR		CHGR_CSO_P	65
	1TO1_DIFFPAIR		CHGR_CSO_N	65

SYNC_MASTER=J31_YONAS SYNC_DATE=08/11/2011

PAGE TITLE

SMC Constraints

Apple Inc.

DRAWING NUMBER 051-9585 SIZE D

REVISION 3.0.0

BRANCH

GDDR5 Frame Buffer Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
GDDR5_45R50SE	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	12.7 MM	=STANDARD	=STANDARD
GDDR5_45SE	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
GDDR5_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GDDR5_CLK	*	=5x_DIELECTRIC	?
GDDR5_CMD	*	=3x_DIELECTRIC	?
GDDR5_DATA	*	=3x_DIELECTRIC	?
GDDR5_EDC	*	=5x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GDDR5_CLK	TOP,BOTTOM	=5x_DIELECTRIC	?
GDDR5_CMD	TOP,BOTTOM	=4x_DIELECTRIC	?
GDDR5_DATA	TOP,BOTTOM	=5x_DIELECTRIC	?
GDDR5_EDC	TOP,BOTTOM	=5x_DIELECTRIC	?

Digital Video Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DP_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
LVDS_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DISPLAYPORT	*	=3x_DIELECTRIC	?
LVDS	*	=3x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DISPLAYPORT	TOP,BOTTOM	=4x_DIELECTRIC	?
LVDS	TOP,BOTTOM	=4x_DIELECTRIC	?

LVDS intra-pair matching should be 0.127 mm. Pairs should be within 0.508mm of entire channel.

DisplayPort/TMDS intra-pair matching should be 0.127mm. Inter-pair matching should be within 2.54cm. Max Length 241.3mm.

DIsplayPort AUX CH intra-pair matching should be 0.127mm. Max length 330.2mm.

Max length of LVDS/DisplayPort/TMDS traces: 13 inches.

SOURCE: Calpella SFF DG Rev 1.5 (407364) and Family GPU DG-04202-001-v04.

GDDR5 FB A Net Properties

	ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING		
	FB_A0_CLK	GDDR5_80D	GDDR5_CLK	FB_A0_CLK_P	77 79
	FB_A0_CLK	GDDR5_80D	GDDR5_CLK	FB_A0_CLK_N	77 79
	FB_A1_CLK	GDDR5_80D	GDDR5_CLK	FB_A1_CLK_P	77 79
	FB_A1_CLK	GDDR5_80D	GDDR5_CLK	FB_A1_CLK_N	77 79
	FB_A0_CMD	GDDR5_45SE	GDDR5_CMD	FB_A0_A<8..0>	6 77 79
	FB_A1_CMD	GDDR5_45SE	GDDR5_CMD	FB_A1_A<8..0>	6 77 79
	FB_A0_CMD	GDDR5_45SE	GDDR5_CMD	FB_A0_ABI_L	6 77 79
	FB_A1_CMD	GDDR5_45SE	GDDR5_CMD	FB_A1_ABI_L	6 77 79
	FB_A0_CMD	GDDR5_45SE	GDDR5_CMD	FB_A0_RAS_L	77 79
	FB_A1_CMD	GDDR5_45SE	GDDR5_CMD	FB_A1_RAS_L	77 79
	FB_A0_CMD	GDDR5_45SE	GDDR5_CMD	FB_A0_CAS_L	77 79
	FB_A1_CMD	GDDR5_45SE	GDDR5_CMD	FB_A1_CAS_L	77 79
	FB_A0_CMD	GDDR5_45SE	GDDR5_CMD	FB_A0_WE_L	77 79
	FB_A1_CMD	GDDR5_45SE	GDDR5_CMD	FB_A1_WE_L	77 79
	FB_A0_CMD_R	GDDR5_45SE	GDDR5_CMD	FB_A0_CKE_L	77 79
	FB_A1_CMD_R	GDDR5_45SE	GDDR5_CMD	FB_A1_CKE_L	77 79
	FB_A0_CMD	GDDR5_45SE	GDDR5_CMD	FB_A0_CS_L	77 79
	FB_A1_CMD	GDDR5_45SE	GDDR5_CMD	FB_A1_CS_L	77 79
	FB_A0_EDC0	GDDR5_45SE	GDDR5_EDC	FB_A0_EDC<0>	6 77 79
I293	FB_A0_EDC1	GDDR5_45SE	GDDR5_EDC	FB_A0_EDC<1>	6 77 79
I294	FB_A0_EDC2	GDDR5_45SE	GDDR5_EDC	FB_A0_EDC<2>	6 77 79
I295	FB_A0_EDC3	GDDR5_45SE	GDDR5_EDC	FB_A0_EDC<3>	6 77 79
I296	FB_A1_EDC0	GDDR5_45SE	GDDR5_EDC	FB_A1_EDC<0>	6 77 79
I297	FB_A1_EDC1	GDDR5_45SE	GDDR5_EDC	FB_A1_EDC<1>	6 77 79
I298	FB_A1_EDC2	GDDR5_45SE	GDDR5_EDC	FB_A1_EDC<2>	6 77 79
	FB_A1_EDC3	GDDR5_45SE	GDDR5_EDC	FB_A1_EDC<3>	6 77 79
	FB_A0_DBI_L0	GDDR5_45SE	GDDR5_DATA	FB_A0_DBI_L<0>	6 77 79
I299	FB_A0_DBI_L1	GDDR5_45SE	GDDR5_DATA	FB_A0_DBI_L<1>	6 77 79
I300	FB_A0_DBI_L2	GDDR5_45SE	GDDR5_DATA	FB_A0_DBI_L<2>	6 77 79
I301	FB_A0_DBI_L3	GDDR5_45SE	GDDR5_DATA	FB_A0_DBI_L<3>	6 77 79
I302	FB_A1_DBI_L0	GDDR5_45SE	GDDR5_DATA	FB_A1_DBI_L<0>	6 77 79
I303	FB_A1_DBI_L1	GDDR5_45SE	GDDR5_DATA	FB_A1_DBI_L<1>	6 77 79
I304	FB_A1_DBI_L2	GDDR5_45SE	GDDR5_DATA	FB_A1_DBI_L<2>	6 77 79
	FB_A1_DBI_L3	GDDR5_45SE	GDDR5_DATA	FB_A1_DBI_L<3>	6 77 79
	FB_A0_WCLK0	GDDR5_80D	GDDR5_CMD	FB_A0_WCLK_P<0>	6 77 79
	FB_A0_WCLK0	GDDR5_80D	GDDR5_CMD	FB_A0_WCLK_N<0>	6 77 79
	FB_A0_WCLK1	GDDR5_80D	GDDR5_CMD	FB_A0_WCLK_P<1>	6 77 79
	FB_A0_WCLK1	GDDR5_80D	GDDR5_CMD	FB_A0_WCLK_N<1>	6 77 79
	FB_A1_WCLK0	GDDR5_80D	GDDR5_CMD	FB_A1_WCLK_P<0>	6 77 79
	FB_A1_WCLK0	GDDR5_80D	GDDR5_CMD	FB_A1_WCLK_N<0>	6 77 79
	FB_A1_WCLK1	GDDR5_80D	GDDR5_CMD	FB_A1_WCLK_P<1>	6 77 79
	FB_A1_WCLK1	GDDR5_80D	GDDR5_CMD	FB_A1_WCLK_N<1>	6 77 79
	FB_A0_DQ_BYTE0	GDDR5_45SE	GDDR5_DATA	FB_A0_DQ<7..0>	6 77 79
	FB_A0_DQ_BYTE1	GDDR5_45SE	GDDR5_DATA	FB_A0_DQ<15..8>	6 77 79
	FB_A0_DQ_BYTE2	GDDR5_45SE	GDDR5_DATA	FB_A0_DQ<23..16>	6 77 79
	FB_A0_DQ_BYTE3	GDDR5_45SE	GDDR5_DATA	FB_A0_DQ<31..24>	6 77 79
	FB_A1_DQ_BYTE0	GDDR5_45SE	GDDR5_DATA	FB_A1_DQ<7..0>	6 77 79
	FB_A1_DQ_BYTE1	GDDR5_45SE	GDDR5_DATA	FB_A1_DQ<15..8>	6 77 79
	FB_A1_DQ_BYTE2	GDDR5_45SE	GDDR5_DATA	FB_A1_DQ<23..16>	6 77 79
	FB_A1_DQ_BYTE3	GDDR5_45SE	GDDR5_DATA	FB_A1_DQ<31..24>	6 77 79
	FB_A0_CMD_R	GDDR5_45SE	GDDR5_CMD	FB_A0_RESET_L	77 79
I323	FB_A1_CMD_R	GDDR5_45SE	GDDR5_CMD	FB_A1_RESET_L	77 79

GDDR5 FB B Net Properties

	ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING		
	FB_B0_CLK	GDDR5_80D	GDDR5_CLK	FB_B0_CLK_P	77 80
	FB_B0_CLK	GDDR5_80D	GDDR5_CLK	FB_B0_CLK_N	77 80
	FB_B1_CLK	GDDR5_80D	GDDR5_CLK	FB_B1_CLK_P	77 80
	FB_B1_CLK	GDDR5_80D	GDDR5_CLK	FB_B1_CLK_N	77 80
	FB_B0_CMD	GDDR5_45SE	GDDR5_CMD	FB_B0_A<8..0>	6 77 80
	FB_B1_CMD	GDDR5_45SE	GDDR5_CMD	FB_B1_A<8..0>	6 77 80
	FB_B0_CMD	GDDR5_45SE	GDDR5_CMD	FB_B0_ABI_L	6 77 80
	FB_B1_CMD	GDDR5_45SE	GDDR5_CMD	FB_B1_ABI_L	6 77 80
	FB_B0_CMD	GDDR5_45SE	GDDR5_CMD	FB_B0_RAS_L	77 80
	FB_B1_CMD	GDDR5_45SE	GDDR5_CMD	FB_B1_RAS_L	77 80
	FB_B0_CMD	GDDR5_45SE	GDDR5_CMD	FB_B0_CAS_L	77 80
	FB_B1_CMD	GDDR5_45SE	GDDR5_CMD	FB_B1_CAS_L	77 80
	FB_B0_CMD	GDDR5_45SE	GDDR5_CMD	FB_B0_WE_L	77 80
	FB_B1_CMD	GDDR5_45SE	GDDR5_CMD	FB_B1_WE_L	77 80
	FB_B0_CMD_R	GDDR5_45SE	GDDR5_CMD	FB_B0_CKE_L	77 80
	FB_B1_CMD_R	GDDR5_45SE	GDDR5_CMD	FB_B1_CKE_L	77 80
	FB_B0_CMD	GDDR5_45SE	GDDR5_CMD	FB_B0_CS_L	77 80
	FB_B1_CMD	GDDR5_45SE	GDDR5_CMD	FB_B1_CS_L	77 80
	FB_B0_EDC0	GDDR5_45SE	GDDR5_EDC	FB_B0_EDC<0>	6 77 80
I305	FB_B0_EDC1	GDDR5_45SE	GDDR5_EDC	FB_B0_EDC<1>	6 77 80
I306	FB_B0_EDC2	GDDR5_45SE	GDDR5_EDC	FB_B0_EDC<2>	6 77 80
I307	FB_B0_EDC3	GDDR5_45SE	GDDR5_EDC	FB_B0_EDC<3>	6 77 80
I310	FB_B1_EDC0	GDDR5_45SE	GDDR5_EDC	FB_B1_EDC<0>	6 77 80
I309	FB_B1_EDC1	GDDR5_45SE	GDDR5_EDC	FB_B1_EDC<1>	6 77 80
I308	FB_B1_EDC2	GDDR5_45SE	GDDR5_EDC	FB_B1_EDC<2>	6 77 80
	FB_B1_EDC3	GDDR5_45SE	GDDR5_EDC	FB_B1_EDC<3>	6 77 80
	FB_B0_DBI_L0	GDDR5_45SE	GDDR5_DATA	FB_B0_DBI_L<0>	6 77 80
I311	FB_B0_DBI_L1	GDDR5_45SE	GDDR5_DATA	FB_B0_DBI_L<1>	6 77 80
I312	FB_B0_DBI_L2	GDDR5_45SE	GDDR5_DATA	FB_B0_DBI_L<2>	6 77 80
I313	FB_B0_DBI_L3	GDDR5_45SE	GDDR5_DATA	FB_B0_DBI_L<3>	6 77 80
I316	FB_B1_DBI_L0	GDDR5_45SE	GDDR5_DATA	FB_B1_DBI_L<0>	6 77 80
I315	FB_B1_DBI_L1	GDDR5_45SE	GDDR5_DATA	FB_B1_DBI_L<1>	6 77 80
I314	FB_B1_DBI_L2	GDDR5_45SE	GDDR5_DATA	FB_B1_DBI_L<2>	6 77 80
	FB_B1_DBI_L3	GDDR5_45SE	GDDR5_DATA	FB_B1_DBI_L<3>	6 77 80
	FB_B0_WCLK0	GDDR5_80D	GDDR5_CMD	FB_B0_WCLK_P<0>	6 77 80
	FB_B0_WCLK0	GDDR5_80D	GDDR5_CMD	FB_B0_WCLK_N<0>	6 77 80
	FB_B0_WCLK1	GDDR5_80D	GDDR5_CMD	FB_B0_WCLK_P<1>	6 77 80
	FB_B0_WCLK1	GDDR5_80D	GDDR5_CMD	FB_B0_WCLK_N<1>	6 77 80
	FB_B1_WCLK0	GDDR5_80D	GDDR5_CMD	FB_B1_WCLK_P<0>	6 77 80
	FB_B1_WCLK0	GDDR5_80D	GDDR5_CMD	FB_B1_WCLK_N<0>	6 77 80
	FB_B1_WCLK1	GDDR5_80D	GDDR5_CMD	FB_B1_WCLK_P<1>	6 77 80
	FB_B1_WCLK1	GDDR5_80D	GDDR5_CMD	FB_B1_WCLK_N<1>	6 77 80
	FB_B0_DQ_BYTE0	GDDR5_45SE	GDDR5_DATA	FB_B0_DQ<7..0>	6 77 80
	FB_B0_DQ_BYTE1	GDDR5_45SE	GDDR5_DATA	FB_B0_DQ<15..8>	6 77 80
	FB_B0_DQ_BYTE2	GDDR5_45SE	GDDR5_DATA	FB_B0_DQ<23..16>	6 77 80
	FB_B0_DQ_BYTE3	GDDR5_45SE	GDDR5_DATA	FB_B0_DQ<31..24>	6 77 80
	FB_B1_DQ_BYTE0	GDDR5_45SE	GDDR5_DATA	FB_B1_DQ<7..0>	6 77 80
	FB_B1_DQ_BYTE1	GDDR5_45SE	GDDR5_DATA	FB_B1_DQ<15..8>	6 77 80
	FB_B1_DQ_BYTE2	GDDR5_45SE	GDDR5_DATA	FB_B1_DQ<23..16>	6 77 80
	FB_B1_DQ_BYTE3	GDDR5_45SE	GDDR5_DATA	FB_B1_DQ<31..24>	6 77 80
I324	FB_B0_CMD_R	GDDR5_45SE	GDDR5_CMD	FB_B0_RESET_L	77 80
I325	FB_B1_CMD_R	GDDR5_45SE	GDDR5_CMD	FB_B1_RESET_L	77 80

MUXGFX Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING		
LVDS_A_CLK	LVDS_85D	LVDS	LVDS_A_CLK_P	86 89
LVDS_A_CLK	LVDS_85D	LVDS	LVDS_A_CLK_N	86 89
LVDS_A_DATA	LVDS_85D	LVDS	LVDS_A_DATA_P<2..0>	86 89
LVDS_A_DATA	LVDS_85D	LVDS	LVDS_A_DATA_N<2..0>	86 89
LVDS_B_CLK	LVDS_85D	LVDS	LVDS_B_CLK_P	86 89
LVDS_B_CLK	LVDS_85D	LVDS	LVDS_B_CLK_N	86 89
LVDS_B_DATA	LVDS_85D	LVDS	LVDS_B_DATA_P<2..0>	86 89
LVDS_B_DATA	LVDS_85D	LVDS	LVDS_B_DATA_N<2..0>	86 89
	LVDS_85D	LVDS	LVDS_CONN_A_CLK_F_P	6 85
	LVDS_85D	LVDS	LVDS_CONN_A_CLK_F_N	6 85
	LVDS_85D	LVDS	LVDS_CONN_B_CLK_F_P	6 85
	LVDS_85D	LVDS	LVDS_CONN_B_CLK_F_N	6 85
	LVDS_85D	LVDS	LVDS_CONN_A_CLK_P	85 86
	LVDS_85D	LVDS	LVDS_CONN_A_CLK_N	85 86
	LVDS_85D	LVDS	LVDS_CONN_A_DATA_P<2..0>	6 85 86
	LVDS_85D	LVDS	LVDS_CONN_A_DATA_N<2..0>	6 85 86
	LVDS_85D	LVDS	LVDS_CONN_B_CLK_P	85 86
	LVDS_85D	LVDS	LVDS_CONN_B_CLK_N	85 86
	LVDS_85D	LVDS	LVDS_CONN_B_DATA_P<2..0>	6 85 86
	LVDS_85D	LVDS	LVDS_CONN_B_DATA_N<2..0>	6 85 86

Kepler Net Properties

	ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING		
	GPU_CLK27M_IN	CLK_SLOW_45S	CLK_SLOW	GPU_OSC_27M_XTALIN	81 82
	GPU_CLK27M_OUT	CLK_SLOW_45S	CLK_SLOW	GPU_OSC_27M_XTALOUT	81 82
I326		CLK_SLOW_45S	CLK_SLOW	GPU_OSC_27M_XTALOUT_R	82
	LVDS_EG_A_CLK	LVDS_85D	LVDS	LVDS_EG_A_CLK_P	81 89
	LVDS_EG_A_CLK	LVDS_85D	LVDS	LVDS_EG_A_CLK_N	81 89
	LVDS_EG_A_DATA	LVDS_85D	LVDS	LVDS_EG_A_DATA_P<2..0>	81 89
	LVDS_EG_A_DATA	LVDS_85D	LVDS	LVDS_EG_A_DATA_N<2..0>	81 89
	LVDS_EG_A_DATA3	LVDS_85D	LVDS	LVDS_EG_A_DATA_P<3>	
	LVDS_EG_A_DATA3	LVDS_85D	LVDS	LVDS_EG_A_DATA_N<3>	
	LVDS_EG_B_DATA	LVDS_85D	LVDS	LVDS_EG_B_DATA_P<2..0>	81 89
	LVDS_EG_B_DATA	LVDS_85D	LVDS	LVDS_EG_B_DATA_N<2..0>	81 89
	LVDS_EG_B_DATA3	LVDS_85D	LVDS	LVDS_EG_B_DATA_P<3>	
	LVDS_EG_B_DATA3	LVDS_85D	LVDS	LVDS_EG_B_DATA_N<3>	
	DP_EXTA_ML	DP_85D	DISPLAYPORT	DP_EXTA_ML_C_P<3..0>	81 87
	DP_EXTA_ML	DP_85D	DISPLAYPORT	DP_EXTA_ML_C_N<3..0>	81 87
	DP_AUX_CH	DP_85D	DISPLAYPORT	DP_EXTA_AUXCH_C_P	86 87
		DP_85D	DISPLAYPORT	DP_EXTA_AUXCH_C_N	86 87
	DP_AUX_CH	DP_85D	DISPLAYPORT	DP_EG_AUX_CH_P	8 81 86
		DP_85D	DISPLAYPORT	DP_EG_AUX_CH_N	8 81 86
I322		DP_85D	DISPLAYPORT	DP_EXTA_ML_P<3..0>	87
I320		DP_85D	DISPLAYPORT	DP_EXTA_ML_N<3..0>	87
I321		DP_85D	DISPLAYPORT	DP_EXTA_AUXCH_P	87
I318		DP_85D	DISPLAYPORT	DP_EXTA_AUXCH_N	87

SYNC_MASTER=K92_MLB SYNC_DATE=08/09/2010

PAGE TITLE

GPU (Kepler) CONSTRAINTS

Apple Inc.

DRAWING NUMBER 051-9585 SIZE D

REVISION 3.0.0

BRANCH

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SENSE_1TO1_55S	*	=1:1_DIFFPAIR	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=1:1_DIFFPAIR	=1:1_DIFFPAIR
THERM_1TO1_55S	*	=1:1_DIFFPAIR	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=1:1_DIFFPAIR	=1:1_DIFFPAIR
DIFFPAIR	*	=1:1_DIFFPAIR	=1:1_DIFFPAIR	=1:1_DIFFPAIR	=1:1_DIFFPAIR	=1:1_DIFFPAIR	=1:1_DIFFPAIR
AUDIODIFF	*	=1:1_DIFFPAIR	0.1 MM	0.1 MM	10 MM	0.1 MM	0.1 MM

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SENSE	*	=2:1_SPACING	?
THERM	*	=2:1_SPACING	?
AUDIO	*	=2:1_SPACING	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
ENETCONN	*	25 MILS	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND	*	=STANDARD	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND_P2MM	*	0.20 MM	1000
PWR_P2MM	*	0.20 MM	1000

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CLK	GND	*	GND_P2MM
MEM_CMD	GND	*	GND_P2MM
MEM_CTRL	GND	*	GND_P2MM
MEM_DQS	GND	*	GND_P2MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CPU_COMP	GND	*	GND_P2MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
ENET_MDI	GND	*	GND_P2MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_PCIE	GND	*	GND_P2MM
PCIE	GND	*	GND_P2MM
SATA	GND	*	GND_P2MM
USB	GND	*	GND_P2MM
CLK_PCIE	SB_POWER	*	PWR_P2MM
SATA	SB_POWER	*	PWR_P2MM
USB	SB_POWER	*	PWR_P2MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
LVDS	GND	*	GND_P2MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_40S OVERRIDE	* OVERRIDE	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	100 MIL OVERRIDE	OVERRIDE	OVERRIDE
MEM_72D OVERRIDE	* OVERRIDE	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	100 MIL OVERRIDE	OVERRIDE	OVERRIDE
PCIE_85D OVERRIDE	* OVERRIDE	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	10 mm OVERRIDE	OVERRIDE	OVERRIDE
USB_85D OVERRIDE	TOP OVERRIDE	OVERRIDE	OVERRIDE	0.1 MM OVERRIDE	500 MIL OVERRIDE	OVERRIDE	OVERRIDE
CPU_27P4S OVERRIDE	BOTTOM OVERRIDE	OVERRIDE	OVERRIDE	0.23 MM OVERRIDE	100 MIL OVERRIDE	OVERRIDE	OVERRIDE

Graphics ,SATA Constraint Relaxations

Alternate diffpair width/gap through BGA fanout areas (95-ohm diff)

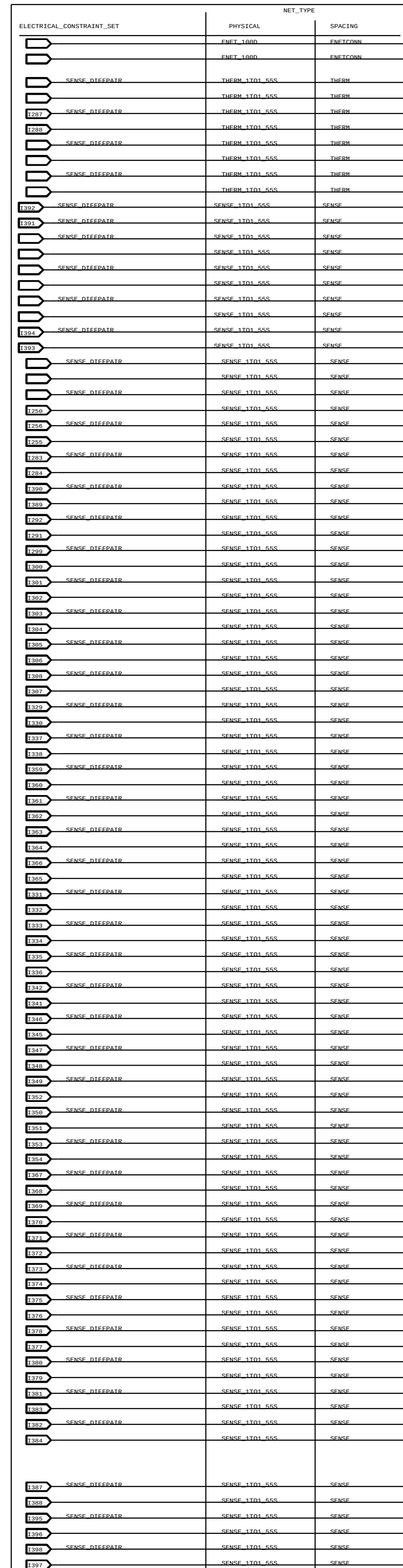
NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
LVDS_85D	BGA	LVDS_85D
DP_85D	BGA	100_DIFF_BGA
SATA_90D	BGA	100_DIFF_BGA
CLK_PCIE_90D	BGA	100_DIFF_BGA

Memory Constraint Relaxations

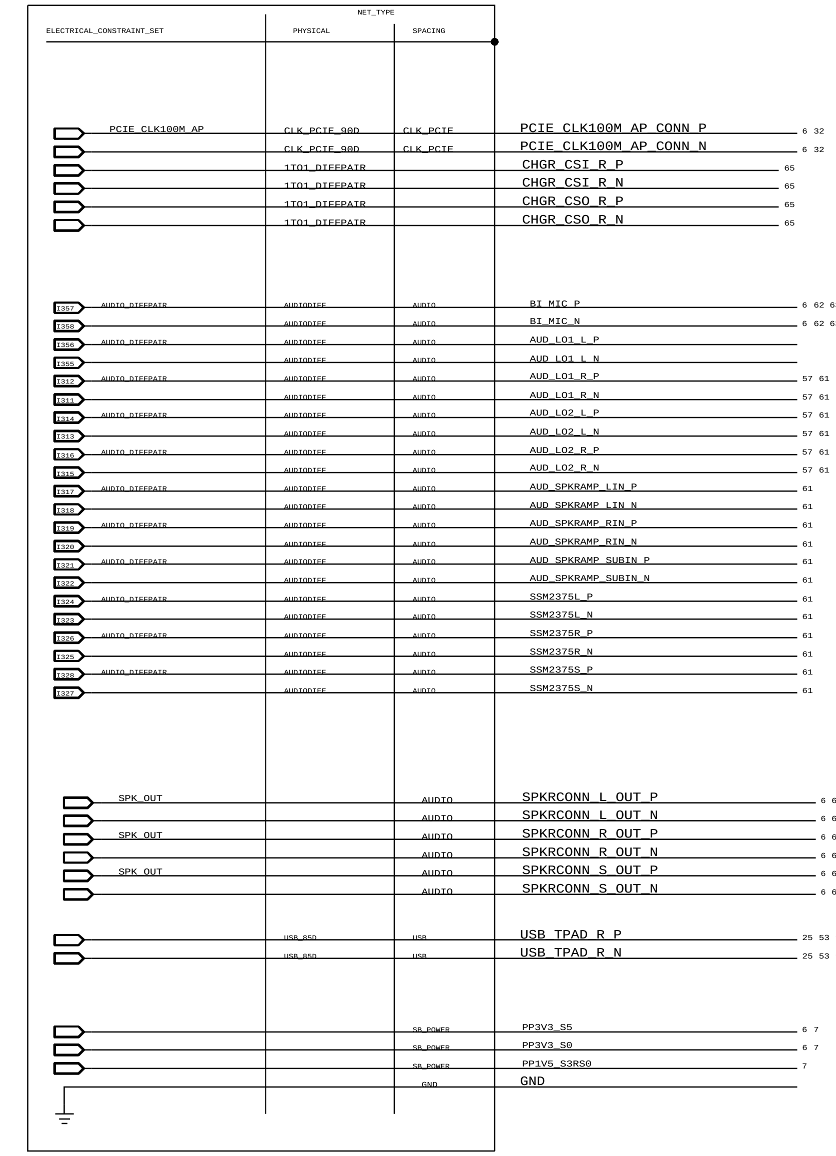
Allow 0.127 mm necks for >0.127 mm lines for ARD fanout.

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_72D	BOTTOM			0.127 MM	6.35 MM		
MEM_85D	TOP			0.1 MM	6.35 MM		

J31 Specific Net Properties



J31 Specific Net Properties



SYNC_MASTER=K18_MLB SYNC_DATE=04/27/2010

PAGE TITLE

Project Specific Constraints

Apple Inc.

DRAWING NUMBER 051-9585 SIZE D

REVISION 3.0.0

BRANCH

J31 Board-Specific Spacing & Physical Constraints

BOARD LAYERS	BOARD AREAS	BOARD UNITS (MIL or MM)	ALLEGRO VERSION
TOP,ISL2,ISL3,ISL4,ISL5,ISL6,ISL7,ISL8,ISL9,ISL10,ISL11,BOTTOM	NO_TYPE,BGA	MM	16.2

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DEFAULT	*	Y	=50_OHM_SE	=50_OHM_SE	10 MM	0 MM	0 MM
STANDARD	*	Y	=DEFAULT	=DEFAULT	10 MM	=DEFAULT	=DEFAULT

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
55_OHM_SE	TOP,BOTTOM	Y	0.090 MM	0.090 MM			
55_OHM_SE	*	Y	0.076 MM	0.076 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
50_OHM_SE	TOP,BOTTOM	Y	0.110 MM	0.095 MM			
50_OHM_SE	*	Y	0.090 MM	0.090 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
45_OHM_SE	TOP,BOTTOM	Y	0.13 MM	0.13 MM			
45_OHM_SE	*	Y	0.099 MM	0.099 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
40_OHM_SE	TOP,BOTTOM	Y	0.165 MM	0.095 MM			
40_OHM_SE	*	Y	0.135 MM	0.090 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
37_OHM_SE	TOP,BOTTOM	Y	0.185 MM	0.095 MM			
37_OHM_SE	*	Y	0.155 MM	0.090 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
27P4_OHM_SE	TOP,BOTTOM	Y	0.310 MM	0.095 MM			
27P4_OHM_SE	*	Y	0.250 MM	0.1 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
72_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
72_OHM_DIFF	ISL3,ISL4,ISL9,ISL10	Y	0.154 MM	0.154 MM		0.200 MM	0.200 MM
72_OHM_DIFF	ISL2,ISL11	Y	0.154 MM	0.154 MM		0.200 MM	0.200 MM
72_OHM_DIFF	TOP,BOTTOM	Y	0.175 MM	0.175 MM		0.200 MM	0.200 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
80_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
80_OHM_DIFF	ISL3,ISL4,ISL9,ISL10	Y	0.105 MM	0.091 MM		0.120 MM	0.080 MM
80_OHM_DIFF	ISL2,ISL11	Y	0.105 MM	0.091 MM		0.120 MM	0.080 MM
80_OHM_DIFF	TOP,BOTTOM	Y	0.135 MM	0.135 MM		0.160 MM	0.160 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
85_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
85_OHM_DIFF	ISL3,ISL4,ISL9,ISL10	Y	0.110 MM	0.090 MM		0.180 MM	0.180 MM
85_OHM_DIFF	ISL2,ISL11	Y	0.110 MM	0.090 MM		0.180 MM	0.180 MM
85_OHM_DIFF	TOP,BOTTOM	Y	0.125 MM	0.090 MM		0.190 MM	0.190 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
90_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
90_OHM_DIFF	ISL3,ISL4,ISL9,ISL10	Y	0.102 MM	0.090 MM		0.220 MM	0.220 MM
90_OHM_DIFF	ISL2,ISL11	Y	0.102 MM	0.090 MM		0.220 MM	0.220 MM
90_OHM_DIFF	TOP,BOTTOM	Y	0.115 MM	0.090 MM		0.230 MM	0.230 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
100_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
100_OHM_DIFF	ISL3,ISL4,ISL9,ISL10	Y	0.080 MM	0.080 MM		0.200 MM	0.200 MM
100_OHM_DIFF	ISL2,ISL11	Y	0.080 MM	0.080 MM		0.200 MM	0.200 MM
100_OHM_DIFF	TOP,BOTTOM	Y	0.089 MM	0.089 MM		0.220 MM	0.220 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
110_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
110_OHM_DIFF	ISL3,ISL4,ISL9,ISL10	Y	0.065 MM	0.065 MM		0.2 MM	0.2 MM
110_OHM_DIFF	ISL2,ISL11	Y	0.065 MM	0.065 MM		0.2 MM	0.2 MM
110_OHM_DIFF	TOP,BOTTOM	Y	0.075 MM	0.075 MM		0.330 MM	0.330 MM

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DEFAULT	*	0.1 MM	?
STANDARD	*	=DEFAULT	?
BGA_P1MM	*	=DEFAULT	?
BGA_P2MM	*	=DEFAULT	?
P072_SPACE	*	0.071 MM	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
*	*	BGA	P072_SPACE

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
2X_DIELECTRIC	*	0.140 MM	?
3X_DIELECTRIC	*	0.210 MM	?
4X_DIELECTRIC	*	0.280 MM	?
5X_DIELECTRIC	*	0.350 MM	?
7X_DIELECTRIC	*	0.490 MM	?
10X_DIELECTRIC	*	0.700 MM	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
1.5:1_SPACING	*	0.15 MM	?
2:1_SPACING	*	0.2 MM	?
2.5:1_SPACING	*	0.25 MM	?
3:1_SPACING	*	0.3 MM	?
4:1_SPACING	*	0.4 MM	?
5:1_SPACING	*	0.5 MM	?

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
1:1_DIFFPAIR	*	Y	=STANDARD	=STANDARD	=STANDARD	0.1 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
90_OHM_DIFF_ALT	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
90_OHM_DIFF_ALT	ISL3,ISL4,ISL9,ISL10	Y	0.099 MM	0.099 MM		0.280 MM	0.280 MM
90_OHM_DIFF_ALT	ISL2,ISL11	Y	0.099 MM	0.099 MM		0.280 MM	0.280 MM
90_OHM_DIFF_ALT	TOP,BOTTOM	Y	0.130 MM	0.130 MM		0.300 MM	0.300 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
100_DIFF_BGA	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF
100_DIFF_BGA	ISL3,ISL4	Y	0.075 MM	0.075 MM		0.125 MM	0.125 MM
100_DIFF_BGA	ISL9,ISL10	Y	0.075 MM	0.075 MM		0.125 MM	0.125 MM

NOTE: 100_DIFF_BGA is 100-ohms differential impedance on outer layers and 95-ohms on inner layers.

SYNC_MASTER=K18_MLB SYNC_DATE=04/27/2010

PAGE TITLE

PCB Rule Definitions

Apple Inc.

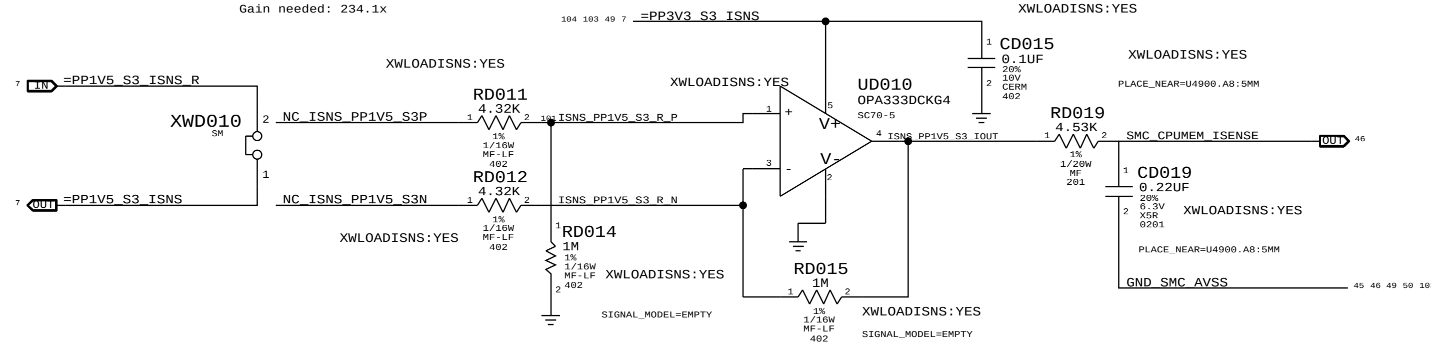
DRAWING NUMBER 051-9585 SIZE D

REVISION 3.0.0

BRANCH

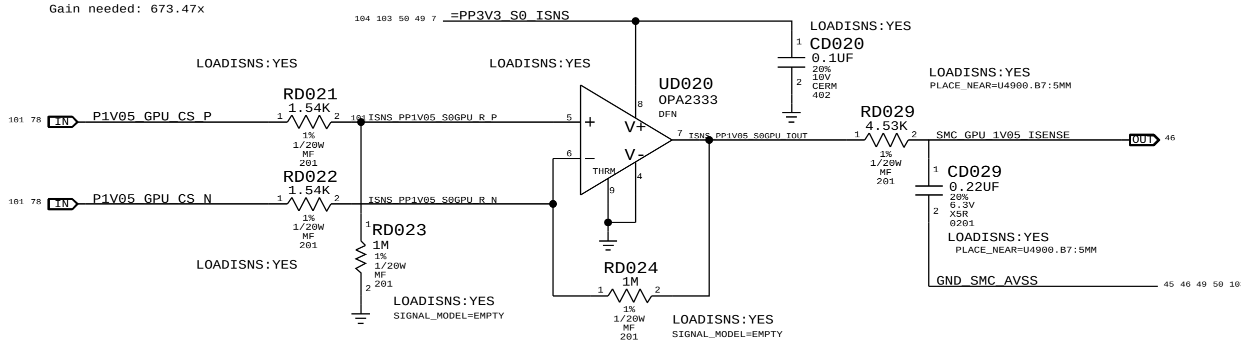
DDR 1.5V S3 (CPU & Memory) Current Sense (IM1C)

Gain: 231.4x, EDP: 14.1 A
Rsense: 0.001 (RD010)
V accross Rsense: 14.1 mV
Gain needed: 234.1x



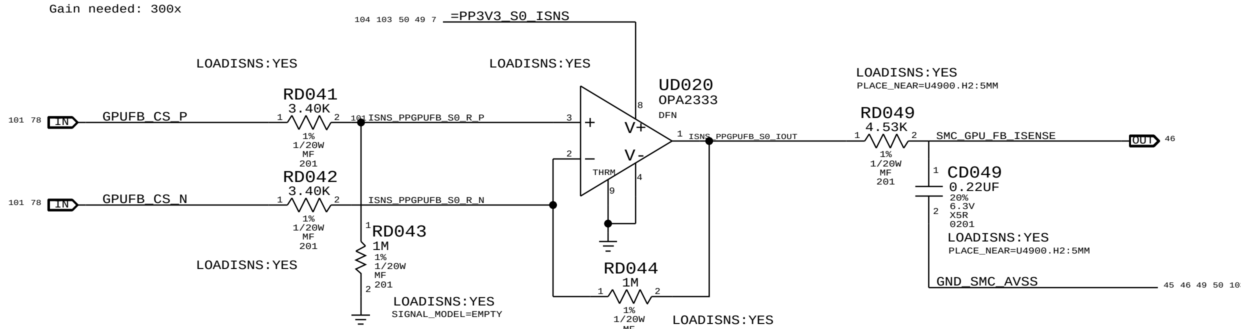
GPU 1.05V Current Sense (IG1C)

Gain: 649.35x, EDP: 4.9 A
Rsense: 0.001 (RD8310)
V accross Rsense: 4.9 mV
Gain needed: 673.47x

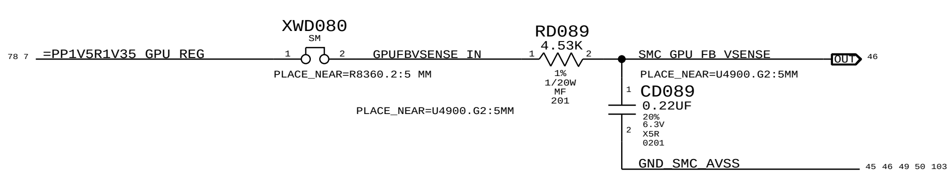


GPU FB (1.35V/1.5V) Current Sense (IG3C)

Gain: 294.12x, EDP: 11 A
Rsense: 0.001 (R8360)
V accross Rsense: 11 mV
Gain needed: 300x

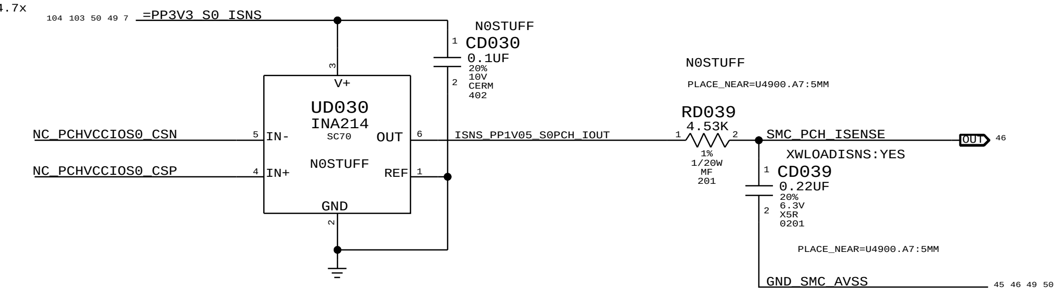


GPU FB (1.35V/1.5V) Voltage Sense (VG3C)



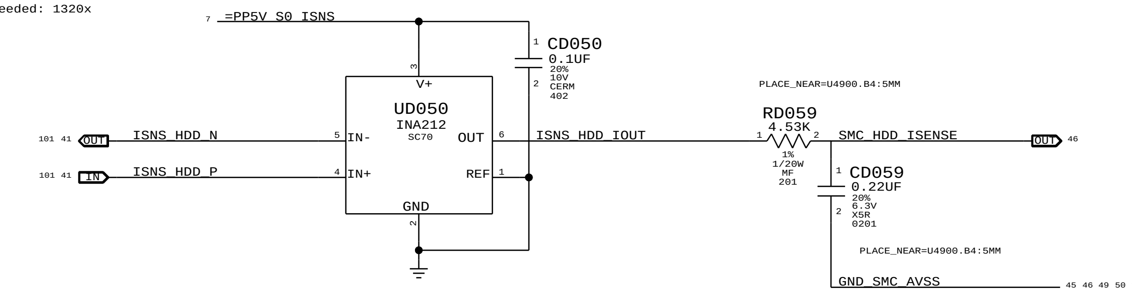
PCH Core (PCH VCCIO) Current Sense (ISBC)

Gain: 100x, EDP: 11.4 A
Rsense: 0.002 (R9840)
V accross Rsense: 22.8 mV
Gain needed: 144.7x



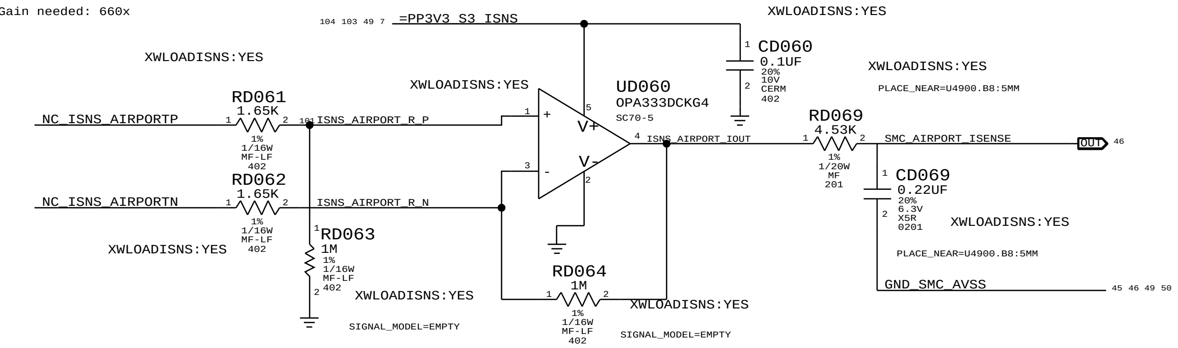
HDD Current Sense (IHDC)

Gain: 1000x, EDP: 2.5 A (12.5 W)
Rsense: 0.001 (R4599)
V accross Rsense: 2.5 mV
Gain needed: 1320x



Airport Current Sense (IAPC)

Gain: 606x, EDP: 1 A
Rsense: 0.005 (R3552)
V accross Rsense: 5 mV
Gain needed: 660x



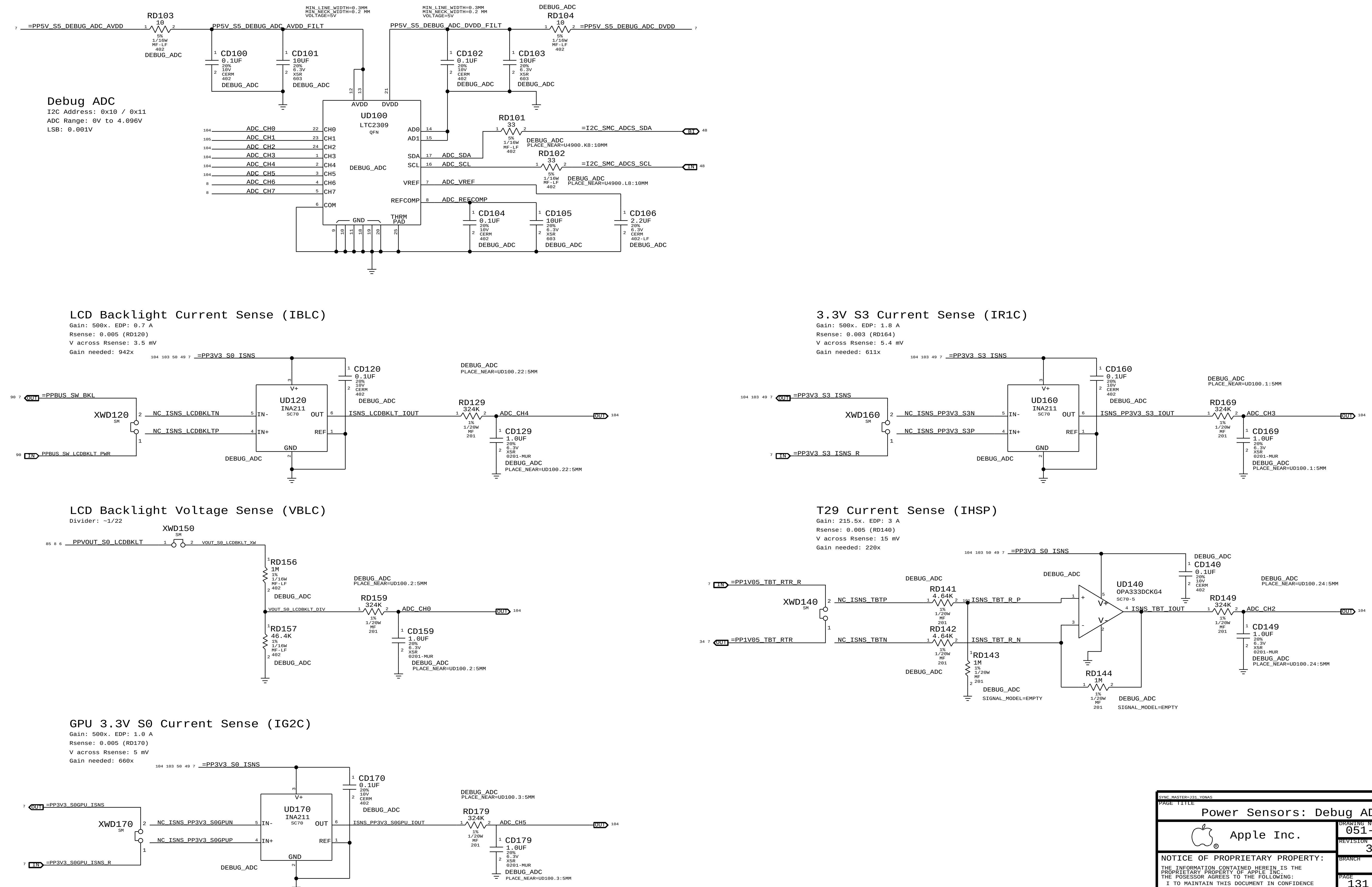
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
117S0008	2	RES,100K,201	CD029,CD049		LOADISNS:NO
117S0008	3	RES,100K,201	CD019,CD039,CD069		XWLOADISNS:NO

SYNC_MASTER=J31_YONAS SYNC_DATE=09/12/2011

PAGE TITLE
Power Sensors: SMC Extended

Apple Inc.

DRAWING NUMBER 051-9585 SIZE D
REVISION 3.0.0
BRANCH



SYNC_MASTER=J31_YONAS SYNC_DATE=09/12/2011

PAGE TITLE

Power Sensors: Debug ADC

Apple Inc.

DRAWING NUMBER 051-9585 SIZE D

REVISION 3.0.0

BRANCH

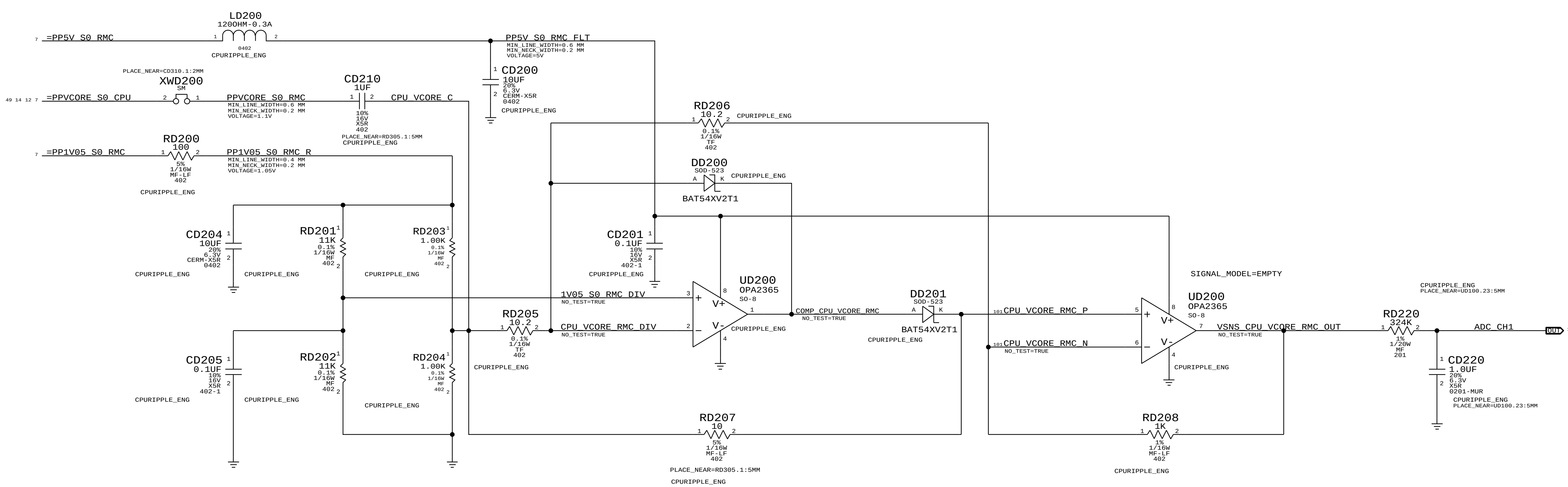
NOTICE OF PROPRIETARY PROPERTY:

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC.
THE POSESSOR AGREES TO THE FOLLOWING:
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE
II NOT TO REPRODUCE OR COPY IT
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART
IV ALL RIGHTS RESERVED

PAGE 131 OF 132

SHEET 104 OF 105

CPU Rippler Voltage Sense (VCRP)



SYNC_MASTER=J31_YONAS SYNC_DATE=08/24/2011

PAGE TITLE

Power Sensors: CPU Ripple

Apple Inc.

DRAWING NUMBER 051-9585 SIZE D

REVISION 3.0.0

BRANCH